Question # 1 of 10 (Start time: 05:18:14 PM, 17 August 2020) **Total Marks:** 1

In North West Corner method, the first step after choosing the appropriate cell in 1st row, we allocate ----------so that the capacity of first row or first column is exhausted.

Select the correct option

- ○ as least as possible
- ◉ as much as possible
- ○ non- negative quantity
- ○ any arbitrary quantity

RIZ MUGHAL

Question # 2 of 10 (Start time: 05:18:41 PM, 17 August 2020)

In two phase method process, first phase _____ the sum of artificial variables.

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Select the correct option

- (●) minimize
- ( ) maximize
- ( ) maximize or minimize depending on the situation
- ( ) non of these

Question # 3 of 10 (Start time: 05:19:05 PM, 17 August 2020) **Total Marks:**

If an LP problem contains large number of constraints and a smaller number of variables then which of the following will reduce the computational burden in finding its solution?

Select the correct option

- ○ M-method
- ○ Two phase method
- ○ Graphical method
- ◉ Duality principle

RIZ MUGHAL

Question # 4 of 10 (Start time: 05:19:30 PM. 17 August 2020)

If initial basic solution is ----------, while solving an LP problem then no further iteration can be performed.

RIZ MUGHAL

Select the correct option

- degenerate
- non-degenerate
- feasible
- infeasible

Question # 5 of 10 (Start time: 05:19:55 PM, 17 August 2020)

Which of the following difficulty may found while attempting an LP problem by M-method?

RIZ MUGHAL

Select the correct option

- ○ It often leads to infeasible solution
- ◉ Computational error due to large value of M
- ○ Degeneracy is inevitable
- ○ Artificial do not leave the basis

Question # 6 of 10 (Start time: 05:20:18 PM, 17 August 2020)

The cost coefficient of artificial variable in Objective function is ----------.

RIZ MUGHAL

Select the correct option

- 0
- M
- 1
- > than 1

Question # 9 of 10 (Start time: 05:21:22 PM, 17 August 2020)

Which of the following will be an example of degenerate basic feasible solution for an LP problem?

RIZ MUGHAL

Select the correct option

- ○ (2,3-1)
- ◉ (0,2,1)
- ○ (2,1,3)
- ○ (-1,2,0)

Question # 10 of 10 (Start time: 05:21:41 PM, 17 August 2020)

In a Transportation Problem, the objective function 'Z' gives ----------.

Select the correct option

- (●) Total Cost of transportation
- ( ) Total Time of transportation
- ( ) Total Profit of transportation
- ( ) Total inventory be supplied in transportation

RIZ MUGHAL

MTH601:Quiz

Question # 1 of 10 (Start time: 09:21:52 PM, 17 August 2020)

In a Transportation Problem, the objective function 'Z' gives ----------.

Select the correct option

- (●) Total Cost of transportation
- ( ) Total Time of transportation
- ( ) Total Profit of transportation
- ( ) Total inventory be supplied in transportation

RIZ MUGHAL

Question # 2 of 10 (Start time: 09:22:13 PM, 17 August 2020) Tota

If an LP problem contains large number of constraints and a smaller number of variables then which of the following will reduc computational burden in finding its solution?

RIZ MUGHAL

Select the correct option

- ○ M-method
- ○ Two phase method
- ○ Graphical method
- ◉ Duality principle

Question # 8 of 10 (Start time: 09:24:24 PM, 17 August 2020) Total Marks

For North West Corner method, in the first row and first column, resource and sink contain '5' and '7' units respectively; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

RIZ MUGHAL

Select the correct option

- ○ (2,2)
- ○ (5,2)
- ○ (1,2)
- ◉ (2,1)

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 Augu

Question # 9 of 10 (Start time: 09:24:41 PM, 17 August 2020) **Total M**

By Simplex method, to minimize 'Z = 2x+9y'of an LP problem, if 'z=A>0' for the initial iteration then for its next improved solution(0<A<100), which of the following would be the next entering variable?

RIZ MUGHAL

Select the correct option

- ○ x<0
- ○ y<0
- ◉ x>0
- ○ y>0

Question # 10 of 10 (Start time: 09:25:00 PM, 17 August 2020)

Dual of a Dual is ------

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Select the correct option

- (●) Primal
- ( ) Dual
- ( ) Primal-dual
- ( ) (Dual)^2

Question # 1 of 10 (Start time: 09:29:31 PM, 17 August 2020)

While solving an LP by two phase method, an objective function of 1st phase is always of---------.

Select the correct option

- maximization
- minimization
- dependent on original objective function
- none of these

RIZ MUGHAL

Question # 2 of 10 (Start time: 09:29:52 PM, 17 August 2020) **Total Mark:**

In two phase method if the minimum value of objective function in the first phase is greater than zero, then the solution of original problem _____.

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Select the correct option

- ○ is uncertain
- ○ exists
- ◉ does not exists
- ○ non of these

MTH601:Quiz

Quiz Start Time: 09:29 PM, 17 August 2020

Question # 8 of 10 (Start time: 09:31:56 PM, 17 August 2020) **Total Marks:**

In the Simplex method to solve an LP problem of maximization, if at the end of iteration, every entry of objective function row is positive then the given problem----------.

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Select the correct option

- needs further improvement
- has been maximized
- can not be optimized
- has no solution

Question # 9 of 10 (Start time: 09:32:16 PM, 17 August 2020)

In the initial iteration of Big M-method, the artificial variables appear in ---------.

Select the correct option

- (●) Basis
- ( ) Non-basic variables' set

RIZ MUGHAL

Question # 10 of 10 (Start time: 09:32:46 PM, 17 August 2020) **Total Mark**

In two phase method, for the phase-I, if the objective function in terms of artificial variables is not minimized then the given problem has----------.

Select the correct option

- ○ Feasible solution
- ○ Optimal solution
- ◉ Infeasible solution
- ○ Degeneracy

RIZ MUGHAL

Question # 5 of 10 (Start time: 09:30:57 PM, 17 August 2020)

Shortcoming of Big M method is that the value of M could be _____.

Select the correct option

- very small
- very large
- negative
- positive

RIZ MUGHAL

MTH601:Quiz — Quiz Start Time: 09:29 PM, 17 August 2020

Question # 3 of 10 (Start time: 09:30:14 PM, 17 August 2020) — **Total Marks:** 1

In the Simplex method to solve an LP problem of minimization, if at the end of iteration, there is a positive coefficient in the objective row then the given problem ----------.

Select the correct option

- ○ can not be optimized
- ○ has no solution
- ○ has been maximized
- ◉ needs further improvement

RIZ MUGHAL

MTH601:Quiz — Quiz Start Time: 09:29 PM, 17 August 2020

Question # 4 of 10 (Start time: 09:30:36 PM, 17 August 2020) — **Total Marks:** 1

For an unbalanced Transportation problem, if the total demand is MORE than total supply then which of the following is true in order to balance the problem?

Select the correct option

- ○ One constraint will have evacuate
- ○ One constraint will have to add
- ○ A dummy sink would have to include with demand equal to the surplus
- ◉ A dummy source would have to include with supply equal to shortage

RIZ MUGHAL

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August 2020

Question # 4 of 10 (Start time: 09:22:56 PM, 17 August 2020) **Total Marks:** 1

Which of the following order pair would minimize the objective function of the linear programming problem; $z = x + 5y$ subject to $x \geq 2, y \geq 0$?

Select the correct option

Reload Math Equations

- (2,3)
- (2,0)
- (0,3)
- (0,0)

RIZ MUGHAL

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August 202

Question # 5 of 10 (Start time: 09:23:19 PM, 17 August 2020) **Total Marks:**

Under which of the following condition to solve an LP by using two phase method, we can't proceed for 2nd phase?

Select the correct option

- Objective function of 1st phase has zero value.
- Objective function of 1st phase has positive value.

RIZ MUGHAL

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August

Question # 6 of 10 (Start time: 09:23:45 PM, 17 August 2020)

Total Ma

A balanced transportation model with '5'number of sources and '7' destinations has -------- number of constraint equations.

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Select the correct option

- 2
- 12
- 35
- 10

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August 2020

Question # 7 of 10 (Start time: 09:24:06 PM, 17 August 2020)

Total Marks: 1

By Simplex method, to maximize 'Z = 2x +9y'of an LP problem, if 'z=0' for the initial iteration then for its next improved solution, which of the following would be the next entering variable?

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Select the correct option

- x>0
- y>0
- x<0
- y<0

MTH601:Quiz

Quiz Start Time: 05:18 PM, 17 August 2020

Question # 7 of 10 (Start time: 05:20:38 PM, 17 August 2020) **Total Marks:** 1

While solving an LP problem by the Simplex method, in the standard table, the element at the intersection of key column and key row is called ---------- element.

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Select the correct option

- ○ Entering
- ○ Leaving
- ○ Slack
- ◉ Pivot

MTH601:Quiz

Quiz Start Time: 05:18 PM, 17 August 2020

Question # 8 of 10 (Start time: 05:20:56 PM, 17 August 2020) **Total Marks:** 1

For finding the maximum profit in an enterprise of selling two products such that 'freezing' the sale of one product and keep selling the other. This scenario is studied under ----------.

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Select the correct option

- ○ Un-boundedness
- ○ Duality
- ◉ Degeneracy
- ○ Artificial variable technique

MTH601:Quiz

Quiz Start Time: 09:29 PM, 17 August 202(

Question # 6 of 10 (Start time: 09:31:15 PM, 17 August 2020) **Total Marks:**

In Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column entries should be zero except-----.

Select the correct option

- ○ 1st row entry
- ◉ key row(pivot)entry
- ○ last row entry
- ○ row with maximum ratio on RHS column

RIZ MUGHAL

MTH601:Quiz

Quiz Start Time: 09:29 PM, 17 August 2020

Question # 7 of 10 (Start time: 09:31:35 PM, 17 August 2020) **Total Marks:** 1

For an unbalanced Transportation problem, if the total demand is LESS than total supply then which of the following is true in order to balance the problem?

Select the correct option

- ◉ A dummy sink would have to include with demand equal to the surplus
- ○ A dummy source would have to include with supply equal to sh[illegible]
- ○ One constraint will have evacuate
- ○ One constraint will have to add

RIZ MUGHAL

QUIZ NO 1 MTH 601 RESEARCH METHODS MCQ

Lectures 10-16

1. Which of the following can NOT be considered to include in shortage cost?

   a. Loss of goodwill of customers due to delay
   b. Cost of expediting
   c. Cost of lost sales (Lecture no 3 Page no 32 Handouts)
   d. Overtime expenses

2. In the Wilison's, an optimum order quantity for one period is found when item holding costs = _____

   a. set up costs
   b. All are equivalent
   c. purchase cost (Lecture no 3 Page no 34 Handouts)
   d. shortage cost

3. If a firm produces its own goods instead of purchasing the same from an outside source, then the cost of resetting the equipment for production may referred to _____

   a. Shortage Cost
   b. Set-up cost (From Google)
   c. Inventory holding cost
   d. Item cost

4. If we increase the inventory, then the shortage cost will _____

   a. remain constant
   b. increase
   c. decrease
   d. uncertain

5. If an item is manufactured then the direct or indirect material, labor, and overhead expenses are referred to _____

   a. Shortage Cost
   b. Set-up cost
   c. Item cost (Lecture no 3 Page no 31 Handouts)
   d. Inventory holding cost

6. In the Purchasing Model with Shortages the Cost Function 'C(S,Q)' can also be expressed as function of _____

   a. (t,D)
   b. (S,t)
   c. (Q,D)
   d. (S,D)

7. Objective of inventory control is to balance between the advantage of having inventories and the _____ of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

   a. profit
   b. demand
   c. cost
   d. transportation

8. The amount of the inventory will always taken to be _____

   a. Positive
   b. Negative

9. Which is the following reason is essential for an enterprise to have an inventory?

   a. It provides adequate service to the customers
   b. It reduces the possibility of duplication of orders
   c. It helps in smooth and efficient running of the business
   d. All above reasons are equivalent

10. Which of the following is true about the Manufacturing model without Shortage?

    a. Demand rate = Replacement rate
    b. Demand rate > Replacement rate
    c. Demand rate < Replacement rate (Page no 40 Handouts)
    d. None of the above

11. If we increase the inventory, then the shortage cost will decrease and the carrying cost will ___

- _remain constant
- **_increase (Lecture no 3 Page no 45 Handouts)**
- _also decrease
- _uncertain

12. If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the number of orders are _____
    The optimum number of orders per year is determined from N* = D/Q* pg 34
    a. 30
    b. 500
    c. 125
    d. **02 (5000/2500) = 2 Ans**

13. Holding cost is _____ proportional to the quantity to be kept in stock and the time to which the inventory is maintained.

a. directly b. Inversely

14. Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

CPM Times (Page no 09 Handouts)
PERT Times

Ref: If we employ more persons or skilled people or given overtime to the workers, the activity could be completed in a reduced duration known as crash duration. But this involves an increased cost in the form of additional resources. With CPM the amount needed to complete the various activities is assumed to be known with certainty. So, the direct costs for the activities increase and hence the cost of the project also increases.

15. Which of the following technique to solve the net work flow diagrams, is activity oriented?

Programme Evaluation and Review Technique
Critical Path Method (Page no 09 Handouts)

Ref:
PERT is an event-oriented technique. By 'event' we mean reaching a certain stage of completion of the project.

CPM, a network technique but it is concerned with obtaining the trade-off, between cost and completion date for large projects.

16. Which of the following cost for an item is independent of the quantity ordered or produced?

a. Set-up cost
b. Item cost
c. Shortage cost (Not sure)
d. Inventory holding cost

17. Under which of the following condition, the Economic Order Quantity (EOQ) of Manufacturing Model with allowed Shortage can be transformed into the same Model without Shortage?

    When Shortage Cost is zero (page 40)
    When Setup Cost is zero
    When Setup Cost is infinity
    When Shortage Cost is infinity

18. In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirement?

    2nd month demand can be included in 1st month
    2nd month demand will have to fulfill in 2nd month
    2nd month demand can be included in 3rd month
    2nd month demand can be included in any month of the year

19. By putting the replacement rate 'R' infinity' in the Economic Order Quantity(EOQ) of Manufacturing Model with allowed Shortage, then the result is ----------.

    EOQ of Purchasing Model without Shortage
    EOQ of Manufacturing Model without Shortage
    EOQ of Purchasing Model with allowed Shortage
    Invariant

20. The Economic Order Quantity in case of Manufacturing Model without Shortage can be transformed into Purchasing Model without Shortage by which of the following substitution?

    When Replacement rate 'R' approaches zero
    When Replacement rate 'R' approaches infinity
    When Demand rate approaches zero
    When Demand rate approaches infinity

21. If the total inventory in one cycle of 't' units of time is '(1/2)Qt', then which of the following is the average inventory at any time?

    Q/t
    Q/2
    (Q/2)t
    2Q

22. In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 120 respectively then which of the following is true about the 2nd month's requirement?

    2nd month demand can be included in 1st month
    2nd month demand will have to fulfill in 2nd month
    2nd month demand can be included in 3rd month
    2nd month demand can be included in any month of the year

23. Wilson's model (Purchasing model with no shortages) is of the ___ type.

 _Probabilistic

 _Deterministic (Page 32 Handout)

Ref: In the deterministic type of inventory control, the parameters like demand, ordering quantity cost, etc are already known or have been ascertained and there is no uncertainty.
In the Probabilistic /stochastic inventory control, the uncertain aspects are taken into account.

24. Which of the following times are not directly related to the activity cost as there is no certainty of time in such activities so their duration cannot be reduced, hence the cost cannot be expressed correctly?

    CPM Times
    PERT Times

25. The parameters like demand and ordering quantity cost are quite --------- in the Probabilistic inventory models.

    a. certain
    b. uncertain

Ref: Probabilistic techniques covers those problem in which all parameters are not known with certainty.

26. Which of the following is the major assumed **difference** between the models of Purchasing and Manufacturing without shortage?

- **_Replacement rate**
- _Shortage
- _Demand
- _Item, Setup and Holding Costs

Model-1 Purchasing model with no shortages(Wilson's model)

The following assumptions are made in deriving the formula for economic order quantity.
(1) Demand (D) is at a constant rate.
(2) Replacement of items is instantaneous (lead time is zero).
(3) The cost coefficients C_1, C_2, and C_3 are constant.
(4) There is no shortage cost or $C_4 = 0$.

Model-3(Manufacturing model with no shortages)

In this model the following assumptions are made:
(1) Demand is at a constant rate (D).
(2) All cost coefficients (C_1, C_2, C_3) are constants.
(3) There is no shortage cost, or $C_4 = 0$.
(4) The replacement rate is finite and greater than the demand rate. This is also called replenishment rate or manufacturing rate, denoted by R.

27. For ABC analysis, in the Random Sample of some items, the ranks to the items are assigned with respect to their ----------.

    lowest annual unit cost
    largest annual unit cost
    lowest annual usage value
    largest annual usage value (Page 47 Handout)

28. The purpose of maintaining the inventory is to balance the inter-related _____

    a. items
    b. costs
    c. demands
    d. times

Ref: Inventory Control: Primarily concerned with the inventory cost control

29. The graph of Wilson's Model (Purchasing Model with no shortage) is of the ---------- shape.

a. rectangular
b. isosceles triangular
c. saw tooth
d. equivalent triangular

Any process allocating objects at regular rate will result in a steady increase in heap consumption, followed by instantaneous drops

30. While estimating the time for a PERT activity, the estimator will give the three times estimates. When these three estimates are plotted on a graph, the probability distribution is closely associated with _____ Distribution curve.

    a. Beta (page 23 handout)
    b. Poisson
    c. Binomial
    d. Chi

31. Which is the following reason is essential for an enterprise to have an inventory?

    a. It helps in minimizing the loss due to deterioration, obsolescence, damage etc.
    b. It helps in maintaining economy by absorbing some of the fluctuations when the demand of an item fluctuates or is seasonal.
    c. Take advantage of price discounts by bulk purchasing.
    d. All above reasons are equivalent.

32. Except the economic order quantity, the value of Q^* will result in _____

    a. zero cost
    b. higher cost
    c. average cost
    d. lower cost

33. Excess of inventories are undesirable due to the following reasons:

    a. It may not lead to high volume of output instead it might hamper the production
    b. It includes the expenses of stores, equipment, personnel, insurance etc.
    c. It locks up the capital of the enterprise.
    d. All above reasons are equivalent.

34. If someone asks you how much time you require to reach your VU campus from your house, you may say that it may take 1 to $1^{1/2}$ hours. Then you meant which of the following variable?

    a. Discrete
    b. Random
    c. Complex
    d. Continuous [Time is a continuous variable]

35. When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to divert the resources to _____

    a. **critical activities (page 29 handout)**
    b. head or tail activities
    c. dummy activities
    d. non-critical activities

Ref: The slack is usually the basis on which priority is accorded to various job. Hence, the jobs, which are most critical, will be scheduled first.

36. The parameters like demand and ordering quantity cost are quite --------- in the Deterministic inventory models.

    a. **Certain (Page 32 Handout)**
    b. uncertain

Question # 2 of 10 (Start time: 10:52:47 AM, 24 December 2021) **Total Marks:** 1

In resource leveling when two or more jobs compete for same resources, first try to allocate an activity, which is of --------- duration and next to the activity which having next --------- duration.

Select the correct option

- maximum, minimum
- most likely, pessimistic
- optimistic, most likely
- minimum , maximum

Click to Save Answer & Move to Next Question

MTH601 - Operations Research (Quiz No 1) Quiz Start Time: 10:51 AM, 24 December 2021

Question # 7 of 10 (Start time: 10:59:43 AM, 24 December 2021) Total Marks: 1

If the demand of an item is occurred in the period of stock out and it is satisfied at the beginning of the next cycle then such type of problem will discuss under which of the following?

Select the correct option

- Purchasing Model with no shortages
- Purchasing Model with shortages

Correct

Click to Save Answer & Move to Next Question

MTH601 - Operations Research (Quiz No 1) Quiz Start Time: 10:51 AM, 24 December 2021

Question # 9 of 10 (Start time: 11:01:19 AM, 24 December 2021) Total Marks: 1

If we decrease the inventory then the shortage cost and the ----------per unit of time will increase.

Select the correct option

- carrying cost
- set up cost

Correct

Click to Save Answer & Move to Next Question

MTH601 - Operations Research (Quiz No 1)

Quiz Start Time: 10:51 AM, 24 December 2021

Question # 10 of 10 (Start time: 11:02:28 AM, 24 December 2021)

Total Marks: 1

In the Dynamic Order Quantity problem, the Demands of all succeeding months can be included in the demand of preceding one provided that _____.

Select the correct option

- Shortage and Item Costs are balanced
- Shortage and Setup Costs are balanced
- Setup and Holding Costs are balanced

Correct

- Item and Holding Costs are balanced

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 11:24:51 AM, 24 December 2021) **Total Marks:** 1

When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to delay the job having ----------,

Select the correct option

- largest free float
- dummy branch
- largest total float

Correct

- largest independent float

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 11:42:12 AM, 24 December 2021)

Which of the following technique to solve the network flow diagrams, is event oriented?

Select the correct option

- Programme Evaluation and Review Technique — Correct
- Critical Path Method

Click to Save Answer & Move to

Question # 2 of 10 (Start time: 11:43:02 AM, 24 December 2021)

For ABC analysis, In the Random Sample of some items, the ranks to the items are assigned with respect to their ------

Select the correct option

○	lowest annual unit cost
○	largest annual usage value
○	largest annual unit cost
○	lowest annual usage value

Correct

MTH601 - Operations Research (Quiz No 1)

Quiz Start Time: 11:42 AM, 24 Decer

Question # 7 of 10 (Start time: 11:49:08 AM, 24 December 2021)

Tot

Which of the following category of items in ABC analysis has the maximum Holding Cost?

Select the correct option

- A category
- C category
- All categories" items have equal Holding Costs
- B category

Correct

Click to Save Answer & Move to Next Qu

Question # 9 of 10 (Start time: 11:51:02 AM, 24 December 2021)

In the graph of the Purchasing Model with shortages, the area under the horizontal axis represents ----------.

Select the correct option

- carrying cost of item
- shortage cost of item

Correct

- set up cost of item
- purchase cost of item

Click to Save Answer & Move to Ne

Question # 26 of 40 (Start time: 11:54:19 AM, 24 December 2021)

In which of the following model, the Carrying Cost per unit of time is reduced in the ratio (1-D/R):1 ?

Select the correct option

- Purchasing Model with allowed Shortage
- Manufacturing Model without Shortage — Correct
- Purchasing Model without Shortage
- Manufacturing Model with allowed Shortage

Type here to search

estion # 5 of 10 (Start time: 11:36:08 AM, 24 December 2021)

n the context of Wilson's model, at time t = 0, the quantity 'Q' of the product is--------- in the inventory.

elect the correct option

- Average
- Minimum
- Maximum
- Zero

Correct

Question # 10 of 10 (Start time: 11:09:37 AM, 24 December 2021)

Which of the following quantity will vary in case of Dynamic Order Quantity Model?

Select the correct option

- Demand
- Item Cost
- Carrying Cost
- Setup Cost

Correct

Question # 6 of 10 (Start time: 11:58:08 AM, 24 December 2021)

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is ---------proportional to the quantity ordered.

Select the correct option

- directly
- inversely

Correct

Type here to search

Which of the following category of items in ABC analysis needs special attention by the management?

Select the correct option

- C category
- B category
- A category
- All categories need equal and fair attention

Correct

Type here to search

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=fa4401da-2477-4ba7-bd7e-72c09078dc70

Question # 7 of 10 (Start time: 11:59:25 AM, 24 December 2021)

If the total inventory in one cycle of 't' units of time is '(1/2)Qt', then which of the following is the average ineventory at any time?

Select the correct option

○	(Q/2)t
○	Q/t
○	Q/2
○	2Q

Correct

Type here to search

Quiz

vulms.vu.edu.pk/Quiz/QuizQuestion.aspx?ver=91e0674d-af60-494a-8261-3849e613ffa3

MTH601 - Operations Research (Quiz No 1) **Quiz Start Time: 11:5**

Question # 8 of 10 (Start time: 12:00:23 PM, 24 December 2021)

In resource leveling when two or more jobs compete for same resources, first try to allocate an activity, which is of --------- duration and next to the activity whi
- duration.

Select the correct option

- optimistic, most likely
- maximum, minimum
- minimum , maximum
- most likely, pessimistic

Correct

Type here to search

19°C

Question # 7 of 10 (Start time: 11:07:45 AM, 24 December 2021)

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is ---------proportional to the quantity ordered.

Select the correct option

- directly
- inversely

Correct

Question # 2 of 10 (Start time: 11:33:28 AM, 24 December 2021)

Which of the following category of items in ABC analysis has the maximum Holding Cost?

Select the correct option

- B category
- A category
- C category
- All categories' items have equal Holding Costs

Correct

Question # 10 of 10 (Start time: 11:40:33 AM, 24 December 2021)

In Wilson's model, Total cost = ----------

Select the correct option

- Item cost + holding cost + shortage cost + set up cost
- Holding cost + shortage cost + set up cost
- Item cost + holding cost + shortage cost
- Item cost + set up cost + holding cost

بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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Quiz No. 02 100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal & Sadiq Ahmed

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https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists Sadiq Ahmed

Sonu Ilyas Mughal

بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz

Question # 1 of 10 (Start time: 06:13:55 PM, 08 February 2021)

An optimal solution of a Linear Programming problem is necessarily a Basic feasible solution.

Select the correct option

- ○ True
- ● False

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz

Quiz Start Time: 06:13 PM, 08

Question # 2 of 10 (Start time: 06:14:22 PM, 08 February 2021)

While solving a Linear Programming problem by Simplex Method, an inequality constraint is transformed into strict equality by the inclusion of ---------- variable.

Select the correct option

- slack
- decision
- artificial
- basic

Click to Save Answer & Move to Nex

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz

Question # 3 of 10 (Start time: 06:14:47 PM, 08 February 2021)

If the primal has an unbounded solution, then the dual has------

Select the correct option

- optimal solution
- infeasible solution
- bounded solution
- no solution

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz Start Time: 06:13 PM, 08 February 20

Question # 4 of 10 (Start time: 06:15:11 PM, 08 February 2021) Total Mark

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, then in a given iteration, how many Basic feasible VARIABLES are possible to exists?

Select the correct option

- ○ 10
- ○ 6
- ◉ 4
- ○ 2

Click to Save Answer & Move to Next Question

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz Start Time: 06:13 PM, 08 Febr

Question # 5 of 10 (Start time: 06:15:29 PM, 08 February 2021) Tota

When we have $\geq$ type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.

Select the correct option

Reload Math Eq

- slack, Big M
- artificial, Two phase
- artificial, Big M
- artificial and slack, Big M or Two phase

Click to Save Answer & Move to Next Qu

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0306

بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz **Quiz Start Time: 06:13 PM, 08 February 2021**

Question # 6 of 10 (Start time: 06:15:52 PM, 08 February 2021) **Total Marks:** 1

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. If the initial basic feasible solution is say;(2,4,1,2), then its corresponding non-basic solution is---------.

Select the correct option

- (0,0)
- (-4,-2)
- (1,2)
- (2,2)

Click to Save Answer & Move to Next Question

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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Question # 7 of 10 (Start time: 06:16:10 PM, 08 February 2021)

Dual of a Dual is ------

Select the correct option

- (●) Primal
- ( ) Dual
- ( ) Primal-dual
- ( ) (Dual)^2

Click t

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz St

Question # 8 of 10 (Start time: 06:16:28 PM, 08 February 2021)

The dual of a dual problem yields the original primal.

Select the correct option

- (●) True
- ( ) False

Click to Save A

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz Start

Question # 9 of 10 (Start time: 06:16:48 PM, 08 February 2021)

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____ .

Select the correct option

- must have a solution
- does not have a solution
- may have a solution
- non of these

Click to Save Ans

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Question # 10 of 10 (Start time: 06:17:12 PM, 08 February 2021)

Once an LP problem is converted to a table, then calculation steps would be same for _____ method(s).

Select the correct option

- simplex
- Big M
- Two phase
- all of above

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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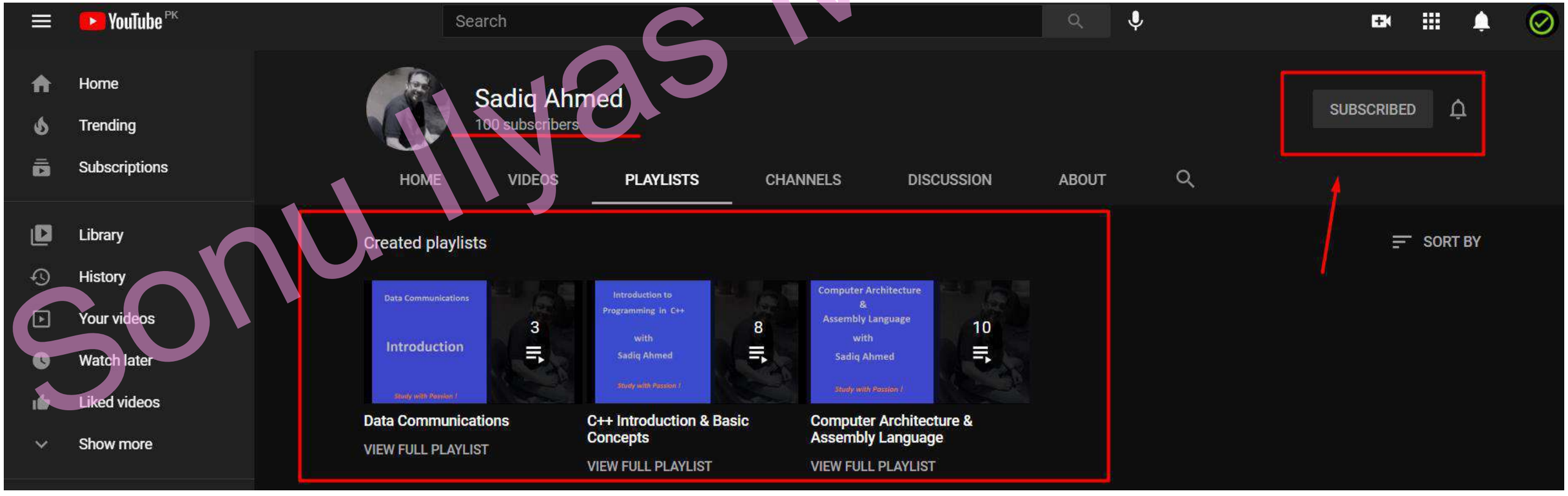
Subscribed MY YOUTUBE channel by Tutor Sadiq Ahmed



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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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Virtual Academy For all Subjects and Projects

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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Quiz No. 03

100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal & Sadiq Ahmed

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz

Question # 1 of 10 (Start time: 01:15:24 AM, 12 February 2021)

The column, which is introduced in the matrix to balance an unbalanced Transportation problem, is known as-------

Select the correct option

- key column
- idle column
- slack column
- dummy Column

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

SONU ILYAS MUGHAL

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MTH601:Quiz Quiz Start Time: 01:15 AM, 12 February 202

Question # 2 of 10 (Start time: 01:15:53 AM, 12 February 2021) **Total Marks:**

For North West Corner method, in the first row and first column, each source and sink contain '5' units ; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

Select the correct option

- (●) Arbitrarily either towards (1,2) or (2,1)
- ( ) Towards (1,2) if 1st row contains more occupied cells than 2nd row
- ( ) Towards (2,1) if 2nd row contains more occupied cells than 1st row
- ( ) Neither towards (1,2) nor (2,1) but move diagonally towards (2,2)

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz

Question # 3 of 10 (Start time: 01:16:14 AM, 12 February 2021)

Which of the following reason to prefer Vogel's approximation method over other two methods?

Select the correct option

- ○ Vogel's method gives the shortest transportation rout
- ○ Vogel's method gives optimal time plan of transportation
- ◉ Vogel's method gives the feasible solution near to the optimal
- ○ Vogel's method allocate minimum amount to the variable with least unit cost

Click

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz

Quiz Start Time: 0

Question # 4 of 10 (Start time: 01:16:33 AM, 12 February 2021)

For a Transportation Problem, if it's initial feasible solution is evaluated by Least Cost Method, then quality of this initial solution is better than ---------.

Select the correct option

- (●) North West Corner Method
- ( ) Vogal's approximation Method

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz Quiz Start Time: 01:15 AM

Question # 5 of 10 (Start time: 01:16:54 AM, 12 February 2021)

Which of the following will be the penalty cost of a row in Vogel's approximation method, if the smallest element is '3' and other elements are 3, 4, 5 ?

Select the correct option

- (●) 0
- ( ) 1
- ( ) 2
- ( ) 3

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MTH601:Quiz

Question # 6 of 10 (Start time: 01:17:16 AM, 12 February 2021)

In a Transportation Problem, the objective function 'Z' gives ----------.

Select the correct option

- (●) Total Cost of transportation
- ( ) Total Time of transportation
- ( ) Total Profit of transportation
- ( ) Total inventory be supplied in transportation

Click to S

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz **Quiz Start Time: 01:15 AM, 12 Feb**

Question # 7 of 10 (Start time: 01:17:36 AM, 12 February 2021) **Tot**

The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if numbers of positive allocations are

Select the correct option

- ○ m+n
- ◉ m+n-1
- ○ m-n+1
- ○ m+n+1

Click to Save Answer & Move to Next Q

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz **Quiz Start Time: 01:15 AM,**

Question # 8 of 10 (Start time: 01:18:05 AM, 12 February 2021)

For an unbalanced Transportation problem, if the total demand is LESS than total supply then which of the following is true in order to balance the problem?

Select the correct option

- (●) A dummy sink would have to include with demand equal to the surplus
- ( ) A dummy source would have to include with supply equal to shortage
- ( ) One constraint will have evacuate
- ( ) One constraint will have to add

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601:Quiz

Question # 9 of 10 (Start time: 01:18:31 AM, 12 February 2021)

The solution of an Assignment problem can be arrived by using which of the following method?

Select the correct option

- [x] Hungarian's method
- [ ] North West corner method
- [ ] Vogel's approximation method
- [ ] Least Cost method

Clic

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz Quiz Start Ti

Question # 10 of 10 (Start time: 01:18:51 AM, 12 February 2021)

For an assignment problem, if the numbers of agents and tasks are _____, then the problem is called _____ assignment.

Select the correct option

- (●) equal, balanced
- ( ) unequal, balanced
- ( ) equal, unbalanced
- ( ) none of these

Click to Save Answe

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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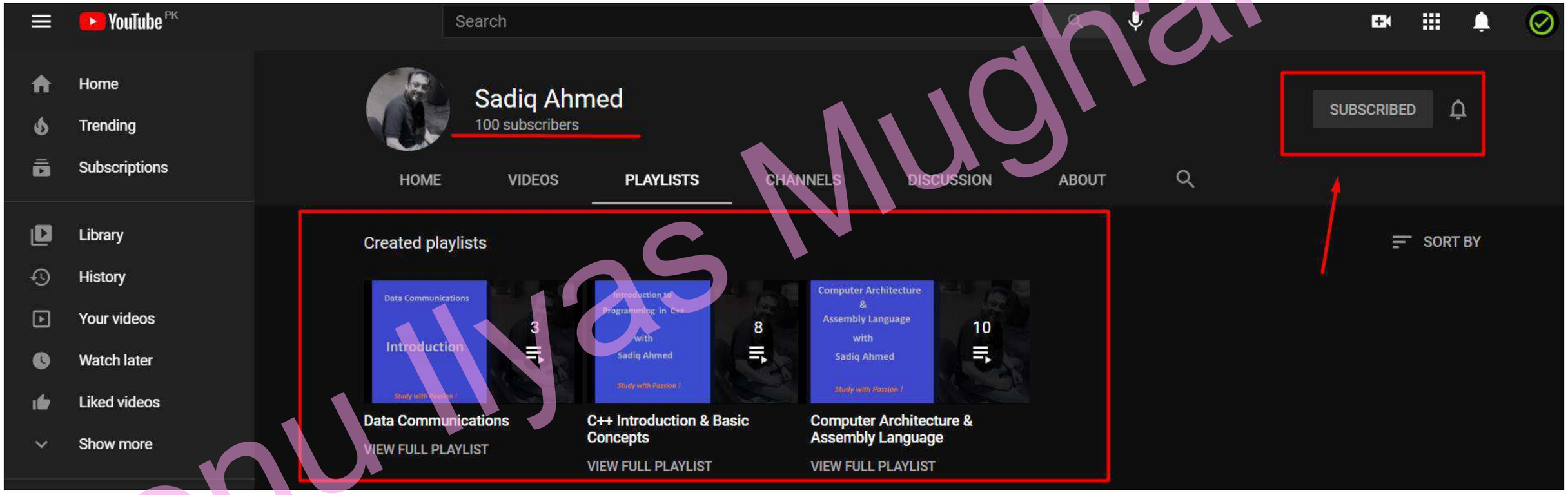

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists

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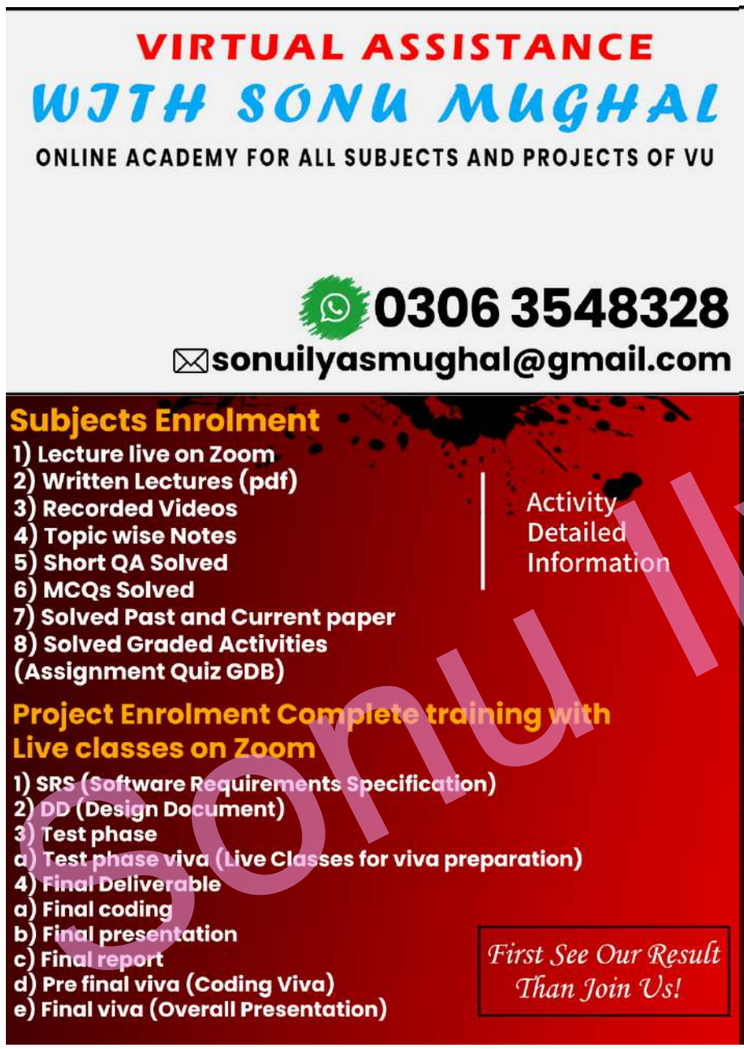
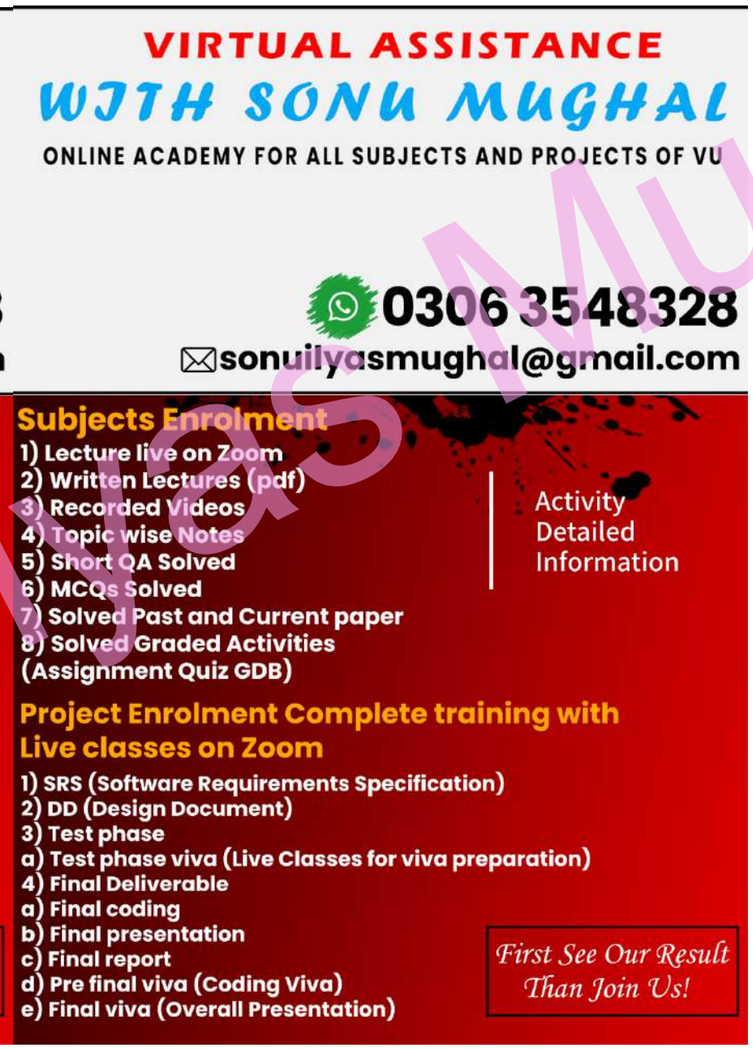
قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ





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Question # 12 of 40 (Start time: 12:36:56 PM, 11 July 2020)

In a network flow diagram, for an activity (i,j) of duration of three days, if its earliest start time is of two days then which of the

Select the correct option

○	Six days
○	One day
○	One and half day
●	Five days

Question # 18 of 40 (Start time: 12:39:38 PM, 11 July 2020)

Under which of the following condition the economic order quantity in case of Purchasing Model with Shortage can be transfo

Select the correct option

○	When shortage Cost tends to zero
■	When Shortag Cost tends to infinity
○	When Carrying Cost tends to zero
○	When Carrying Cost tends to infinity

Question # 17 of 40 (Start time: 12:39:06 PM, 11 July 2020)

While solving a Linear programming problem, we find ----------number of basic feasible solution.

Select the correct option

○	even
○	odd
○	infinite
■	only finite

Question # 20 of 40 (Start time: 12:40:26 PM, 11 July 2020)

In PERT, the possible variation in activity times is measured from Standard Deviation which is ----------of the difference betw

Select the correct option

- [x] one sixth
- [ ] one fourth
- [ ] one third
- [ ] one fifth

Question # 3 of 40 (Start time: 12:32:55 PM, 11 July 2020)

Which of the following property assumed that the coefficients of decision variables in a Linear Programming model are known

Select the correct option

●	Deterministic
○	Additivity
○	Proportionality
○	Divisibility

Question # 1 of 40 (Start time: 12:31:44 PM, 11 July 2020)

The parameters like demand and ordering quantity cost are quite --------- in the Probabilistic inventory models.

Select the correct option

○	certain
●	uncertain

Question # 19 of 40 (Start time: 12:39:58 PM, 11 July 2020)

In the PERT, the expected time is classified as a---------- variable.

Select the correct option

○	complex valued
○	fix valued (constant)
●	random
○	continuous

Question # 16 of 40 (Start time: 12:38:37 PM, 11 July 2020)

For any activity (i,j), if , a) Earliest start time of 'i' = Latest finish time of 'i', b) Earliest start time of 'j' = Latest finish time of 'j', time to complete the job, then the activity (i,j) is said to be -------.

Select the correct option

○	dummy
●	critical
○	non-Critical
○	non of the above

Question # 14 of 40 (Start time: 12:37:50 PM, 11 July 2020)

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously -- is completed.

Select the correct option

●	(m,n)
○	(l,n)
○	(m,l)
○	(l,m)

Question # 8 of 40 (Start time: 12:35:06 PM, 11 July 2020)

Which of the following correspond to the practical limitations of available resources while modeling a real life situation into a li

Select the correct option

○	Objective function
●	Constraints
○	Decision variables
○	Linear function

Question # 15 of 40 (Start time: 12:38:16 PM, 11 July 2020)

Duration of dummy activity is _____.

Select the correct option

○	latest start time
●	zero
○	early start time
○	1

Question # 4 of 40 (Start time: 12:33:25 PM, 11 July 2020)

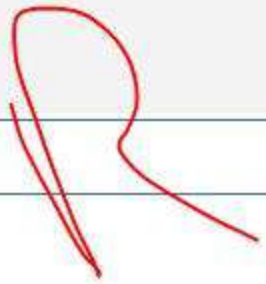
In the Wilson's Model, an optimum order quantity for one period is found when item holding cost = ----------.

Select the correct option

○	shortage cost
■	purchase cost
○	set up cost
○	All are equivalent

Question # 5 of 40 (Start time: 12:33:49 PM, 11 July 2020)

Excess of inventories are undesirable due to the following reasons;



Select the correct option

○	It locks up the capital of the enterprise.
○	It includes the expenses of stores, equipment, personnel, insurance etc..
○	It may not lead to high volume of output instead it might hamper the production.
●	All above reasons are equivalent

Question # 9 of 40 (Start time: 12:35:34 PM, 11 July 2020)

Which of the following would be the objective of the daily loss of heat in a heating system?

Select the correct option

○	Maximization
●	Minimization
○	Inflection
○	Average

Question # 13 of 40 (Start time: 12:37:21 PM, 11 July 2020)

In a development project, if an activity (i,j) of duration three days finish earlier on the tenth day then which of the following will

Select the correct option

●	Seventh day
○	Third day
○	First day
○	Fifth day

Question # 6 of 40 (Start time: 12:34:16 PM, 11 July 2020)

For a development project, if its Standard Normal variable = 1.38, expected and the scheduled durations of the project are 10

Select the correct option

○	-7.246
○	-52.50
●	52.50
○	7.246

Question # 10 of 40 (Start time: 12:36:03 PM, 11 July 2020)

Which of the following would be the objective of the revenue in a production of cell phones?

Select the correct option

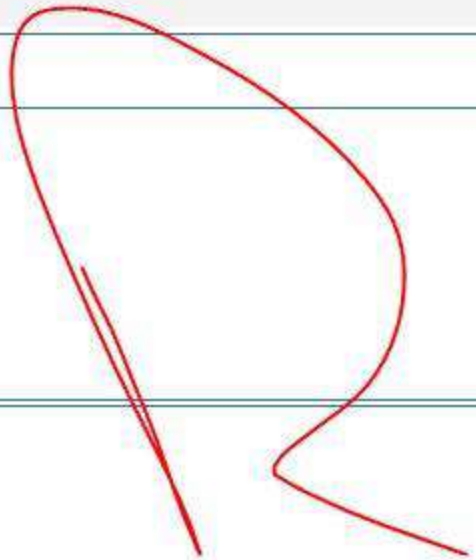
○	Inflection
○	Average
■	Maximization
○	Minimization

Question # 11 of 40 (Start time: 12:36:27 PM, 11 July 2020)

The amount of the inventory will always taken to be ----------

Select the correct option

○	Negative
●	Positive



Question # 7 of 40 (Start time: 12:34:42 PM, 11 July 2020)

Which of the following phases are performed to find the critical path in CPM?

Select the correct option

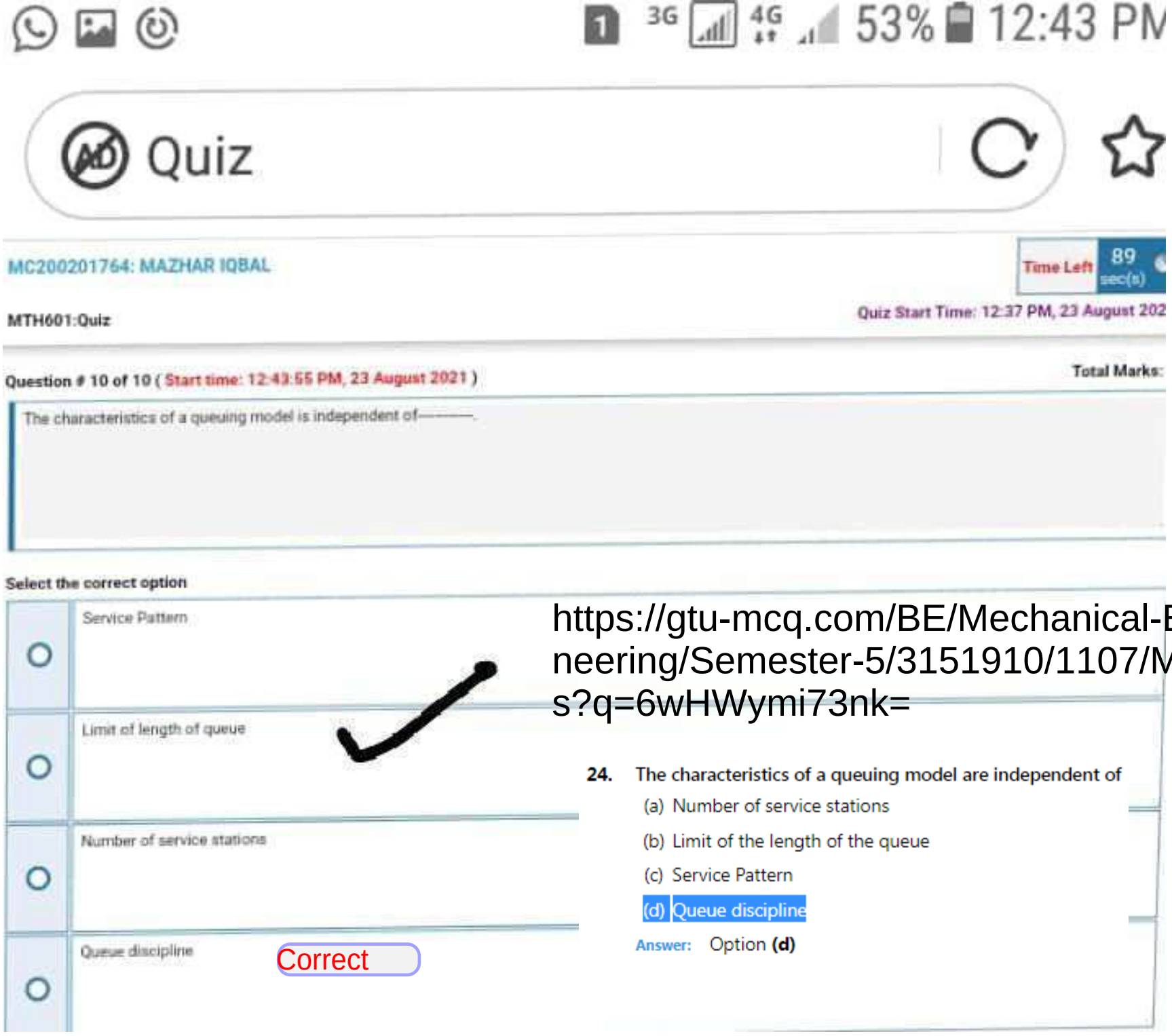
●	Forward and Backward(Early and Late finish program)
○	Program Evaluation and Review
○	Resource Scheduling and Allocation
○	Precedence and succession

Question # 2 of 40 (Start time: 12:32:18 PM, 11 July 2020)

Convex set is the collection of all ----------.

Select the correct option

○	optimal solutions
■	feasible solutions
○	non-feasible solutions
○	negative solutions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Quiz
MC200201764: MAZHAR IQBAL
Time Left 89 sec(s)
MTH601:Quiz
Quiz Start Time: 12:37 PM, 23 August 202
Question # 10 of 10 (Start time: 12:43:55 PM, 23 August 2021)
Total Marks:
The characteristics of a queuing model is independent of———.
Select the correct option
Service Pattern
Limit of length of queue
Number of service stations
Queue discipline
Correct
https://gtu-mcq.com/BE/Mechanical-Engineering/Semester-5/3151910/1107/MCQs?q=6wHWymi73nk=
24. The characteristics of a queuing model are independent of
(a) Number of service stations
(b) Limit of the length of the queue
(c) Service Pattern
(d) Queue discipline
Answer: Option (d)
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Quiz

MC200201764: MAZHAR IQBAL

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 12:37 PM, 23 August 2021

Question # 9 of 10 (Start time: 12:43:01 PM, 23 August 2021)

Total Marks: 1

For an assignment problem, if the numbers of agents and tasks are _____, then the problem is called _____ assignment.

Select the correct option

- none of these
- unequal, balanced
- unequal, unbalanced

Correct

Handout pg 146

- equal, unbalanced

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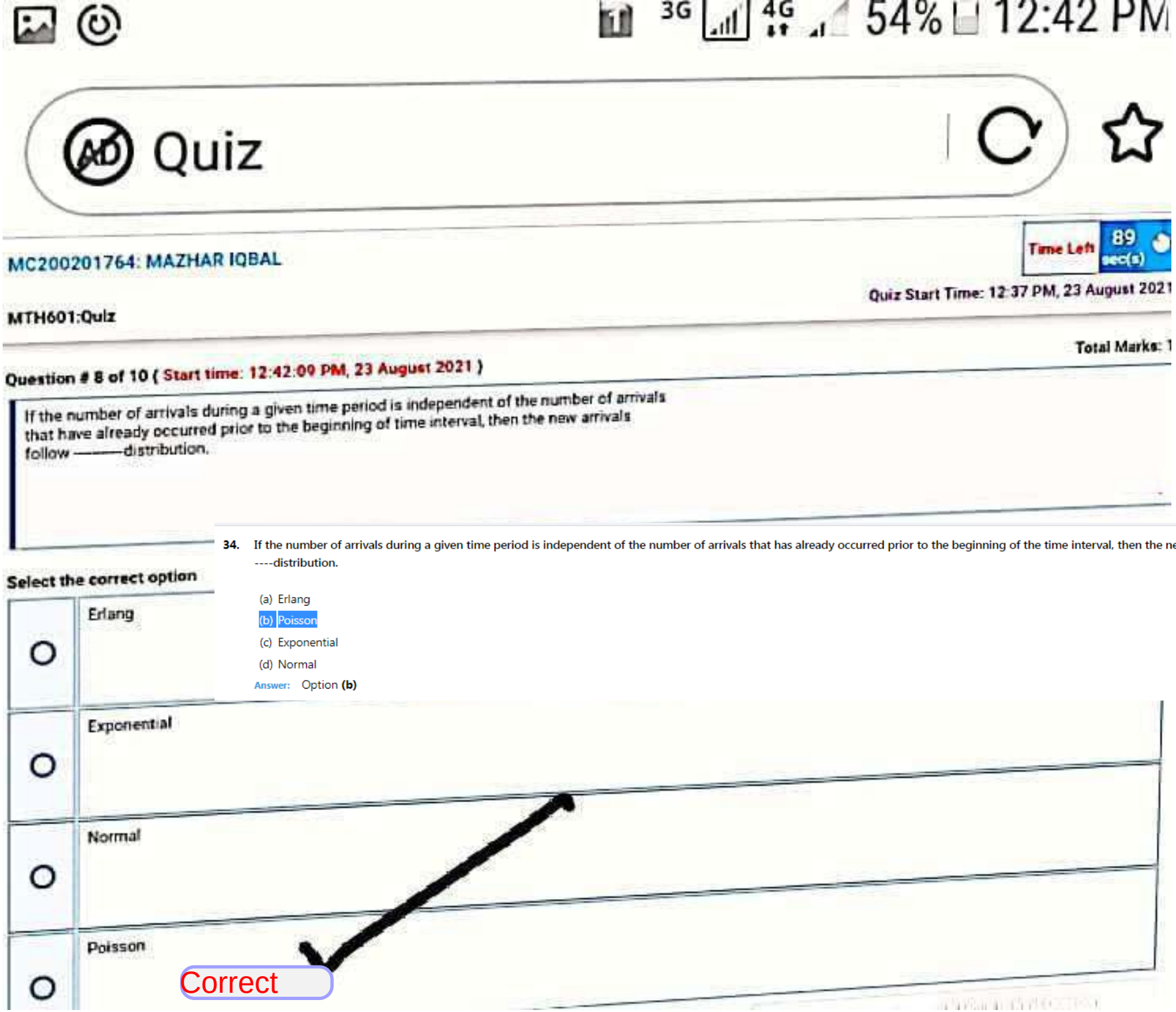
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Quiz
MC200201764: MAZHAR IQBAL
Time Left
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sec(s)
MTH601:Quiz
Quiz Start Time: 12:37 PM, 23 August 2021
Total Marks:
Question # 8 of 10 (Start time: 12:42:09 PM, 23 August 2021)
If the number of arrivals during a given time period is independent of the number of arrivals that have already occurred prior to the beginning of time interval, then the new arrivals follow ———distribution.
34. If the number of arrivals during a given time period is independent of the number of arrivals that has already occurred prior to the beginning of the time interval, then the new arrivals follow the ---------distribution.
(a) Erlang
(b) Poisson
(c) Exponential
(d) Normal
Answer: Option (b)
Select the correct option
Erlang
Exponential
Normal
Poisson
Correct
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Quiz

MC200201764: MAZHAR IQBAL

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 12:37 PM, 23 August 2021

Question # 7 of 10 (Start time: 12:41:27 PM, 23 August 2021)

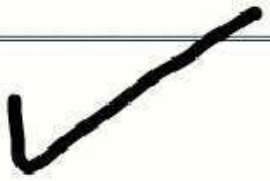
Total Marks: 1

If the total demand is equal to total supply as per requirement of a balanced transportation problem i-e "$a_1 + a_2 + \cdots + a_n = b_1 + b_2 + \cdots + b_n$" then which of the following is true?

Select the correct option

Reload Math Equations

- "$a_1 = b_1, a_2 = b_2, \cdots, a_n = b_n$" is necessarily implied
- "$a_1 = b_1, a_2 = b_2, \cdots, a_n = b_n$" is NOT necessarily implied



Correct

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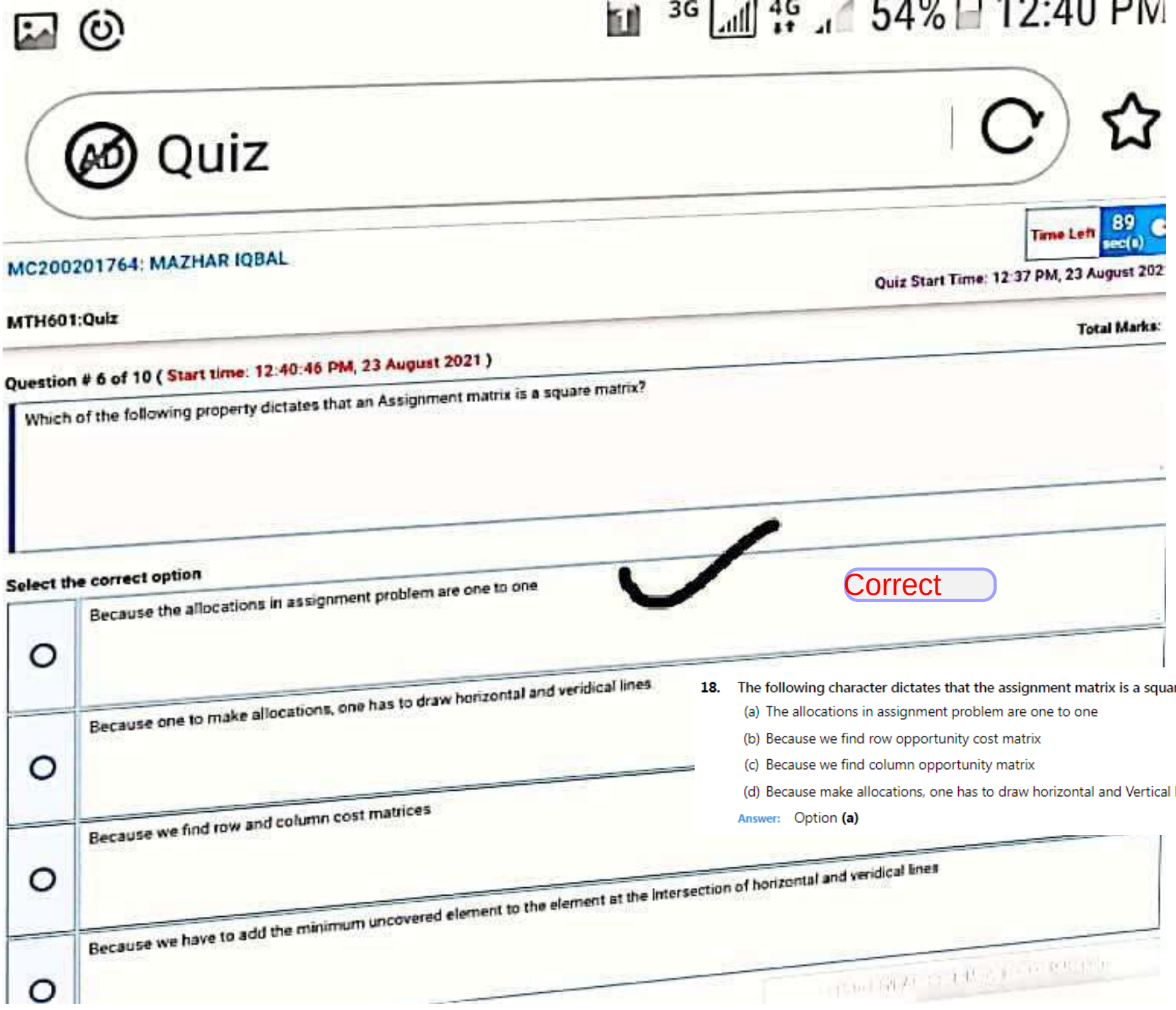
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Quiz
MC200201764: MAZHAR IQBAL
Time Left 89 sec(s)
Quiz Start Time: 12:37 PM, 23 August 202
MTH601:Quiz
Total Marks:
Question # 6 of 10 (Start time: 12:40:46 PM, 23 August 2021)
Which of the following property dictates that an Assignment matrix is a square matrix?
Select the correct option
Because the allocations in assignment problem are one to one
Correct
Because one to make allocations, one has to draw horizontal and veridical lines
Because we find row and column cost matrices
Because we have to add the minimum uncovered element to the element at the intersection of horizontal and veridical lines
18. The following character dictates that the assignment matrix is a square matrix
(a) The allocations in assignment problem are one to one
(b) Because we find row opportunity cost matrix
(c) Because we find column opportunity matrix
(d) Because make allocations, one has to draw horizontal and Vertical lines
Answer: Option (a)
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Quiz

MC200201764: MAZHAR IQBAL

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 12:37 PM, 23 August 2021

Question # 5 of 10 (Start time: 12:39:55 PM, 23 August 2021) Total Marks: 1

In the Least Cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?

Select the correct option

- Only column
- Only row
- Either of row or column
- Both row and column

Correct

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Question # 4 of 10 (Start time: 12:39:11 PM, 23 August 2021)

Total Marks: 1

To balance Assignment matrix we have to———.

Select the correct option

- open a Dummy row
- you cannot balance the assignment matrix
- open either a dummy row or column depending on the situation

Correct

- open a Dummy column

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Quiz

MC200201764: MAZHAR IQBAL

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 12:37 PM, 23 August 202

Question # 3 of 10 (Start time: 12:38:38 PM, 23 August 2021)

Total Marks:

Which of the following cost is irrelevant to replacement analysis?

Select the correct option

- ○ Machine hour rate of the machine — Correct
- ○ Purchase cost of the machine
- ○ Maintenance cost of the machine
- ○ Operating cost of the machine

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Quiz

MC200201764: MAZHAR IQBAL

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 12:37 PM, 23 August 2021

Question # 2 of 10 (Start time: 12:37:59 PM, 23 August 2021)

Total Marks: 1

The solution of an Assignment problem can be arrived by using which of the following method?

Select the correct option

- Least Cost method
- Vogel's approximation method
- Hungarian's method ✓ Correct
- North West corner method

12. The Assignment Problem is solved by
(a) Complex method
(b) Graphical method
(c) Vector method
(d) Hungarian method

Answer: Option **(d)**

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Quiz

MC200201764: MAZHAR IQBAL

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 12:37 PM, 23 August 2021

Question # 1 of 10 (Start time: 12:37:03 PM, 23 August 2021) Total Marks: 1

If the total cost of the assignment for all the tasks is equal to the sum of the costs for each agent, then the problem is called _____.

Select the correct option

- unbalanced assignment
- linear assignment ✓ Correct
- simplex
- optimization

If the total cost of the assignment for all tasks is equal to the sum of the costs for each agent (or the sum of the costs for each task, which is the same thing in this case), then the problem is called **linear assignment**.

https://en.wikipedia.org › wiki › Assignment_problem

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Question # 4 of 10 (Start time: 10:10:29 PM, 22 August 2021)

For the project of a firm, if '5' sales persons are assigned '5' different sales territories, then in how many ways a single territory can be assigned to a single sale person?

Select the correct option

○	10
○	25
○	120
○	5

5! = 5 * 4 * 3 * 2 * 1 = 120

Correct

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MC200203588: MUHAMMAD SOHAIL

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 11:08 PM, 22 August 202

Question # 4 of 10 (Start time: 11:11:46 PM, 22 August 2021)

Total Marks:

For North West Corner method, in the first row and first column, each resource and sink contain '5' units; then after allocating the appropriate amount 'x11' in the cell (1,1), in which of the following cell the next allocation will be zero?

Select the correct option

- Arbitrarily chosen either in (1,2) or in (2,1)
- In cell (1,2) only
- Neither in (1,2) nor in (2,1) but in along the diagonal in (2,2)
- In cell (2,1) only

Correct

From	To City-1	City-2	City-3	City-4	Supply
Plant-1	8 35	6	10	9	35
Plant-2	9 10	12	13	7	~~50~~/40
Plant-3	14	9	16	5	40
Demand	~~45~~ / ~~10~~	20	30	30	

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MC200203588: MUHAMMAD SOHAIL

Time Left 84 sec(s)

MTH601:Quiz

Quiz Start Time: 11:08 PM, 22 August 202

Question # 2 of 10 (Start time: 11:10:13 PM, 22 August 2021)

Total Marks:

There always exists a non-degenerate Feasible solution of a Balanced Transportation Problem.

Select the correct option

- ○ True ✓ Correct
- ○ False

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Today 10:14 pm

MTH601 Quiz

Quiz Start Time: 10:11 PM, 29 August 2021

Total Marks:

If there is infinite number of servers then all the customers are served ———— on arrival.

the correct option

- instantaneously

Correct

- randomly

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uestion # 10 of 10 (Start time: 09:52:31 PM, 22 August 2021)

Which of the following cost is irrelevant to replacement analysis?

elect the correct option

- Machine hour rate of the machine
- Purchase cost of the machine
- Maintenance cost of the machine
- Operating cost of the machine

Correct

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MC200203649 NAZAM FAROOQ

Time Left 87 sec(s)

MTH601.Quiz

Quiz Start Time: 10:11 PM, 29 August 202

Question # 5 of 10 (Start time: 10:15:05 PM, 29 August 2021)

Total Marks:

Queuing models measure the effect of———.

Select the correct option

- Random arrivals
- Effect of uncertainty on the behavior of the queuing system
- Length of queue

Correct

- Random service

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Question # 5 of 10 (Start time: 10:11:47 PM, 22 August 2021) **Total Marks: 1**

During the substitution process of determining the Transportation Cost 'xij' in a balanced Transportation problem, no——— operation is used as the coefficients of 'xij' are '0' or '1'. Due to this property, every basic feasible solution have only non-negative integral values.

Select the correct option

- ○ Addition
- ○ Division

Correct

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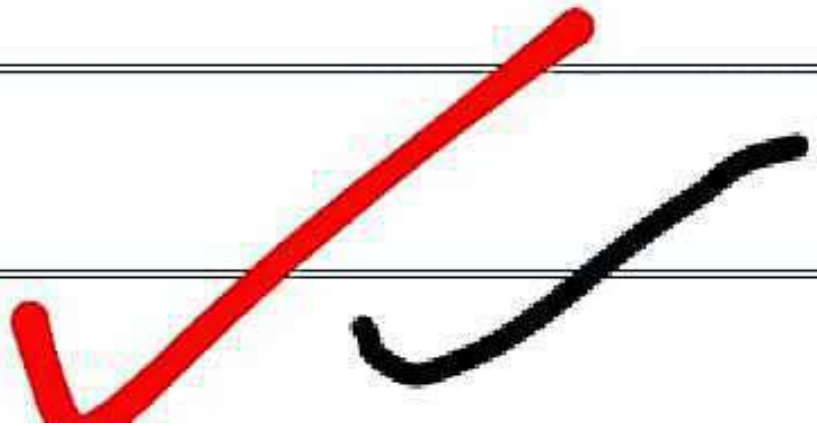
Question # 7 of 10 (Start time: 10:14:23 PM, 22 August 2021)

In an Assignment problem, while applying Hungarian's method, if in the final modified matrix any row or column does not have single zero, then which of the following is true?

Select the correct option

○	None of the above
○	Unique assignment will be made
○	No assignment can be made
○	Multiple assignments can be made



Correct

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Question # 9 of 10 (Start time: 10:16:57 PM, 22 August 2021)

The North - West Corner Rule

Select the correct option

○	Is based on the concept of minimizing opportunity cost
○	None of above
○	Is used to find an initial feasible solution
○	Is used to find optimal solution

Handout pg 130

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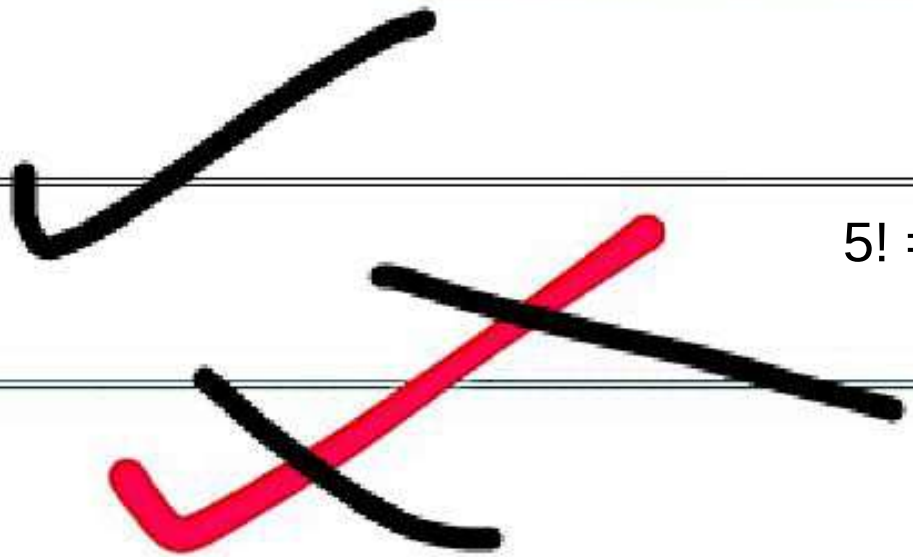
Question # 4 of 10 (Start time: 10:10:29 PM, 22 August 2021)

For the project of a firm, if '5' sales persons are assigned '5' different sales territories, then in how many ways a single territory can be assigned to a single sale person?

Select the correct option

- 10
- 25
- 120
- 5



5! = 5 * 4 * 3 * 2 * 1 = 120

Correct

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Question # 8 of 10 (Start time: 02:19:36 PM, 12 February 2021)

In the assignment problem, the cost matrix is ----------.

Select the correct option

- rectangular
- square
- non-singular
- singular

RIZ MUGHAL

Correct

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Question # 10 of 10 (Start time: 02:20:06 PM, 12 February 2021)

The assignment problem is unbalanced if the cost matrix is not a ———— matrix

Select the correct option

- [x] square — Correct
- [ ] rectangular
- [ ] non-singular
- [ ] singular

RIZ MUGHAL

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Question # 3 of 10 (Start time: 10:09:42 PM, 22 August 2021)

For a Transportation Problem, if it's initial feasible solution is evaluated by Least Cost Method, then quality of this initial solution is better than ———.

Select the correct option

- ○ Vogal's approximation Method
- ○ North West Corner Method

I think it should be NWC, as LCM is 2nd best and Vogel is the best

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Question # 6 of 10 (Start time: 10:12:59 PM, 22 August 2021)

Despite of minimizing the cost of Transportation, if we want to maximize the profit then which of the following modification would have to do?

Select the correct option

○	Add '1' from all the cost coefficients in Transportation matrix
○	Subtract '1' from all the cost coefficients in Transportation matrix
○	Multiply '-1' to all the cost coefficients in Transportation matrix
○	Divide all the cost coefficients in Transportation matrix by minimum cost coefficient

Correct

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10:17

20.9 K/s

quiz.vu.edu.pk/QuizQ

MC200203649 NAZAM FAROOQ

MTH601:Quiz

Quiz Start Time: 10:11 PM, 20 August 202

Question # 8 of 10 (Start time: 10:17:28 PM, 28 August 2021)

Total Marks:

Which of the following statement applies to both transportation model and assignment model?

Select the correct option

- Both are tested by Steping Stone method for optimality
- Both use Vogel's approximation for getting basic feasible solution
- The inequalities of both problems are related to one type of resource.
- Both have objective function, structural constraint and non-negativity constraints

Correct

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In North West Corner Method, the first step after choosing appropriate cell in 1st row, we allocate ____ so that the capacity of first row or first column is exhausted.

- as least as possible
- as much as possible ✓
- non-negative quantity
- any arbitrary quantity

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In vogel's approximation Method for solving a Transportation problem. Penality measure for any row or column, is given by which of following?

- Difference between largest unit cost to the next largest cost in same row(column)
- Difference between the smallest unit cost to the next smallest cost in same row(column)
- Difference between the largest unit cost to the smallest unit cost in the same row(column)
- Difference between any two arbitrary costs in the same row(column)

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10:16
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MC200203449 NAZAM FAROOQ
Time Left
MTH601.Quiz
Quiz Start Time: 10:11 PM, 29 August 202
Question # 6 of 10 (Start time: 10:16:09 PM, 29 August 2021)
Total Marks:

Replacement of an item will become necessary when ————.

Select the correct option:

- your opponent changes his machine in his unit
- the company has surplus funds to spend
- your operator desires to work on a new machine
- old item becomes too expensive to operate or maintain

Correct

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For an assignment problem, if the number of agents and tasks are ____ then the problem is called ____ assignment.

- equal, balanced ✓
- Unequal, balanced
- equal, unbalanced
- None

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9:53 PM

Quiz

MC200205167, MUHAMMAD MUAWIA

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 09:45 PM, 20 August 2021

Question # 5 of 10 (Start time: 09:53:47 PM, 20 August 2021)

Total Mark: 1

Which of the following method to find the solution of a Transportation Problem can be observed as improved version of Least Cost Method?

Select the correct option

- North West Corner Method
- Vogel's approximation Method

Correct

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10:22 PM

Quiz

MC200202956: SHAHZAD AHMAD

MTH601:Quiz

Question # 9 of 10 (Start time: 10:21:59 PM, 20 August 2021)

In the Least Cost Method, allocations are made in a cell by assigning ———to cell with smallest unit cost.

Select the correct option

- average of maximum and minimum
- any arbitrary amount
- maximum

Correct

- minimum

5.4.2 Least Cost Method

STEP 1: Determine the least cost among all the rows of the transportation table.

STEP 2:Identify the row and allocate the maximum feasible quantity in the cell corresponding to the least cost in the row. Then eliminate that row (column) when an allocation is made.

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Which of the following will be penalty cost of a row in Vogel's approximation method, if the smallest element is 3 and other elements are 3, 4, 5 ?

- 0 ✓
- 1
- 2
- 3

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10:09 PM 3G 16.6 K/s 78

Quiz

MC200202956: SHAHZAD AHMAD

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:09 PM, 20 August 2021

Question # 1 of 10 (Start time: 10:09:08 PM, 20 August 2021)

Total Mark : 1

Which of the following statement applies to both transportation model and assignment model?

Select the correct option

- Both have objective function, structural constraint and non-negativity constraints

Correct

- Both use Vogel's approximation for getting basic feasible solution
- Both are tested by Steping Stone method for optimality
- The inequalities of both problems are related to one type of resource

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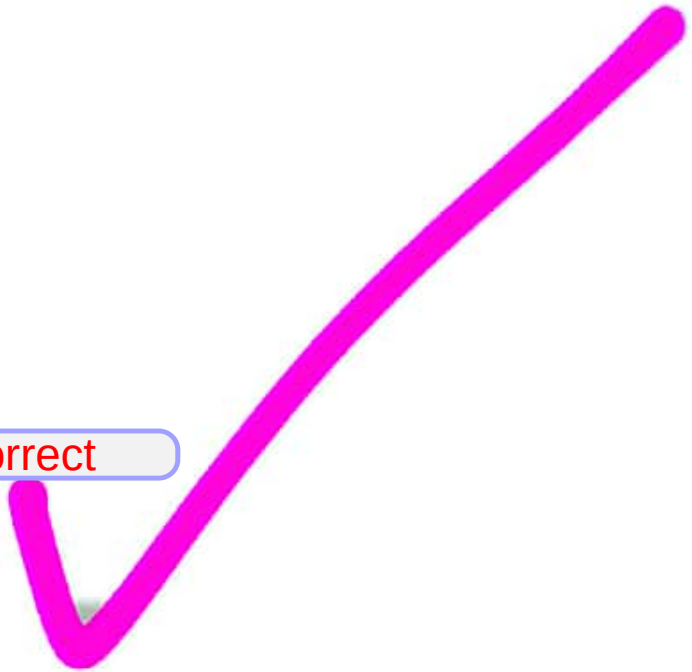
In which of following method to solve a Transportation problem, we have to calculate one more row and more column?

- North West Corner Method
- Least Cost Method
- Vogel's approximation Method

Correct

- Simplex Method



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For an unbalanced Transportation problem, if the total demand is less than total supply then which of the following is true in order to balance the problem?

Correct

- A dummy sink would have to include with demand equal to supply
- A dummy source would have to include with supply equal to shortage.
- One constraint will have evacute
- One constraint will have to add

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In a Transportation problem, the objective function 'Z' gives—

- Total Cost of transportation ✓ Correct
- Total Time of transportation
- Total Profit of transportation
- Total inventory be supplied in transportation.

Scanned with CamScanner

The column, which is introduced in the matrix to balance an unbalanced Transportation problem, is known as ——

- Key column
- idle column
- slack column
- dummy column

Correct

Scanned with CamScanner

For an assignment problem, if the number of agents and tasks are ___ then the problem is called ___ assignment.

- equal, balanced
- uncqual, balanced
- equal, unbalanced
- None

Correct

Scanned with CamScanner

The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if number of positive allocations are

- $m+n$
- $m+n-1$ ✓ Correct
- $m-n+1$
- $m+n+1$

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Which of the following will be the penalty cost of a row in Vogel's approximation method, if the smallest element is '3' and other elements are 3, 4, 5 ?

Select the correct option

☑	0 Correct
○	1
○	2
○	3

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Quiz 3 Solved by Riz Mughal Fall 2020/21

Question # 9 of 10 (Start time: 02:19:50 PM, 12 February 2021)

Time

In which of the following method to solve a Transportation problem, we have to calculate one more row and one more column?

Select the correct option

- North West Corner method
- Least Cost Method
- Vogel's approximation Method
- Simplex Method

RIZ MUGHAL

Correct

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Question # 1 of 10 (Start time: 03:51:44 PM, 12 February 2021) Total Marks:

Under which of the following condition in the Least Cost Method, after first allocation in cell (1,1), next allocation along the diagonal in cell (2,2) can be made?

RIZ MUGHAL

Select the correct option

○	Total Supply = Total Demand
○	If in the second row and second column, each resource and sink contain equal units
●	If after crossed out 1st row & 1st column, cell (2,2) is of minimum cost
○	If Total Supply is not equal to Total Demand

Correct

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In the Least Cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?

- only row
- only column
- Both row and column — Correct
- Either of row or column

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quiz.vu.edu.pk/QuizQu 2

MC200204552: ZEESHAN ALI HAIDER

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 11:43 AM, 23 August 202

Question # 10 of 10 (Start time: 11:50:02 AM, 23 August 2021)

Total Marks:

Hungarian Algorithm is used to solve _____problem.

Select the correct option

- o both a and b
- o transportation
- o assignment

Correct

- o none of these

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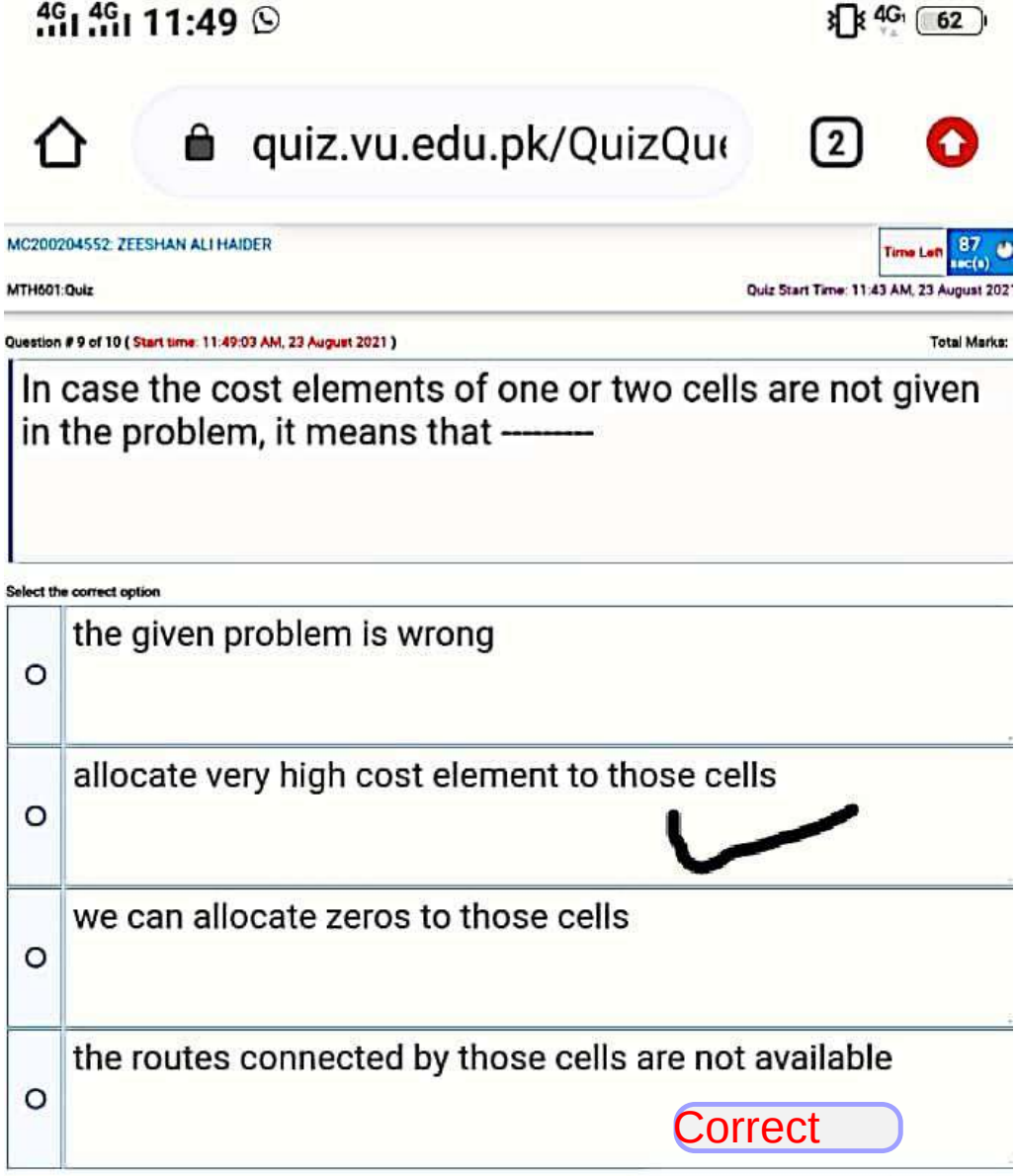

D

14. In case the cost elements of one or two cells are not given in the problem, it means: (a) The given problem is wrong (b) We can allocate zeros to those cells (c) Allocate very high cost element to those cells (d) To assume that the route connected by those cells are not available.

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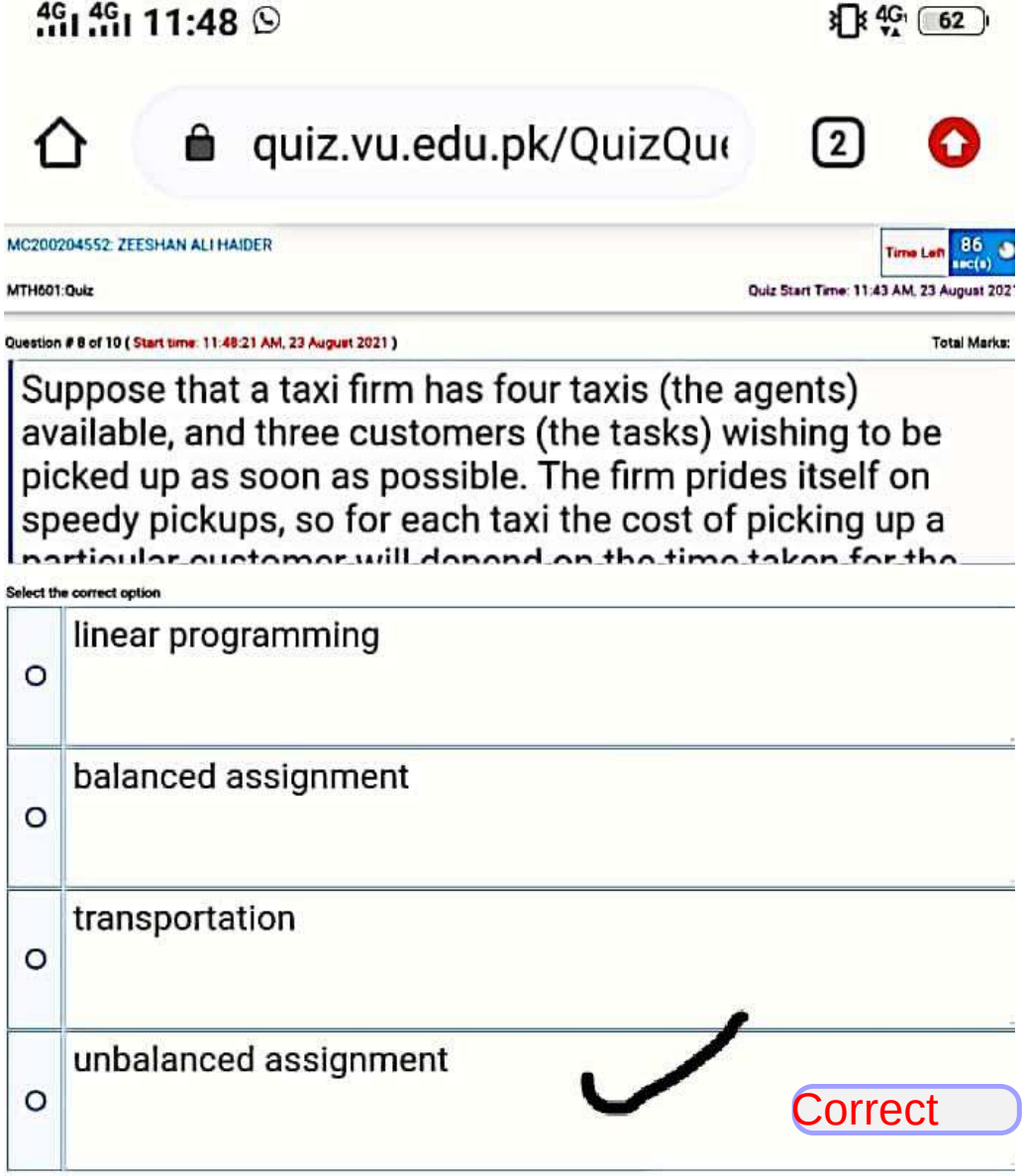

Suppose that a taxi firm has four taxis (the agents) available, and three customers (the tasks) wishing to be picked up as soon as possible. The firm prides itself on speedy pickups, so for each taxi the "cost" of picking up a particular customer will depend on the time taken for the taxi to reach the pickup point

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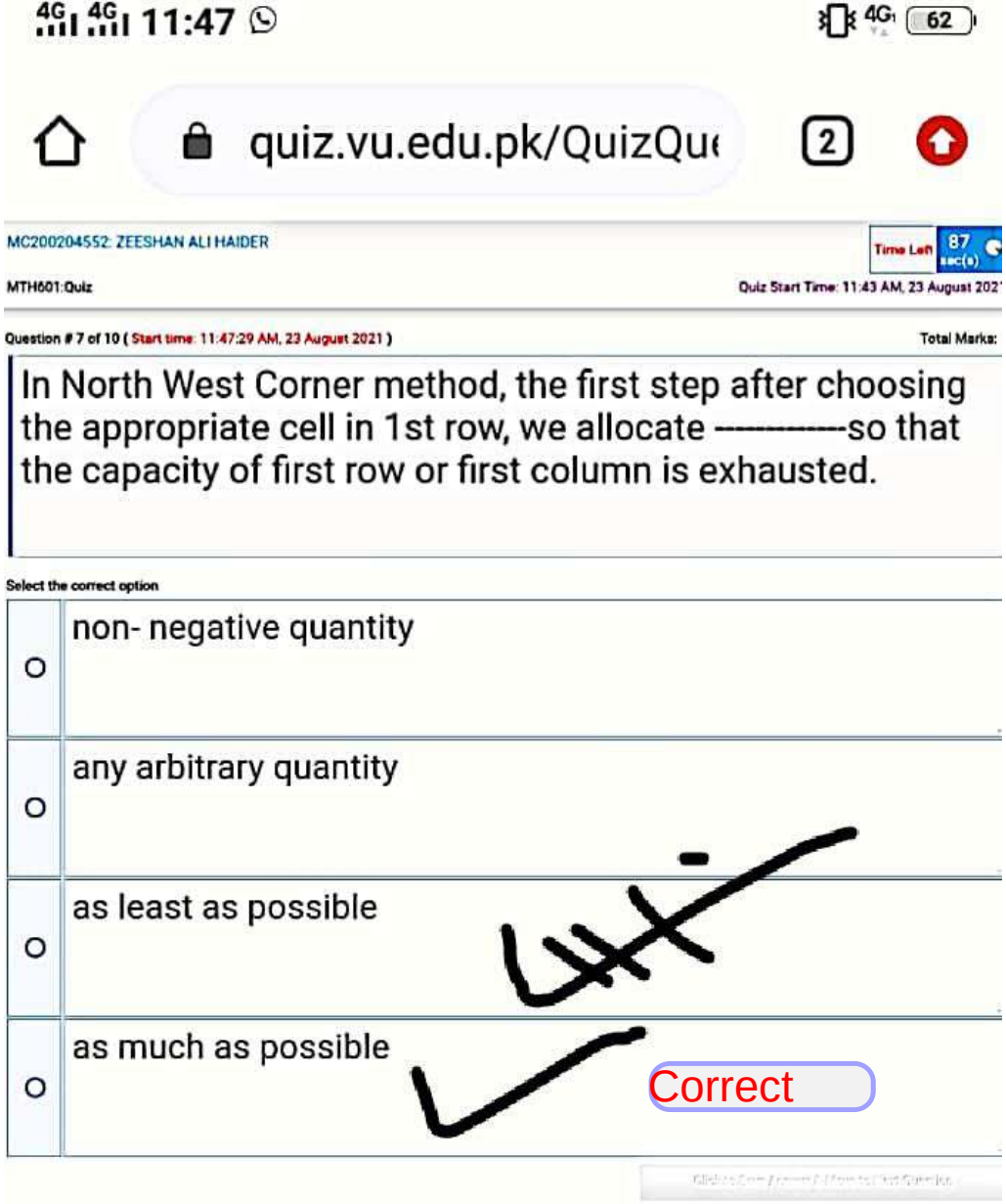

5.4.1 North West Corner Rule

STEP 1 Start with the cell in the upper left hand corner (North West Corner).

STEP 2 Allocate the maximum feasible amount.

STEP 3 Move one cell to the right if there is any remaining supply. Otherwise, move one cell down. If both are impossible, stop or go to step (2).

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MC200204552: ZEESHAN ALI HAIDER

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 11:43 AM, 23 August 202

Question # 6 of 10 (Start time: 11:46:53 AM, 23 August 2021)

Total Marks:

If a balanced Transportation problem with 'm' sources and 'n' sinks then degeneracy will arise only if there are less than ——— independent allocations in the solution.

Select the correct option

- o m−n−1 ✓ Correct
- o m−n−3
- o m−n−4
- o m−n−2

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MC200204552: ZEESHAN ALI HAIDER

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 11:43 AM, 23 August 2021

Question # 5 of 10 (Start time: 11:46:02 AM, 23 August 2021)

Total Marks:

A balanced transportation model with '5'number of sources and '7' destinations has ------- number of constraint equations.

Select the correct option

o	12 ✓ 5+7=12 Correct
o	2
o	35
o	10

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MC200204552: ZEESHAN ALI HAIDER

Time Left 90 sec(s)

MTH601:Quiz

Quiz Start Time: 11:43 AM, 23 August 202

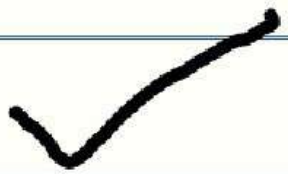
Question # 4 of 10 (Start time: 11:45:38 AM, 23 August 2021)

Total Marks:

While solving a Transportation problem by Vogel's approximation method, if the row has two elements of the same magnitude say 'x' as the smallest element then the opportunity(penalty) cost of that row or column is ----------.

Select the correct option

- o x
- o 2x
- o −x
- o Zero



Correct

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MC200204552: ZEESHAN ALI HAIDER

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MTH601:Quiz

Quiz Start Time: 11:43 AM, 23 August 2021

Question # 3 of 10 (Start time: 11:45:01 AM, 23 August 2021)

Total Marks:

There always exists a non-degenerate Feasible solution of a Balanced Transportation Problem.

Select the correct option

- ○ True — Correct
- ○ False

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MC200204552: ZEESHAN ALI HAIDER

MTH601 Quiz

Time Left 89 sec(s)

Quiz Start Time: 11:43 AM, 23 August 202

Question # 2 of 10 (Start time: 11:44:09 AM, 23 August 2021)

Total Marks:

For North West Corner method, in the first row and first column, each source and sink contain '5' units ; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

Select the correct option

- Towards (1,2) if 1st row contains more occupied cells than 2nd row
- Arbitrarily either towards (1,2) or (2,1)
- Towards (2,1) if 2nd row contains more occupied cells than 1st row
- Neither towards (1,2) nor (2,1) but move diagonally towards (2,2)

Correct

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MC200204552: ZEESHAN ALI HAIDER

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 11:43 AM, 23 August 202

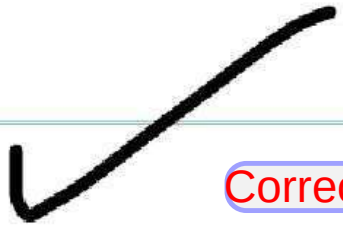
Question # 1 of 10 (Start time: 11:43:34 AM, 23 August 2021)

Total Marks:

When the total allocations in a transportation model of m*n size is not equals to "m + n – 1", the situation is known as:

Select the correct option

- Unbalanced situation
- Tie situation
- None of the above
- Degeneracy



Correct

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Question # 9 of 10 (Start time: 09:33:43 AM, 23 August 2021)

In North West Corner method, the first step after choosing the appropriate cell in 1st row, we allocate ———— so that the capacity of first row or first column is exhausted

Select the correct option

- as much as possible ✓ Correct
- non- negative quantity
- any arbitrary quantity
- as least as possible

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:200200462: SABILA KAUSAR

H601:Quiz

estion # 8 of 10 (Start time: 09:33:09 AM, 23 August 2021)

n North West Corner Rule, we start initially from which of the following cell in the standard transportation tab

elect the correct option

- Upper right
- From any arbitrary cell in first row
- Upper left
- Upper middle

Correct

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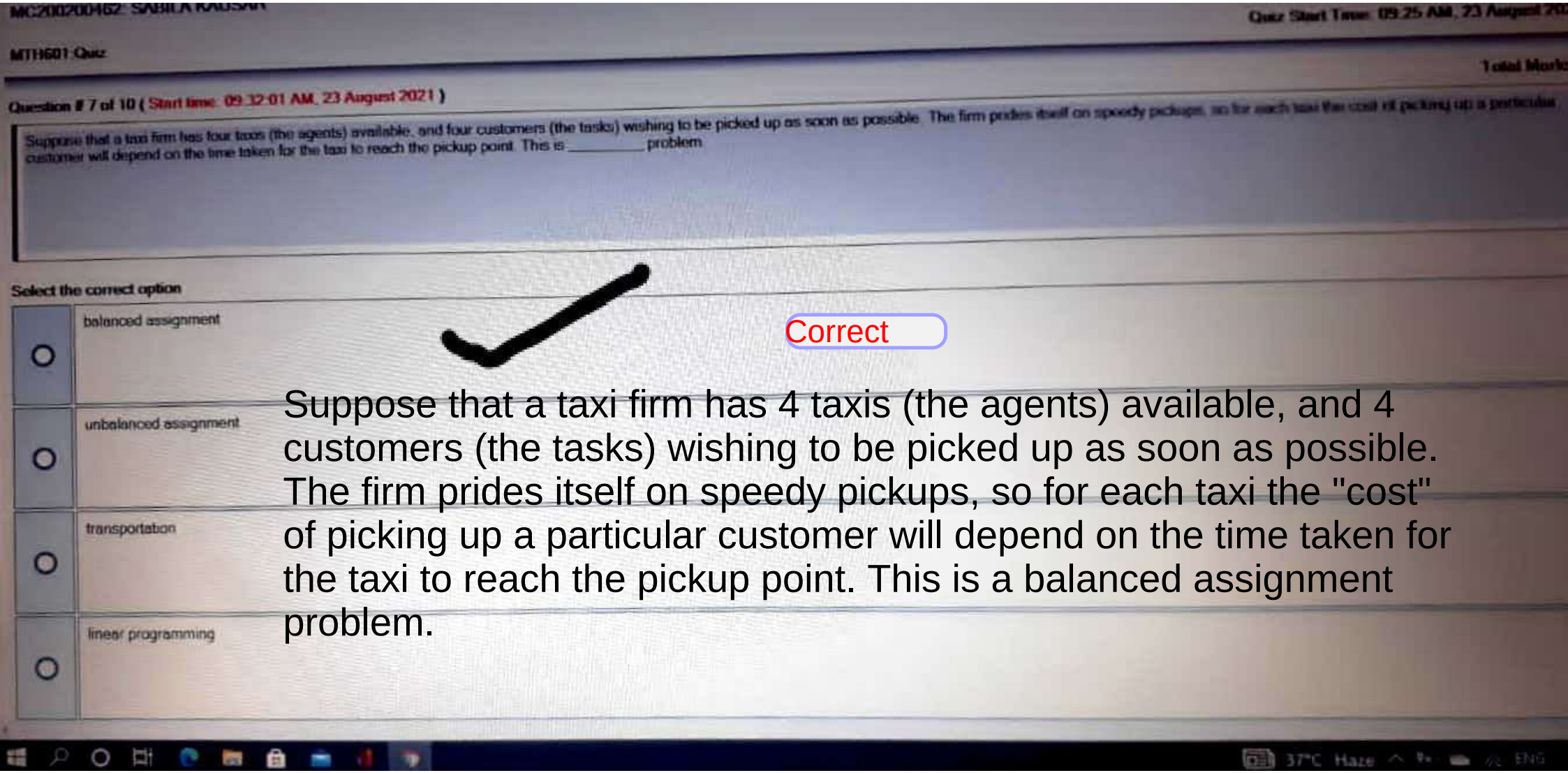

Suppose that a taxi firm has 4 taxis (the agents) available, and 4 customers (the tasks) wishing to be picked up as soon as possible. The firm prides itself on speedy pickups, so for each taxi the "cost" of picking up a particular customer will depend on the time taken for the taxi to reach the pickup point. This is a balanced assignment problem.

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Question # 6 of 10 (Start time: 09:31:01 AM, 23 August 2021)

If a balanced Transportation problem with 'm' sources and 'n' sinks then degeneracy will arise only if there are less than ——— independent allocations in the solution

Select the correct option

- m+n-4
- m+n-3
- m+n-1
- m+n-2

Correct

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Question # 5 of 10 (Start time: 09:30:08 AM, 23 August 2021)

The Transportation Problem is one of the sub-class of ----------.

Select the correct option

- Stochastic Problem
- Inventory Control Problem
- Network Flow Problem
- Linear Programming Problem ✓

Correct

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MC200200462: SABILA KAUSAR

MTH601 Quiz

Quiz Start Time: 09:25 AM, 23 Aug

Question # 4 of 10 (Start time: 09:29:04 AM, 23 August 2021)

Tota

While solving an Assignment problem by Hunganan's method, in the modified cost matrix if the minimum number of horizontal and vertical lines to cover zeros are not equal to the number of rows (or columns), then which of followag operation is done?

Select the correct option

- Add smallest element of the uncovered cells to the elements to the line
- Subtract the smallest element from the next highest number in the element
- Subtract the smallest element from the element at the intersection of horizontal and vertical line
- Subtract smallest element of uncovered rows from all other elements of uncovered cells

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Quiz Start Time: 09:25 AM, 23 A

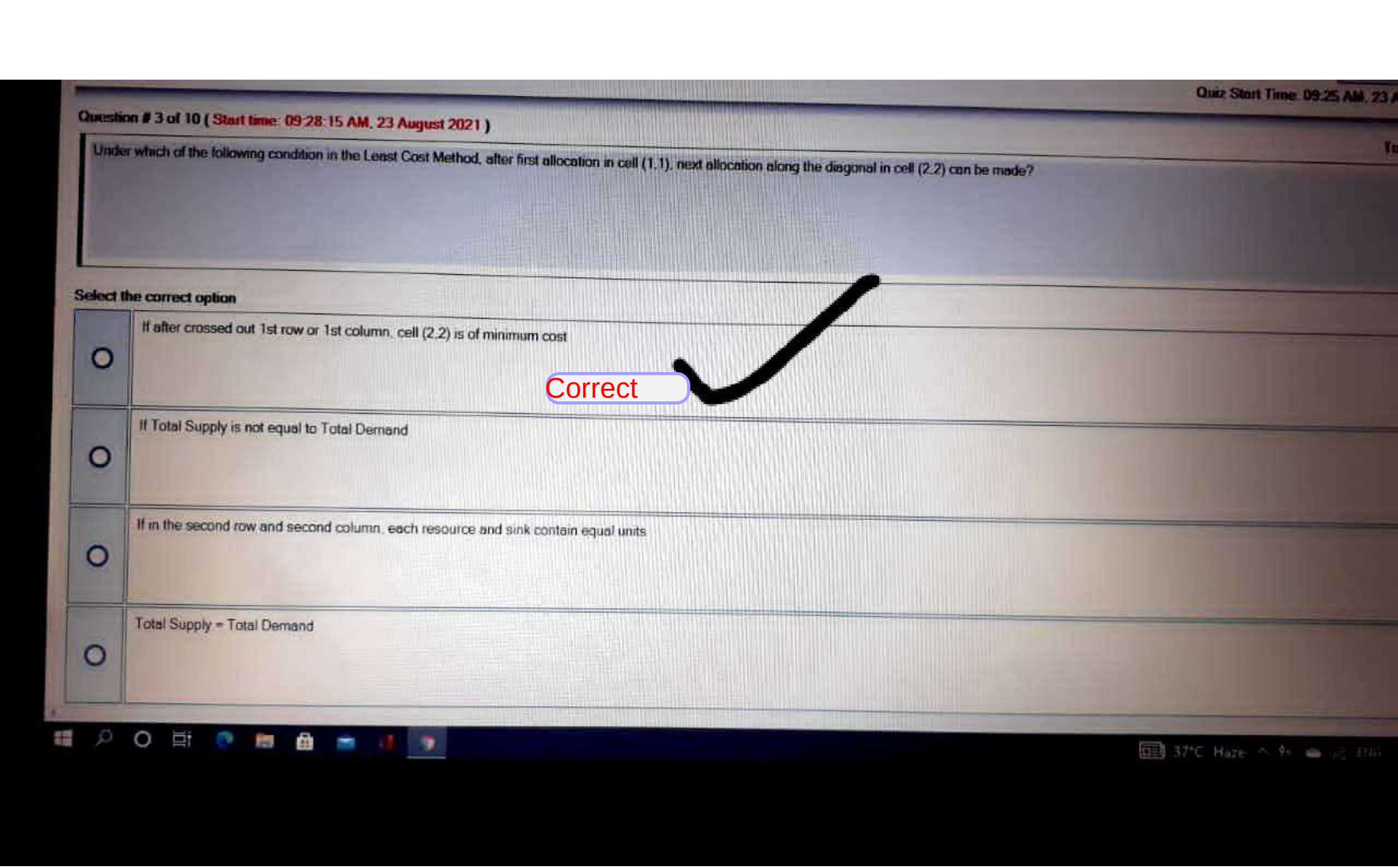
Question # 3 of 10 (Start time: 09:28:15 AM, 23 August 2021)

Under which of the following condition in the Least Cost Method, after first allocation in cell (1,1), next allocation along the diagonal in cell (2,2) can be made?

Select the correct option

- If after crossed out 1st row or 1st column, cell (2,2) is of minimum cost
- If Total Supply is not equal to Total Demand
- If in the second row and second column, each resource and sink contain equal units
- Total Supply = Total Demand

Correct



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Question # 2 of 10 (Start time: 09:26:45 AM, 23 August 2021)

When a doctor attends to an emergency case leaving his regular service is called

Select the correct option

- Reneging
- Non-Pre-Emptive queue discipline

Correct

- Balking
- Pre-emptive queue discipline

31. When a doctor attends to an emergency case leaving his regular service is called:

(a) Reneging

(b) Balking

(c) Pre-emptive queue discipline

(d) Non-Pre-Emptive queue discipline

Answer: Option **(d)**

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9:11 AM

Quiz

For North West Corner method, in the first row and first column, each resource and sink contain '5' units; then after allocating the appropriate amount 'x11' in the cell (1,1), in which of the following cell the next allocation will be zero?

- In cell (1,2) only
- In cell (2,1) only
- Neither in (1,2) nor in (2,1) but in along the diagonal in (2,2)

Correct

- Arbitrarily chosen either in (1,2) or in (2,1)

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9:10 AM

Quiz

For the project of a firm, if '5' sales persons are assigned '5' different sales territories, then in how many ways a single territory can be assigned to a single sale person?

- 120 ✓ Correct
- 10
- 5
- 25

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9:10 AM

Quiz

MTH603 Quiz

The solution of an Assignment problem can be arrived by using which of the following method?

Select the correct option

- Least Cost method
- Vogel's approximation method
- North West corner method
- Hungarian's method

Correct

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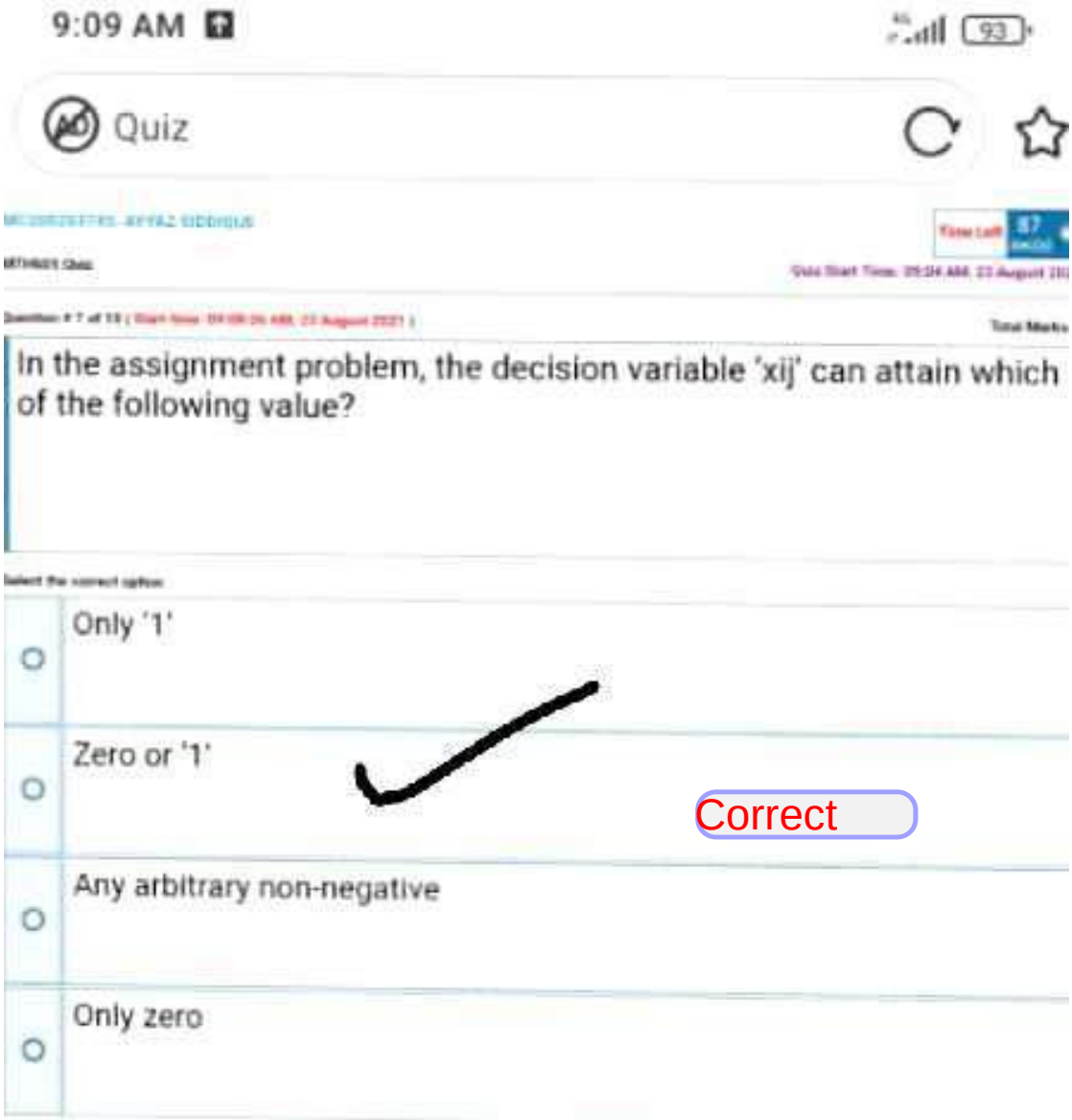

and $x_{1j} + x_{2j} + \ldots + x_{nj} = 1, \quad \text{for } j = 1, 2, \ldots, n$

When compared with a transportation problem, we see that $a_i = 1$ and $b_j = 1$ for all rows and columns, $x_{ij} = 0$ *or* 1.

We shall not attempt simplex algorithem or the transportation algorithm to get a solution to an assignment problem. Certain systematic procedure has been devised so as to obtain the solution to the problem with ease.

6.3 Solution of an Assignment Problem

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9:09 AM

Quiz

During the substitution process of determining the Transportation Cost 'xij' in a balanced Transportation problem, no——— operation is used as the coefficients of 'xij' are '0' or '1'. Due to this property, every basic feasible solution have only non-negative integral values.

- Addition
- Division

Correct

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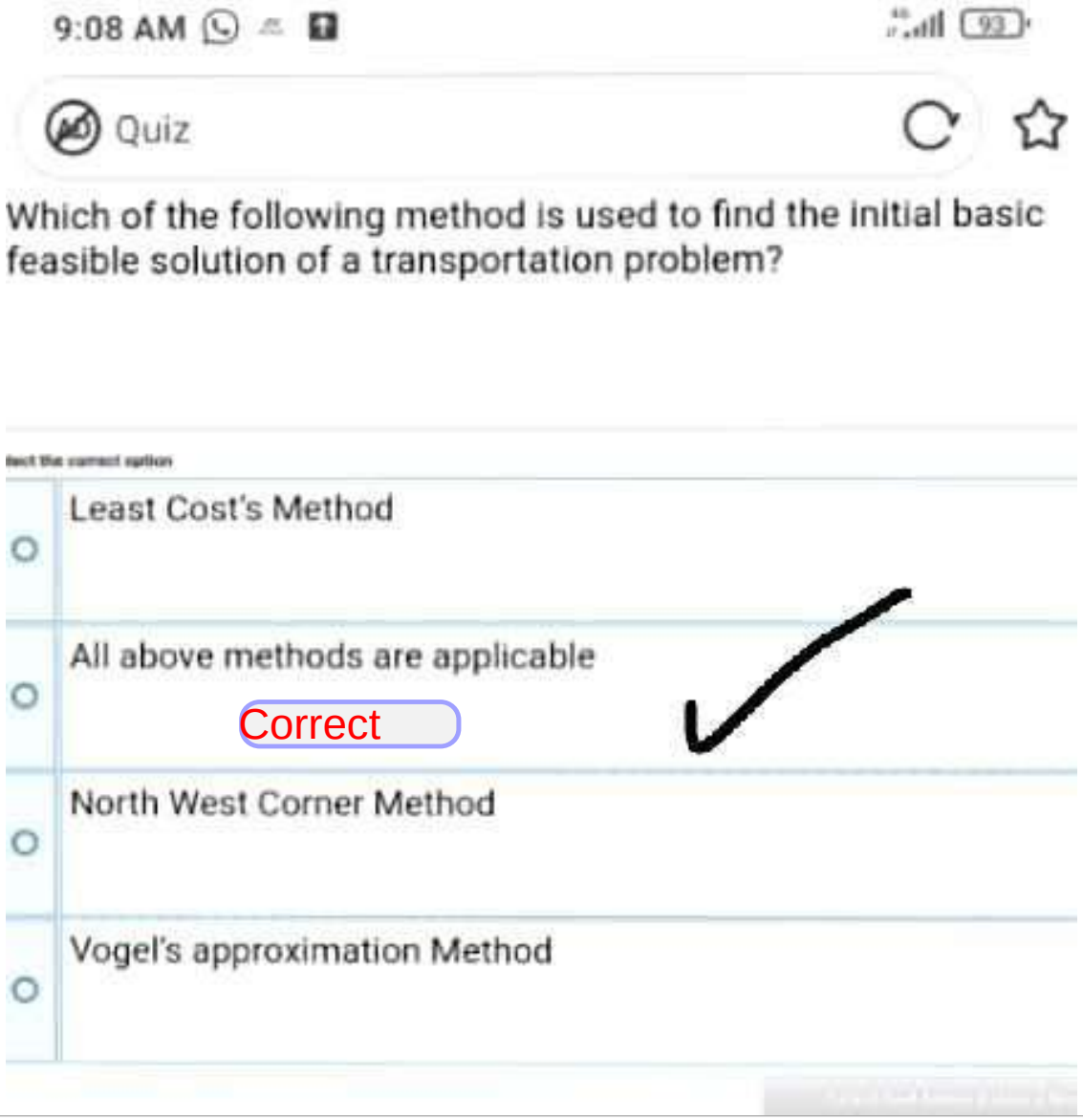

5.4 Initial Basic Feasible Solution

An initial basic feasible solution to a transportation problem can be found by any one of the three following methods:

(i) North west corner rule (NWC)
(ii) Least cost method (LCM)
(iii) Vogel's approximation method (VAM)

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9:08 AM

Quiz

The row, which is introduced in the matrix to balance an unbalnced Transportation problem, is known as ———.

- key row
- idle row
- slack row
- dummy row

Correct

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9:07 AM

Quiz

In the Least Cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?

- Only row
- Either of row or column
- Both row and column

Correct

- Only column

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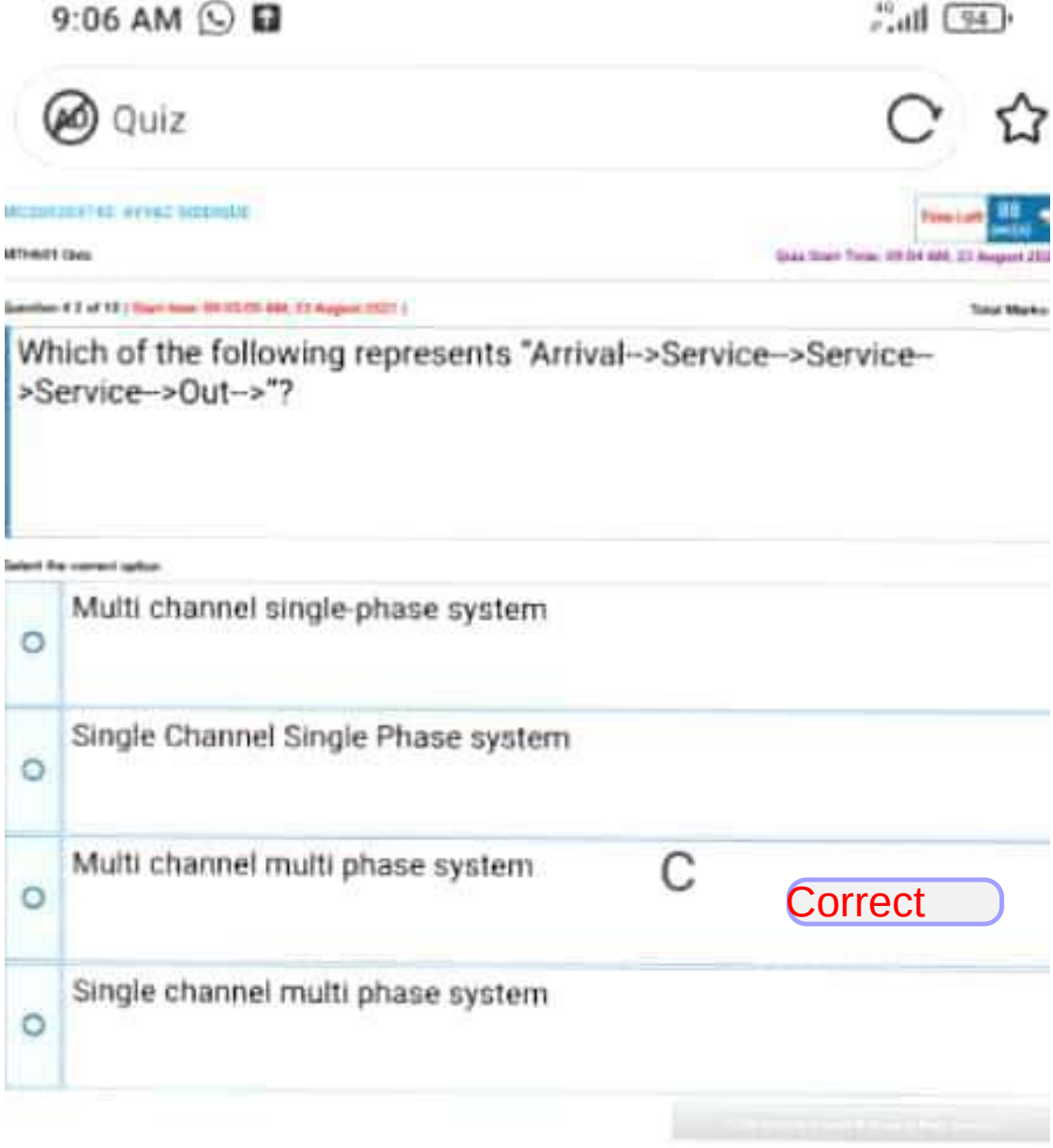

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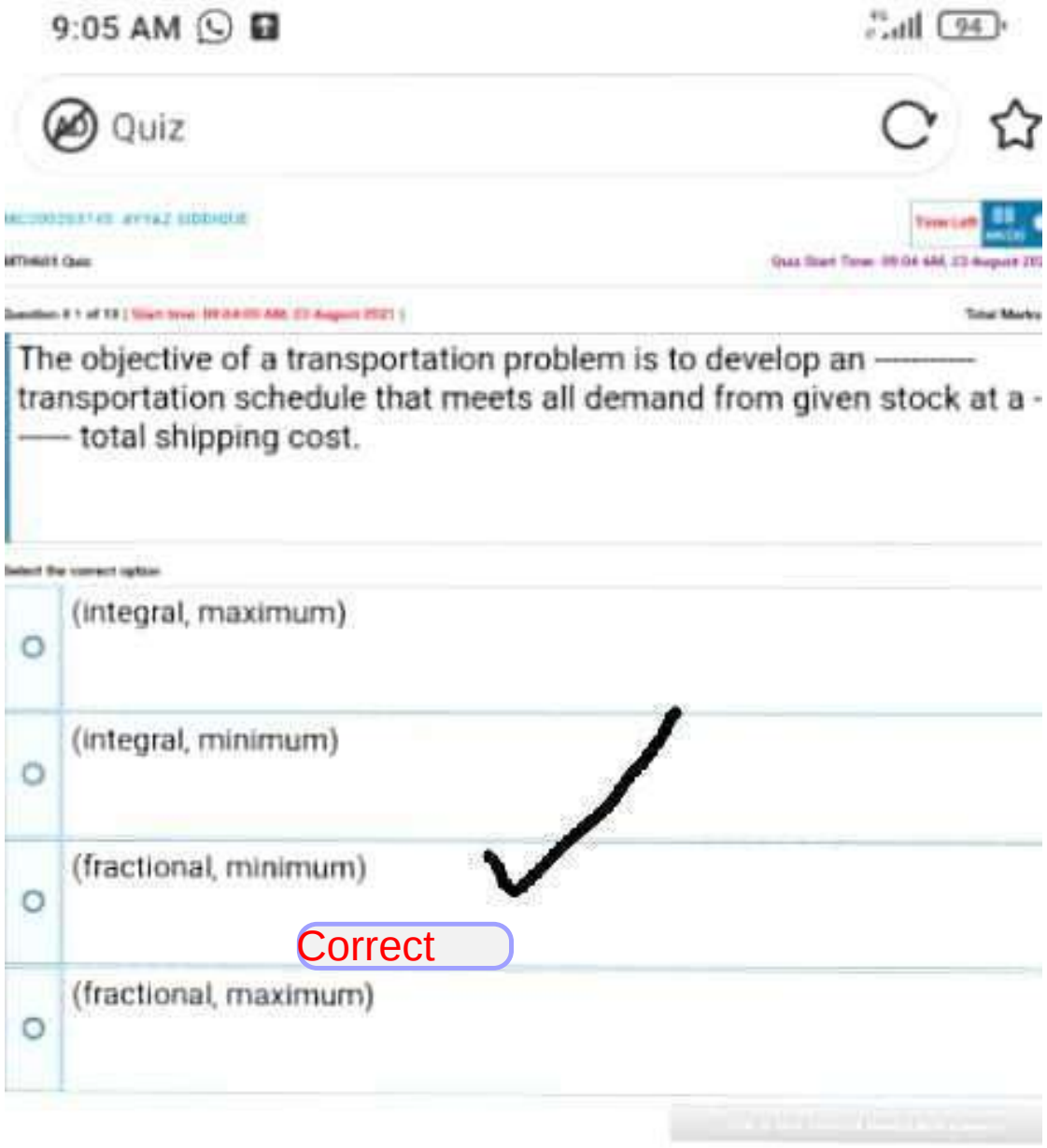

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601 - Operations Research

Quiz No. 03

100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal & Sadiq Ahmed

Sindh, Pakistan

03063548328

sonuilyasmughal@gmail.com

https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists_Sadiq Ahmed

Sonu Ilyas Mughal

بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz

Question # 1 of 10 (Start time: 01:15:24 AM, 12 February 2021)

The column, which is introduced in the matrix to balance an unbalanced Transportation problem, is known as-------

Select the correct option

- key column
- idle column
- slack column
- dummy Column

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz **Quiz Start Time: 01:15 AM, 12 February 202**

Question # 2 of 10 (Start time: 01:15:53 AM, 12 February 2021) **Total Marks:**

For North West Corner method, in the first row and first column, each source and sink contain '5' units ; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

Select the correct option

- (●) Arbitrarily either towards (1,2) or (2,1)
- ( ) Towards (1,2) if 1st row contains more occupied cells than 2nd row
- ( ) Towards (2,1) if 2nd row contains more occupied cells than 1st row
- ( ) Neither towards (1,2) nor (2,1) but move diagonally towards (2,2)

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz

Question # 3 of 10 (Start time: 01:16:14 AM, 12 February 2021)

Which of the following reason to prefer Vogel's approximation method over other two methods?

Select the correct option

- ○ Vogel's method gives the shortest transportation rout
- ○ Vogel's method gives optimal time plan of transportation
- ◉ Vogel's method gives the feasible solution near to the optimal
- ○ Vogel's method allocate minimum amount to the variable with least unit cost

Click

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz

Quiz Start Time: 0

Question # 4 of 10 (Start time: 01:16:33 AM, 12 February 2021)

For a Transportation Problem, if it's initial feasible solution is evaluated by Least Cost Method, then quality of this initial solution is better than ---------.

Select the correct option

- (●) North West Corner Method
- ( ) Vogal's approximation Method

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz Start Time: 01:15 AM

Question # 5 of 10 (Start time: 01:16:54 AM, 12 February 2021)

Which of the following will be the penalty cost of a row in Vogel's approximation method, if the smallest element is '3' and other elements are 3, 4, 5 ?

Select the correct option

- (●) 0
- ( ) 1
- ( ) 2
- ( ) 3

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz

Question # 6 of 10 (Start time: 01:17:16 AM, 12 February 2021)

In a Transportation Problem, the objective function 'Z' gives ----------.

Select the correct option

- (●) Total Cost of transportation
- ( ) Total Time of transportation
- ( ) Total Profit of transportation
- ( ) Total inventory be supplied in transportation

Click to S

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz **Quiz Start Time: 01:15 AM, 12 Feb**

Question # 7 of 10 (Start time: 01:17:36 AM, 12 February 2021) **Tot**

The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if numbers of positive allocations are

Select the correct option

- ○ m+n
- ◉ m+n-1
- ○ m-n+1
- ○ m+n+1

Click to Save Answer & Move to Next Q

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz Quiz Start Time: 01:15 AM,

Question # 8 of 10 (Start time: 01:18:05 AM, 12 February 2021)

For an unbalanced Transportation problem, if the total demand is LESS than total supply then which of the following is true in order to balance the problem?

Select the correct option

- (●) A dummy sink would have to include with demand equal to the surplus
- ( ) A dummy source would have to include with supply equal to shortage
- ( ) One constraint will have evacuate
- ( ) One constraint will have to add

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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Question # 9 of 10 (Start time: 01:18:31 AM, 12 February 2021)

The solution of an Assignment problem can be arrived by using which of the following method?

Select the correct option

- [x] Hungarian's method
- [ ] North West corner method
- [ ] Vogel's approximation method
- [ ] Least Cost method

Clic

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MTH601:Quiz Quiz Start Ti

Question # 10 of 10 (Start time: 01:18:51 AM, 12 February 2021)

For an assignment problem, if the numbers of agents and tasks are _____, then the problem is called _____ assignment.

Select the correct option

- (●) equal, balanced
- ( ) unequal, balanced
- ( ) equal, unbalanced
- ( ) none of these

Click to Save Answe

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Subscribed MY YOUTUBE channel by Tutor Sadiq Ahmed

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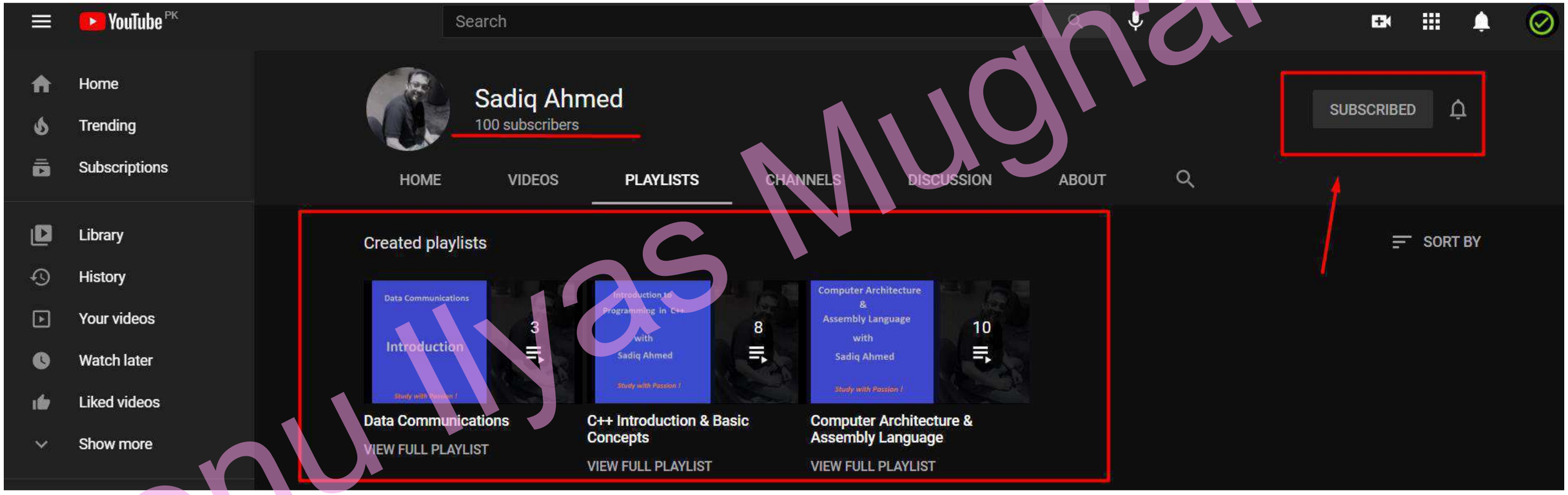

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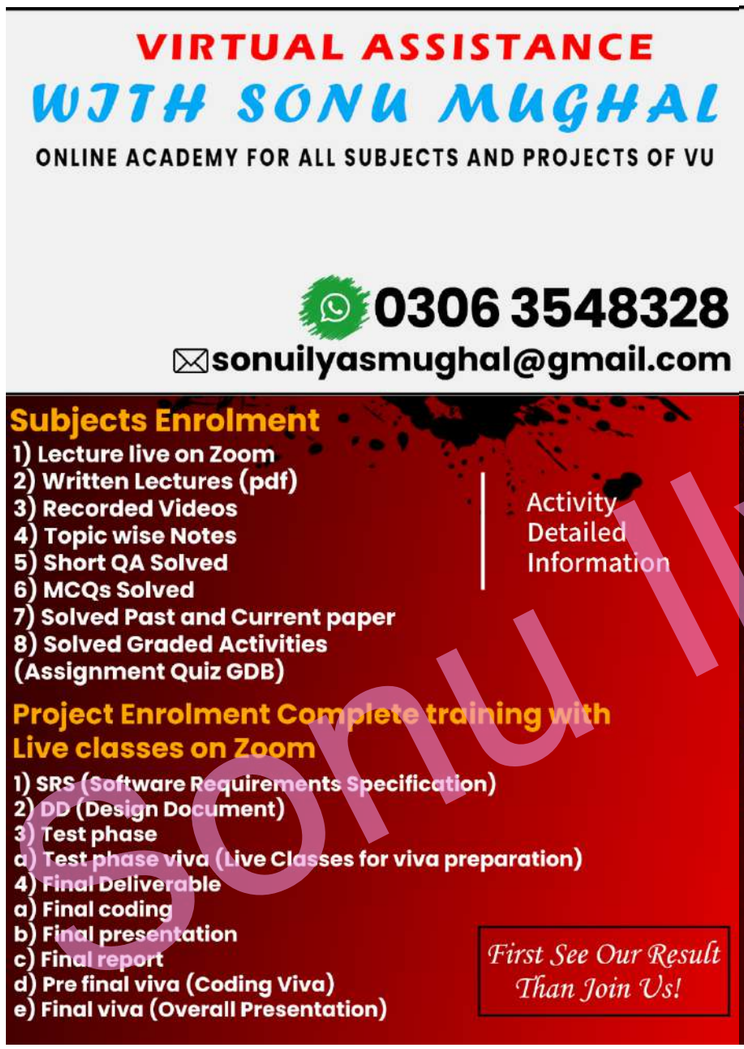

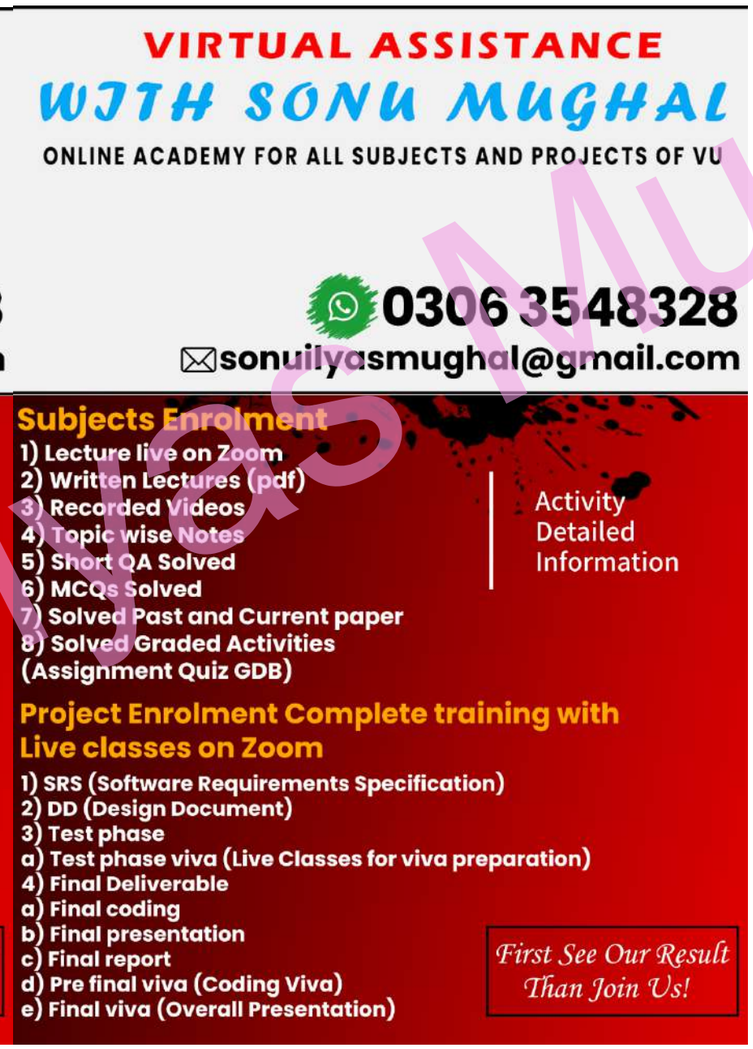

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Quiz No. 02 100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal & Sadiq Ahmed

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MTH601:Quiz

Question # 1 of 10 (Start time: 06:13:55 PM, 08 February 2021)

An optimal solution of a Linear Programming problem is necessarily a Basic feasible solution.

Select the correct option

- ○ True
- ● False

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MTH601 - Operations Research

MTH601:Quiz Quiz Start Time: 06:13 PM, 08

Question # 2 of 10 (Start time: 06:14:22 PM, 08 February 2021)

While solving a Linear Programming problem by Simplex Method, an inequality constraint is transformed into strict equality by the inclusion of ---------- variable.

Select the correct option

- slack
- decision
- artificial
- basic

Click to Save Answer & Move to Nex

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MTH601:Quiz Quiz

Question # 3 of 10 (Start time: 06:14:47 PM, 08 February 2021)

If the primal has an unbounded solution, then the dual has------

Select the correct option

- optimal solution
- infeasible solution
- bounded solution
- no solution

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz Start Time: 06:13 PM, 08 February 20

Question # 4 of 10 (Start time: 06:15:11 PM, 08 February 2021) Total Mark

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, then in a given iteration, how many Basic feasible VARIABLES are possible to exists?

Select the correct option

- 10
- 6
- 4
- 2

Click to Save Answer & Move to Next Question

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MTH601:Quiz Quiz Start Time: 06:13 PM, 08 Febr

Question # 5 of 10 (Start time: 06:15:29 PM, 08 February 2021) Tota

When we have $\geq$ type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.

Select the correct option

Reload Math Eq

- slack, Big M
- artificial, Two phase
- artificial, Big M
- artificial and slack, Big M or Two phase

Click to Save Answer & Move to Next Qu

Sonu Ilyas Mughal

0306

بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz **Quiz Start Time: 06:13 PM, 08 February 2021**

Question # 6 of 10 (Start time: 06:15:52 PM, 08 February 2021) **Total Marks:** 1

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. If the initial basic feasible solution is say;(2,4,1,2), then its corresponding non-basic solution is---------.

Select the correct option

- (0,0)
- (-4,-2)
- (1,2)
- (2,2)

Click to Save Answer & Move to Next Question

Sonu Ilyas Mughal

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz

Question # 7 of 10 (Start time: 06:16:10 PM, 08 February 2021)

Dual of a Dual is ------

Select the correct option

- (●) Primal
- ( ) Dual
- ( ) Primal-dual
- ( ) (Dual)^2

Click t

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz St

Question # 8 of 10 (Start time: 06:16:28 PM, 08 February 2021)

The dual of a dual problem yields the original primal.

Select the correct option

- (●) True
- ( ) False

Click to Save A

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz Quiz Start

Question # 9 of 10 (Start time: 06:16:48 PM, 08 February 2021)

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____ .

Select the correct option

- must have a solution
- does not have a solution
- may have a solution
- non of these

Click to Save Ans

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MTH601:Quiz Quiz Sta

Question # 10 of 10 (Start time: 06:17:12 PM, 08 February 2021)

Once an LP problem is converted to a table, then calculation steps would be same for _____ method(s).

Select the correct option

- simplex
- Big M
- Two phase
- all of above

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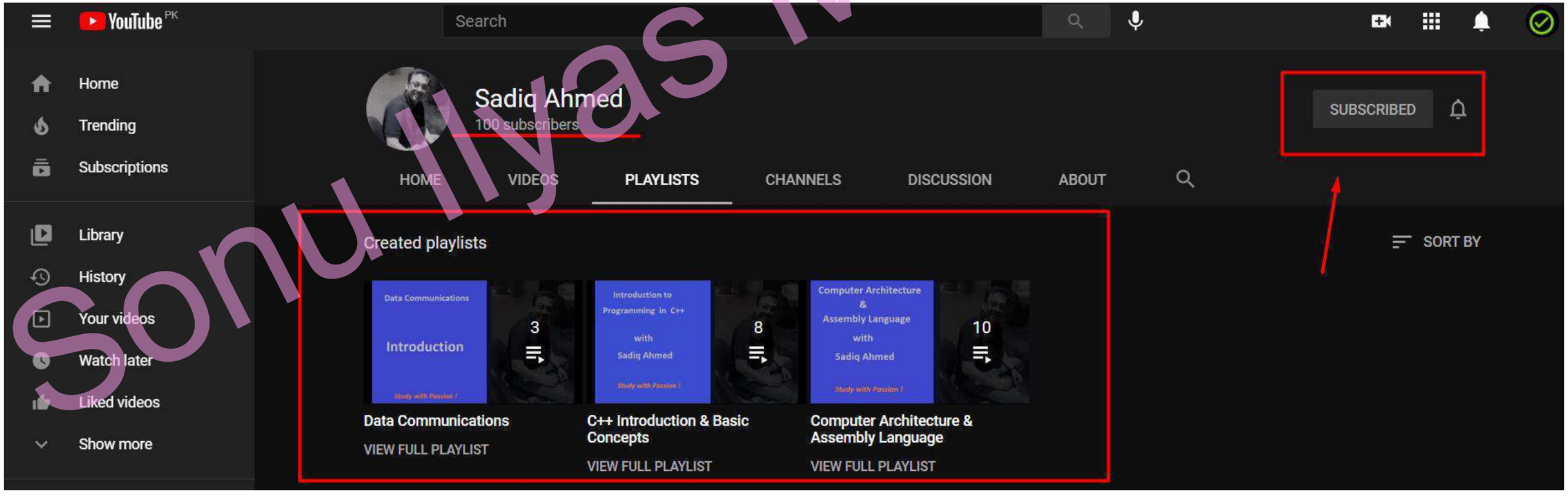

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بسم الله الرحمن الرحيم

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بسم الله الرحمن الرحيم

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MTH601 - Operations Research

Quiz No. 02 100% Correct Solve (Alhamdulillah)

BY Virtual Assistance Sonu Ilyas Mughal & Sadiq Ahmed

Sindh, Pakistan

03063548328

sonuilyasmughal@gmail.com

https://www.youtube.com/channel/UCjboEX65C_yN7as5di1OuIQ/playlists Sadiq Ahmed

Sonu Ilyas Mughal

بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz

Question # 1 of 10 (Start time: 06:13:55 PM, 08 February 2021)

An optimal solution of a Linear Programming problem is necessarily a Basic feasible solution.

Select the correct option

- ○ True
- ● False

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz Quiz Start Time: 06:13 PM, 08

Question # 2 of 10 (Start time: 06:14:22 PM, 08 February 2021)

While solving a Linear Programming problem by Simplex Method, an inequality constraint is transformed into strict equality by the inclusion of ---------- variable.

Select the correct option

- slack
- decision
- artificial
- basic

Click to Save Answer & Move to Nex

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz

Question # 3 of 10 (Start time: 06:14:47 PM, 08 February 2021)

If the primal has an unbounded solution, then the dual has------

Select the correct option

- optimal solution
- infeasible solution
- bounded solution
- no solution

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

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MTH601:Quiz Quiz Start Time: 06:13 PM, 08 February 20

Question # 4 of 10 (Start time: 06:15:11 PM, 08 February 2021) Total Mark

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, then in a given iteration, how many Basic feasible VARIABLES are possible to exists?

Select the correct option

- 10
- 6
- 4
- 2

Click to Save Answer & Move to Next Question

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قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz Quiz Start Time: 06:13 PM, 08 Febr

Question # 5 of 10 (Start time: 06:15:29 PM, 08 February 2021) Tota

When we have $\geq$ type constraints, then we convert it to equality constraints by introducing _____ variable(s) and solve it by _____ method.

Select the correct option

Reload Math Eq

- slack, Big M
- artificial, Two phase
- artificial, Big M
- artificial and slack, Big M or Two phase

Click to Save Answer & Move to Next Qu

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بسم الله الرحمن الرحيم

قُلِ اللَّهُمَّ مَالِكَ الْمُلْكِ تُؤْتِي الْمُلْكَ مَن تَشَاءُ وَتَنزِعُ الْمُلْكَ مِمَّن تَشَاءُ وَتُعِزُّ مَن تَشَاءُ وَتُذِلُّ مَن تَشَاءُ ۖ بِيَدِكَ الْخَيْرُ ۖ إِنَّكَ عَلَىٰ كُلِّ شَيْءٍ قَدِيرٌ

MTH601 - Operations Research

MTH601:Quiz **Quiz Start Time: 06:13 PM, 08 February 2021**

Question # 6 of 10 (Start time: 06:15:52 PM, 08 February 2021) **Total Marks:** 1

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables. If the initial basic feasible solution is say;(2,4,1,2), then its corresponding non-basic solution is---------.

Select the correct option

- (0,0)
- (-4,-2)
- (1,2)
- (2,2)

Click to Save Answer & Move to Next Question

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MTH601 - Operations Research

MTH601:Quiz

Question # 7 of 10 (Start time: 06:16:10 PM, 08 February 2021)

Dual of a Dual is ------

Select the correct option

- (●) Primal
- ( ) Dual
- ( ) Primal-dual
- ( ) (Dual)^2

Click t

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MTH601:Quiz Quiz St

Question # 8 of 10 (Start time: 06:16:28 PM, 08 February 2021)

The dual of a dual problem yields the original primal.

Select the correct option

- (●) True
- ( ) False

Click to Save A

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MTH601:Quiz Quiz Start

Question # 9 of 10 (Start time: 06:16:48 PM, 08 February 2021)

If any or all of the artificial variables do not leave the basis in the final solution, then this indicates that the problem _____ .

Select the correct option

- must have a solution
- does not have a solution
- may have a solution
- non of these

Click to Save Ans

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MTH601:Quiz Quiz Sta

Question # 10 of 10 (Start time: 06:17:12 PM, 08 February 2021)

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- Two phase
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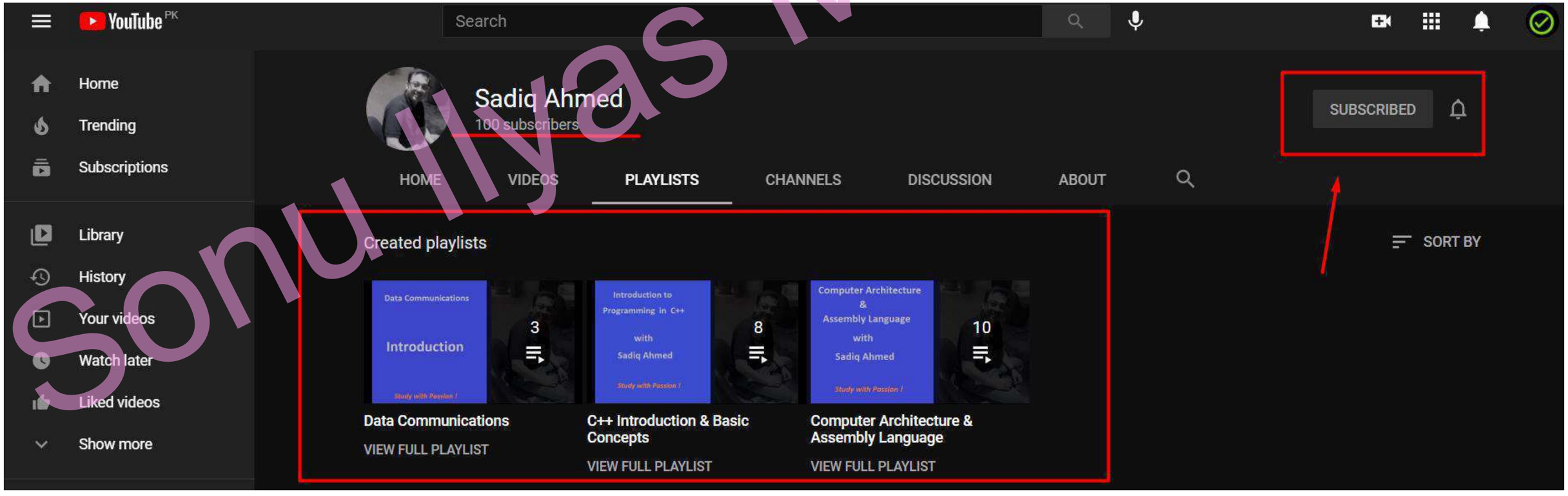

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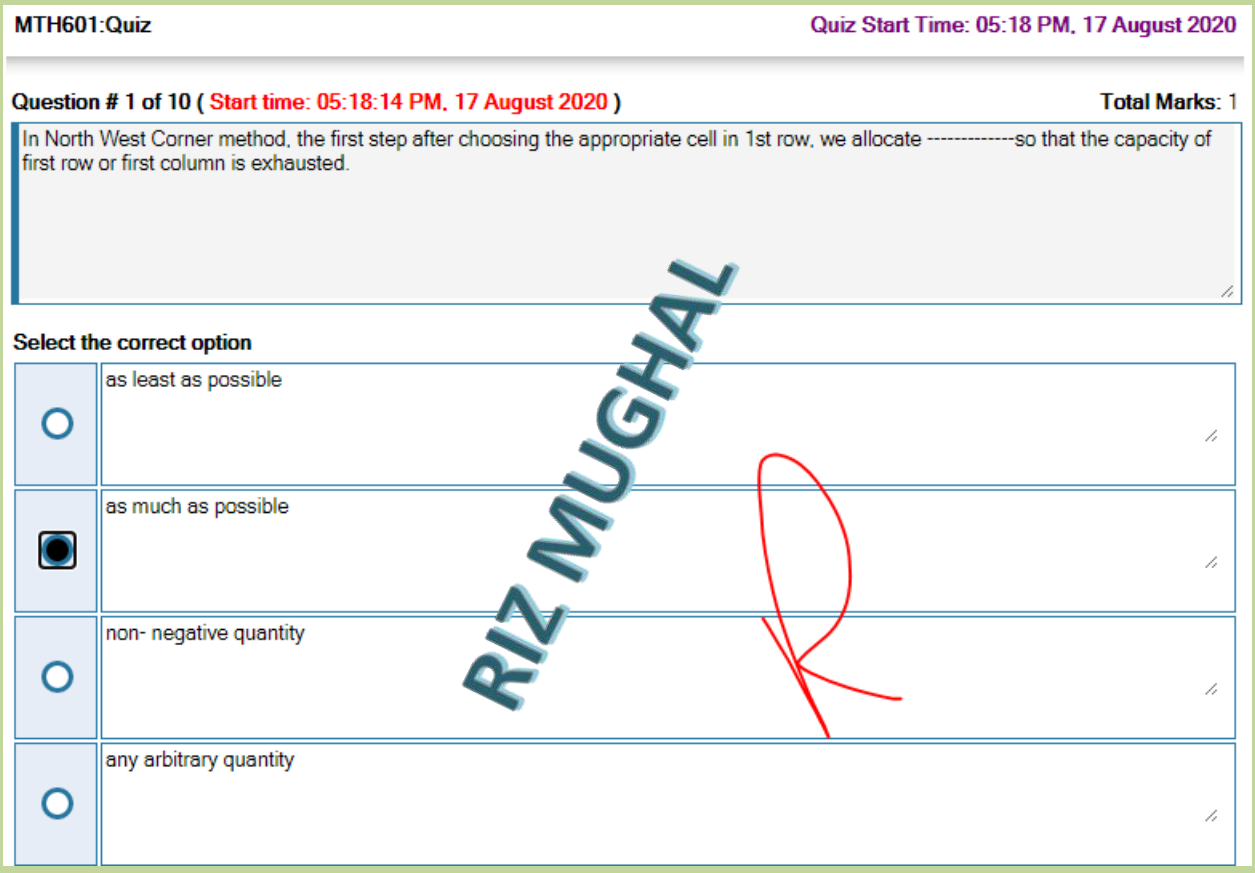
MTH601 Quiz

Solved by Rizwan Qadeer (Riz Mughal)
BSCS Completed
SQA Engineer
I can solve any quiz of any book from any Department
With 100% result
Guarantee marks

Youtube link:

https://www.youtube.com/channel/UCINsFwDiB62SValCcPDZbRQ/playlists



MTH601:Quiz Quiz Start Time:

Question # 2 of 10 (Start time: 05:18:41 PM, 17 August 2020)

In two phase method process, first phase _____ the sum of artificial variables.

Select the correct option

- (●) minimize
- ( ) maximize
- ( ) maximize or minimize depending on the situation
- ( ) non of these

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MTH601:Quiz

Quiz Start Time: 05:18 PM, 17 August 2020

Question # 3 of 10 (Start time: 05:19:05 PM, 17 August 2020) **Total Marks:**

If an LP problem contains large number of constraints and a smaller number of variables then which of the following will reduce the computational burden in finding its solution?

Select the correct option

- ○ M-method
- ○ Two phase method
- ○ Graphical method
- ◉ Duality principle

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MTH601:Quiz Quiz Start Time: 05:18 PM, 17

Question # 4 of 10 (Start time: 05:19:30 PM, 17 August 2020)

If initial basic solution is ----------, while solving an LP problem then no further iteration can be performed.

Select the correct option

- ○ degenerate
- ○ non-degenerate
- ○ feasible
- ◉ infeasible

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MTH601:Quiz Quiz Start Time: 05:18 PM

Question # 5 of 10 (Start time: 05:19:55 PM, 17 August 2020)

Which of the following difficulty may found while attempting an LP problem by M-method?

Select the correct option

- It often leads to infeasible solution
- Computational error due to large value of M
- Degeneracy is inevitable
- Artificial do not leave the basis

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MTH601:Quiz Quiz Start Time:

Question # 6 of 10 (Start time: 05:20:18 PM, 17 August 2020)

The cost coefficient of artificial variable in Objective function is ----------.

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Select the correct option

- 0
- M
- 1
- > than 1

MTH601:Quiz Quiz Start Time: 05:18 PM, 17 August 2020

Question # 7 of 10 (Start time: 05:20:38 PM, 17 August 2020) **Total Marks:** 1

While solving an LP problem by the Simplex method, in the standard table, the element at the intersection of key column and key row is called ---------- element.

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Select the correct option

- ○ Entering
- ○ Leaving
- ○ Slack
- ◉ Pivot

MTH601:Quiz Quiz Start Time: 05:18 PM, 17 August 2020

Question # 8 of 10 (Start time: 05:20:56 PM, 17 August 2020) **Total Marks:** 1

For finding the maximum profit in an enterprise of selling two products such that 'freezing' the sale of one product and keep selling the other. This scenario is studied under ----------.

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Select the correct option

- ○ Un-boundedness
- ○ Duality
- ◉ Degeneracy
- ○ Artificial variable technique

MTH601:Quiz

Quiz Start Time: 05:18

Question # 9 of 10 (Start time: 05:21:22 PM, 17 August 2020)

Which of the following will be an example of degenerate basic feasible solution for an LP problem?

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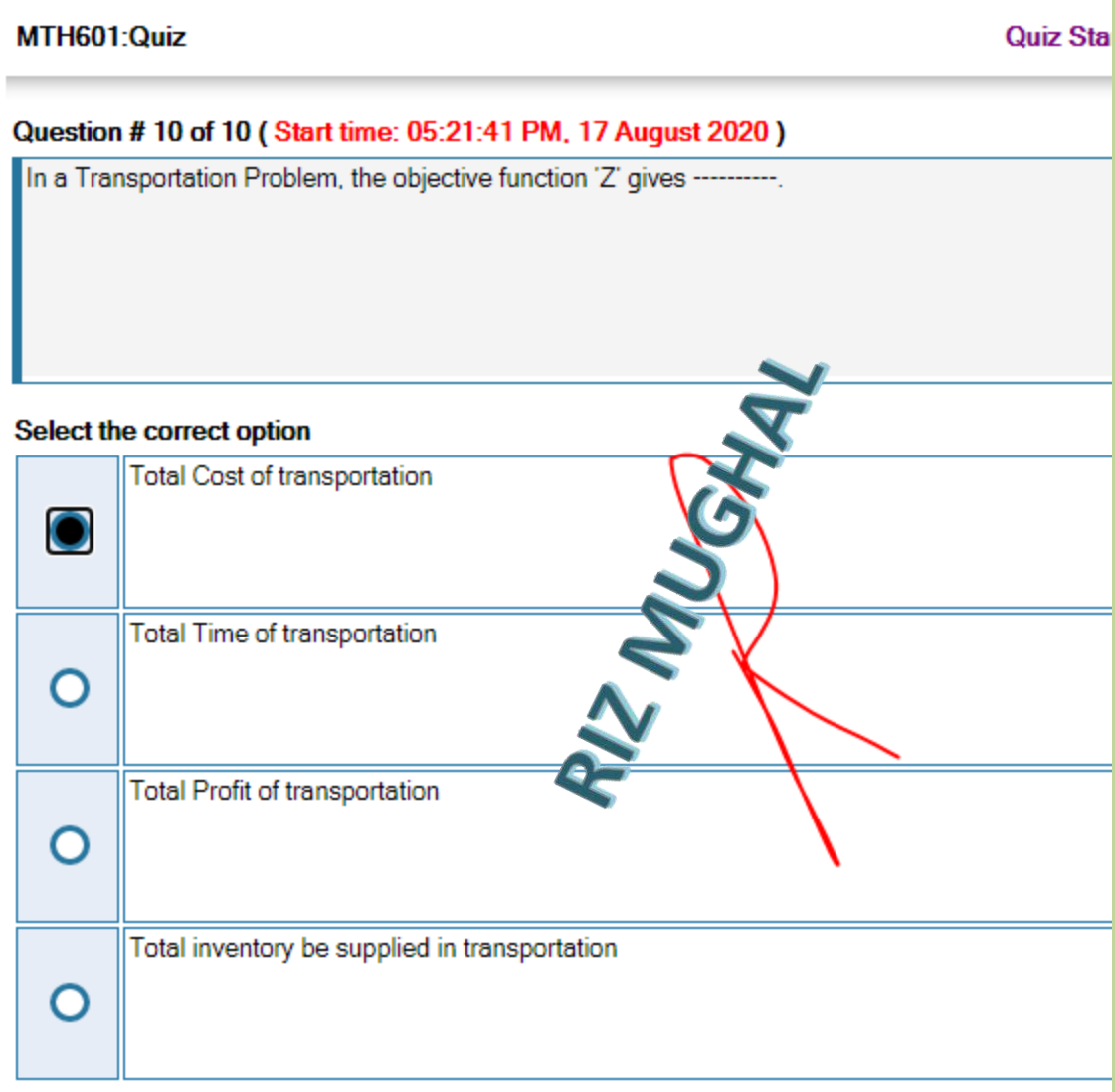
Select the correct option

- (2,3-1)
- (0,2,1)
- (2,1,3)
- (-1,2,0)


MTH601:Quiz
Quiz Sta
Question # 10 of 10 (Start time: 05:21:41 PM, 17 August 2020)
In a Transportation Problem, the objective function 'Z' gives ----------.
Select the correct option
Total Cost of transportation
Total Time of transportation
Total Profit of transportation
Total inventory be supplied in transportation
RIZ MUGHAL

2nd account

MTH601:Quiz

Question # 1 of 10 (Start time: 09:21:52 PM, 17 August 2020)

In a Transportation Problem, the objective function 'Z' gives ----------.

Select the correct option

- (●) Total Cost of transportation
- ( ) Total Time of transportation
- ( ) Total Profit of transportation
- ( ) Total inventory be supplied in transportation

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MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 Au

Question # 2 of 10 (Start time: 09:22:13 PM, 17 August 2020) Tota

If an LP problem contains large number of constraints and a smaller number of variables then which of the following will reduc computational burden in finding its solution?

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Select the correct option

- ○ M-method
- ○ Two phase method
- ○ Graphical method
- ◉ Duality principle

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August 2020

Question # 4 of 10 (Start time: 09:22:56 PM, 17 August 2020) **Total Marks:** 1

Which of the following order pair would minimize the objective function of the linear programming problem: $z = x + 5y$ subject to $x \geq 2, y \geq 0$?

Select the correct option

Reload Math Equations

- (2,3)
- (2,0) ◉
- (0,3)
- (0,0)

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MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August 202

Question # 5 of 10 (Start time: 09:23:19 PM, 17 August 2020) **Total Marks:**

Under which of the following condition to solve an LP by using two phase method, we can't proceed for 2nd phase?

Select the correct option

- Objective function of 1st phase has zero value.
- Objective function of 1st phase has positive value. ◉

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MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August

Question # 6 of 10 (Start time: 09:23:45 PM, 17 August 2020) **Total Ma**

A balanced transportation model with '5'number of sources and '7' destinations has -------- number of constraint equations.

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Select the correct option

- ○ 2
- ◉ 12
- ○ 35
- ○ 10

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 August 2020

Question # 7 of 10 (Start time: 09:24:06 PM, 17 August 2020) **Total Marks:** 1

By Simplex method, to maximize 'Z = 2x +9y'of an LP problem, if 'z=0' for the initial iteration then for its next improved solution, which of the following would be the next entering variable?

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Select the correct option

- ○ x>0
- ◉ y>0
- ○ x<0
- ○ y<0

MTH601:Quiz Quiz Start Time: 09:21 PM, 17 August 20

Question # 8 of 10 (Start time: 09:24:24 PM, 17 August 2020) **Total Marks**

For North West Corner method, in the first row and first column, resource and sink contain '5' and '7' units respectively; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?

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Select the correct option

- ○ (2,2)
- ○ (5,2)
- ○ (1,2)
- ◉ (2,1)

MTH601:Quiz

Quiz Start Time: 09:21 PM, 17 Augu

Question # 9 of 10 (Start time: 09:24:41 PM, 17 August 2020) **Total M**

By Simplex method, to minimize 'Z = 2x+9y'of an LP problem, if 'z=A>0' for the initial iteration then for its next improved solution(0<A<100), which of the following would be the next entering variable?

Select the correct option

- ○ x<0
- ○ y<0
- ◉ x>0
- ○ y>0

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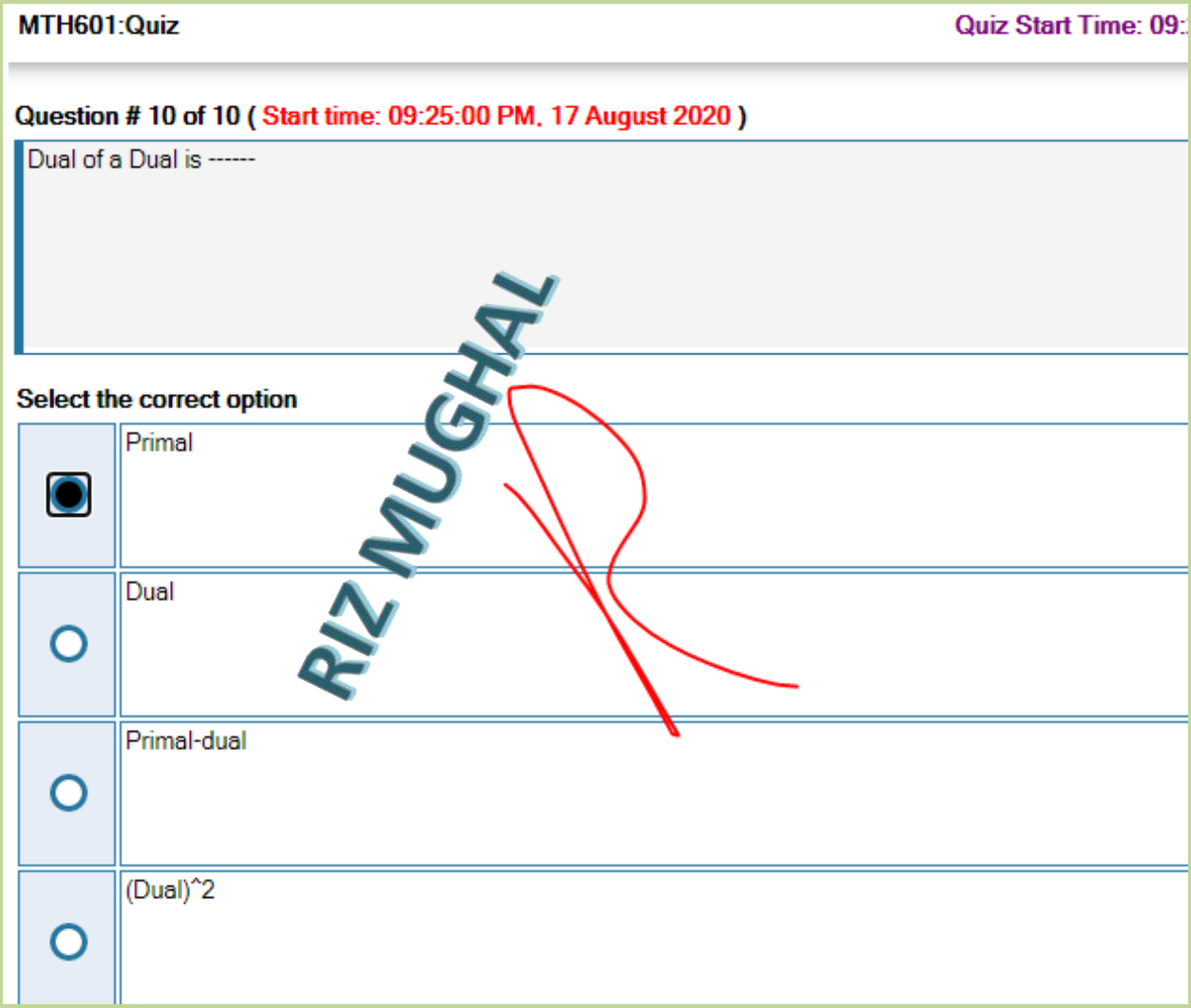
MTH601:Quiz
Quiz Start Time: 09:
Question # 10 of 10 (Start time: 09:25:00 PM, 17 August 2020)
Dual of a Dual is ------
Select the correct option
Primal
Dual
Primal-dual
(Dual)^2
RIZ MUGHAL

3rd account

MTH601:Quiz Quiz Start Time: 09:2

Question # 1 of 10 (Start time: 09:29:31 PM, 17 August 2020)

While solving an LP by two phase method, an objective function of 1st phase is always of---------.

Select the correct option

- maximization
- minimization
- dependent on original objective function
- none of these

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MTH601:Quiz Quiz Start Time: 09:29 PM, 17 August 20

Question # 2 of 10 (Start time: 09:29:52 PM, 17 August 2020) Total Mark

In two phase method if the minimum value of objective function in the first phase is greater than zero, then the solution of original problem _____.

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Select the correct option

- is uncertain
- exists
- does not exists
- non of these

MTH601:Quiz Quiz Start Time: 09:29 PM, 17 August 2020

Question # 3 of 10 (Start time: 09:30:14 PM, 17 August 2020) **Total Marks:** 1

In the Simplex method to solve an LP problem of minimization, if at the end of iteration, there is a positive coefficient in the objective row then the given problem ----------.

Select the correct option

- ○ can not be optimized
- ○ has no solution
- ○ has been maximized
- ◉ needs further improvement

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MTH601:Quiz Quiz Start Time: 09:29 PM, 17 August 2020

Question # 4 of 10 (Start time: 09:30:36 PM, 17 August 2020) **Total Marks:** 1

For an unbalanced Transportation problem, if the total demand is MORE than total supply then which of the following is true in order to balance the problem?

Select the correct option

- ○ One constraint will have evacuate
- ○ One constraint will have to add
- ○ A dummy sink would have to include with demand equal to the surplus
- ◉ A dummy source would have to include with supply equal to shortage

RIZ MUGHAL

MTH601:Quiz Quiz Start T

Question # 5 of 10 (Start time: 09:30:57 PM, 17 August 2020)

Shortcoming of Big M method is that the value of M could be _____.

Select the correct option

- very small
- very large
- negative
- positive

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MTH601:Quiz

Quiz Start Time: 09:29 PM, 17 August 202(

Question # 6 of 10 (Start time: 09:31:15 PM, 17 August 2020) **Total Marks:**

In Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column entries should be zero except-----.

Select the correct option

- ○ 1st row entry
- ◉ key row(pivot)entry
- ○ last row entry
- ○ row with maximum ratio on RHS column

RIZ MUGHAL

MTH601:Quiz

Quiz Start Time: 09:29 PM, 17 August 2020

Question # 7 of 10 (Start time: 09:31:35 PM, 17 August 2020) **Total Marks:** 1

For an unbalanced Transportation problem, if the total demand is LESS than total supply then which of the following is true in order to balance the problem?

Select the correct option

- ◉ A dummy sink would have to include with demand equal to the surplus
- ○ A dummy source would have to include with supply equal to sh[illegible]
- ○ One constraint will have evacuate
- ○ One constraint will have to add

RIZ MUGHAL

MTH601:Quiz

Quiz Start Time: 09:29 PM, 17 August 2020

Question # 8 of 10 (Start time: 09:31:56 PM, 17 August 2020) **Total Marks:**

In the Simplex method to solve an LP problem of maximization, if at the end of iteration, every entry of objective function row is positive then the given problem----------.

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Select the correct option

- needs further improvement
- has been maximized
- can not be optimized
- has no solution

MTH601:Quiz Quiz S

Question # 9 of 10 (Start time: 09:32:16 PM, 17 August 2020)

In the initial iteration of Big M-method, the artificial variables appear in ---------.

Select the correct option

- (●) Basis
- ( ) Non-basic variables' set

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MTH601:Quiz

Quiz Start Time: 09:29 PM, 17 August 2(

Question # 10 of 10 (Start time: 09:32:46 PM, 17 August 2020) **Total Mark**

In two phase method, for the phase-I, if the objective function in terms of artificial variables is not minimized then the given problem has----------.

Select the correct option

- ○ Feasible solution
- ○ Optimal solution
- ◉ Infeasible solution
- ○ Degeneracy

RIZ MUGHAL

MTH601 VUSS SOLVED WITH REFERENCE

1.

Destinations					
	S. No.	**D1**	**D2**	**D3**	**Supply**
Source	S1	5	1	2	**25**
	S2	7	2	4	**10**
	S3	3 10	3	5	**15**
	Demand	**10**	**20**	**20**	

Using Vogel's Approximation Method:
These below are the three cells which can be next cell:

I. S3 → D2
II. S2 → D2
III. S1 → D3

Which one is the correct option?

a. I, II and III
b. I only
c. II only
d. III only
e. I and II only

2. If a basic feasible solution contains less than "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

a. Degenerate
b. Multiple Solutions
c. Non degenerate
d. Alternative Optima

3. In the big-M method, if the introduced _____ variables do not leave the basis in the final iteration, then this indicates that the give linear programming problem cant be optimized.

a. entering
b. positive slack
c. negative slack
d. artificial

4. If a company manufacture ‘x’ units of product ‘A’ and ‘y’ units of ‘B’ with associated profits of Rs.5 and Rs.3 then which of the following is the objective function to maximize is the profit?

a. $z = 15xy$
b. $z = 5x – 3y$
c. $z = 3x – 5y$
d. $z = 5x + 3y$

5. Any set of non negative allocation ($Xij>0$) which satisfy the row and column sum is called a _____ solution

a. feasible
b. non basic feasible
c. basic feasible
d. optimal

6. The inequality $2x + 3y \geq 18$ is equivalent to

a. $2x + 3y \leq -18$
b. $- 2x – 3y \geq 18$
c. $- 2x – 3y \geq -18$
d. $- 2x – 3y \leq -18$

7. If the mean arrival and mean service rates are 4 and 7 respectively in a queue then expected waiting time in the system is

a. 1/3

b. 3

c. 28

d. 7/4

8. The cost matrix in assignment problem is always

a. identity matrix

b. zero matrix

c. diagonal matrix

d. square matrix

9. For the linear programming problem;

MaxZ = 2x + 3y

Subject to

$x \geq 2$ } $x - s_1 + A = 2$ }

$y \leq 3$ } $\rightarrow y + s_2 = 3$ }

$x,y \geq 0$} $x,y,s1,s2,A \geq 0$ }

Which of the following is associated objective function of the 1st phase?

a. MaxZ = 2x + 3y + A
b. MinZ = 2x + 3y + A
c. MaxZ = A
d. MinZ = A

10. Standard Deviation (S.D.) = _____MAD

a. $\sqrt{(2\pi/3)}$
b. $\sqrt{(2/\pi)}$
c. $\sqrt{(\pi/2)}$
d. $\sqrt{(3\pi/2)}$

11. If ”Ni” be the Number of replacement made at the end of the i^{th}week and “Pj” be the probability of failure during the i^{th} week, then N_1 = _____.

a. N_0P_1
b. N_1P_1
c. N_0P_0
d. N_1P_2

12. Product of’item cost’ and ‘ordered item’ is

a. Crash cost
b. Cost period
c. Set up cost
d. Shortage cost

13. In PERT, activity time estimates are distributed according to

a. Beta Distribution
b. Normal Distribution
c. Poisson distribution
d. Binomial Distribution

14. For any activity backward pass computations provide its

a. Earliest start times
b. Latest start times
c. Moderate start times
d. Completion time

15. The present worth of a rupee to be spent after a year is denoted by v and given by

a. v= (1 + r)
b. v= (1 / r)
c. v = (1 + r) / 10
d. v = 1/ (1 + r)

16. Formula for a geometric series “$x + vx + v^2x + ….. + v^{n-1}x$” is

a. $P(n) = x (1-v^n)/(1-v)$
b. $P(n) = (1-v^n)/(1-v)$
c. $P(n) = x (1-v)/(1-v^n)$
d. $P(n) = (1-v)/(1-v^n)$

17. Zero valued artificial variables may appear as _____ variables in the final solution, when one or more of the original constraints equations is redundant.

a. non basic
b. basic
c. slacks
d. surplus
e. artificial

18. MAD = _____S.D.

a. $\sqrt{(2/\pi)}$
b. $\sqrt{(\pi/2)}$
c. $\sqrt{(2\pi/3)}$
e. $\sqrt{(3\pi/2)}$

19. To convert the transportation problem into a maximization model we have to _____.

a. write the inverse of the coefficient matrix
b. multiply the feasibility condition by -1
c. multiply the coefficient matrix by -1
d. We can't convert the transportation problem into a maximization problem, as it is basically a minimization problem.

20. We go in probabilistic replacement model when period between installation and failure is

a. varying exponentially
b. varying linearly
c. Constant
d. is not constant (varying arbitrarily)

21. Which of the following binary operation in assignment problem among all the elements in the given profit matrix from the highest element in the matrix

a. Subtraction
b. Division
c. Multiplication
d. Addition

22. Best possible time estimate that a given activity would take under normal conditions which often exist, is called

a. Most Likely time estimate
b. Pessimistic time estimate
c. Smallest time estimate
d. Activity time estimate

23. If the money carries a rate of interest of 12% per year, the present worth factor of one rupee due in one year is

a. 0.08333
b. 0.89285
c. 0.0769
d. 13

24. In which of the following models, Simplex algorithm is not preferred to use due to laborious computations?

a. Transportations models
b. Degenerate Linear models
c. Non-degenerate Linear models
d. Dual or unbounded linear models

25.A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the mean service time "μ" is

a. ½ = 0.5
b. 4
c. ¼ = 0.25
d. 2

26. In sequencing problem, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machine problems

a. 3n
b. 3n!
c. 2*3=6
d. 2

27. During a replacement if the value of money decreases at the rate of 3% then the present worth factor of unit amount to be spent after one year is given by

a. 0.25
b. 0.333
c. 0.9708
d. 4

28. Consider the following cost table:

Destinations					
		D1	**D2**	**D3**	**Supply**
Source	**S1**	5	1	2	25
	S2	7	2	4	10
	S3	3	3	5	15
	Demand	10	20	20	

a. 10
b. 15
c. 25
d. 5

29.A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the average rate "λ" is

a. 1/8 = 0.125
b. 8
c. 4
d. ¼ = 0.25

30. Degeneracy in a 5x6 transportation problem occurs when the number of occupied cell is less than

a. 20 but greater than 10
b. 10
c. Zero
d. Infinity

31. The _____ variable is chosen by examining the cost coefficients in the objective function.

a. entering
b. leaving
c. positive slack
d. negative slack

32. Transportation models consist of _____ like the production centers and _____ which may be the sales centers.

a. (sinks, sources)
b. (sources, sinks)
c. (origins, sources)
d. (sinks, destinations)

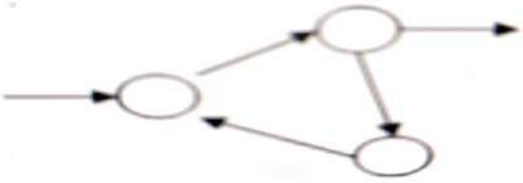
33. In a bank, every 15 minutes one customer arrives for cashing the cheque. The staff in the payment counter takes only 10 minutes for serving a customer on an average, then the service rate " μ" = _____

a. 6 per hour
b. 4 per hour
c. 10 per hour
d. 1/6 per hour

34. Solution region for the constraint $y \leq 0$ is the

a. Half plane below y-axis
b. Half plane below the line: y = 0
c. Set of all those points where ordinates are non-positive
d. All are equivalent

35. The following network is an example of



a. Redundancy
b. Dangling
c. Cycling
d. Dummy

36. In a linear Programming Problem (LPP), which of the following must be hold?

a. Only objective function is linear
b. Both objective function and constraints are linear
c. Only constraints needs to be linear
d. At least one of the objective function or constraint should be linear

37. While solving a linear programming problem by big M – Method, traditionally the _____ variables are chosen in the initial basic feasible solution.

a. neagtive slack
b. positive slack
c. entering
d. artificial

38. By using Two phase method to solve a linear programming problem, in phase I, a new objective function is formed by assigning on left hand side, zero to every original variable (including slack and surplus variables) and ____ to each of the artificial variables.

a. M
b. –M
c. +1
d. -1

39. Under the individual replacement policy an item is replaced on its

a. Success
b. **Failure**
c. Maximum utilization
d. Minimum utilization

40. The name of the method in getting the optimum assignment is

a. Simplex method
b. North-West corner method
c. Vogel's approximation method
d. Hungarian method

41. Transportation technique or the simplex method cannot be used to solve the assignment problem because of

a. Degeneracy
b. Involvement of dummy activities
c. Looping among the entries of assignment matrix
d. Inequilibrium between demand and supply

42. The representation of reality in some physical form or in some form of mathematical equation may be discussed under the topic entitled as

a. **Transportation problems**
b. Game theory
c. Simulation
d. Replacement problems

43. Search and cutting methods are used to solve following types of problems

a. Transportation

b. Assignment

c. **Integer Programming**

d. Queuing problems

44. CPM can help you figure out except:

A. How long your complex project will take to complete

B. Which activities are “critical” meaning that they have to be done on time or else the whole project will take longer

C. You can try to speed up the project

D. What is the most costly way to speed up the project

45. The Vogel Approximation method is an iterative procedure for computing a _____ solution of the transportation problem

a. Optimal

b. Basic feasible

c. Non basic feasible

d. Basic infeasible

46. In assignment problem, each job requires exactly one resource.

a. True

b. False

c. May not be true

47. If “Ni” be the Number of replacement made at the end of the i^{th} week and “Pj” be the probability of failure during the i^{th} week, then N_2 = ----------.

a. $N_1P_2 + N_1P_1$

b. $N_0P_2 + N_1P_1$

c. $N_0P_1 + N_1P_2$

d. $N_0P_2 + N_1P_2$

48. In replacement Models, Capital Cost = (Item / Machine Costs) – (----------)

a. Resale Cost

b. Running Cost

c. Maintenance Cost

d. Operating Cost

49. In replacement Models, Capital Cost = _____ - Resale cost

a. (Item / Machine Costs)

b. Running Cost

c. Maintenance Cost

d. Operating Cost

50. Let FS = Free Slack, TS = Total Slack, INDS = Independent Slack, then which relation is true

- TS ≤ FS
- **INDS ≤ FS**
- **FS ≤ TS**

Then which relation is true

a. Only I

b. Only II

c. Only III

d. Only I & II

e. **Only II & III** (as free slack can never exceed total slack)

51. Which of the following give the excess of available time over the activity time when all jobs start as early as possible?

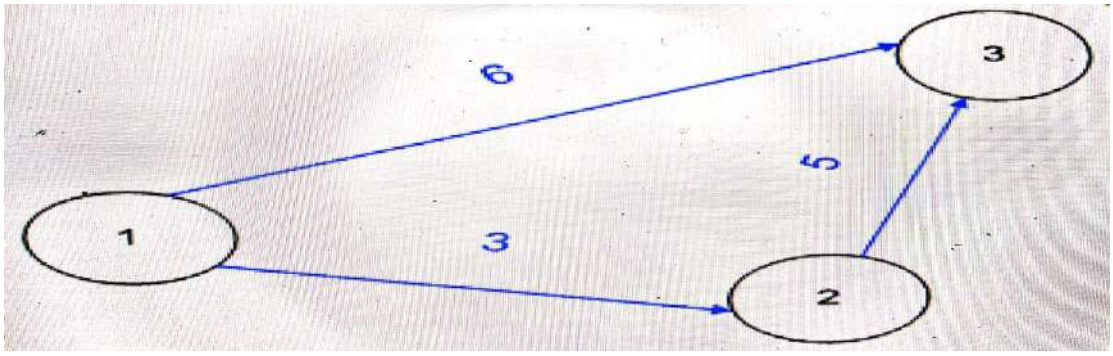
a. Total float

b. Free float

c. Independent float

d. Early finish time

52. Early finish time of the event ‘3’ in the following project is ----------.



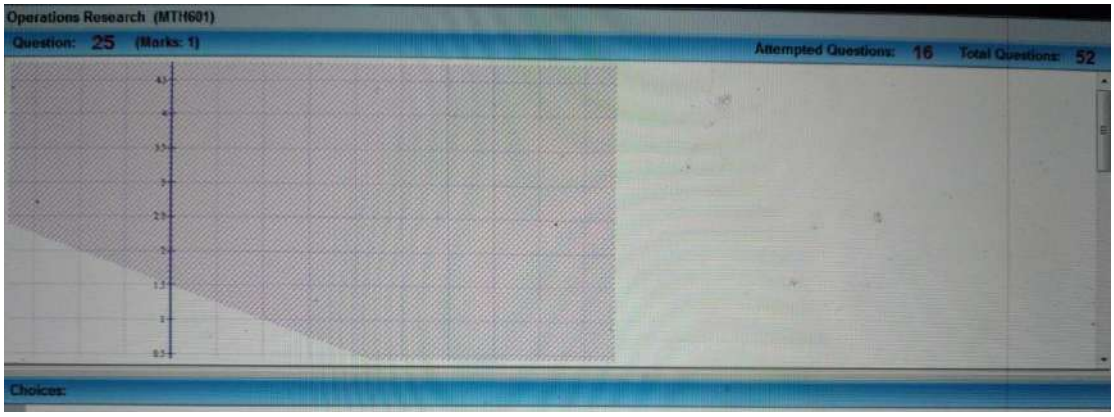
a. 14 days

b. 2 days

c. 8 days

d. 6 days

53. **(Important Note this question is not complete this is overseas student mcq)**



a. $x + 2y \leq 3$

b. $x + 2y \geq 3$

c. $x - 2y \leq 3$

d. $x + 2y \geq 3$

54. If shortage cost is infinity, then

a. No shortages are allowed

b. No inventory carrying cost is allowed

c. Ordering cost is zero

d. Purchase cost = Carrying cost

55. It is a property of simplex method that there always exist infinite number of basic feasible solution.

a. True

b. False

56. If the objective function of a linear programming problem needs further improvement then which of the following will have to proceed?

a. A new decision variable to be entered and other decision variable to leave the basis

b. A new slack variable to be entered and other slack to leave the basis

c. A new non-degenerate variable to be entered and other no-degenerate to leave the basis

d. A new basic variable to be included and other basic variable to leave the basis

57. The constrained $3x_1+2x_2 \leq 18$ in standard form can be as:
(Where R is artificial variable and S_1 is slack variable)

a. $3x_1+2x_2+S_1+R=18$

b. $3x_1+2x_2+S_1-R=18$

c. $3x_1+2x_2-S_1+R=18$

d. $3x_1+2x_2+S_1=18$

58. If no _____ variable appears in the basis and the optimality conditions are satisfied, then the current solution will be optimal one.

a. artificial

b. non basic

c. slack

d. surplus

59. Min. Z = 0 and no artificial variable appears in the basic variables _____ solution to original problem has been found and we proceed to phase II method.

a. A basic feasible

b. An optimal

c. An infeasible

60. If the problem has a_____ solution such that the minimum value of the objective function is zero along with zero values of artificial variables. Then proceed to phase II method.

a. Unbounded

b. Feasible

c. Non feasible

d. De-generate feasible solution

61. If the degeneracy arises in the initial stage, then which of the following is true?

a. One of the basic variables is zero

b. More than one variable is eligible to leave the basis

c. Tie the ratio of each row, by taking right hand side of each row and dividing by the corresponding element of the key column.

d. Iterative Cycling of basic solution exists without reaching the optimality

62. For a linear programming problem, the Unboundedness is related to which of the following?

a. Solution space only

b. Objective value only

c. Neither solution space nor to objective value

d. Both solution space and objective value

63. Consider the following simplex table corresponding to maximization problem

Basic	x_1	x_2	x_3	S_1	S_2	S_3	R.H.S
Z	-4	-3	-2	0	0	0	0
x_1	0	0	1	1	0	0	8
S_2	-2	1	3	0	1	0	9
S_3	0	2	1	0	0	1	3

Then this is the indication that

a. Problem has Unbounded Solutions

b. Problem has Infeasible Solutions

c. Problem has temporarily degenerate

d. Problem has Multiple Optimal Solutions

64. Consider the following simplex table corresponding to maximization problem

Basic	x_1	x_2	x_3	x_4	A_1	A_2	R.H.S
Z	0	0	M + 1/2	M	16M - 11	0	-0.6M – 5.7
x_1	1	0	5	0	-1	0	3
x_2	0	1	-5	0	3	0	9
A_2	0	2	-1	-1	-0.6	1	0.6

Then this is the indication that problem has _____

a. infeasible Solution

b. temporarily Degenerate Solution

c. unbounded Solution

d. degenerate Solution

65. In the Simplex table, if the coefficient of non-basic variable says x_1 in the z-row is zero, then it indicates that value of z _____.

a. Increase

b. Decreases

c. Does not change

d. Becomes infinity

66. A primal's constraint equation produces _____ dual variable.

a. Non basic

b. An unrestricted

c. Artificial

d. Basic

67. Transportation problem is a special class of linear programming problem in which transport a commodity from the source to a destination in such a way that the total transportation cost is _____.

a. Zero

b. Maximum

c. Minimum

d. Average

68. Many decision making problems involve a process that takes place in multiple stage in such a way that at each stage, the process is dependent on the strategy chosen. Such types or problems are called.

a. Linear Programming problems

b. Integer programming problems

c. Dynamic programming problems

d. Assignment problems

69. A problem in which the decisions have to be made at successive stages, is called a multistage decision problem.

a. True (From Book Page no 390)

b. False

5. Transportation model was given and one block has x we have to find the value of x..... 3 marks

6. Values were given and we need to tell the capital cost for each year ..... 3 marks

7. In the context of pert and CPM summarize the project planning techniques

8. One stage problem is given find the two stage problem

s	Fi*(s)	Xi
8	25	10
9	35	10

9. Table was given and asked that check initial solution is feasible or not..... 5 marks

10. Question no 1 was again appeared as 5 marks question

11. The cost of the new machine is 5000. Maintenance during the nth year is given by Mn = 500 rps (n-1) when n = 1, 2, 3 ....if discount rate per year is 0.05 calculate the present worth ..... 5 marks

12. Graph was given and question was construct the table relation show between events and activities. 5 mark s

FINALTERM EXAMINATION

Fall 2012

MTH601 - Operations Research

Ref No: 3886822

Time: 120 min

Question No: 1 (Marks: 1) - Please choose one

Which of the following is a characteristic of critical path scheduling techniques?

▶The jobs or tasks are dependent on each other

▶Interruptions may be permitted if they are documented

Selected answer The jobs or tasks must be well defined, and their completion marks the end of the project

▶The jobs or tasks do not necessarily have to be in a predetermined sequence

Question No: 2 (Marks: 1) - Please choose one

During a replacement if the value of money decreases at the rate of **3**% then the present worth factor of unit amount to be spent after one year is given by

▶0.25

▶0.333

Selected answer 0.9708

▶4

Question No: 3 (Marks: 1) - Please choose one

If the money carries a rate of interest of **12**% per year, the present worth factor of one rupee due in one year is

Selected answer 0.08333

▶0.89285

▶0.0769

▶13

Question No: 4 (Marks: 1) - Please choose one

Search and cutting methods are used to solve following types of problems

▶Transportation

▶Assignment

Selected answer Integer Programming

▶Queuing problems

Question No: 5 (Marks: 1) - Please choose one

A feasible solution is called a _____ solution if the number of non negative allocation is equal to “m + n – 1” (Here m is the number of rows, n is the number of columns in transportation problem)

Selected answer basic feasible

▶non basic feasible

▶basic infeasible

▶optimal

Question No: 6 (Marks: 1) - Please choose one

If a basic feasible solution contains less than “m + n – 1” (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

▶Degenerate

▶Multiple Solutions

▶Non degenerate

Selected answer Alternative Optima

Question No: 7 (Marks: 1) - Please choose one

Consider the following cost table

	Destinations				
Source		**D1**	**D2**	**D3**	**Supply**
	S1	5	1	2	**25**
	S2	7	2	4	**10**
	S3	3	3	5	**15**
	Demand	**10**	**20**	**20**	

Using Vogel's Approximation Method, which one is the starting cell?

Selected answer S1 à D1

▶ S3 à D1

▶ S2 à D2

► S2 à D1

Question No: 8 (Marks: 1) - Please choose one

		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10
	S3	3 10	3	5	15
	Demand	10	20	20	

Using Vogel's Approximation Method:
These below are the three cells which can be next cell:

I. S3 → D2
II. S2 → D2
III. S1 → D3

Which one is the correct option?

►I, II and III

►I only

►II only

Selected answer III only

▶I and II only

Question No: 9 (Marks: 1) - Please choose one

	D1	D2	D3	Supply
S1	5 10	1 15	2	25
S2	7	2 5	4 5	10
S3	3	3	5 15	15
Demand	10	20	20	

Cost = 10 (5) + 15 (1) + 5 (2) + 5 (4) + 15 (5) = 170

This is the initial basic feasible solution of a question by North – West Corner Method.

Finding optimal basic feasible solution: Consider U1 = 0 and V1 = 5, what is the value of V2?

▶

▶

▶

Selected answer

Question No: 10 (Marks: 1) - Please choose one

Transportation problem is a special class of linear programming problem in which transport a commodity from the source to a destination in such a way that the total transportation cost is

▶Zero

▶Maximum

Selected answer Minimum

▶Average

Question No: 11 (Marks: 1) - Please choose one

We solve a assignment problem by subtracting least number in each row and we get the table below:

Cost Table	Dist Center D	Dist Center E
Plant A	0	135
Plant B	0	8
Plant C	34	0

After doing next step, which table should be next?

Cost Table	Dist Center D	Dist Center E
Plant A	0	135
Plant B	0	8
Plant C	34	0

▶

Cost Table	Dist Center D	Dist Center E
Plant A	0	127
Plant B	0	0
Plant C	0	0

▶

Cost Table	Dist Center D	Dist Center E
Plant A	135	0
Plant B	8	0
Plant C	0	34

▶

Cost Table	Dist Center D	Dist Center E
Plant A	0	0
Plant B	34	8
Plant C	0	135

Selected answer

Question No: 12 (Marks: 1) - Please choose one

The cost matrix in assignment problem is always

Selected answer identity matrix

▶zero matrix

▶diagonal matrix

▶square matrix

Question No: 13 (Marks: 1) - Please choose one

If the arrival rate is 5 per hour and the service rate is 10 per hour, then the average system time is:

▶1/5

Selected answer 5

▶1

▶2

Question No: 14 (Marks: 1) - Please choose one

If the arrival rate is 5 per hour and the service rate is 10 per hour, then ρ (traffic intensity or system utilization) =?

▶□□□

Selected answer □□□

▶□□□

▶□□□

Question No: 15 (Marks: 1) - Please choose one

Formula for the expected number in the system L is:

$$L=\frac{\lambda}{\mu-\lambda}$$

▶

$$L=\frac{\lambda}{\mu+\lambda}$$

▶

$$L=\frac{\lambda}{\lambda-\mu}$$

▶

$$L=\frac{\mu}{\mu-\lambda}$$

Selected answer

Question No: 16 (Marks: 1) - Please choose one

If the arrival rate is 5 per hour and the service rate is 10 per hour, then the expected number in the queue or average queue length is:

▶

Selected answer

▶

▶

Question No: 17 (Marks: 1) - Please choose one

Total cost per period = Item cost + Order cost + Holding cost + _____.

Selected answer Shortage cost

▶Optimum Shortage (S*)

▶Economic Oreder Quantity. (Q*)

▶Maximum Inventory. ($I_{max.}$)

Question No: 18 (Marks: 1) - Please choose one

In Manufacturing Model with no shortage, the replacement rate is finite and _____ the demand rate.

▶greater than

Selected answer less than

▶equal to

Question No: 19 (Marks: 1) - Please choose one

If "N*i*" be the Number of replacement made at the end of the i^{th} week and "P*j*" be the probability of failure during the i^{th} week, then N_1 = ----------.

▶N_0P_1

Selected answer N_1P_1

▶N_0P_0

▶N_1P_2

Question No: 20 (Marks: 1) - Please choose one

If "N*i*" be the Number of replacement made at the end of the i^{th} week and "P*j*" be the probability of failure during the i^{th} week, then N_2 = ----------.

▶$N_1P_2 + N_1P_1$

▶$N_0P_2 + N_1P_1$

▶$N_0P_1 + N_1P_2$

▶$N_0P_2 + N_1P_2$

Question No: 21 (Marks: 1) - Please choose one

The present worth of a rupee to be spent after a year is denoted by *v* and given by

- ▶*v* = (1 + *r*)
- ▶*v* = (1 / *r*)
- ▶*v* = (1 + *r*) / 10

Selected answer *v* = 1/ (1 + *r*)

Question No: 22 (Marks: 1) - Please choose one

Formula for a geometric series "$x + vx + v^2x + ... + v^{n-1}x$" is

$$P(n) = \frac{1-v^n}{1-v}$$

Selected answer

- ▶ $P(n) = x\frac{1-v}{1-v^n}$
- ▶ $P(n) = \frac{1-v}{1-v^n}$
- ▶ $P(n) = x\ \frac{1-v^n}{1-v}$

Question No: 23 (Marks: 1) - Please choose one

In replacement Models, Capital Cost = (Item / Machine Cots) – (----------).

▶Resale Cost

▶Running Cost

Selected answer Maintenance Cost

▶Operating Cost

Question No: 24 (Marks: 1) - Please choose one

Formula of EOQ in manufacturing model with shortages is

▶ $Q^* = \sqrt{2C_2D/C_3(1-D/R)}.\sqrt{C_3+C_4/C_4}$

$Q^* = \sqrt{2C_2D/C_3}.\sqrt{R/R-D}$

Selected answer

▶ $Q^* = \sqrt{2C_2D/C_3}$

▶Non of them

Question No: 25 (Marks: 1) - Please choose one

To identify and maintain the proper precedence relationship between activities those are not connected by event, we introduce

▶Parallel Activity

Selected answer Dummy Activity

▶Sequential Activity

▶None of these

Question No: 26 (Marks: 1) - Please choose one

PERT is based on

▶Deterministic times

Selected answer Probabilistic Times

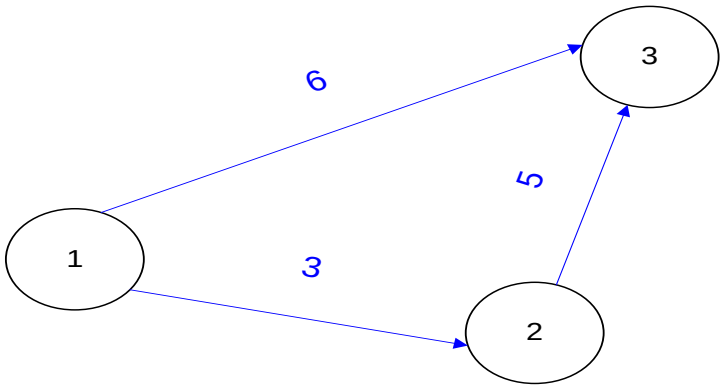
▶Dummy Times

▶Estimated times

Question No: 27 (Marks: 1) - Please choose one

Non-critical activities in the following network are -------

www.ranapk.com



▶Activity (1,2)

▶Activity (2,3)

Selected answer Dummy Actvity

▶Activity (1,3)

Question No: 28 (Marks: 1) - Please choose one

If $t_0 = 10$, $t_m = 10$ and $t_p = 16$, then *S.D* = _____.

▶

▶

▶

Selected answer

Question No: 29 (Marks: 1) - Please choose one

Which of the following is an essential characteristic of a linear programming problem?

▶The relationship between variables and constraints must be linear

▶The model must have an objective function

▶The model must have structural constraints

Selected answer All are equivalent

Question No: 30 (Marks: 1) - Please choose one

In the simplex table for a linear programming problem, we select the leaving basic variable corresponds to ----------

▶Maximum non-negative ratio

▶Maximum negative ratio

Selected answer Minimum non-negative ratio

▶Minimum negative ratio

Question No: 31 (Marks: 1) - Please choose one

A technique for solving a linear programming problems in which artificial variables are included with coefficients of very large number say equal 10 times of any cost coefficient of decision variables is known as ---------

▶Big M – Method.

▶Least Cost Method

♦♏●♏♍♦♏♎ ♋■♦♦♏❒ Hungarian Method

🕓 Branch and Bound Method

Question No: 32 (Marks: 1) - Please choose one

If the _____ variables appear in the final basic set, then the LP (Linear Program) has no solution.

▶slack

▶surplus

▶non – basic

Selected answer artificial

Question No: 33 (Marks: 1) - Please choose one

If no _____ variable appears in the basis and the optimality conditions are satisfied, then the current solution will be optimal one.

▶artificial

Selected answer non basic

▶slack

▶surplus

Question No: 34 (Marks: 1) - Please choose one

Min. Z = 0 and no artificial variable appears in the basic variables. _____ solution to original problem has been found and we proceed to phase II method.

▶A basic feasible

▶An optimal

Selected answer An infeasible

Question No: 35 (Marks: 1) - Please choose one

In Two – Phase method, if all the _____ variables become zero, we stop the phase I and proceed to phase-II.

▶artificial

Selected answer basic

▶non basic

▶positive slack

▶negative slack

Question No: 36 (Marks: 1) - Please choose one

The inequality $2x+3y\leq 18$ is equivalent to

▶ $2x+3y\leq -18$

Selected answer $-2x-3y\geq 18$

▶ $-2x-3y\geq -18$

▶ $-2x-3y\leq -18$

Question No: 37 (Marks: 1) - Please choose one

Which of the following order pair would minimize the objective function of the linear programming problem; z = x + y subject to x≥3, y≥0 ?

►(3,3)

Selected answer (3,0)

►(0,3)

►(0,0)

Question No: 38 (Marks: 1) - Please choose one

In which of the following models, Simplex algorithm is not preferred to use due to laborious computations?

►Transportations models

►Degenerate Linear models

►Non-degenerate Linear models

Selected answer Dual or unbounded linear models

Question No: 39 (Marks: 1) - Please choose one

Transportations models consist of ------- like the production centers and --------- which may be the sales centers.

►(sinks, sources)

►(sources, sinks)

Selected answer (origins, sources)

►(sinks, destinations)

Question No: 40 (Marks: 1) - Please choose one

To convert the transportation problem into a maximization model we have to----------.

►write the inverse of the coefficient matrix

►multiply the feasibility condition by –1

►multiply the coefficient matrix by –1

Selected answer We can't convert the transportation problem into a maximization problem, as it is basically a minimization problem.

Muhammad Rizwan Ashraf
BS(CS)
Virtual University of Pakistan
MTH 601 Final term Data

Note:- Please confirm the answers by your own too
I made it with a lot of hard work as there was not much data on web for this subject
Please duaon main yad rakhiay ga

While solving an LP problem by the Simplex method, in the standard table, the element at the intersection of key column and key row is called ---------- element.
Entering
Leaving
Slack
Pivot

In the initial table of Simplex method, the objective function should not contain the terms involving----------variables
Artificial
Degenerate
Basic
Non basic

If a balanced Transportation problem with 'm' sources and 'n' sinks then degeneracy will arise only if there are less than ---------- independent allocations in the solution.
m–n–1
m+n-1
(In this question all options are in (-) negative range like m–n–1 , m–n–2 m–n–3, m–n–4) and there was no one true. correct was m+n–1 .however i select m–n–1).

Which of the following type of Elementary matrix operations are performed while solving an Assignment problem by Hungarian's method?
Row operations
Column operations
Both of the above
None of the above
Check Not sure

While solving an Assignment problem by Hungarian's method, in the modified cost matrix if the minimum number of horizontal and vertical lines to cover zeros are not equal to the number of rows (or columns), then which of the following operation is done?
Subtract smallest element of uncovered rows from all other elements of uncovered cells

Which of the following is a characteristic of critical path scheduling techniques?
The jobs or tasks are dependent on each other

Made By Rizwan Ashraf

Interruptions may be permitted if they are documented
The jobs or tasks must be well defined, and their completion marks the end of the project
The jobs or tasks do not necessarily have to be in a predetermined sequence

Search and cutting methods are used to solve following types of problems
Transportation
Assignment
Integer Programming
Queuing problems

A feasible solution is called a _____ solution if the number of non negative allocation is equal to "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem)
basic feasible
non basic feasible
basic infeasible
optimal

Consider the following cost table

		Destinations			
Source		**D1**	**D2**	**D3**	**Supply**
	S1	5	1	2	**25**
	S2	7	2	4	**10**
	S3	3	3	5	**15**
	Demand	**10**	**20**	**20**	

Using Vogel's Approximation Method, which one is the starting cell?
S1 à D1
S3 à D1
S2 à D2
S2 à D1

Using Vogel's Approximation method:
These below are the three cells which can be next cell:
Which one is the correct option?

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		Destinations			
		D1	D2	D3	Supply
Source	S1	5	1	2	25
	S2	7	2	4	10
	S3	3 10	3	5	15
	Demand	10	20	20	

Using Vogel's Approximation Method:
These below are the three cells which can be next cell:

I. S3 → D2
II. S2 → D2
III. S1 → D3

Which one is the correct option?

I, II and III
I only
II only
III only
I and II only

This is the initial basic feasible solution of a question by North-West Corner Method. Finding optimal basic feasible solution: Consider U1=0 and V1=5, what is the value of V2?

	D1	D2	D3	Supply
S1	5 10	1 15	2	25
S2	7	2 5	4 5	10
S3	3	3	5 15	15
Demand	10	20	20	

Cost = 10 (5) + 15 (1) + 5 (2) + 5 (4) + 15 (5) = 170

This is the initial basic feasible solution of a question by North – West Corner Method.

Finding optimal basic feasible solution: Consider U1 = 0 and V1 = 5, what is the value of V2?

10
15
25
5

Transportation problem is a special class of linear programming problem in which transport a commodity from the source to a destination in such a way that the total transportation cost is _____.
Zero
Maximum
Minimum
Average

We solve a assignment problem by subtracting least number in each row and we get the table below:

Cost Table	Dist Center D	Dist Center E
Plant A	0	135
Plant B	0	8
Plant C	34	0

After doing next step, which table should be next?

Cost Table	Dist Center D	Dist Center E
Plant A	0	135
Plant B	0	8
Plant C	34	0

►

Cost Table	Dist Center D	Dist Center E
Plant A	0	127
Plant B	0	0
Plant C	0	0

▶

Cost Table	Dist Center D	Dist Center E
Plant A	135	0
Plant B	8	0
Plant C	0	34

▶

Cost Table	Dist Center D	Dist Center E
Plant A	0	0
Plant B	34	8
Plant C	0	135

Selected answer

If the arrival rate is 5 per hour and the service rate is 10 per hour, then the average system time is:

1/5
5
1
2

If the arrival rate is 5 per hour and the service rate is 10 per hour, then ρ (traffic intensity or system utilization) =?

If the arrival rate is 5 per hour and the service rate is 10 per hour, then the expected number in the queue or average queue length is:

Formula for the expected number in the system L is:

$L=\frac{\lambda}{\mu-\lambda}$

$L=\frac{\lambda}{\mu+\lambda}$

$L=\frac{\lambda}{\lambda-\mu}$

$L=\frac{\mu}{\mu-\lambda}$

Total cost per period = Item cost + Order cost + Holding cost + _____.
Shortage cost
Optimum Shortage (S*)
Economic Oreder Quantity. (Q*)
Maximum Inventory. ($I_{max.}$)

In Manufacturing Model with no shortage, the replacement rate is finite and _____ the demand rate
greater than
less than
equal to

Made By Rizwan Ashraf

Formula of EOQ in manufacturing model with shortages is

$$Q^* = \sqrt{2C_2D/C_3(1-D/R)}.\sqrt{C_3+C_4/C_4}$$

$$Q^* = \sqrt{2C_2D/C_3}.\sqrt{R/R-D}$$

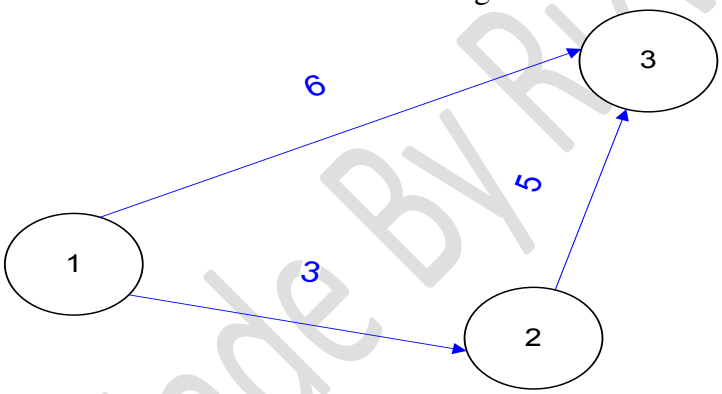
$$Q^* = \sqrt{2C_2D/C_3}$$

None of them

To identify and maintain the proper precedence relationship between activities those are not connected by event, we introduce
Parallel Activity
Dummy Activity
Sequential Activity
None of these

PERT is based on
Deterministic times
Probabilistic Times
Dummy Times
Estimated times

Non-critical activities in the following network are -------



Activity (1,2)
Activity (2,3)
Dummy Actvity
Activity (1,3)

If $t_0 = 10$, $t_m = 10$ and $t_p = 16$, then S.D = _____.
Answer will be 1

Which of the following is an essential characteristic of a linear programming problem?
The relationship between variables and constraints must be linear
The model must have an objective function
The model must have structural constraints

Made By Rizwan Ashraf

All are equivalent

In the simplex table for a linear programming problem, we select the leaving basic variable corresponds to ----------
Maximum non-negative ratio
Maximum negative ratio
Minimum non-negative ratio
Minimum negative ratio

A technique for solving a linear programming problems in which artificial variables are included with coefficients of very large number say equal 10 times of any cost coefficient of decision variables is known as ---------
Big M – Method.
Least Cost Method
Hungarian Method
Branch and Bound Method
Check

If the _____ variables appear in the final basic set, then the LP (Linear Program) has no solution.
slack
surplus
non – basic
artificial

In Two – Phase method, if all the _____ variables become zero, we stop the phase I and proceed to phase-II.
artificial
basic
non basic
positive slack
negative slack

In two phase method, for the phase-I, a new objective function is expressed as -----.
sum of slack variables
difference of slack variables
difference of artificial variables
sum of artificial variables
Page # 133 – My Ok

If "MaxZ=x–y, subject to x>3, x<2, x,y>0" is solved by two phase method, then which of the following would be the objective function of 1st phase?
Max Z =A

In two phase method, for the phase-I, the problem has infeasible solution if the minimum value of objective function

Made By Rizwan Ashraf

zero
greater than zero
less than zero

In two phase method, for the phase-I, a new objective function in terms of artificial is to be
minimized
Maximized

In two phase method, which of the following will be taken as starting solution for 2nd phase?
Basic solution of 1st phase containing non-zero artificial variables
Non-basic solution of 1st phase containing non-negative slacks
Optimum feasible solution of 1st phase
Infeasible solution of 1st phase

Which of the following order pair would minimize the objective function of the linear programming problem; z = x + y subject to x≥3, y≥0 ?
(3,3)
(3,0)
(0,3)
(0,0)

In a Transportation Problem, the objective function 'Z' gives ----------.
Total Cost of transportation
Total Time of transportation
Total Profit of transportation
Total inventory be supplied in transportation

For North West Corner method, in the first row and first column, resource and sink contain '5' and '7' units respectively; then after allocating the appropriate amount 'x11' in the cell (1,1), we will move towards which of the following cell?
missed

To convert an Assignment problem into a maximization problem, which of the following operations would have to apply?
Deduct all elements of the row from highest element in that row

If the total demand is equal to total supply as per requirement of a balanced transportation problem i-e "a1+a2+---+an = b1+b2+---+bn" then which of the following is true?
"a1=b1, a2=b2,---, an=bn" is necessarily implied (not sure)

In case the cost elements of one or two cells are not given in the problem, it means that ---------
The routes connected by those cells are not available

If the cost matrix in an Assignment problem is not square then which the following modification will be made to balance the given problem?
Add a dummy row(column) with negative cost elements (not sure)

Made By Rizwan Ashraf

The cost coefficient of artificial variable in Objective function is ----------.
M (sure)

While solving a Linear Programming problem by Simplex Method, an inequality says:'2x-3y<5' is transformed into strict equality of the form---------, where 's>=0'.
2x-3y +s =5

In the Simplex method to solve an LP problem, Gauss Jordan Elimination method demands that all the key column's entries should be --------- except key row(pivot) entry.
strictly negative

While solving an LP problem by Simplex method, the inclusion of slacks in the constraints' inequalities helps in finding ---------- variables
Check for options

In a feasible region, if the end points of the line segment are basic solutions, then all the points between these two will ----------.
also be the basic

In a feasible region, if the end points of a line segment are basic solutions which BOTH also give the optimal solution, then ---------- will also serve for optimal.
the point which divides segment in ratio 1:2

After converting constraints into the respective Standard equalities, we have an LP problem of '4' equations in '6' variables, if the initial basic feasible solution is say;(2,4,0,2), then it is ---------- solution.
degenerate feasible

The positive slack variables indicate ----------.
excess resource available (sure)

If in a LP problem, the objective is to maximize: z = 2x+3y, with all associated constraints of '=<' type, then in the 1st iteration, z = ---------.
0

In M-method, which of the following is the standard equality corresponds to the constraint:"3x+5y=2"?
3x+2y +A=2
3x+2y–A=2
3x+2y–MA=2
3x+2y+MA=2

In the final iteration of M-method to obtain the optimal solution, the artificial variable must------- the basis.

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add in each variable of
subtract from each variable of
leave
include in

If Primal Problem is a maximization problem, then the dual will be -------
minimization Problem sure

Graphically, in case of optimal degenerate solution,feasible solution always exists at corner points ----------.
lie on any axis

The number of variables in the Primal will be the number of -------- in Dual and vice versa.
constraints sure

When the total allocations in a transportation model of m*n size is not equals to "m + n – 1", the situation is known as:
Unbalanced situation

In North West Corner method, the first step after choosing the appropriate cell in 1st row, we allocate ----------so that the capacity of first row or first column is exhausted.
as much as possible

The row, which is introduced in the matrix to balance an unbalnced Transportation problem, is known as ----------.
dummy row (conform)

The solution of a transportation problem with m rows (supplies) and n (destinations) is feasible if numbers of positive allocations are
m+n-1 (conform)

The amounts shipped from a dummy source represent shortages at the receiving destinations
True (conform)
False

In the Vogel's approximation Method for solving a Transportation problem, Penalty measure for any row or column, is given by which of the following?
Difference between the smallest unit cost to the next smallest cost in the same row(column)

While solving a LP problem by Simplex method in a given iteration, the new basic variable is ---------- and the variable remove from the basis is called ---------
Entering Variable, Leaving variable
Page # 113 - My Ok

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Due to which of the following reason, Simplex method is not preferred to solve a Transportation problem
Since it contains large number of decision variables ‘xij’s’ so that it becomes complicated
Since Transportation problem contains constraints of ‘=’ type
Since Transportation problem does not contain positive slacks
Since Transportation problem does not contain negative slacks

If a variable in the Primal is unrestricted in sign, then the corresponding constraint in the dual will be of -------- type and vice versa
>=
<=
=
None

In M-method, which of the following is true about the coefficient (M) of artificial variable(A) in the objective function
M ->+infinitie
M->-infinite
M-> zero
M-> optimal solution

For finding the maximum profit in an enterprise of selling two products such that ‘freezing’ the sale of one product and keep selling the other
Degeneracy
Duality

In Hungarian method of solving assignment problem, the cost matrix is obtained by----------.
Dividing each row by the elements of the row above it
Subtracting the elements of the row from the elements of the row above it
Subtracting the smallest element from all other elements of the row
Subtracting all the elements of the row from the highest element in the matrix
Page # 204 Or **Click Here** – My Ok

Which of the following statement applies to both transportation model and assignment model?
The inequalities of both problems are related to one type of resource.
Both have objective function, structural constraint and non-negativity constraints
Both use Volgel’s approximation for grtting basic feasible solution
Both are tested by Steping Stone for optimality
Click Here – My Ok

The similarity between Assignment problem and Transportation problem is _____
Both are rectangular matrices
Both are square matrices
Both can be graphical method
Both have objective function and non negativity constraints

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Click Here – My Ok

For the project of a firm, if '5' sales persons are assigned '5' different sales territories, then in how many ways a single territory can be assigned to a single sale person?
120
25
10
5

In which of the following stage, the machine operates at highest efficiency and its production rate will be high and hence no need of replacement?
Infant stage
Youth stage
Old stage
None of the above

When money value changes with time at Ten percent(10%), then Power Worth Factor(PWT) for first year is.
1
0.909
0.852
0.9

Which of the following is the correct assumption for replacement policy when money value does not change with time
No Capital cost
No scrap value
Constant scrap value
Zero maintenance cost
My Ok - google Books

Which of the following department is more responsible for the development of queuing theory.
Railway station
Municipal office
Telephone department
Health department
My Ok - google books

This department is responsible for the development of queuing theory:
Railway station
Municipal office
Telephone department
Health department
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If the number of arrivals during a given time period is independent of the number of arrivals that have already occurred prior to the beginning of time interval, then the new arrivals follow -------- distribution.
Erlang
Poisson
Exponential
Normal
My Ok - google books

The arrival rate of customers in a Queue is an example of _____ variable
Continuous
Complex
Stochastic
Discrete

Primal of a Primal is-------.
Primal
Dual
Primal Primal
Primal dual
My Ok

Primal of a Dual is-------.
Primal
Dual
Primal Primal
Primal dual
My Ok

The dual of a dual problem yields the original primal.
True
False
My Ok - Note

Dual of a dual is
Primal
Dual
Primal Primal
Primal dual
My Ok

1. Primal of a Prima is Primal
2. Dual of a Dual is Primal.
3. Primal of a Dual is Primal.
4. Dual of a Primal is Dual.

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5. Dual of a Dual of a Dual is Primal.

An unrestricted primal variable will result in an equality dual constraint. Conversely a primal equation produces an unrestricted dual variable.
True (I think)
False

If the primal (either) problem has an unbound solution, then the dual has no solution
Optimal solution
Infeasible solution
Bounded solution
No solution
My Ok – Google books

If the primal has an unbound solution, then the dual has no solution
Optimal solution
Infeasible solution
Bounded solution
No solution
My Ok – Google books

In two phase method, for the Phase-I, if the objective function has zero value with all vanishing artificial variables then we _____
Necessarily have optimal solution
Infeasible solution
Can't proceed for 2nd phase
Proceed for 2nd phase
Page # 133 – My Ok

In M-method, if the given LP problem has the feasible solution then the optimization algorithm enforces artificial variable to
+infinitie
-infinite
zero
optimal solution
Check

Which of the following is the standard form of objective function corresponding to,MinZ=2x–11y, subject to x=5 and y=7? Where As are artificial
Z=2x-11y-MA1-MA2
Z=2x-11y+MA1+MA2
Z=2x-11y-MA1+MA2
Z=2x-11y+MA1-MA2
Not confirm – check

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The insensitivity of the solution relative to the original decision variables in an LP problem which is solved by M-method is overcome by ----------
Simplex method
Graphical method
Two phase method
Duality principle
Page # 133 – My Ok

In Simplex method, which of the following is the standard equality corresponding to the constraint"3x+5y>=2"?
3x+2y+S+A=2
3x+2y–S+A=2
3x+2y–S–A=2
3x+2y+S–A=2

In the initial iteration of Big M-method, the artificial variables appear in ---------.
Basis
Non-basic variables' set

In two phase method, for the phase-I, if the given problem has feasible solution then----------
both objective function and artificial are zero
objective function is zero but artificial may arbitrary
objective function is arbitrary but artificial is zero
both objective and artificial can have arbitrary values

Under which of the following condition to solve an LP by using two phase method, we can't proceed for 2nd phase?
Objective function of 1st phase has zero value
Objective function of 1st phase has positive value

In replacement analysis, the maintenance cost is a function of _____ My Quiz
Time
Final investment
Initial investment
Resale value

Replacement of an item will become necessary when_____
The company has surplus funds to spend
Old item becomes too expensive to operate or maintain
Your opponent changes his machine in his unit
Your operator desires to work on a new machine

Which of the following type of failure that usually occurs in old age of the machine and hence become a reason of replacement? My Quiz
Random failure
Early failure

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Wear-out failure
Chance failure
My Ok – Google Books

The type of failure that usually occurs in old age of the machine is
(*a*) Random failure
(*b*) Early failure
(*c*) Chance failure
(*d*) Wear - out failure
My Ok – Google Books

Group replacement policy applies to _____
Irreparable items
Reparable items
Items that fails partially
Items that fail completely
My Ok – Google Books

Which of the following Replacement policy is imposed on an item irrespective of its failure?
Group replacement
Individual replacement
Repair spare replacement
Successive replacement
My Ok – Google Books

The replacement policy that is imposed on an item irrespective of its failure is
(*a*) Group replacement
(*b*) Individual replacement,
(*c*) Repair spare replacement,
(*d*) Successive replacement. ( )
My Ok – Google Books

Which of the following cost is irrelevant to replacement analysis?
Purchase cost of the machine
Operating cost of the machine
Maintenance cost of the machine
Machine hour rate of the machine
My Ok – Google Books

Which cost of the following is irrelevant to replacement analysis?
(*a*) Purchase cost of the machine,
(*b*) Operating cost of the machine,
(*c*) Maintenance cost of the machine,
(*d*) Machine hour rate of the machine. ( )
My Ok – Google Books

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In a queue, the arrival pattern can be expressed in terms of --------.
Probabilities
Exact continuous variables
Customer behavior
Number of servers
Page # 230 – My Ok

If there is infinite number of servers then all the customers are served _____ on arrival.
Randomly
Instantaneously

To convert the transportation problem into a maximization model we have to----------
write the inverse of the coefficient matrix
multiply the feasibility condition by –1
multiply the coefficient matrix by –1
We can't convert the transportation problem into a maximization problem, as it is basically a minimization problem.
My Ok – Google Books

Transportations models consist of ------- like the production centers and --------- which may be the sales centers.
(sinks, sources)
(sources, sinks) Most
(origins, sources)
(sinks, destinations)

In which of the following models, Simplex algorithm is not preferred to use due to laborious computations?
Transportations models
Degenerate Linear models (Waqar Sidhu)
Non-degenerate Linear models
Dual or unbounded linear models

The inequality $2x+3y\leq 18$ is equivalent to
$2x+3y\leq -18$

$-2x-3y\geq 18$

$-2x-3y\geq -18$

$-2x-3y\leq -18$

The inequality $2x+3y\geq 18$ is equivalent to
$2x+3y\leq -18$

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$-2x-3y \geq 18$

$-2x-3y \geq -18$

$-2x-3y \leq -18$

MaxZ=2x+3y
Subject to

$$\left.\begin{array}{l} x \geq 2 \\ y \leq 3 \\ x, y \geq 0 \end{array}\right\} \left.\begin{array}{l} x - s_1 + A = 2 \\ \Rightarrow y + s_2 = 3 \\ x, y, s_1, s_2, A \geq 0 \end{array}\right\}$$

Which of the following is associated objective function of the 1*st phase*?
Which of the following is associated objective function of the 1st phase?
MazZ=2x+3y+A
MinZ=2x+3y+A
MaxZ=A
MinZ=A

By using two phase method to solve a linear programming problem, in Phase I, a new objective function is formed by assigning on left hand side, zero to every original variable (including slack and surplus variables) and _____ to each of the artificial variables.
M
-M
+1
-1

Zero valued artificial variables may appear as _____ variables in the final solution, when one or more of the original constraints equations is reduendant.
Non basic
Basic
Slacks
Surplus
Artificial

In the big-M method, if the introduced _____ variables do not leave the basis in final iteration, then this indicates that the give linear programming problem can't be optimized.
Entering
Positive slack
Negative slack
Artificial

While solving a linear programming problem by using M-Method, traditionally the _____ variables are chosen in the initial basic feasible solution.
Negative slack
Positive slack
Entering

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Artificial

The _____ variable is chosen by examining the cost coefficients in the objective function.
Entering
Leaving
Positive slack
Negative slack

If a company manufacture 'x' units of product 'A' and 'y' units of 'B' with associated profits of Rs.5 and Rs.3 then which of the following is the objective function to maximize is the profit?
Z=15xy
Z=5x-3y
Z=3x-5y
Z=5x+3y

In a linear Programming problem (LPP), which of the following must be hold?
Only objective function is linear
Both objective function and constraints are linear
Only constraints needs to be linear
At least one of objective function or constraint should be linear

Solution region for constraint $y \leq 0$ is the
Half plane below y-axis
Half plane below the line y=0
Set of all those points where ordinates are non-positive
All are equivalent

In PERT, activity time estimates are distributed according to
Beta distribution
Normal Distribution
Poisson Distribution
Binomial Distribution
Page # 35, 36 – My Ok

Best possible time estimate that a given activity would take under normal conditions which often exist, is called
Most likely time estimate
Pessimistic time estimate
Smallest time estimate
Activity time estimate
Page # 35 – My Ok

For any activity backward pass computations provide its
Earliest start times

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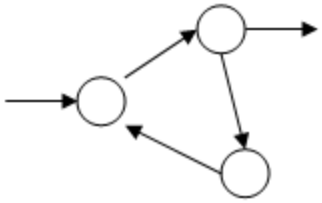
Latest start time
Moderate start times
Completion time
Page # 32 – My Ok

The following network is an example of



Redundancy
Dangling
Cycling
Dummy

MAD=_____ S.D

$\frac{\sqrt{\frac{2}{\pi}}}{\sqrt{\frac{\pi}{2}}}$

$\sqrt{\frac{2\pi}{3}}$

$\sqrt{\frac{3\pi}{2}}$

Standard Deviation (S.D) = _____ MAD

$\frac{\sqrt{\frac{\pi}{2}}}{\sqrt{\frac{2}{\pi}}}$

$\sqrt{\frac{2\pi}{3}}$

$\sqrt{\frac{3\pi}{2}}$

Product of 'item cost' and 'ordered item' is
Crash cost

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Cost period
Set up cost
Shortage cost
Page # 51 – My Ok

Formula for geometric series
Formula for geometric series "x+vx+v^2x+… is

$$P(n) = x\frac{1-v^n}{1-v}$$

$$P(n) = \frac{1-v^n}{1-v}$$

$$P(n) = x\frac{1-v}{1-v^n}$$

$$P(n) = \frac{1-v}{1-v^n}$$

Page # 255 - My Ok

The present worth of a rupee to be spent after a year is denoted by v and given by
v= (1 + r)
v= (1 / r)
v = (1 + r) / 10
v = 1/ (1 + r)

If "Ni" be the Number of replacement made at the end of the i^{th} week and "Pj" be the probability of failure during the i^{th} week, then N_1 = ----------.
N_0P_1
N_1P_1
N_0P_0
N_1P_2

If "Ni" be the Number of replacement made at the end of the i^{th} week and "Pj" be the probability of failure during the i^{th} week, then N_2 = ----------.
$N_1P_2 + N_1P_1$
$N_0P_2 + N_1P_1$
$N_0P_1 + N_1P_2$
$N_0P_2 + N_1P_2$
Check

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the mean service time " μ " is
½=0.5
4
¼=0.25

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2

A repairman services three machines. For each machine the time between service requirements is 8 hours following exponential distribution. The time of repair also has the same distribution with a mean of 2 hours. Then the average rate " λ " is

1/8=0.125

8

¼=0.25

4

A duplicating machine maintained for office use is used and operated by people in the office who need to make copies. Since the work to be copied varies in length (number of pages of the original) and copies required, the service rate is randomly distributed, the arrival rate is 5 per hour and the service rate is 10 per hour then the equipment utilization " ρ " is equal to

0.50

0.20

5

2

In a bank, every 15 minutes one customer arrives for cashing the cheque. The staff in the payment counter takes only 10 minutes for serving a customer on an average, then the service rate " μ " =_____

6 per hour

4 per hour

10 per hour

1/6 per hour

If the mean arrival and mean service rates are 4 and 7 respectively in a queue then expected waiting time in system is

1/3

3

28

7/4

Page # 237 – My Ok

We go in probabilistic replacement model when period between installation and failure is Varying exponentially

Varying linearly

Constant

Is not constant (varying arbitrarily)

If a basic feasible solution contains less than "m + n – 1" (Here m is the number of rows, n is the number of columns in transportation problem) non negative allocation, then it is said to be

Degenerate (Waqar Sidhu)

Multiple Solutions

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Non degenerate
Alternative Optima

Any set of non negative allocation ($X_{ij}>0$) which satisfy the row and column sum is called a _____ solution
Feasible (Waqar Siddhu)
Non basic feasible
Basic infeasible
Optimal

Degeneracy in a 5x6 transportation problem occurs when the number of occupied cell is less than
20 but greater than 10
10
Zero
Infinity
Check

In sequencing problem, the Johnson's algorithm in finding the optimal ordering of n jobs through 3 machines can be applied, if the problem is converted into following number of machine problems
3n
3n!
2*3=6
2

During a replacement if the value of money decreases at the rate of **3**% then the present worth factor of unit amount to be spent after one year is given by
0.25
0.333
0.9708
4

If the money carries a rate of interest of **12**% per year, the present worth factor of one rupee due in one year is
0.08333
0.89285 (Waqar Sidhu)
0.0769
13

Which of the following binary operation in assignment problem among all the elements in the given profit matrix from the highest element in the matrix
Subtraction
Division
Multiplication
Addition

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The cost matrix in assignment problem is always diagonal matrix
identity matrix
zero matrix
diagonal matrix
square matrix (Waqar Sidhu)

The cost matrix in assignment problem is always
identity matrix
zero matrix
diagonal matrix
square matrix (Waqar Sidhu)

The important characteristic of Cost matrix associated with Assignment problem, while solving it by Hungarian's method is _____
It will have zero as element of one diagonal
It will have zero as the element of both diagonals
It will have at least one zero in each column and each row
It will not have zero as its element

In which of the following age, the Replacement decision is very much common?
Infant stage
Old age
Youth
In all the above

The assignment problem is unbalanced if the cost matrix is not a ____ matrix.
Square
Rectangle
Non-singular
Singular

In the assignment problem, the cost matrix is _____
Rectangular
Square
Non-Singular
Singular

Which of the following replacement policy is considered to be dynamic in nature? My Quiz
Time is continuous variable and the money value does not change with time.
When money value does not changes with time and time is a discrete variable.
When money value changes with time.
When money value remains constant for some time and then goes on changing with time.
My Ok – Google books

To balance Assignment matrix we have to:
(*a*) Open a Dummy row,

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(*b*) Open a Dummy column,
(*c*) Open either a dummy row or column depending on the situation,
(*d*) You cannot balance the assignment matrix
My Ok – Google books

To balance the assignment matrix we have to:
(*a*) Open a Dummy row,
(*b*) Open a Dummy column,
(*c*) Open either a dummy row or column depending on the situation,
(*d*) You cannot balance the assignment matrix.
My Ok – Google books
Note google book waly sary answers dobara sy confirm krny hain lazmi

In an assignment problem, while applying Hungarian's method, if in the final modified matrix any row or column does not have single zero, then which of the following is true?
No assignment can be made
Unique assignment will be made
Multiple assignments can be made
None of the above

For a Transportation Problem, if it's initial feasible solution is evaluated by Least cost method, the quality of this initial solution is better than_____
North West Corner Method
Vogal's approximation Method

For an unbalanced Transportation problem, if the total demand is MORE than total supply then which of the following is true in order to balance the problem?
One constraint will have evacuate
One constraint will have to add
A dummy sink would have to include with demand equal to surplus
A dummy source would have to include with supply equal to shortage
Check

In the assignment problem, the decision variable 'xij' can attain which of the following value?
Only '1'
Only zero
Any arbitrary non-negative
Zero or 1

In Least cost Method, for any cell both demand and supply are satisfied then which of the following will be crossed out?
Only row
Only column
Both row and column
Either of row or column

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Which of the following method is used to find the basic feasible solution of a transportation problem?
North West Corner Method
Least Cost's Method
Vogel's approximation Method
All above methods are applicable

If shortage cost is infinity, then _____
No shortages are allowed
No inventory carrying cost is allowed
Ordering cost is zero
Purchase cost = Carrying cost

Many decision making problems involve a process that takes place in multiple stage in such a way that at each stage, the process is dependent on the strategy chosen. Such types or problems are called.
Linear Programming problems
Integer programming problems
Dynamic programming problems
Assignment problems

A primal's constraint equation produces _____ dual variable
Non basic
An unrestricted
Artificial
Basic

In the Simplex table, if the coefficient of non-basic variable says x_1 in the Z-row is zero, then it indicates that value of z_____
Increase
Decrease
Does not change
Becomes infinity

Consider the following simplex table corresponding to maximization problem

Basic	x_1	x_2	x_3	x_4	A_1	A_2	R.H.S
Z	0	0	M+1/2	M	16M-11	0	-0.6M-5.7
x_1	1	0	5	0	-1	0	3
x_2	0	1	-5	0	3	0	9
A_2	0	0	-1	-1	-0.6	1	0.6

This is the indication that problem has _____.
Infeasible Solution
Temporarily Degenerate Solution
Unbounded Solution
Degenerate Solution

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Consider the following simplex table corresponding to maximization problem

Basic	x_1	x_2	x_3	S_1	S_2	S_3	R.H.S
Z	-4	-3	-2	0	0	0	0
x_1	0	0	1	1	0	0	8
x_2	-2	1	3	0	1	0	9
A_2	0	2	1	0	0	1	3

This is the indication that problem has _____.
Problem has Unbounded Solutions
Problem has Infeasible Solutions
Problem has temporarily degenerate
Problem has Multiple Optimal Solutions

For a linear programming problem, the Unboundedness is related to which of the following?
Solution space only
Objective value only
Neither solution space nor to objective value
Both solution space and objective value

If the degeneracy arises in the initial stage, then which of the following is true?
One of the basic variables is zero
More than one variable is eligible to leave the baiss
Tie the ratio of each row, by taking right hand side of each row and dividing by the corresponding element of the key column.
Iterative Cycling of basic solution exists without reaching the optimality

If the problem has a _____ solution such that the minimum value of the objective function is zero along with zero values of artificial variables. Then proceed to phase II method.
Unbounded
Feasible
Non feasible
De-generate feasible solution
Page # 133 – My Ok

Min. Z = 0 and no artificial variable appears in the basic variables. _____ solution to original problem has been found and we proceed to phase II method.
A basic feasible
An optimal
An infeasible

If no _____ variable appears in the basis and the optimality conditions are satisfied, then the current solution will be optimal one.
artificial
non basic
slack
surplus

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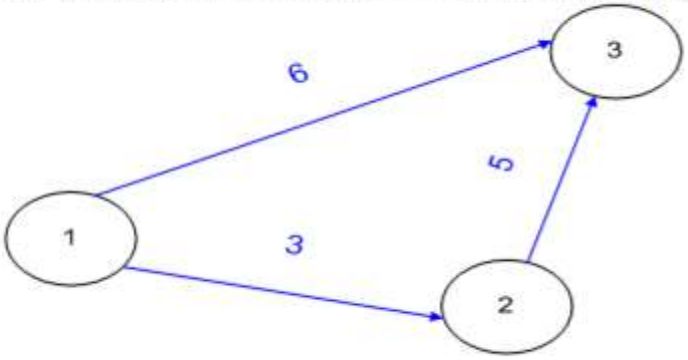
The constrained $3x_1+2x_2 \leq 18$ in standard form can be as:
(Where R is artificial variable and S_1 is slack variable)
$3x_1+2x_2+S_1+R=18$
$3x_1+2x_2+S_1-R=18$
$3x_1+2x_2-S_1+R=18$
$3x_1+2x_2+S_1 =18$

If the objective function of a linear programming problem needs further improvement then which of the following will have to proceed?
A new decision variable to be entered and other decision variable to leave the basis
A new s;ack variable to be entered and other slack to leave the basis
A new non-degenerate variable to be entered and other no-degenerate to leave the basis
A new basic variable to be included and other basic variable to leave the basis

It is a property of simplex method that there always exist infinite number of basic feasible solution.
True
False

Early finish time of the event '3' in the following is _____

Early finish time of the event '3' in the following project is -------.



14 days
2 days
8 days
6 days

Which of the following give the excess of available time over the activity time when all jobs start as early as possible?
Total float
Free float
Independent float
Early finish time

Let FS = Free Slack, TS = Total Slack, INDS = Independent Slack, then which relation is true

I. $TS \leq FS$
II. $INDS \leq FS$
III. $FS \leq TS$

Then which relation is true
Only I
Only II
Only III
Only I & II
Only II & III

In replacement Models, Capital Cost = (Item / Machine Cots) – (----------)
Resale Cost
Running Cost
Maintenance Cost
Operating Cost

In assignment problem, each job requires exactly one resource.
True
False
May not be true

The name of the method in getting the optimum assignment is
Simplex method
North-West corner method
Vogel's approximation method
Hungarian method

The Vogel Approximation method is an iterative procedure for computing a _____ solution of the transportation problem
Basic feasible
Non basic feasible
Basic infeasible

CPM can help you figure out EXCEPT:
How long your complex project will take to complete
Which activities are "critical" meaning that they have to be done on time or else the whole project will take longer
You can try to speed up the project
What is the most costly way to speed up the project

Transportation technique or the simplex method cannot be used to solve the assignment problem because of
Degeneracy
Involvement of dummy activities
Looping among the entries of assignment matrix

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Inequilibrium between demand and supply

The representation of reality in some physical form or in some form of mathematical equation may be discussed under the topic entitled as
Transportation problems
Game theory
Simulation
Replacement problems

Under the individual replacement policy an item is replaced on its
Success
Failure
Maximum utilization
Minimum utilization

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Note this is a very important link from google books jahan sy is subject ky past quizzes hain
https://books.google.com.pk/books?id=u-eRHBbJW7AC&pg=PA263&lpg=PA263&dq=transportation+model+and+assignment+model+The+inequalities+of+both+problems+are+related+to+one+type+of+resource.&source=bl&ots=93F4IylWQB&sig=e84VsaivWDsZYmMAVJO0T1ZvBeE&hl=en&sa=X&ved=0ahUKEwjC9qvDu5fcAhUHOJoKHRfrB0oQ6AEILjAB#v=onepage&q=transportation%20model%20and%20assignment%20model%20The%20inequalities%20of%20both%20problems%20are%20related%20to%20one%20type%20of%20resource.&f=false

Note Yeh MCQs Quiz k hain lakin in k options nhi hain

Which of the following represents "Arrival-->Service-->Service-->Service-->Out-->"?
(*a*) Single Channel Single Phase system
(*b*) Multi channel single-phase system
(*c*) Single channel multi phase system
(*d*) Multi channel multi phase system. ( )

4. Distribution of service time in Queuing Model is ----------exponential distribution.
5. Which of the following replacement model is said to be probabilistic model?
6. Queuing models measure the effect of----------.

In Simplex standard table to solve an LP problem of Minimization, we choose the candidate for entering variable in _____
Objective function with most positive coefficient
Objective function with most negative coefficient
Constraint with most positive coefficient
Constraint with most negative coefficient

In Simplex standard table to solve an LP problem of Maximization, we choose the candidate for entering variable in _____
Objective function with most positive coefficient
Objective function with most negative coefficient
Constraint with most positive coefficient
Constraint with most negative coefficient

If an artificial variable is not included while converting a constraint of type'>=' into equation, then we will have _____ solution
Feasible
Infeasible
Degenerate
Non-degenerate

By Simplex method, to minimize 'Z=2x+9y' of an LP problem, if 'z=A>0' for the initial iteration then for its next improved solution (0<A<100), which of the following would be the next entering variable
X<0

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Y<0
x>0
y>0

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Transportation problem is basically a
Maximization model
Minimization model
Transshipment problem
Iconic model

The column, which is introduced in the matrix to balance the rim requirements, is known as:
Key column
Idle column
Slack column
Dummy Column

The row, which is introduced in the matrix to balance the rim requirement, is known as:
Key row
Idle row
Dummy row
Slack row

One of the differences between the Resource allocation model and Transportation Model is:
The coefficients of problem variables in Resource allocation model may be any number and in transportation model it must be either zeros or ones. (I Think)
The coefficients of problem variable in Resource allocation model must be either zeros or ones and in Transportation model they may be any number
In both models they must be either zeros or ones only
In both models they may be any number
Check

To convert the transportation problem into a maximization model we have to
To write the inverse of the matrix
To multiply the rim requirements by –1
To multiply the matrix by –1 (I think)
We cannot convert the transportation problem in to a maximization problem, as it is basically a minimization problem
Check

In a transportation problem where the demand or requirement is equals to the available resource is known as
Balanced transportation problem (I think)
Regular transportation problem,
Resource allocation transportation problem
Simple transportation model
Check

The total number of allocation in a basic feasible solution of transportation problem of $m \times n$ size is equal to:
$m \times n$

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$(m / n) - 1$
$m + n + 1$
$m + n - 1$

When the total allocations in a transportation model of $m \times n$ size is not equals to $m + n - 1$ the situation is known as:
Unbalanced situation
Tie situation
Degeneracy (I Think)
None of the above
Check

The opportunity cost of a row in a transportation problem is obtained by:
Deducting the smallest element in the row from all other elements of the row,
Adding the smallest element in the row to all other elements of the row,
Deducting the smallest element in the row from the next highest element of the row
Deducting the smallest element in the row from the highest element in that row.

In Northwest corner method the allocations are made
Starting from the left hand side top corner,
Starting from the right hand side top corner
Starting from the lowest cost cell
Starting from the lowest requirement and satisfying first

VAM stands for:
Value added method
Value assessment method
Vogel Adam method
Vogel's approximation method

MODI stands for
Modern distribution
Mendel's distribution method
Modified distribution method
Model index method

In the optimal solution, more than one empty cell have their opportunity cost as zero, it indicates
The solution is not optimal
The problem has alternate solution
Something wrong in the solution
The problem will cycle. ( )

In case the cost elements of one or two cells are not given in the problem, it means:
The given problem is wrong
We can allocate zeros to those cells
Allocate very high cost element to those cells

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To assume that the route connected by those cells are not available

To solve degeneracy in the transportation problem we have to:
Put allocation in one of the empty cell as zero
Put a small element epsilon in any one of the empty cell
Allocate the smallest element epsilon in such a cell, which will not form a closed loop with other loaded cells.
Allocate the smallest element epsilon in such a cell, which will form a closed loop with other loaded cells

A problem where the produce of a factory is stored in warehouses and then they are transported to various demand point as and when the demand arises is known as:
Transshipment problem
Warehouse problem
Storing and transport problem
None of the above ( )

Implied Cost in transportation problem sets (in the existing program):
The lowest limit for the empty cell beyond which it is not advisable to include in the programme,
The highest limit for the empty cell beyond which it is not advisable to include in the programme,
The opportunity cost of the empty cell,
None of the above

In transportation model, the opportunity cost is given by
Implied cost + Actual cost of the cell
Actual cost of the cell – Implied cost,
Implied cost – Actual cost of the cell (I think)
Implied cost × Actual cost of the cell
Check

If ui and *vj* are row and column numbers respectively, then the implied cost is given by:
ui + vj
ui – *vj*
ui × *vj*
ui /*vj* ( )

If a transportation problem has an alternate solution, then the other alternate solutions are derived by:
(Given that the two matricides of alternate solutions are A and B, and d is any positive fraction number)
A* + (1 – *d*) × *B
A (1 – *d*) + *B*
dA + *dB*
dA* + (1 – *d*) × *B
Check

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Assignment Problem is basically a
Maximization Problem
Minimization Problem
Transportation Problem
Primal problem

The Assignment Problem is solved by
Simplex method
Graphical method
Vector method
Hungarian method

In Index method of solving assignment problem
The whole matrix is divided by smallest element
The smallest element is subtracted from whole matrix
Each row or column is divided by smallest element in that particular row or column
The whole matrix is multiplied by – 1.

In Hungarian method of solving assignment problem, the row opportunity cost matrix is obtained by:
Dividing each row by the elements of the row above it,
By subtracting the elements of the row from the elements of the row above it
By subtracting the smallest element from all other elements of the row
By subtracting all the elements of the row from the highest element in the matrix

In Flood's technique of solving assignment problem the column opportunity cost matrix is obtained by:
Dividing each column by the elements of a column which is right side of the column
By subtracting the elements of a column from the elements of the column which is right side of the column
By subtracting the elements of the column from the highest element of the matrix
By subtracting the smallest elements in the column from all other elements of the column

The property of total opportunity cost matrix is
It will have zero as elements of one diagonal
It will have zero as the elements of both diagonals
It will have at least one zero in each column and each row
It will not have zeros as its elements

The horizontal and vertical lines drawn to cover all zeros of total opportunity matrix must be:
Equal to each other,
Must be equal to $m \times$ n (where m and n are number of rows and columns)
$m + n$ (m and n are number of rows and columns)
Number of rows or columns

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The assignment matrix is always is a
Rectangular matrix
Square matrix
Identity matrix
None of the above.

To balance the assignment matrix we have to:
Open a Dummy row
Open a Dummy column
Open either a dummy row or column depending on the situation
You cannot balance the assignment matrix

In cyclic traveling salesman problem the elements of diagonal from left top to right bottom are
Zeros
All negative elements
All are infinity
all are ones

In cyclic traveling salesman problem the elements of diagonal from left top to right bottom are
Zeros
All negative elements
All infinity
All ones

To convert the assignment problem into a maximization problem
Deduct smallest element in the matrix from all other elements.
All elements of the matrix are deducted from the highest element in the matrix.
Deduct smallest element in any row form all other elements of the row.
Deduct all elements of the row from highest element in that row. ( )

The similarity between Assignment Problem and Transportation problem is:
Both are rectangular matrices
Both are square matrices,
Both can be solved by graphical method
Both have objective function and nonnegativity constraints

The following statement applies to both transportation model and assignment model
The inequalities of both problems are related to one type of resource.
Both use VAM for getting basic feasible solution
Both are tested by MODI method for optimality
Both have objective function, structural constraint and non-negativity constraints

To test whether allocations can be made or not (in assignment problem), minimum number of horizontal and vertical lines are drawn. In case the lines drawn is not equal to the number of rows (or columns), to get additional zeros, the following operation is done:
Add smallest element of the uncovered cells to the elements to the line

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Subtract smallest element of uncovered rows from all other elements of uncovered cells
Subtract the smallest element from the next highest number in the element.
Subtract the smallest element from the element at the intersection of horizontal and vertical line

The total opportunity cost matrix is obtained by doing:
Row operation on row opportunity cost matrix,
By doing column operation on row opportunity cost matrix,
By doing column operation on column opportunity cost matrix
None of the above

The total opportunity cost matrix is obtained by doing:
Row operation on row opportunity cost matrix,
Column operation on row opportunity cost matrix,
Column operation on column opportunity cost matrix,
None of the above

Flood's technique is a method used for solving
Transportation problem
Resource allocation model
Assignment mode.
Sequencing model

The assignment problem will have alternate solutions when total opportunity cost matrix has
At least one zero in each row and column,
When all rows have two zeros,
When there is a tie between zero opportunity cost cells,
If two diagonal elements are zeros

The following character dictates that assignment matrix is a square matrix:
The allocations in assignment problem are one to one
Because we find row opportunity cost matrix
Because we find column opportunity matrix
Because one to make allocations, one has to draw horizontal and veridical lines. ( )

When we try to solve assignment problem by transportation algorithm the following difficulty arises:
There will be a tie while making allocations
The problem will get alternate solutions,
The problem degenerate and we have to use epsilon to solve degeneracy
We cannot solve the assignment problem by transportation algorithm

The objective of sequencing problem is:
To find the order in which jobs are to be made
To find the time required for completing all the jobs on hand
To find the sequence in which jobs on hand are to be processed to minimize the total time required for processing the jobs

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To maximize the effectiveness

The time required for printing of four books *A*, *B*, *C* and *D* is 5, 8, 10 and 7 hours. While its data entry requires 7, 4, 3 and 6 hours respectively. The sequence time that minimizes total elapsed time is
ACBD
ABCD
ADCB
CBDA

If there are '*n*' jobs and '*m*' machines, there will be ----------sequences of doing the jobs.
$n \times m$
$m \times n$
n m
(*n* !) *m* ( )

In general sequencing problem will be solved by using …….
Hungarian Method
Simplex method
Johnson and Bellman method
Flood's technique

In solving 2 machine and '*n*' jobs, the following assumption is wrong:
No passing is allowed
Processing times are known
Handling time is negligible
The time of processing depends on the order of machining

The following is the assumption made in processing of '*n*' jobs on 2 machines
The processing time of jobs is exactly known and is independent of order of processing
The processing times are known and they depend on the order of processing the job
The processing time of a job is unknown and it is to be worked out after finding the sequence
The sequence of doing jobs and processing times are inversely proportional

The following is one of the assumptions made while sequencing '*n*' jobs on 2 machines
Two jobs must be loaded at a time on any machine
Jobs are to be done alternatively on each machine
The order of completing the jobs has high significance
Each job once started on a machine is to be performed up to completion on that machine

This is not allowed in sequencing of '*n*' jobs on two machines:
Passing
Loading
Repeating the job
Once loaded on the machine it should be completed before removing from the machine.

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Write the sequence of performing the jobs for the problem given below:
Jobs: *A B C D E*
Time of machining
On Machine X: 6 8 5 9 1
They can be processed in any order.
As there is only one machine, sequencing cannot be done
This is not a sequencing problem.
None of the above

Johnson Bellman rule states that:
If smallest processing time occurs under first machine, do that job first
If the smallest processing time occurs under the second machine, do that job first
If the smallest processing time occurs under first machine, do that job last
If the smallest processing time occurs under second machine keep the processing pending

To convert '*n*' jobs and 3-machine problem into '*n*' jobs and 2-machine problem, the following rule must be satisfied.
All the processing time on second machine must be same.
The maximum processing time of 2nd machine must be ≤ to minimum processing times of first and third machine.
The maximum processing time of 1st machine must be ≤ to minimum processing time of other two machines.
The minimum processing time of 2nd machine must be ≤ to minimum processing times of first and third machine

If two jobs *J*1 and *J*2 have same minimum process time under first machine but processing time of *J*1 is less than that of *J*2 under second machine, then *J*1 occupies:
First available place from the left
Second available place from left
First available place from right
Second available place from right

If Job A and B have same processing times under machine I and Machine II, then prefer
Job A
Job B
Both A and B
Either A or B

The given sequencing problem will have multiple optimal solutions when the two jobs have same processing times under:
First Machine,
Under both machines,
Under second machine
None of the above

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The given sequencing problem will have multiple optimal solutions when the two jobs have same processing times under:
First machine
Both machines,
Second machine
None of the above

If a job is having minimum processing time under both the machines, then the job is placed in:
Any one (first or last) position,
Available last position,
Available first position,
Both first and last position

FIFO is most applicable to sequencing of
One machine and '*n*' jobs,
2 machines and '*n*' jobs,
3 machine '*n*' jobs,
'*n*' machines and 2 jobs. ( )

At a petrol Bunk, when '*n*' vehicle are waiting for service then this service rule is used:
LIFO
FIFO
Service in Random Order
Service by highest profit rule

Consider the following sequencing problem, and write the optimal sequence:
Jobs: 1 2 3 4 5
Processing M/C X 1 5 3 10 7
Time in Hrs.
M/C Y 6 2 8 4 9
1 2 3 4 5
1 3 5 4 2
5 4 3 2 1
1 4 3 5 2 (I think)
Check

In a 3 machine and 5 jobs problem, the least of processing times on machine *A, B* and *C* are 5, 1, and 3 hours and the highest processing times are 9, 5, and 7 respectively, then Johnson and Bellman rule is applicable if order of the machine is:
B-A-C
A-B-C
C - B – A
Any order

In maximization case of sequencing problem of 2 machines and '*n*' jobs, the job is placed at available left first position if it has ---------- process time under machine ----------.

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Least, first
highest, first
least, second
highest, second

The fundamental assumption of Johnson's method of sequencing is:
No Passing rule
Passing rule
Same type of machines are to be used
Non zero process time

If a job has zero process time for any machine, the job must be
Possess first position only
Possess last position only
Possess extreme position
Be deleted from the sequencing

The assumption made in sequencing problems *i.e.* No passing rule means:
A job once loaded on a machine should not be removed until it is completed
A job cannot be processed on second machine unless it is processed on first machine
A machine should not be started unless the other is ready to start
No job should be processed unless all other are kept ready to start

The technological order of machine to be operated is fixed in a problem having:
1 machine and '*n*' jobs
2 machines and '*n*' jobs
3 machine and '*n*' jobs
'*n*' machines and 2 jobs

A sequencing problem is infeasible in case of:
1 machine and '*n*' jobs
2 machines and '*n*' jobs
3 machines and '*n*' jobs
2 jobs and '*n*' machines

In a 2 jobs and '*n*' machine problem a lie at 45° represents:
Job 2 is idle
Job 1 is idle
Both jobs are idle
both jobs are under processing

In a 2 jobs and '*n*' machine problem, the elapsed time for job 1 is calculated as (Job 1 is represented on X -axis).
Process time for Job 1 + Total length of vertical line on graph
Process time for Job 2 + Idle time for Job 1
Process time for job 1 + Total length of horizontal line on graph

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Process time for job 2 – Idle time for job 1

In a 2 jobs and '*n*' machine-sequencing problem the horizontal line on a graph indicates:
Processing time of Job I
Idle time of Job I,
Idle time of both jobs,
Processing time of both jobs

In a 2 job, '*n*' machine sequencing problem, the vertical line on the graph indicates:
Processing time of Job 1
Processing time of Job 2
Idle time of Job 2
Idle time of both jobs

In a 2 job and '*n*' machine sequencing problem we find that:
Sum of processing time of both the jobs is same,
Sum of idle time of both the jobs is same,
Sum of processing time and idle time of both the jobs is same,
Sum of processing time and idle time of both the jobs is different

One of the important basic objective of Inventory management is
To calculate EOQ for all materials in the organization
To go in person to the market and purchase the materials
To employ the available capital efficiently so as to yield maximum results
Once materials are issued to the departments, personally check how they are used

The best way of improving the productivity of capital is
Purchase automatic machines
Effective labour control
To use good financial management
Productivity of capital is to be increased through effective materials management

Materials management is a body of knowledge, which helps manager to
Study the properties of materials
Search for needed material
Increase the productivity of capital by reducing the cost of material
None of the above

The stock of materials kept in the stores in anticipation of future demand is known as
Storage of materials
Stock of materials
Inventory
Raw materials

The stock of animals reared in anticipation of future demand is known as:
Live stock Inventory

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Animal inventory
Flesh inventory
None of the above

The working class of human beings is a class of inventor known as:
Live stock
Human inventory
Population
Human resource inventory

In general the percentage of materials cost in product is approximately equals to:
40 to 50 %
5 to 10 %
2 to 3 %
90 to 95%

Materials management bring about increased productivity of capital by:
Very strict control over use of materials
Increasing the efficiency of workers
Preventing large amounts of capital locked up for long periods in the form of inventory
To apply the principles of capital management

We can reduce the materials cost by
Using systematic inventory control techniques,
Using the cheap material,
Reducing the use of materials,
Making hand to mouth purchase

The basis for ABC analysis is
Interests of Materials manager
Interests of the top management
Pareto's 80-20 rule
None of the above

ABC analysis depends on the
Quality of materials
Cost of materials
Quantity of materials used
Annual consumption value of materials

'*A*' class materials consumes:
10 % of total annual inventory cost
30% of total annual inventory cost
70 to 75% of total inventory cost
90 % of total annual inventory cost

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'*B*' class of materials consumes ---------% of annual inventory cost.
60 to 70%
20 to 25%
90 to 95%
5 to 8%

'*C*' class materials consume -------% of annual inventory cost.
5 to 10 %
20 to 30%
40 to 50%
70 to 80%

The rent for the stores where materials are stored falls under:
Inventory carrying cost
Ordering cost,
Procurement cost
Stocking cost

Insurance charges of materials cost falls under:
Ordering cost
Inventory carrying cost
Stock out cost
Procurement cost

As the volume of inventory increases, the following cost will increase:
Stock out cost
Ordering cost
Procuring cost
Inventory carrying cost

As the order quantity increases, this cost will reduce:
Ordering cost
Insurance cost
Inventory carrying cost
Stock out cost

Procurement cost may be clubbed with:
Inventory carrying charges
Stock out cost
Loss due to deterioration
Ordering cost

The penalty for not having materials when needed is:
Loss of materials cost
Loss of order cost,
Stock out cost

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General losses

Losses due to deterioration, theft and pilferage comes under,
Inventory Carrying charges
Losses due to theft,
Does not come under any cost
Consumption cost

Economic Batch Quantity is given by: (where, $C1$ = Inventory carrying cost, $C3$ = Ordering cost, r = Demand for the product)
$(2C1 / C3)1/2$
$(2 C3 / C1r)1/2$
$2C3r / C1$
$(2C3r / C1)1/2$

If λ is the annual demand, $C1$ = Inventory carrying cost, i = rate of inventory carrying charges, p = unit cost of material in Rs., then EOQ =
$(2C \quad ip)1/2$
$2C \quad ip,$
$(2 C3 / ip$
$C3\ ip)1/2, ( )$

If $C1$ = Carrying cost, $C3$ is the ordering cost, r = demand for the product, then the optimal period for placing an order is given by:
$(2 C3/C1r)1/2$
$(2C1\ C3/r\ )1/2$
$(\ 2C3r/C1)1/2$
$(2C1C3r)1/2$

When $C1$ = Inventory carrying cost, $C3$ = ordering cost, r = demand for the product, the total cost of inventory is given by:
$(2C1C3r)$
$(2C1C3)1/2$
$(2C3r/C1)1/2$
$(2C1C3r)1/2$

When λ is the annual demand for the material, p = unit price of the material in Rs., $C3$ is the ordering cost, q = order quantity, then the total cost including the material cost is given by:
$(q/2)\ ip \quad q\ C \quad p$
$2C \quad ip \quad p$
$(q/2)\ ip \quad p$
$(2C3q \quad ip)1/2$

In VED analyses, the letter V stands for:
Very important material
Viscous material

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Weighty materials
Vital materials

In VED analysis, the letter D strands for:
Dead stock
Delayed material
Deserved materials
Diluted materials

The VED analysis depends on:
Annual consumption cost of materials
Unit price of materials
Time of arrival of materials
Criticality of materials

In FSN analysis the letter S stands for:
Slack materials
Stocked materials,
Slow moving materials
Standard materials

In FSN analyses, the letter N stands for:
Non moving materials
Next issuing materials,
No materials
None of the above

FSN analysis depends on:
Weight of the material
Volume of material,
Consumption pattern
Method of moving materials

MRP stands for:
Material Requirement Planning
Material Reordering Planning,
Material Requisition Procedure
Material Recording Procedure

A system where the period of placing the order is fixed is known as:
q – system
Fixed order system
***p* – system**
Fixed quantity system

A system in which quantity for which order is placed is constant is known as:

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***q* – System**
p - system,
Period system
Bin system

LOB stands for:
Lot of Bills
Line of Batches
Lot of Batches
Line of Balance

High reliability spare parts in inventory are known as:
Reliable spares
Insurance spares
Capital spares
Highly reliable spares

The property of capital spares is:
They have very low reliability;
These can be purchased in large quantities, as the price is low,
These spares have relatively higher purchase cost than the maintenance spares.
They are very much similar to breakdown spares.

Re-usable spares are known as:
Multi use spares
Repeated useable stores,
Scrap materials
Rotable spares

JIT stands for:
Just in time Purchase
Just in time production,
Just in time use of materials
Just in time order the material

The cycle time, selected in balancing a line must be:
Must be greater than the smallest time element given in the problem,
Must be less than the highest time element given in the problem,
Must be slightly greater than the highest time element given in the problem,
Left to the choice of the problem solver

The cycle time, selected in balancing a line must be:
Greater than the smallest time element given in the problem,
Less than the highest time element given in the problem,
Slightly greater than the highest time element given in the problem,
Left to the choice of the problem solver

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The lead-time is the time:
To place orders for materials,
Time of receiving materials,
Time between receipt of material and using materials
Time between placing the order and receiving the materials

The lead-time is the time:
To place orders for materials,
Of receiving materials,
Between receipt of material and using materials,
Between placing the order and receiving the materials

The PQR classification of inventory depends on:
Unit price of the material
Annual consumption value of material,
Criticality of material
Shelf life of the materials

The PQR classification of inventory depends on:
Unit price of the material
Annual consumption value of the material,
Criticality of the material
Shelf life of the materials

The classification made on the weight of the materials is known as:
PQR analysis
VED analysis
XYZ analysis
FSN analysis

At *EOQ*
Annual purchase cost = Annual ordering cost
Annual ordering cost = Annual carrying cost
Annual carrying cost = annual shortage cost
Annual shortage cost = Annual purchase cost

If shortage cost is infinity,
No shortages are allowed
No inventory carrying cost is allowed,
Ordering cost is zero
Purchase cost = Carrying cost

The most suitable system for a retail shop is
***FSN* Analysis**
ABC analysis,

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VED analysis
GOLF analysis

The inventory maintained to meet unknown demand changes is known as
Pipeline inventory
Anticipatory inventory
De coupling inventory
Fluctuatory inventory

The most suitable inventory system for a Petrol bunk is
P- System
2 Bin system,
***Q*- System**
Probabilistic model

The water consumption from a water tank follows
***P* – system**
PQ -system,
Q – System
EOQ System

Which of the following inventories is maintained to meet expected demand fluctuations?
Fluctuatory Inventory
Buffer stock,
De-coupling inventory
Anticipatory inventory

Which of the following inventory is maintained to meet expected demand fluctuations?
Fluctuatory Inventory
Buffer stock
De- coupling inventory
Anticipatory inventory

Which of the following increases with quantity ordered per order
Carrying cost
Ordering cost,
Purchase cost
Demand

The ordering cost per order and average unit carrying cost are constant, and demand suddenly falls by 75 % then EOQ will:
Decreases by 50 %
Does not change
Increases by 50 %
Decreases by 40%

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In JIT system, the following is assumed to be zero.
Ordering cost
Transportation cost
Carrying cost
Purchase cost

Which of the following analyses neither considers cost nor value?
ABC
XYZ
HML
VED

Which of the following analysis neither considers cost nor value:
ABC
XYZ
HML
VED

As per queue discipline the following is not a negative behaviour of a customer:
Balking
Reneging
Boarding
Collusion

The expediting or follow up function in production control is an example of
LIFO
FIFO
SIRO
Pre emptive

The expediting or follow up function in production control is an example of
LIFO
FIFO
SIRO
Preemptive

In M/M/S: N/FIFO the following does not apply
Poisson arrival
Limited service
Exponential service
Single server

The dead bodies coming to a burial ground is an example of:
Pure Birth Process
Pure death Process
Birth and Death Process

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Constant rate of arrival

The system of loading and unloading of goods usually follows:
LIFO
FIFO
SIRO
SBP

A steady state exist in a queue if:

$\leq \mu$
$\geq \mu$
Note this question font should be symbol

If the operating characteristics of a queue are dependent on time, then is said to be:
Transient state
Busy state
Steady state
Explosive state

A person who leaves the queue by losing his patience to wait is said to be: ( )
Reneging
Balking
Jockeying
Collusion

The characteristics of a queuing model is independent of:
Number of service stations
Limit of length of queue
Service Pattern
Queue discipline

The unit of traffic intensity is:
Poisson
Markow
Erlang
Kendall

In (M /M /1) : (∞ / FCFS) model, the length of the system *L*s is given by:
*p*2/1/*p*
***p*/1–**
–
–
Note this question font should be symbol

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In (*M* / *M* / 1): (∞□/ FIFO) model, 1 / (μ . □□) represents:
Ls, Length of the system
Lq length of the queue
***Wq* Waiting time in queue**
Ws Waiting time in system. ( )

The queue discipline in stack of plates is:
SIRO
Non-Pre-Emptive
FIFO
LIFO

Office filing system follows:
LIFO
FIFO
SIRO
SBP

SIRO discipline is generally found in:
Loading and unloading
Office filing
Lottery draw
Train arrivals at platform

The designation of Poisson arrival, Exponential service, single server and limited queue selected randomly is represented by:
(M / E / S) : (∞□/ SIRO)
(M / M / 1) : (∞□/ SIRO)
(M / M / S) : (N / SIRO)
(M / M / 1) : (N / SIRO)

For a simple queue (*M* / *M* / 1), □□□□□□/□□is known as:
Poisson busy period
Random factor,
Traffic intensity
Exponential service factor

With respect to simple queuing model which on of the given below is wrong:
Lq *Wq*
β
***Ws* = *Wq* + μ**
Ls = *Lq* + β

When a doctor attends to an emergency case leaving his regular service is called:
Reneging
Balking

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Pre-emptive queue discipline
Non-Pre-Emptive queue discipline ( )

A service system, where customer is stationary and server is moving is found with:
Buffet Meals
Out patient at a clinic
Person attending the breakdowns of heavy machines
Vehicle at Petrol bunk

In a simple queuing model the waiting time in the system is given by:
(*Lq*. □ □) + (1/μ)

Wq + μ ( )

This department is responsible for the development of queuing theory:
Railway station
Municipal office
Telephone department
Health department.

If the number of arrivals during a given time period is independent of the number of arrivals that have already occurred prior to the beginning of time interval, then the new arrivals follow ----------distribution.
Erlang
Poisson
Exponential
Normal

Arrival →□Service →□Service →□Service →□Out →□
The figure given represents:
Single Channel Single Phase system
Multi channel single-phase system
Single channel multi phase system
Multi channel multi phase system

In queue designation *A*/*B*/*S* : (*d*/*f*), what does *S* represents:
Arrival Pattern
Service Pattern
Number of service channels
Capacity of the system

When the operating characteristics of the queue system dependent on time, then it is said to be:
Steady state
Explosive state
Transient state

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Any one of the above

The distribution of arrivals in a queuing system can be considered as a:
Death Process
Pure Birth Process
Pure live process
Sick process

Queuing models measure the effect of:
Random arrivals
Random service
Effect of uncertainty on the behaviour of the queuing system
Length of queue

Traffic intensity is given by:
Mean arrival rate/Mean service rate,
μ,

Number present in the queue / Number served

Variance of queue length is :
□□□□□□/μ
–
–
–

If the value of the game is zero, then the game is known as:
Fair strategy
Pure strategy
Pure game
Mixed strategy

The games with saddle points are:
Probabilistic in nature
Normative in nature
Stochastic in nature
Deterministic in nature

When Minimax and Maximin criteria matches, then
Fair game is exists
Unfair game is exists,
Mixed strategy exists
Saddle point exists

When the game is played on a predetermined course of action, which does not change throughout game, then the game is said to be

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Pure strategy game
Fair strategy game
Mixed strategy game
Unsteady game

If the losses of player *A* are the gins of the player *B*, then the game is known as:
Fair game
Unfair game
Non- zero sum game
Zero sum game

Identify the wrong statement:
Game without saddle point is probabilistic
Game with saddle point will have pure strategies
Game with saddle point cannot be solved by dominance rule.
Game without saddle point uses mixed strategies

In a two person zero sum game, the following does not hold correct:
Row player is always a loser
Column Player is always a winner.
Column player always minimizes losses
If one loses, the other gains

If a two person zero sum game is converted to a Linear Programming Problem,
Number of variables must be two only
There will be no objective function
If row player represents Primal problem, Column player represent Dual problem
Number of constraints is two only

In case, there is no saddle point in a game then the game is
Deterministic game
Fair game,
Mixed strategy game
Multi player game

When there is dominance in a game then
Least of the row $\geq$ highest of another row
Least of the row $\leq$ highest of another row
Every element of a row $\geq$ corresponding element of another row.
Every element of the row $\leq$ corresponding element of another row. ( )

When the game is not having a saddle point, then the following method is used to solve the game:
Linear Programming method
Minimax and maximin criteria
Algebraic method

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Graphical method

Consider the matrix given, which is a pay off matrix of a game. Identify the dominance in it.
B
X Y Z
P 1 7 3
A Q 5 6 4
R 7 2 0
P dominates Q
Y dominates Z
Q dominates R
Z dominates Y

Identify the unfair game:
C D C D
(*a*) A 0 0 (*b*) A 1 –1
B 0 0 B –1 1
C D C D
(*c*) A –5 +5 **(*d*) A 1 0**
B + 10 – 10 B 0 1

If there are more than two persons in a game then the game is known as:
Non zero sum game
Open game
Multiplayer game
Big game

For the pay of matrix the player *A* always uses:
B
I II
I –5 –2
A
II 10 5
First strategy
Mixed strategy of both II and I
Does not play game
Second strategy

For the pay off matrix the player prefers to play
B
I II
I –7 6
A
II –10 8
Second strategy
First strategy

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Keep quite
Mixed strategy

For the game given the value is:
B
I II
1 2 3
A
II –5 5
3
–5
5
2

In the game given the saddle point is:
B
I II III
I 2 –4 6
A II 0 –3 –2
III 3 –5 4
–2
0
–3
2

A competitive situation is known as:
Competition
Marketing
Game
None of the above

One of the assumptions in the game theory is:
All players act rationally and intelligently
Winner alone acts rationally
Loser acts intelligently,
Both the players believe luck

A play is played when:
The manager gives green signal
Each player chooses one of his courses of action simultaneously
The player who comes to the place first says that he will start the game
When the latecomer says that he starts the game. ( )

The list of courses of action with each player ……
Is finite
Number of strategies with each player must be same

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Number of strategies with each player need not be same
None of the above

A game involving '*n*' persons is known as:
Multi member game
Multi player game
***n* - person game**
not a game

Theory of games and economic behavior is published by:
John Von Neumann and Morgenstern
John Flood
Bellman and Neumann
Mr. Erlang

In the matrix of a game given below the negative entries are:
B
I II
I 1 –1
A
II –1 1
Payments from *A* to *B*
Payments from *B* to *A*
Payment from players to organizers
Payment to players from organizers

In Dynamic Programming Problems, the decisions are made in
Single stage
2-stages
Multi-stages
No decision making process

In dynamic programming problems, the main problem is divided into subproblems. Each sub-problem is known as:
Part
Stage
State
Mini problem

The technique of Dynamic Programming problem is developed by:
Taylor
Gilberth
Richard Bellman
Bellman and Clarke

Another name used to Dynamic Programming is:

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Multistage problem
Recursive optimization
State problems
No second name

If the outcome at any decision stage is unique and known for the problem, then the Dynamic programming problem is known as:
Probabilistic dynamic programming problem
Stochastic dynamic programming problem
Static dynamic programming problem
Deterministic dynamic programming problem

The possible decisions at any stage are known as:
States
Steps
Parts
None

The rule which determines the decision at each stage is known as
State
Stage
Policy
Decision

The conclusion of a process designed to weigh the relative utilities of a set of available alternatives to select most preferred one is known as:
Concluding session
Conclusion
End of the process
Decision

The body of knowledge that deals with the analysis of making of decisions is known as
Decisions
Knowledge base
Decision theory
Decision analysis

Decisions that are meant to solve repetitive and well structured problems are known as:
Repetitive decisions
Structured decisions,
Programmed decisions
Linear programming

Decisions that handle non-routine, novel, and ill structured problems are known as:
Non-programmed decisions
Programmed decisions,

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Ill-structured decisions
Non-linear programming

Operations Research is the outcome of
National emergency
Political problems
Combined efforts of talents of all fields
Economics and Engineering

O.R. came into existence during
World War I
India and Pakistan War,
World War II
None of the above

The name of the subject Operations Research is due to the fact that
Problems can be solved by war approach
The researchers do the operations
The war problems are generally known as operations and inventing a new way of solving such problems.
Mathematical operations are used in solving the problems

The first country to use Operations Research method to solve problems is
India
China,
U.K
U.S.A

The name Operations Research is first coined in the year
1945
1935,
1940
1950

The person who coined the name Operations Research is:
Bellman
Newman,
McClosky and Trefrhen
None of the above

O.R. Society of India is founded in the year
1965
1970,
1959
1972

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The objective of Operations Research is:
To find new methods of solving Problems,
To derive formulas
Optimal utilization of existing resources
To utilize the services of scientists

Operations Research is the art of giving
Good answers for war problems,
Bad answers to war problems,
Bad answers to problems where otherwise worse answers are given.
Good answers to problems where otherwise bad answers are given

Operations Research is
Independent thinking approach
Group thinking approach
Inter-disciplinary team approach
None of the above

The first step in solving Operations Research problem is
Model building
Obtain alternate solutions,
Obtain basic feasible solutions
Formulation of the problem

The model, which gives physical or visual representation of the problem, is
Analogue model
Static model,
Iconic model
Symbolic model

One of the properties of O.R. model is
Model should be complicated,
Model is structured to suit O.R. techniques,
Model should be structured to suit all the problems we come across,
Model should be easy to derive

The problem, which is used to disburse the available limited resources to activities, is known as
O.R. Model
Resources Model
Allocation Model
Activities model

A wide class of allocation models can be solved by a mathematical technique know as:
Classical model
Mathematical Model,
Descriptive model

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Linear Programming model

One of the properties of Linear Programming Model is
It will not have constraints,
It should be easy to solve,
It must be able to adopt to solve any type of problem,
The relationship between problem variables and constraints must be linear

The constraints of Maximisation problem are of
Greater than or equal type
Less than or equal type
Less than type
Greater than type

The slack variables indicate
Excess resource available
Shortage of resource available,
Nil resources
Idle resource

In graphical solution of solving Linear Programming problem to convert inequalities into equations, we
Use Slack variables,
Use Surplus variables
Use Artificial surplus variables,
Simply assume them to be equations

To convert ≤ type of inequality into equations, we have to
Assume them to be equations
Add surplus variables,
Subtract slack variables
Add slack variables

To convert ≥ type of inequality into equations, we have to
Add slack variable
Subtract slack Variable
Subtract surplus variable
Add surplus variable

In Graphical solution of maximisation problem, the line, which we move from origin to the extreme point of the polygon is :
Any one side of the polygon
Iso cost line,
Iso profit line
An imaginary line

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The key row indicates
Incoming variable
outgoing variable
Slack variable
Surplus variable

The key column indicates
Outgoing variable
Incoming variable
Independent variable
Dependent variable

The penalty for not taking correct decision is known as
Fine
Loss,
Cost
Opportunity cost.

To transfer the key row in simplex table we have to
Add the elements of key row to key number
Subtract the elements of key row from topmost no key row
Divide the elements of key row by key number
None of the above.

The solution of the Linear programming problem in graphical solution lies in
First quadrant
Second quadrant,
Third quadrant
Fourth quadrant

When we solve maximization problem by simplex method the elements of net evaluation row of optimal solution must be (when we use opportunity cost concept
Either zeros or positive numbers
Either zeros or negative numbers
All are negative numbers
All are zeros

When all the elements of replacement ratio column are equal, the situation is known as
Tie
Degeneracy
Break
None of the above

When the elements of net evaluation row of simplex table are equal, the situation is known as
Tie
Degeneracy

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Break
Shadow price

The number at the intersection of key row and key column is known as
Column number
Row number,
Key number
Cross number

Dual of a Duel is
Primal
Dual
Prima dual
None of the above

Primal of a Primal is :
Primal
Dual
Prima primal
duo primal

Dual of a Dual of Dual is
Dual
Primal
Double dual
Single dual

Primal of a dual is
Primal
Dual,
Prime dual
Prime primal

If Dual has a solution, then the primal will
Not have a solution
Have only basic feasible solution,
Have a solution
None of the above

If Primal Problem is a maximisation problem, then the dual will be
Maximisation Problem
Minimisation Problem
Mixed Problem
None of the above

To get the Replacement ration column elements we have to

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Divide Profit column elements by key number,
The first column elements of identity is divided by key number
Divide the capacity column elements by key number
Answer mentioned there is c option but there is no C option

The cost coefficient of slack variable is
Zero
One,
> than one
< than one, ( )

The cost coefficient of artificial surplus variable is
0
1
M
> than 1

If the primal has an unbounded solution, then the dual has
Optimal solution
No solution
Bound solution
None of the above

One of the important basic objectives of Inventory management is:
To calculate EOQ for all materials in the organisation,
To go in person to the market and purchase the materials,
To employ the available capital efficiently so as to yield maximum results
Once materials are issued to the departments, personally check how they are used

The best way of improving the productivity of capital is:
Purchase automatic machines
Effective labour control
To use good financial management
Productivity of capital is to be increased through effective materials management

Materials management is a body of knowledge, which helps manager to:
Study the properties of materials
Search for needed material
Increase the productivity of capital by reducing the cost of material
None of the above

The stock of materials kept in the stores in anticipation of future demand is known as:
Storage of materials
Stock of materials,
Inventory
Raw materials

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The stock of animals reared in anticipation of future demand is known as:
Live stock inventory
Animal inventory,
Flesh inventory
None of the above

The working class of human beings is a class of inventor known as:
Live stock
Human inventory,
Population
Human resource inventory

In general, the percentage of materials cost in product is approximately equal to:
40 to 50 %
5 to 10 %
2 to 3 %
90 to 95%

Materials management brings about increased productivity of capital by:
Very strict control over use of materials,
Increasing the efficiency of workers,
Preventing large amounts of capital locked up for long periods in the form of inventory
To apply the principles of capital management

We can reduce the materials cost by:
Using systematic inventory control techniques
Using the cheap material,
Reducing the use of materials,
Making hand to mouth purchase, ( )

The basis for ABC analysis is
Interests of Materials manager
Interests of the top management,
Pareto's 80-20 rule
None of the above

ABC analysis depends on the:
Quality of materials,
Cost of materials,
Quantity of materials used,
Annual consumption value of materials

'*A*' class materials consume:
10% of total annual inventory cost
30% of total annual inventory cost,

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70 to 75% of total inventory cost
90% of total annual inventory cost. ( )

'*B*' class of materials consumes .....% of annual inventory cost.
60 to 70%
20 to 25%
90 to 95%
5 to 8% ( )
'*C*' class of materials consume .....% of annual inventory cost.
5 to 10 %
20 to 30%
40 to 50%
70 to 80% ( )

The rent for the stores where materials are stored falls under:
Inventory carrying cost
Ordering cost,
Procurement cost
Stocking cost

Insurance charges of materials cost fall under:
Ordering cost
Inventory carrying cost
Stock out cost
Procurement cost

As the volume of inventory increases, the following cost will increase:
Stock out cost
Ordering cost,
Procuring cost
Inventory carrying cost

As the order quantity increases, this cost will reduce:
Ordering cost
Insurance cost,
Inventory carrying cost
Stock out cost

Procurement cost may be clubbed with:
Inventory carrying charges
Stock out cost
Loss due to deterioration
Ordering cost

The penalty for not having materials when needed is:
Loss of materials cost

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Loss of ordering cost,
Stock out cost
General losses

When load is the annual demand for the material, p = unit price of the material in Rs., $C3$ is the ordering cost, q = order quantity, then the total cost including the martial cost is given by:
(q/2) ip q C p
2*C* *ip* *p*
(*q*/2) *ip* *p*
(2*C*3 *q* *ip*) 1/2 ( )

Contractual maintenance or agreement maintenance with manufacturer is suitable for equipment, which is
In its infant state
When machine is old one,
Scrapped
None of the above

When money value changes with time at 10 %, then PWF for first year is :
1
0.909
0.852
0.9

When money value changes with time at 20%, the discount factor for 2nd year is:
1
0.833
0
0.6955 ( )

Which of the following maintenance policy is not used in old age stage of a machine? My Quiz
Operate up to failure and do corrective maintenance
Reconditioning
Replacement
Scheduled preventive maintenance

Which of the following maintenance policies is not used in old age stage of a machine?
Operate up to failure and do corrective maintenance,
Reconditioning,
Replacement,
Scheduled preventive maintenance

Which of the following replacement policies is considered to be dynamic in nature?
Time is continuous variable and the money value does not change with time,
When money value does not change with time and time is a discrete variable,
When money value changes with time

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When money value remains constant for some time and then goes on changing with time

Which of the following replacement policy is considered to be dynamic in nature?
Time is continuous variable and the money value does not change with time.
When money value does not changes with time and time is a discrete variable.
When money value changes with time
When money value remains constant for some time and then goes on changing with time

When the probability of failure reduces gradually, the failure mode is said to be:
Regressive
Retrogressive
Progressive
Recursive

The following replacement model is said to be probabilistic model:
When money value does not change with time and time is a continuous variable
When money value changes with time
When money value does not change with time and time is a discrete variable
Preventive maintenance policy (I think)
Check

A machine is replaced with average running cost_____ My Quiz
Is not equal to current running cost,
Till current period is greater than that of next period,
If current period is greater than that of next period,
If current period is less than that of next period

The curve used to interpret machine life cycle is
Bath tub curve
Time curve,
Product life cycle
Ogive curve

Decreasing failure rate is usually observed in ……. stage of the machine
Infant
Youth
Old age
Any time in its life

Which cost of the following is irrelevant to replacement analysis?
Purchase cost of the machine,
Operating cost of the machine,
Maintenance cost of the machine,
Machine hour rate of the machine

The type of failure that usually occurs in old age of the machine is

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Random failure
Early failure,
Chance failure
Wear-out failure

Group replacement policy is most suitable for: My Quiz
Trucks
Infant machines,
Street light bulbs
New cars

The chance failure that occur on a machine are commonly found on a graph of time Vs Failure rate (on X and Y axis respectively as
Parabolic
Hyperbolic
Line nearly parallel to X axis
Line nearly parallel to Y-axis

The chance failure that occurs on a machine is commonly found on a graph of time Vs failure rate (on *X* and *Y* axes respectively as
Parabolic
Hyperbolic,
Line nearly parallel to X-axis
Line nearly parallel to Y-axis

Replacement of an item will become necessary when
Old item becomes too expensive to operate or maintain
When your operator desires to work on a new machine
When your opponent changes his machine in his unit
When company has surplus funds to spend

The production manager will not recommend group replacement policy My Quiz
When large number of identical items are to be replaced
In case Low cost items are to be replaced, where record keeping is a problem
For items that fail completely
For Reparable items

Which of the following is the correct assumption for replacement policy when money value does not change with time? My Quiz
No Capital cost
No scrap value,
Constant scrap value
Zero maintenance cost

Which one of the following does not match the group?
Present Worth Factor (PWF)

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Discounted rate (DR),
Depreciation value (DV)
Mortality Tables (MT)

Reliability of an item is
Failure Probability
1 / Failure probability,
1 - failure probability
Life period / Failure rate

The following is not discussed in group replacement policy: My Quiz
Failure Probability
Cost of individual replacement,
Loss due to failure
Present worth factor series

It is assumed that maintenance cost mostly depends on:
Calendar age
Manufacturing date
Running age
User's age

Group replacement policy applies to:
Irreparable items
Reparable items
Items that fail partially
Items that fail completely

If a machine becomes old, then the failure rate expected will be:
Constant
Increasing
Decreasing
We Cannot be said

Replacement is said to be necessary if
Failure rate is increasing
Failure cost is increasing,
Failure probability is increasing
Any of the above

In this stage, the machine operates at highest efficiency and its production rate will be high.
Infant stage
Youth stage
Old age
None of the above

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Replacement decision is very much common in this stage:
Infant stage
Old age
Youth
In all the above

The replacement policy that is imposed on an item irrespective of its failure is My Quiz
Group replacement
Individual replacement
Repair spare replacement
Successive replacement

When certain symptoms indicate that a machine is going to fail and to avoid failure if maintenance is done it is known as:
Symptoms maintenance
Predictive maintenance
Repair maintenance
Scheduled maintenance

In retrogressive failures, the failure probability ---------- with time
Increases
Remains constant,
Decreases
None of the above

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10:48 AM 84%
quiz.vu.edu.pk/QuizQue
MC200203075: MUHAMMAD FAISAL
Time Left 87 sec(s)
MTH601:Quiz
Quiz Start Time: 10:40 AM, 16 May 2021
Question # 10 of 10 (Start time: 10:48:05 AM, 16 May 2021)
Total Marks: 1

Which of the following is the objective of Project Management by using PERT and CPM methods, for any project subject to resource constraints?

Select the correct option

- To minimize the project time
- To maximize the total project profit
- To minimize the total project cost
- To minimize the resource constraints

10:47 AM 84%
quiz.vu.edu.pk/QuizQue
MC200203075: MUHAMMAD FAISAL
Time Left 85 sec(s)
MTH601:Quiz
Quiz Start Time: 10:40 AM, 16 May 2021
Question # 9 of 10 (Start time: 10:47:27 AM, 16 May 2021)
Total Marks: 1

PERT is an acronym for _____ .

Select the correct option

- (●) Program Evaluation and Review Technique
- ( ) Performance Evaluation and Review Technique Observation
- ( ) Program Execution and Revenue Technique
- ( ) Program Evaluation and Review Teacher

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 10:45:15 AM, 16 May 2021) Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Observation
- Simplex method

10:44 AM 85%
quiz.vu.edu.pk/QuizQue
MC200203075: MUHAMMAD FAISAL
Time Left 83 sec(s)
MTH601:Quiz
Quiz Start Time: 10:40 AM, 16 May 2021

Question # 7 of 10 (Start time: 10:44:02 AM, 16 May 2021) Total Marks: 1

Operations research (OR) is the application of quantitative methods to _____ making.

Select the correct option

- formula
- decision
- observation
- all of these

Click to Save Answer & Move to Next Question

4G 3G 10:43 AM 85%
quiz.vu.edu.pk/QuizQue
1

MC200203075: MUHAMMAD FAISAL

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:40 AM, 16 May 2021

Question # 6 of 10 (Start time: 10:43:51 AM, 16 May 2021)

Total Marks: 1

If for path 1-3-4 time is 5+3=8 days, for path 1-2-3-4 time is 6+7+3=16 days and for path 1-2-4 time is 6+6=12 days, then which of the followings is true.

Select the correct option

- (●) 16 days
- ( ) 8 days
- ( ) 12 days
- ( ) 36 days

Click to Save Answer & Move to Next Question

10:43 AM 85%
quiz.vu.edu.pk/QuizQue
MC200203075: MUHAMMAD FAISAL
Time Left 85 sec(s)
MTH601:Quiz
Quiz Start Time: 10:40 AM, 16 May 2021
Question # 5 of 10 (Start time: 10:43:11 AM, 16 May 2021)
Total Marks: 1

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, the computations are proceeded from ---------- to the final event.

Select the correct option

- bottom to top
- right to left
- left to right
- top to bottom

4G 3G 10:42 AM 85%

quiz.vu.edu.pk/QuizQue:

MC200203075: MUHAMMAD FAISAL

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:40 AM, 16 May 2021

Question # 4 of 10 (Start time: 10:42:36 AM, 16 May 2021)

Total Marks: 1

In a project, a network diagram shows the precedence relations of inter related activities along with their corresponding activity ---------.

Select the correct option

- times
- cost
- profit
- quantity

10:41 AM 85%
quiz.vu.edu.pk/QuizQue
MC200203075: MUHAMMAD FAISAL
Time Left 86 sec(s)
MTH601:Quiz
Quiz Start Time: 10:40 AM, 16 May 2021

Question # 3 of 10 (Start time: 10:41:49 AM, 16 May 2021) Total Marks: 1

Dummy activity is represented by _____ .

Select the correct option

- (●) doted line
- ( ) doted line with right to left arrow
- ( ) doted line with left to right arrow
- ( ) left to right arrow

Click to Save Answer & Move to Next Question

10:41 AM 85%
quiz.vu.edu.pk/QuizQue
MC200203075: MUHAMMAD FAISAL
Time Left 88 sec(s)
MTH601:Quiz
Quiz Start Time: 10:40 AM, 16 May 2021

Question # 2 of 10 (Start time: 10:41:13 AM, 16 May 2021) Total Marks: 1

In a network flow diagram, a/an ---------- is shown by dotted arrows and these consume no time or resources.

Select the correct option

- activity
- branch
- dummy
- nodes

Click to Save Answer & Move to Next Question

4G 3G 10:40 AM 86%
quiz.vu.edu.pk/QuizQue
1

MC200203075: MUHAMMAD FAISAL

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 10:40 AM, 16 May 2021

Question # 1 of 10 (Start time: 10:40:40 AM, 16 May 2021)

Total Marks: 1

There are _____ kinds of time calclulations for a project.

Select the correct option

- 1
- 2
- 3
- 4

Click to Save Answer & Move to Next Question

10:35 AM
Quiz

MTH601:Quiz

Question # 10 of 10 (Start time: 10:34:39 AM, 16 May 2021)

Operations research (OR) is the application of quantitative methods _____ making.

Select the correct option

- formula
- decision
- observation
- all of these

Click to Save Answer & Move to Next Question

10:30

quiz.vu.edu.pk/QuizQue:

MC200203677: MUHAMMAD AWAIS

Time Left 85 sec(s)

MTH601:Quiz

Quiz Start Time: 10:26 AM, 16 May 2021

Question # 7 of 10 (Start time: 10:30:16 AM, 16 May 2021)

Total Marks: 1

In a network flow diagram, events are just the points in ---------be represented by circles.

Select the correct option

- xy-plane
- space only
- time only
- space-time

10:34 AM
Quiz

MC200203745: AYYAZ SIDDIQUE

MTH601:Quiz

Question # 9 of 10 (Start time: 10:34:17 AM, 16 May 2021)

If for path 1-3-4 time is 5+3=8 days, for path 1-2-3-4 time is 6+7+3=16 days and for path 1-2-4 time is 6+6=12 days, then which of the followings is true.

Select the correct option

- 1-3-4 is critical path
- 1-2-4 is critical path
- 1-2-3-4 is critical path
- 1-4 is critical path

Click to Save Answer & Move to Next Question

10:34 AM

Quiz

MC200203745: AYYAZ SIDDIQUE

MTH601:Quiz

Question # 8 of 10 (Start time: 10:33:44 AM, 16 May 2021)

In a network flow diagram, for an activity (i,j) of duration of three days, if its earliest start time is of two days then which of the following will be its early finish time?

Select the correct option

- Six days
- One day
- One and half day
- Five days

Click to Save Answer & Move to Next Question

10:33 AM

Quiz

Question # 7 of 10 (Start time: 10:33:18 AM, 16 May 2021)

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Observation
- Simplex method

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 10:30:18 AM, 16 Ma

In Project Management, Program Evaluation and Review Technique (PERT) is based on ---------- times.

Select the correct option

- deterministic
- probabilistic
- stochastic
- serial

Click to Save Answer & Move to Next Question

10:33 AM
Quiz

Question # 6 of 10 (Start time: 10:32:45 AM, 16 May 2021)

Which of the following is not a categories of the operations research (OR) techniques.

Select the correct option

- Linear mathematical programming technique
- Feasible solution
- Inventory techniques
- Network techniques

Click to Save Answer & Move to Next Question

10:32 493 B/s 55
quiz.vu.edu.pk/QuizQue:

MC200203677: MUHAMMAD AWAIS

Time Left 85 sec(s)

MTH601:Quiz

Quiz Start Time: 10:26 AM, 16 May 2021

Question # 10 of 10 (Start time: 10:32:38 AM, 16 May 2021)

Total Marks: 1

Which one of the following is not an operations research problem solving steps?

Select the correct option

- observation
- definition of the problems
- variable selections
- model construction

LMS

Question # 5 of 10 (Start time: 10:31:42 AM, 16 May 2021)

CPM is abbreviation of _____ .

Select the correct option

- Constant Process Method
- Critical Path Method
- Continuous Path Membership
- Critical Performance Method

Click to Save Answer & Move to Next Question

10:31

590 B/s

quiz.vu.edu.pk/QuizQue:

MC200203677: MUHAMMAD AWAIS

Time Left 81 sec(s)

MTH601:Quiz

Quiz Start Time: 10:26 AM, 16 May 2021

Question # 9 of 10 (Start time: 10:31:28 AM, 16 May 2021)

Total Marks: 1

Which of the following is not a problem of probabilistic technique.

Select the correct option

- Linear programming
- Decision analysis
- Game theory
- Forecasting

Click to Save Answer & Move to Next Question

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Simplex method
- Problem definition

10:30 AM
Quiz

Question # 3 of 10 (Start time: 10:29:41 AM, 16 May 202

In 1957, Churchmen Ackoff and Arnoff defined OR as "OR is the application of _____, techniques and tools to problems involving the operations of systems so as to provide those in control of the operations with optimum solutions to the problem."

Select the correct option

- ○ computational methods
- ○ scientific methods
- ○ graphical methods
- ○ none of these

Click to Save Answer & Move to Next Question

Select the correct option

- Resource Scheduling
- Resource Allocation
- PERT
- CPM

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

In a network flow diagram, which of the following method through computations provides, i) start and completion times for each activity, ii) critical and non critical activities and iii) total and free slacks?

Select the correct option

- Resource Scheduling
- Resource Allocation
- PERT

10:29 AM
Quiz

Question # 2 of 10 (Start time: 10:29:03 AM, 16 May 2021)

In a network flow diagram, a/an ---------- is shown by dotted arrows and these consume no time or resources.

Select the correct option

- activity
- branch
- dummy
- nodes

Click to Save Answer & Move to Next Question

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

In a network flow diagram, events are just the points in ----------be represented by circles.

Select the correct option

- xy-plane
- space only
- time only
- space-time

10:28

quiz.vu.edu.pk/QuizQue:

MC200203677: MUHAMMAD AWAIS

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:26 AM, 16 May 2021

Question # 3 of 10 (Start time: 10:28:32 AM, 16 May 2021)

Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Observation
- Simplex method

10:29 AM

Quiz

MTH601:Quiz

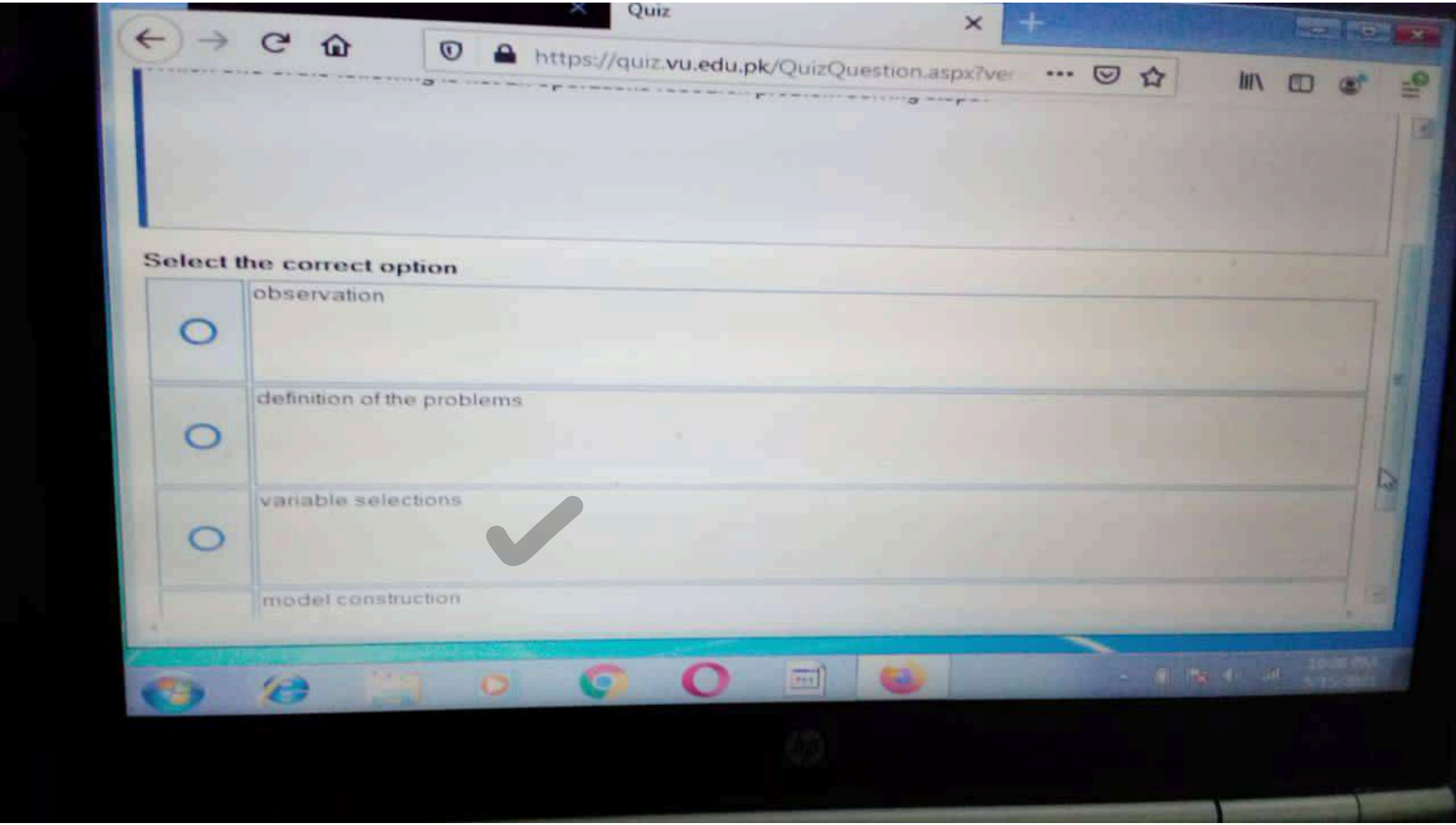
Question # 1 of 10 (Start time: 10:28:35 AM, 16 May 2021)

PERT is an acronym for _____ .

Select the correct option

- Program Evaluation and Review Technique
- Performance Evaluation and Review Technique Observation
- Program Execution and Revenue Technique
- Program Evaluation and Review Teacher

Click to Save Answer & Move to Next Question


Quiz
https://quiz.vu.edu.pk/QuizQuestion.aspx?ver
Select the correct option
observation
definition of the problems
variable selections
model construction

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

Which one of the following is not an operations research problem solving steps?

Select the correct option

- observation
- definition of the problems
- variable selections
- model construction

10:27

539 B/s

quiz.vu.edu.pk/QuizQue:

MC200203677: MUHAMMAD AWAIS

Time Left 86 sec(s)

MTH601:Quiz

Quiz Start Time: 10:26 AM, 16 May 2021

Question # 2 of 10 (Start time: 10:27:34 AM, 16 May 2021)

Total Marks: 1

The first step in the application of _____ is to develop a network representation of the project plan.

Select the correct option

- probability
- CPM
- linear programming
- none of these

Save and Next Move to next Question

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

In a network flow diagram, for an activity (i,j) of duration of three days, if its earliest start time is of two days then which of the following will be its early finish time?

Select the correct option

- Six days
- One day
- One and half day
- Five days

Solved

10:27

741 B/s

quiz.vu.edu.pk/QuizQue:

MC200203677: MUHAMMAD AWAIS

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:26 AM, 16 May 2021

Question # 1 of 10 (Start time: 10:26:59 AM, 16 May 2021)

Total Marks: 1

In Project Management, Critical Path method is based on ---------- times.

Select the correct option

- deterministic
- probabilistic
- stochastic
- serial

Save & Next Answer & Move to next Question

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver=

The length of a path in a network is the total _____ it takes to travel along the path.

Select the correct option

- length
- days
- time
- months

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, the computations are proceeded from ---------- to the final event.

Select the correct option

- bottom to top
- right to left
- left to right
- top to bottom

be its early start day?

Select the correct option

- Seventh day
- Third day
- First day
- Fifth day

Jazz
Zong CMPak
1.4K/s 4G 2G 50% 10:25
Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 10 of 10 (Start time: 10:25:29 AM, 16 May 2021)

Total Marks: 1

In 1957, Churchmen Ackoff and Arnoff defined OR as "OR is the application of _____, techniques and tools to problems involving the operations of systems so as to provide those in control of the operations with optimum solutions to the problem."

Select the correct option

- computational methods
- scientific methods
- graphical methods
- none of these

Click to Save Answer & Move to Next Question

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

In a development project, if an activity (i,j) of duration three days finish earlier on the tenth day then which of the following will be its early start day?

Select the correct option

- Seventh day
- Third day
- First day

Question # 10 of 10 (Start time: 10:25:02 AM, 16 May 2021) Total Marks: 1

CPM is abbreviation of _____ .

Select the correct option

- Constant Process Method
- Critical Path Method
- Continuous Path Membership
- Critical Performance Method

Click to Save Answer & Move to Next Question

Select the correct option

- node
- event
- project
- activity

Jazz
Zong CMPak
1.7K/s 4G 2G 50% 10:24
Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 9 of 10 (Start time: 10:24:43 AM, 16 May 2021)

Total Marks: 1

A set of nodes connected by lines which begin at the initial node of a network and end at the terminal node is called _____.

Select the correct option

- Critical path
- Path
- Node
- Network

Click to Save Answer & Move to Next Question

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver=

The task which is executed by the usage of resources and time is called ---------.

Select the correct option

- node
- event
- project

activity

Question # 9 of 10 (Start time: 10:24:28 AM, 16 May 2021) Total Marks: 1

In a network flow diagram, a/an ---------- is shown by dotted arrows and these consume no time or resources.

Select the correct option

- activity
- branch
- dummy
- nodes

Click to Save Answer & Move to Next Question

Jazz
Zong CMPak
538B/s 4G 2G 50% 10:24
Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 8 of 10 (Start time: 10:24:10 AM, 16 May 2021)

Total Marks: 1

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, for the first node(event), we start with time ----------.

Select the correct option

- t = infinity
- t = 0
- t = a (arbitrary)
- with strict positive value

Click to Save Answer & Move to Next Question

Quiz List

Quiz

https://quiz.vu.edu.pk/QuizQuestion.aspx?ver

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Observation
- Simplex method

Question # 8 of 10 (Start time: 10:23:41 AM, 16 May 2021)
Total Marks: 1

A/An ---------- is the collection of inter related activities to be performed in a particular sequence to completion.

Select the correct option

- node
- event
- project
- branch

Click to Save Answer & Move to Next Question

Jazz
Zong CMPak
380B/s 4G 2G 50% 10:23

Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 7 of 10 (Start time: 10:23:36 AM, 16 May 2021)

Total Marks: 1

In a network flow diagram, events are just the points in ---------be represented by circles.

Select the correct option

- xy-plane
- space only
- time only
- space-time

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 10:23:06 AM, 16 May 2021) Total Marks: 1

Duration of dummy activity is _____.

Select the correct option

- latest start time
- zero
- early start time
- 1

Click to Save Answer & Move to Next Question

Jazz
Zong CMPak
2.6K/s 4G 2G 50% 10:22
Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 83 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 6 of 10 (Start time: 10:22:47 AM, 16 May 2021)

Total Marks: 1

Operations research (OR) is the application of _____ methods to decision making.

Select the correct option

- engineering
- quantitative
- computing
- all of these

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 10:22:32 AM, 16 May 2021) Total Marks: 1

If both jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously on 4th day, then 'n' can start earlier on ----------day.

Select the correct option

- 8th
- 11th
- 15th
- 12th

Click to Save Answer & Move to Next Question

Jazz
Zong CMPak
17.9K/s 4G 2G 49% 10:22

Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 5 of 10 (Start time: 10:22:05 AM, 16 May 2021)

Total Marks: 1

In 1948, Blackett defined OR as "OR is a _____ of providing executives with any _____ and objective basis for decisions"

Select the correct option

- computational method, control
- scientific method, analytical
- graphical method, home
- none of these

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 10:22:03 AM, 16 May 2021) Total Marks: 1

In Project Management, Critical Path method is based on ---------- times.

Select the correct option

- deterministic
- probabilistic
- stochastic
- serial

WhatsApp 1:22 pm
Message from +92 306 9713773 @ MSc M@TH [|||]
quiz.vu.edu.pk/QuizQue:

MC200203325: KASHMALA ASIF

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 01:20 PM, 15 May 2021

Question # 2 of 10 (Start time: 01:22:15 PM, 15 May 2021)

Total Marks: 1

The task which is executed by the usage of resources and time is called ----------.

Select the correct option

- node
- event
- project
- activity

Jazz
Zong CMPak
747B/s 4G 2G 49% 10:21
Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 4 of 10 (Start time: 10:21:30 AM, 16 May 2021)

Total Marks: 1

The length of a path in a network is the total _____ it takes to travel along the path.

Select the correct option

- length
- days
- time
- months

Click to Save Answer & Move to Next Question

10:27 AM | 16.9KB/s 56%

quiz.vu.edu.pk/QuizQuestion 3

MC240205135: MUHAMMAD MAZHAR

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 4 of 10 (Start time: 10:21:36 AM, 16 May 2021) Total Marks: 1

There are _____ kinds of time calclulations for a project.

Select the correct option

- 1
- 2
- 3
- 4

LMS

Question # 3 of 10 (Start time: 10:21:01 AM, 16 May 2021) Total Marks: 1

In a network flow diagram, the precedence relationship among the activitiees are indicated through ----------.

Select the correct option

- project
- branch
- dummy
- nodes

Click to Save Answer & Move to Next Question

Jazz
Zong CMPak
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Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 86 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 3 of 10 (Start time: 10:20:59 AM, 16 May 2021)

Total Marks: 1

Operations research is based on _____.

Select the correct option

- social science
- statistics
- engineering
- all of these

Click to Save Answer & Move to Next Question

Jazz
Zong CMPak
1.3K/s 4G 2G 48% 10:19
Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 87 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 2 of 10 (Start time: 10:19:21 AM, 16 May 2021)

Total Marks: 1

In 1958, Saaty defined OR as "The art of giving _____ answer to problems to which, otherwise, _____ answers are given."

Select the correct option

- bad, worse
- worse, bad
- good, excellent
- excellent, worse

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 10:19:08 AM, 16 May 2021)

Total Marks: 1

In a network flow diagram, for an activity (i,j) of duration of three days, if its earliest start time is of two days then which of the following will be its early finish time?

Select the correct option

- Six days
- One day
- One and half day
- Five days

Click to Save Answer & Move to Next Question

Jazz
Zong CMPak
43.1K/s 4G 2G 48% 10:18

Quiz
quiz.vu.edu.pk

MC200201616: SIRDA RANI

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:18 AM, 16 May 2021

Question # 1 of 10 (Start time: 10:18:23 AM, 16 May 2021)

Total Marks: 1

Which of the following is not a categories of the operations research (OR) techniques.

Select the correct option

- Linear mathematical programming technique
- Feasible solution
- Inventory techniques
- Network techniques

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 10:18:22 AM, 16 May 2021) Total Marks: 1

PERT is an acronym for _____ .

Select the correct option

- Program Evaluation and Review Technique
- Performance Evaluation and Review Technique Observation
- Program Execution and Revenue Technique
- Program Evaluation and Review Teacher

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 10:13:49 AM, 16 May 2021) Total Marks: 1

In a network flow diagram, two jobs 'a(i,k)' and 'b(j,k)' of durations '4' and '5' days respectively, enter an event 'k' then which of the following will be earliest start day of 'k' provided that (i,k) and (j,k) have started earlier at 3rd and 4th day respectively?

Select the correct option

- 7th
- 8th
- 9th
- 6th

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 10:13:12 AM, 16 May 2021) Total Marks: 1

Which one of the following is not an operations research problem solving steps?

Select the correct option

- observation
- definition of the problems
- variable selections
- model construction

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 10:12:24 AM, 16 May 2021) Total Marks: 1

In a development project, if an activity (i,j) of duration three days finish earlier on the tenth day then which of the following will be its early start day?

Select the correct option

- Seventh day
- Third day
- First day
- Fifth day

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 10:11:50 AM, 16 May 2021) Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Observation
- Simplex method

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 10:11:15 AM, 16 May 2021) Total Marks: 1

The task which is executed by the usage of resources and time is called ————.

Select the correct option

- node
- event
- project
- activity

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 10:10:42 AM, 16 May 2021) Total Marks: 1

In 1946, Morse & Kimbel has defined operations research as "OR is a _____ of providing executive departments with a quantitative basis for decision regarding the operations under their_____"

Select the correct option

- scientific method, control
- computational method, control
- graphical method, analytical
- none of these

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 10:10:04 AM, 16 May 2021)

Total Marks: 1

Which of the following is not a categories of the operations research (OR) techniques.

Select the correct option

- Linear mathematical programming technique
- Feasible solution
- Inventory techniques
- Network techniques

Click to Save Answer & Move to Next Question

Question # 3 of 10 (Start time: 10:09:25 AM, 16 May 2021) Total Marks: 1

Choose the correct one for the project times that are used in project management.

Select the correct option

- Latest start time, Latest finish time, Earliest start time, Earliest finish time
- Latest start time, Latest finish time, Earliest start time, CPM
- Latest start time, Latest finish time, PERT, Earliest finish time
- PERT, CPM

Click to Save Answer & Move to Next Question

Question # 2 of 10 (Start time: 10:08:45 AM, 16 May 2021)

Total Marks: 1

If for path 1-3-4 time is 5+3=8 days, for path 1-2-3-4 time is 6+7+3=16 days and for path 1-2-4 time is 6+6=12 days, then which of the followings is true.

Select the correct option

- 16 days
- 8 days
- 12 days
- 36 days

Click to Save Answer & Move to Next Question

Time Left sec(s)

MTH601:Quiz

Quiz Start Time: 10:08 AM, 16 May 2021

Question # 1 of 10 (Start time: 10:08:07 AM, 16 May 2021)

Total Marks: 1

CPM is abbreviation of _____ .

Select the correct option

- Constant Process Method
- Critical Path Method
- Continuous Path Membership
- Critical Performance Method

Click to Save Answer & Move to Next Question

Telenor Jazz
69% 10:08 AM
quiz.vu.edu.pk/QuizQue:

MC200200170: NAFEES AHMAD

Time Left 79 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 10 of 10 (Start time: 10:07:48 AM, 16 May 2021)

Total Marks: 1

Choose the correct one for the project times that are used in project management.

Select the correct option

- Latest start time, Latest finish time, Earliest start time, Earliest finish time
- Latest start time, Latest finish time, Earliest start time, CPM
- Latest start time, Latest finish time, PERT, Earliest finish time
- PERT, CPM

Click to Save Answer & Move to Next Question

Telenor
Jazz
70% 10:07 AM
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MC200200170: NAFEES AHMAD

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 9 of 10 (Start time: 10:07:11 AM, 16 May 2021)

Total Marks: 1

PERT is an acronym for _____ .

Select the correct option

- Program Evaluation and Review Technique
- Performance Evaluation and Review Technique Observation
- Program Execution and Revenue Technique
- Program Evaluation and Review Teacher

Click to Save Answer & Move to Next Question

Question # 10 of 10 (Start time: 10:05:31 AM, 16 May 2021) Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Simplex method
- Problem definition

Telenor
Jazz
70% 10:06 AM
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Question # 8 of 10 (Start time: 10:06:37 AM, 16 May 2021) Total Marks: 1

In 1948, Morse defined OR as "The term OR, has here-to fore been used to connote various attempts to study operations of war by _____. From a more general point of view, OR can be considered to be an attempt to study those operations of modern society which involved organizations of men or men and machines."

Select the correct option

- computational methods
- graphical methods
- scientific methods
- none of these

LMS

Telenor
Jazz
70% 10:06 AM
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MC200200170: NAFEES AHMAD

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 7 of 10 (Start time: 10:06:01 AM, 16 May 2021)

Total Marks: 1

In 1957, Churchmen Ackoff and Arnoff defined OR as "OR is the application of _____, techniques and tools to problems involving the operations of systems so as to provide those in control of the operations with optimum solutions to the problem."

Select the correct option

- computational methods
- scientific methods
- graphical methods
- none of these

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 10:05:41 AM, 16 May 2021) Total Marks: 1

A set of nodes connected by lines which begin at the initial node of a network and end at the terminal node is called _____.

Select the correct option

- Critical path
- Path
- Node
- Network

Telenor Jazz
71% 10:05 AM
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MC200200170: NAFEES AHMAD

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 6 of 10 (Start time: 10:05:31 AM, 16 May 2021)

Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Model construction
- Linear programming
- Networking
- Simplex method

Click to Save Answer & Move to Next Question

MTH601:Quiz

Quiz Start Time: 09:38 AM, 16 May 2021

Question # 8 of 10 (Start time: 10:04:53 AM, 16 May 2021)

Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Model construction
- Linear programming
- Networking
- Simplex method

Telenor
Jazz
71% 10:05 AM
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MC200200170: NAFEES AHMAD

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 5 of 10 (Start time: 10:05:00 AM, 16 May 2021)

Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Implementation
- Simplex method

Click to Save Answer & Move to Next Question

Telenor
Jazz
71% 10:04 AM
quiz.vu.edu.pk/QuizQue

MC200200170: NAFEES AHMAD

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 4 of 10 (Start time: 10:04:25 AM, 16 May 2021)

Total Marks: 1

In Project Management, Critical Path method is based on ---------- times.

Select the correct option

- deterministic
- probabilistic
- stochastic
- serial

Click to Save Answer & Move to Next Question

Question # 7 of 10 (Start time: 10:03:34 AM, 16 May 2021) Total Marks: 1

In a network flow diagram, the precedence relationship among the activitiees are indicated through ----------.

Select the correct option

- project
- branch
- dummy
- nodes

Telenor
Jazz
71% 10:03 AM
quiz.vu.edu.pk/QuizQue:

MC200200170: NAFEES AHMAD

Time Left 89 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 3 of 10 (Start time: 10:03:44 AM, 16 May 2021)

Total Marks: 1

About which of the following it is true that, “they only signify with the passage of time, the beginning and ending of some activities under no consumption of resources” ?

Select the correct option

- project
- nodes
- dummy
- branch

Click to Save Answer & Move to Next Question

Telenor Jazz
71% 10:03 AM
quiz.vu.edu.pk/QuizQue:

MC200200170: NAFEES AHMAD

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 2 of 10 (Start time: 10:03:17 AM, 16 May 2021)

Total Marks: 1

Operations research is based on _____.

Select the correct option

- social science
- statistics
- engineering
- all of these

Click to Save Answer & Move to Next Question

Question # 6 of 10 (Start time: 10:02:41 AM, 16 May 2021) Total Marks: 1

Which of the following is an operations research (OR) process?

Select the correct option

- Linear programming
- Networking
- Observation
- Simplex method

Telenor
Jazz
72% 10:02 AM
quiz.vu.edu.pk/QuizQue:

MC200200170: NAFEES AHMAD

Time Left 88 sec(s)

MTH601:Quiz

Quiz Start Time: 10:02 AM, 16 May 2021

Question # 1 of 10 (Start time: 10:02:39 AM, 16 May 2021)

Total Marks: 1

In 1948, Blackett defined OR as "OR is a _____ of providing executives with any _____ and objective basis for decisions"

Select the correct option

- computational method, control
- scientific method, analytical
- graphical method, home
- none of these

Click to Save Answer & Move to Next Question

Question # 5 of 10 (Start time: 10:02:05 AM, 16 May 2021) Total Marks: 1

The length of a path in a network is the total _____ it takes to travel along the path.

Select the correct option

- length
- days
- time
- months

Question # 4 of 10 (Start time: 10:01:20 AM, 16 May 2021)
Total Marks: 1

In Project Management, Program Evaluation and Review Technique (PERT) is based on ---------- times.

Select the correct option

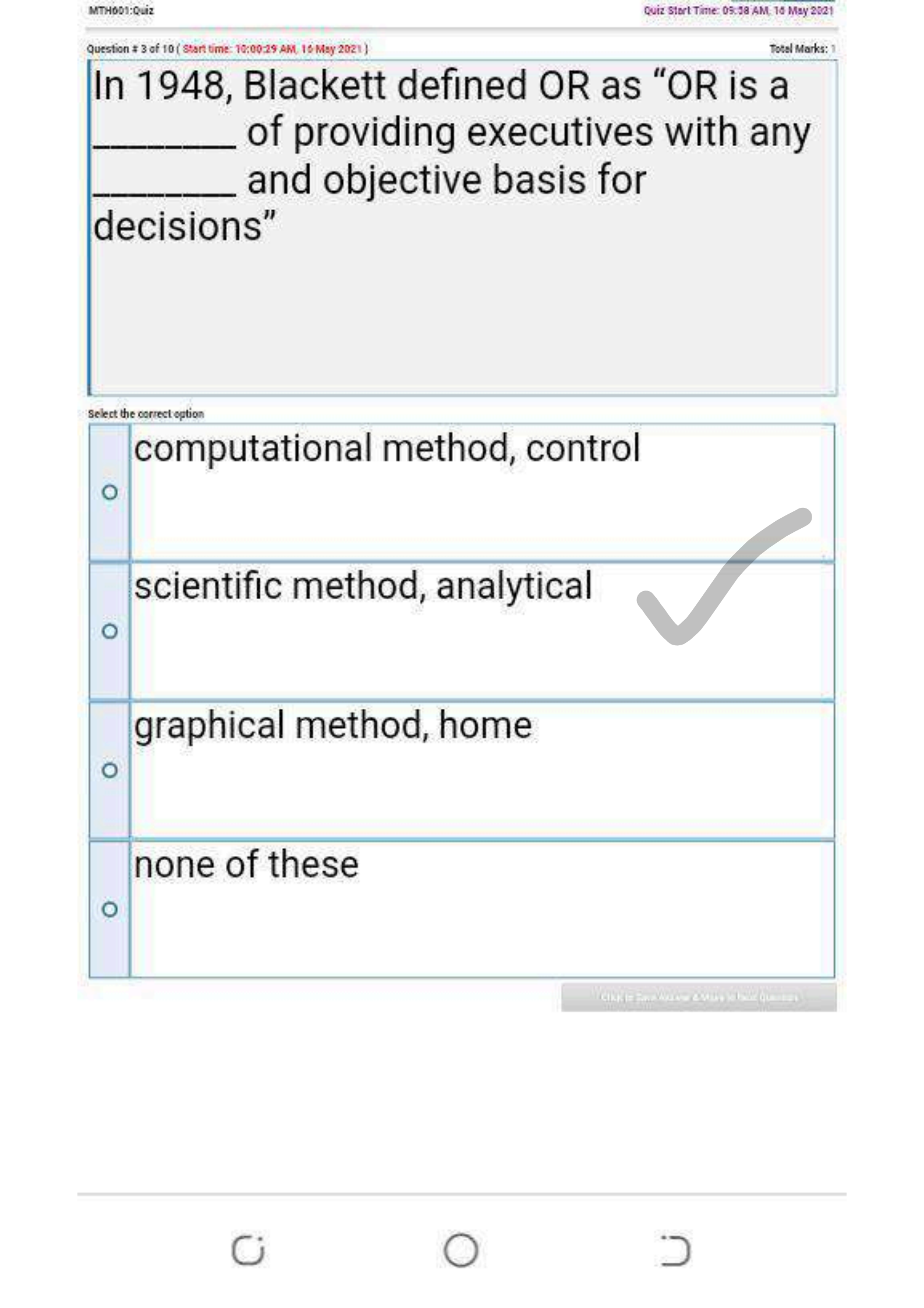
- deterministic
- probabilistic
- stochastic
- serial

Question # 3 of 10 (Start time: 10:00:29 AM, 16 May 2021)
Total Marks: 1

In 1948, Blackett defined OR as "OR is a _____ of providing executives with any _____ and objective basis for decisions"

Select the correct option

- computational method, control
- scientific method, analytical
- graphical method, home
- none of these



0B/s 100% 9:59 AM

quiz.vu.edu.pk/Qu

The Operational Research Society of U.K. defines OR as "Operational Research is the application of the methods of _____to complex problems arising in the direction and management of large systems of men, machines, materials and money in industry, business, government

Select the correct option

- arts
- statistics
- mathematics
- science

Click to view Answer & Save Your Answer

Question # 1 of 10 (Start time: 09:38:17 AM, 16 May 2021) Total Marks: 1

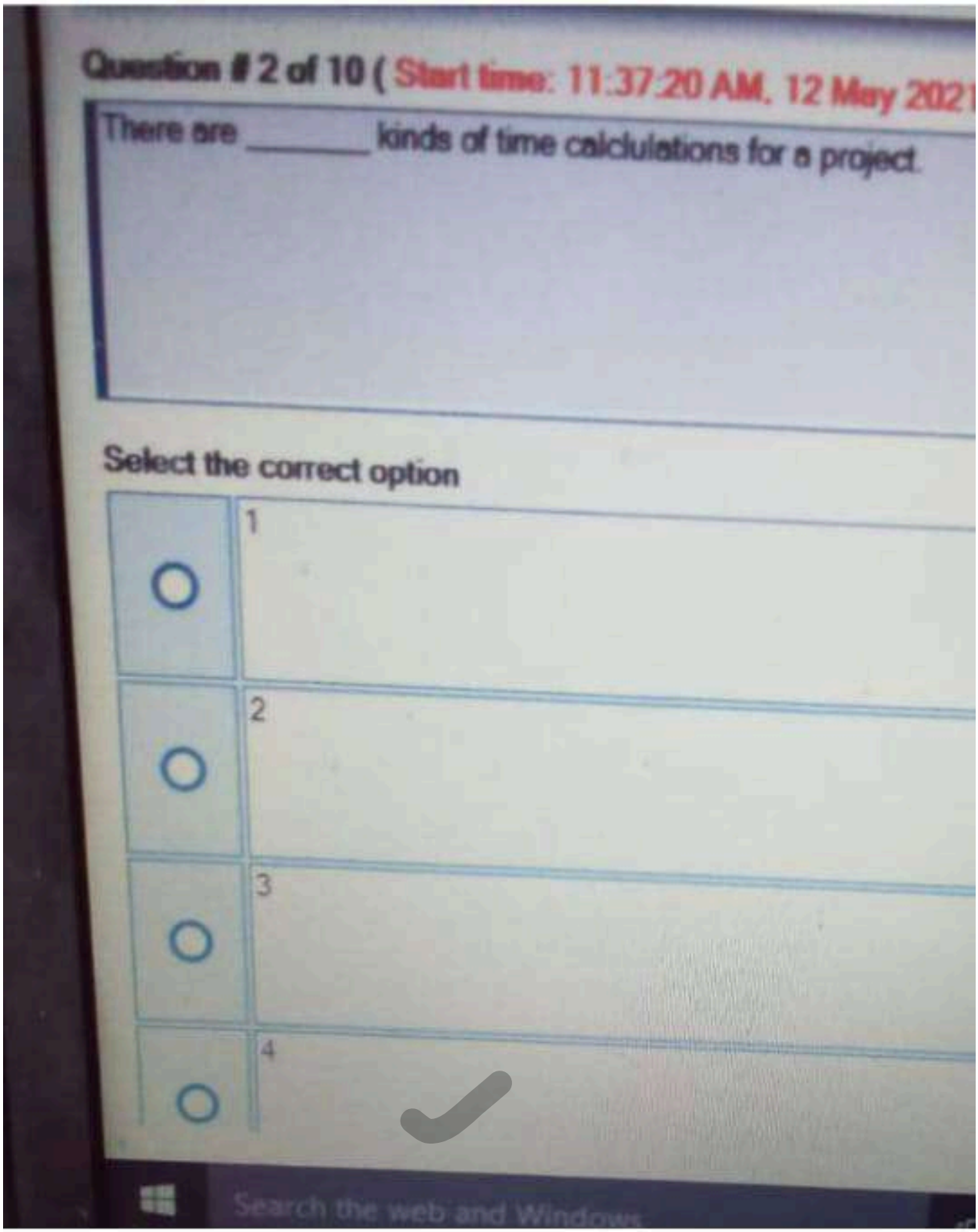
The first step in the application of _____ is to develop a network representation of the project plan.

Select the correct option

- probability
- CPM
- linear programming
- none of these

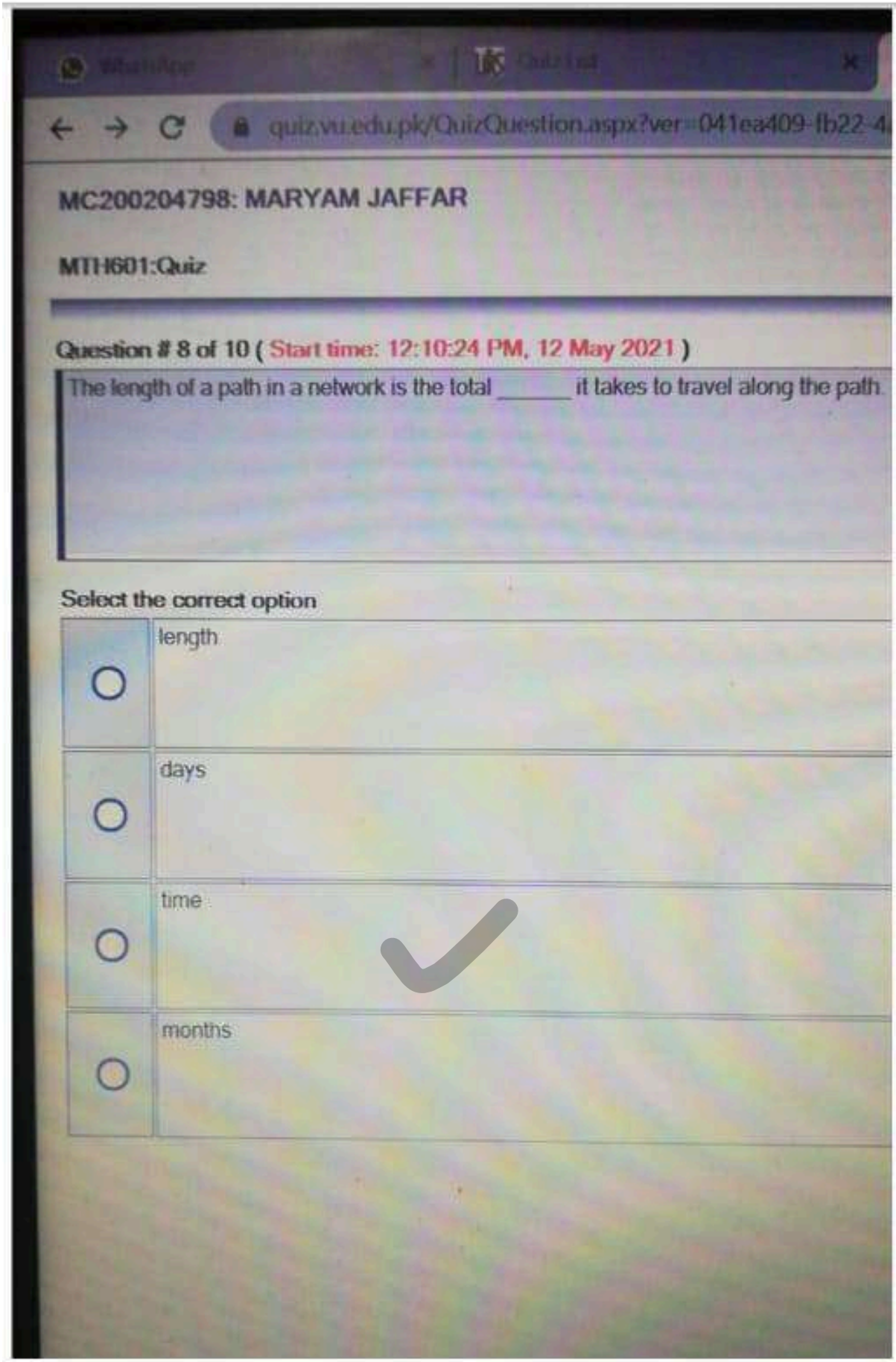
Telenor 9:25 AM

MTH 601 (2021)...



Telenor 9:29 AM

MTH 601 (2021)...



Telenor 9:35 AM

MTH601:Quiz

Question # 3 of 10 (Start time: 12:32:25 PM, 12 May 2021)

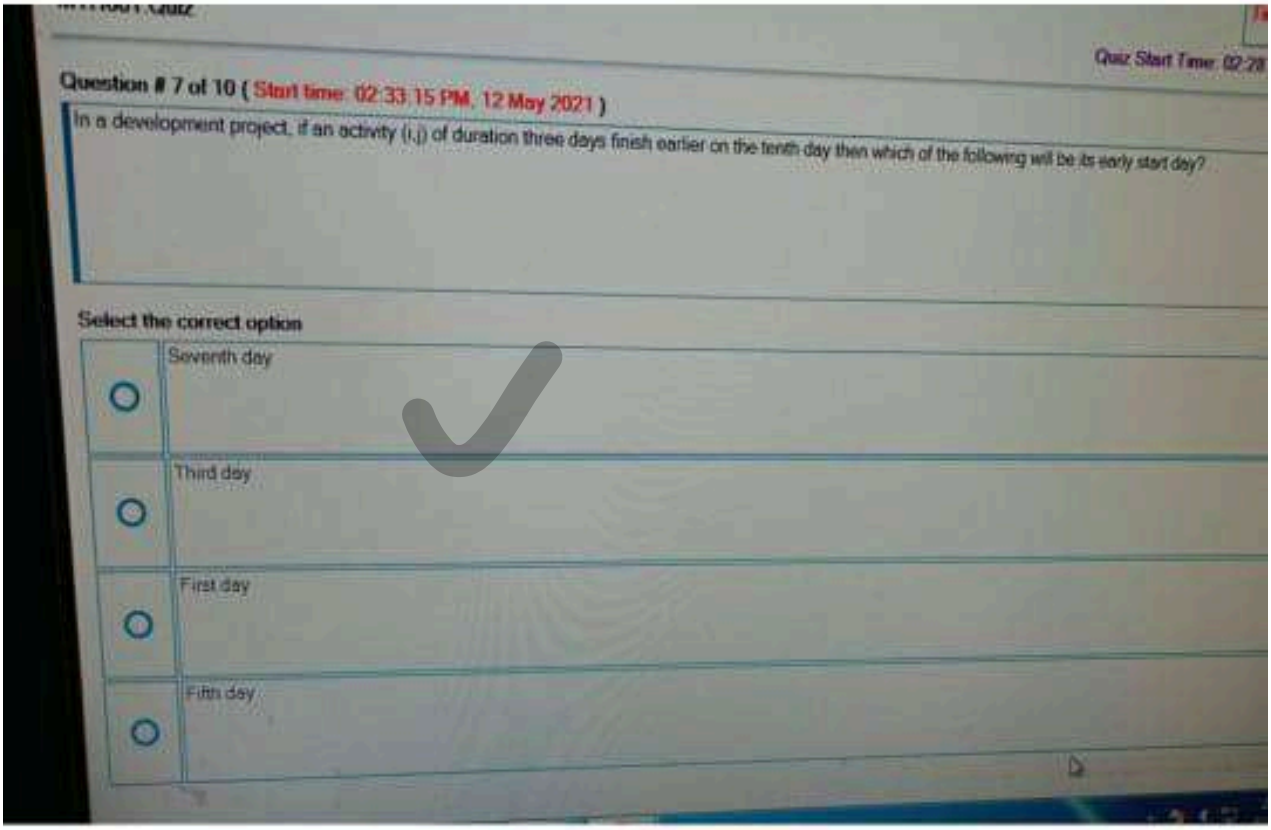
There are _____ kinds of time calclulations for a project.

Select the correct option

- 1
- 2
- 3
- 4

Telenor 9:38 AM

MTH 601 (2021)...



(Start time: 09:43:22 AM, 16 May 2021)

Which of the following is not a categories of the operations research (OR) techniques.

Select the correct option

- Linear mathematical programming technique
- Feasible solution
- Inventory techniques
- Network techniques

Question # 8 of 10 (Start time: 09:42:39 AM, 16 May 2021)

Dummy activity is represented by _____ .

Select the correct option

- ○ doted line
- ○ doted line with right to left arrow
- ○ doted line with left to right arrow
- ○ left to right arrow

Question # 7 of 10 (Start time: 09:41:48 AM, 16 May 2021)

Qu

a development project, if an activity (i,j) of duration three days finish earlier on the tenth day then which of the following will be

Select the correct option

- [x] Seventh day
- [ ] Third day
- [ ] First day
- [ ] Fifth day

Click to Save Answ

Question # 6 of 10 (Start time: 09:41:09 AM, 16 May 2021)

In the phase of Early Start and Early Finish, to find the Critical Path in a network flow diagram, for the first node(event), we start with tim

Select the correct option

- t = infinity
- t = 0
- t = a (arbitrary)
- with strict positive value

Quiz Start Time: 09:38 AM, 16

Question # 5 of 10 (Start time: 09:40:18 AM, 16 May 2021)

Tota

In a network flow diagram, if an activity (ij) starts earlier after two days and finish earlier on fifth day, then which of the following will be its completion time?

Select the correct option

- (●) Three days
- ( ) Five days
- ( ) Two days
- ( ) Ten days

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 09:39:33 AM, 16 May 2021)

In a network flow diagram, events are just the points in ———be represented by circles.

Select the correct option

- xy-plane
- space only
- time only
- space-time

Question # 3 of 10 (Start time: 09:38:56 AM, 16 May 2021)

Operations research (OR) is the application of _____ methods to decision making.

Select the correct option

- ○ engineering
- ○ quantitative
- ○ computing
- ● all of these

Question # 1 of 10 (Start time: 09:38:14 AM, 16 May 2021)

Duration of dummy activity is _____.

Select the correct option

- latest start time
- zero
- early start time
- 1

Question # 2 of 30 (Start time: 01:13:23 PM, 31 December 2020) Total Marks: 1

When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to delay the job having ----------.

Select the correct option

- dummy branch
- largest free float
- largest total float
- largest independent float

1:12 PM ... 50.0 K/s 43

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 90 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 1 of 30 (Start time: 01:12:48 PM, 31 December 2020)

Total Marks: 1

While solving a network flow problem by using PERT, the area under the Normal Distribution curve gives ----------.

Select the correct option

- Variance of the project length
- Standard Deviation of project length
- Probability of completion
- Expected time of completion

Click to Save Answer & Move to Next Question

1:14 PM
Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 3 of 30 (Start time: 01:14:45 PM, 31 December 2020)

Total Marks: 1

Non-feasible solution———— associated with a given linear programming problem.

Select the correct option

- satisfy all the constraints
- does not satisfy all the constraints
- satisfy least one of the constraint
- does not satisfy least one of the constraint

Click to Save Answer & Move to Next Question

1:16 PM
Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 4 of 30 (Start time: 01:16:12 PM, 31 December 2020)

Total Marks: 1

Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

Select the correct option

- CPM Times
- PERT Times

Click to Save Answer & Move to Next Question

1:16 PM 16.7 K/s 42

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 5 of 30 (Start time: 01:16:46 PM, 31 December 2020)

Total Marks: 1

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

- Maximization
- Minimization
- Inflection
- Average

Click to Save Answer & Move to Next Question

1:17 PM ••• 16.6 K/s 42

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 6 of 30 (Start time: 01:17:53 PM, 31 December 2020)

Total Marks: 1

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is --------proportional to the quantity ordered.

Select the correct option

- directly
- inversely

Click to Save Answer & Move to Next Question

1:21 PM
Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 9 of 30 (Start time: 01:21:46 PM, 31 December 2020)

Total Marks: 1

While solving a LP by Simplex method, which of the following corresponds to corner points of the feasible region in Graphical method?

Select the correct option

- non-feasible solution
- basic feasible solution
- non-basic feasible solution
- non-negative solutions

Click to Save Answer & Move to Next Question

1:20 PM ••• 279 B/s 42

Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 8 of 30 (Start time: 01:20:26 PM, 31 December 2020)

Total Marks: 1

The objective function and ——— together form a Linear Programming problem.

Select the correct option

- a set of constraints
- unrestricted variables
- feasible region
- optimal solution

Click to Save Answer & Move to Next Question

◻ ○ ◁

1:19 PM ••• 14.0 K/s 42

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 7 of 30 (Start time: 01:19:10 PM, 31 December 2020)

Total Marks: 1

If we increase the inventory, then the shortage cost will ------ ------.

Select the correct option

- remain constant
- increase
- decrease
- uncertain

Click to Save / Answer & Move to Next Question

1:24 PM

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 12 of 30 (Start time: 01:24:14 PM, 31 December 2020)

Total Marks: 1

In the Purchasing Model with Shortages the Cost Function 'C(S,Q)' can also be expressed as function of ----------.

Select the correct option

- (t,D)
- (Q,D)
- (S,t)
- (S,D)

Click to Save / Answer & Move to Next Question

1:23 PM ••• 4.4 K/s 41

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 11 of 30 (Start time: 01:23:35 PM, 31 December 2020)

Total Marks: 1

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously on 4th day, then the next activity containing 'n' as head event can't start until the entering activity ---------- is completed.

Select the correct option

- (m,n)
- (l,n)
- (m,l)
- (l,m)

Click to Save Answer & Move to Next Question

◻ ○ ◁

1:29 PM ••• 15.6 K/s 40

Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 16 of 30 (Start time: 01:29:21 PM, 31 December 2020)

Total Marks: 1

Which of the following is true about the Manufacturing model without Shortage?

Select the correct option

- Demand rate = Replacement rate
- Demand rate > Replacement rate
- Demand rate < Replacement rate
- None of the above choices

Click to Save Answer & Move to Next Question

1:30 PM
Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 17 of 30 (Start time: 01:30:14 PM, 31 December 2020)

Total Marks: 1

In the context of Wilson's model, at time t = 0, the quantity 'Q' of the product is-------- in the inventory.

Select the correct option

- Minimum
- Average
- Maximum
- Zero

Click to Save / Answer & Move to Next Question

1:30 PM ••• 0 B/s 39

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 82 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 18 of 30 (Start time: 01:30:44 PM, 31 December 2020)

Total Marks: 1

Convex set is the collection of all ————.

Select the correct option

- ○ optimal solutions
- ● feasible solutions
- ○ non-feasible solutions
- ○ negative solutions

Click to Save Answer & Move to Next Question

1:33 PM ••• 12.9 K/s 39

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 23 of 30 (Start time: 01:33:51 PM, 31 December 2020)

Total Marks: 1

Which of the following event has no immediate successor in any network flow diagram?

Select the correct option

- Head Event
- Critical Event
- Non-Critical Event
- Tail Event

Click to Save Answer & Move to Next Question

1:33 PM
15.7 K/s 39
Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 22 of 30 (Start time: 01:33:18 PM, 31 December 2020)

Total Marks: 1

For an industrialist, optimization means ----------.

Select the correct option

- maximizing its profit only
- minimizing its cost only
- both maximizing its profit and minimizing its cost
- change the business strategy and looking forward new customers

Click to Save / Answer & Move to Next Question

1:32 PM 423 B/s 39

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 20 of 30 (Start time: 01:32:09 PM, 31 December 2020)

Total Marks: 1

The solution set of the constraint: 2x+3y>12, will lie in ——— bisected by the line:2x+3y=12.

Select the correct option

- origin side of half plane
- opposite to the origin side in half plane
- both half planes
- first quadrant only

Click to Save Answer & Move to Next Question

1:34 PM
144 K/s 39
Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 24 of 30 (Start time: 01:34:20 PM, 31 December 2020)

Total Marks: 1

The Simplex method is a———— for solving Linear Programming problem.

Select the correct option

- graphical procedure
- analytical procedure
- probabilistic procedure
- computational procedure(Algorithm)

Click to Save Answer & Move to Next Question

1:38 PM ••• 102 B/s 38

Quiz
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MC180403689: SALMAN ALI

Time Left 79 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 27 of 30 (Start time: 01:37:49 PM, 31 December 2020)

Total Marks: 1

Which of the type of path will be critical in network flow diagram?

Select the correct option

- ○ Shortest path from starting node to end node
- ○ Any path from starting node to end node
- ○ Longest path from starting node to end node
- ○ Any path from starting node to end node with minimum cost

Click to Save Answer & Move to Next Question

1:37 PM
Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 26 of 30 (Start time: 01:37:06 PM, 31 December 2020)

Total Marks: 1

In the PERT, the expected time is classified as a———— variable.

Select the correct option

- complex valued
- fix valued (constant)
- random
- continuous

Click to Save Answer & Move to Next Question

1:35 PM ••• 14.4 K/s 38

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 25 of 30 (Start time: 01:35:50 PM, 31 December 2020)

Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the -------- of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

- demand
- profit
- cost
- transportation

Click to Save Answer & Move to Next Question

1:39 PM ••• 199 B/s 37

Quiz
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MC180403689: SALMAN ALI

Time Left 83 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 28 of 30 (Start time: 01:39:15 PM, 31 December 2020)

Total Marks: 1

Under which of the following condition, a mathematical Program would be non-linear?

Select the correct option

- If least Objective function is non-linear
- If both Objective function and constraints are non-linear

Click to Save Answer & Move to Next Question

1:40 PM
234 B/s 37
Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 29 of 30 (Start time: 01:40:29 PM, 31 December 2020)

Total Marks: 1

While finding the Critical Path, in the phase of Late Start and Late Finish, we start from ---------- node, the computations are performed from right to left up to the beginning event.

Select the correct option

- first
- arbitrary
- middle
- last

Click to Save Answer & Move to Next Question

1:41 PM ••• 10.7 K/s 37

Quiz
quiz.vu.edu.pk

MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 30 of 30 (Start time: 01:41:20 PM, 31 December 2020)

Total Marks: 1

By which of the following method, any complex linear programming problem can be handled?

Select the correct option

- Graphical method
- Simplex method
- Dual method
- Degenerate method

Click to Save Answer & Move to Next Question

Question # 3 of 30 (Start time: 01:44:36 PM, 31 December 2020)

Total Marks: 1

By putting the Replacement Rate 'R?Infinity' and Shortage Cost 'C4?Infinity' in the Economic Order Quantity(EOQ) of Manufacturing Model with allowed Shortage, then the result is ----------

Select the correct option

- EOQ of Wilson's Model
- EOQ of Manufacturing Model without Shortage
- EOQ of Purchasing Model with allowed Shortage
- Invariant

Question # 2 of 30 (Start time: 01:43:21 PM, 31 December 2020) Total Marks: 1

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirment?

Select the correct option

- 2nd month demand can be included in 1st month
- 2nd month demand will have to fulfill in 2nd month
- 2nd month demand can be included in 3rd month

Question # 4 of 30 (Start time: 01:45:35 PM, 31 December 2020) Total Marks: 1

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Question # 5 of 30 (Start time: 01:46:56 PM, 31 December 2020)

Total Marks: 1

If the variance of the project length = 25, expected and scheduled times are 40 and 36 respectively, then the Standard Normal Variable of the given project is----------.

Select the correct option

- 0.80
- -0.80
- 0.16
- -0.16

Question # 8 of 30 (Start time: 01:51:51 PM, 31 December 2020)

Total Marks: 1

Excess of inventories are undesirable due to the following reasons;

Select the correct option

- It locks up the capital of the enterprise.
- It includes the expenses of stores, equipment, personnel, insurance etc..
- It may not lead to high volume of output instead it might hamper the production.
- All above reasons are equivalent

MC190401605: SIDRA ATTA

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 7 of 30 (Start time: 01:50:24 PM, 31 December 2020)

Total Marks: 1

If the cost function in terms of economic order quantity is C = Q^2-400Q, then the minimum cost of inventory per unit of time is ------- units.

Select the correct option

- 800
- 400
- 200
- 398

Question # 6 of 30 (Start time: 01:48:38 PM, 31 December 2020)

Total Marks: 1

Which of the following would be the objective of the yield per minute in a chemical process which depends upon the temperature ‘x’ and pressure ‘y’?

Select the correct option

- Maximization
- Minimization
- Inflection
- Average

MC190401605: SIDRA ATTA

Time Left 76 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 9 of 30 (Start time: 01:52:58 PM, 31 December 2020) Total Marks: 1

In a Project Management, if the Critical activities of a network are delayed then --------.

Select the correct option

- ○ project finish time will have to extend
- ○ project cost will increase
- ○ more resources have to employed
- ● All above choices are equivalent

Question # 10 of 30 (Start time: 01:54:31 PM, 31 December 2020)

Total Marks: 1

A Critical Path in a network flow diagram ------.

Select the correct option

- is unique
- may at most two
- depends on number of dummies
- may be multiple

Question # 11 of 30 (Start time: 01:55:39 PM, 31 December 2020)

Total Marks: 1

In the ABC analysis, the items are classified into three categories with respect their----------.

Select the correct option

- cost value
- demand value
- turn over value
- mark up value

MC190401605: SIDRA ATTA

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 13 of 30 (Start time: 01:59:19 PM, 31 December 2020)

Total Marks: 1

Which of the following method is used to find an optimal solution of a LP by assuming an initial solution, which later on modified through well defined steps?

Select the correct option

- Graphical
- Algebraic(Simplex)

Click to Save Answer & Move to Next Question

Question # 12 of 30 (Start time: 01:57:34 PM, 31 December 2020) Total Marks: 1

Which of the following event has no immediate successor in any network flow diagram?

Select the correct option

- Head Event
- Critical Event
- Non-Critical Event
- Tail Event

Question # 14 of 30 (Start time: 02:00:21 PM, 31 December 2020) Total Marks: 1

Which of the following is the is major assumed difference between the models of Purchasing and Manufacting without shortage ?

Select the correct option

- Replacement rate
- Shortage
- Demand
- Item, Setup and Holding Costs

Question # 15 of 30 (Start time: 02:01:10 PM, 31 December 2020)

Total Marks: 1

If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the time between the consecutive orders is -------- months.

Select the correct option

- 02
- 2.50
- 0.50
- 1.50

Question # 16 of 30 (Start time: 02:02:31 PM, 31 December 2020) Total Marks: 1

While solving a LP by Simplex method, which of the following corresponds to corner points of the feasible region in Graphical method?

Select the correct option

- non-feasible solution
- basic feasible solution
- non-basic feasible solution
- non-negative solutions

MC190401605: SIDRA ATTA

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 17 of 30 (Start time: 02:03:36 PM, 31 December 2020)

Total Marks: 1

Which of the following is the meaning of 'Linear'?

Select the correct option

- Relationship between the variables is a straight line
- Relationship between the variables is a curvilinear
- Relationship between the variables is discrete line
- Relationship between the variables is finite or infinite

Question # 18 of 30 (Start time: 02:04:58 PM, 31 December 2020) Total Marks: 1

While solving a Linear Programming problem by graphical method, the non-negativity constraints will restrict the feasible region(solution space) in --------quadrant.

Select the correct option

- 1st
- 2nd
- 3rd
- 4th

Question # 20 of 30 (Start time: 02:07:54 PM, 31 December 2020) Total Marks: 1

Time by which an activity can expand with out affecting subsequent activity is _____ .

Select the correct option

Reload Math Equations

- O F=$J_E - i_E - D$
- O F=$J_E - i_E + D$
- O F=$J_E + i_E - D$
- O F=$-J_E + i_E + D$

Click to Save Answer & Move to Next Question

Question # 19 of 30 (Start time: 02:07:01 PM, 31 December 2020) Total Marks: 1

For any activity (i,j), if , a) Earliest start time of 'i' = Latest finish time of 'i', b) Earliest start time of 'j' = Latest finish time of 'j', c) difference of Earliest start times of events 'i' and "j = difference of Latest finish times of events 'i'and 'j' = time to complete the job, then the activity (i,j) is said to be -------.

Select the correct option

- dummy
- critical
- non-Critical
- non of the above

Question # 22 of 30 (Start time: 02:09:45 PM, 31 December 2020)

Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the ------ of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

- demand
- profit
- cost
- transportation

MC190401605: SIDRA ATTA

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 21 of 30 (Start time: 02:08:42 PM, 31 December 2020)

Total Marks: 1

Constraints: x>2, x=2 and x<2 are equivalent to ----------.

Select the correct option

- |x|=2
- x=2
- |x|= −2
- x= −2

Question # 23 of 30 (Start time: 02:11:18 PM, 31 December 2020) Total Marks: 1

The purpose of maintaining the inventory is to balance the inter-related ----------.

Select the correct option

- times
- costs
- demands
- items

Question # 25 of 30 (Start time: 02:12:46 PM, 31 December 2020) Total Marks: 1

While finding the critical path in a network flow diagram, if an activity (i,j) starts and finish late on 3rd and 9th day respectively, then it will be completed in ---------days.

Select the correct option

- Three days
- Twelve days
- Fifteen days
- Six days

Question # 1 of 30 (Start time: 02:11:47 PM, 31 December 2020) Total Marks: 1

In a development project, if an activity (m,n) of seven days duration, starts late on 3rd day then which of the following will be its latest finish time?

Select the correct option

- 3rd day
- 4nd day
- 10th day
- 21th day

MC190401605: SIDRA ATTA

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 24 of 30 (Start time: 02:11:53 PM, 31 December 2020)

Total Marks: 1

The amount of the inventory will always taken to be ---------

Select the correct option

- Negative
- Positive

Click to Save Answer & Move to Next Question

Question # 2 of 30 (Start time: 02:12:58 PM, 31 December 2020) Total Marks: 1

Which of the following distribution gives the probability of completing the assigned project by considering the expected time of any activity as random variable?

Select the correct option

- Binomial
- Poisson
- Beta
- Normal

Click to Save Answer & Move to Next Question

Question # 26 of 30 (Start time: 02:13:46 PM, 31 December 2020) Total Marks: 1

Which of the following quantity will vary in case of Dynamic Order Quantity Model?

Select the correct option

- Setup Cost
- Carrying Cost
- Item Cost
- Demand

MC190401605: SIDRA ATTA

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 27 of 30 (Start time: 02:14:51 PM, 31 December 2020)

Total Marks: 1

Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

Select the correct option

- CPM Times
- PERT Times

Click to Save Answer & Move to Next Question

Question # 4 of 30 (Start time: 02:16:17 PM, 31 December 2020) Total Marks: 1

Which of the following is first step to model a linear programming problem?

Select the correct option

- Identifying the objective function
- Identifying the non-negative constraints
- Identifying the unknown decision variables
- Identifying all the restrictions

Click to Save Answer & Move to Next Question

Question # 28 of 30 (Start time: 02:15:45 PM, 31 December 2020) Total Marks: 1

If we increase the inventory, then the shortage cost will ————————.

Select the correct option

- remain constant
- increase
- decrease
- uncertain

Question # 3 of 30 (Start time: 02:14:15 PM, 31 December 2020) Total Marks: 1

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Click to Save Answer & Move to Next Question

Question # 29 of 30 (Start time: 02:47:39 PM, 31 December 2020) Total Marks: 1

In a development project, if an activity (m,n) of seven days duration, starts late on 5rd day then which of the following will be its latest finish time?

Select the correct option

- 5th day
- 7th day
- 12th day
- 5th or 7th day

Click to Save Answer & Move to Next Question

Question # 28 of 30 (Start time: 02:45:41 PM, 31 December 2020)
Total Marks: 1

In which of the following model, the Carrying Cost per unit of time is reduced in the ratio (1-D/R):1 ?

Select the correct option

- Purchasing Model without Shortage
- Manufacturing Model without Shortage
- Purchasing Model with allowed Shortage
- Manufacturing Model with allowed Shortage

Click to Save Answer & Move to Next Question

Question # 30 of 30 (Start time: 02:48:32 PM, 31 December 2020)
Total Marks: 1

While solving a network flow problem by PERT, which of the following type of time will be used to measure the length of Critical Path?

Select the correct option

- Pessimistic
- Expected
- Most Likely
- Optimistic

Click to Save Answer & Move to Next Question

Question # 25 of 30 (Start time: 02:42:30 PM, 31 December 2020)
Total Marks: 1

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is --------- proportional to the quantity ordered.

Select the correct option

- directly
- inversely

Click to Save Answer & Move to Next Question

Question # 26 of 30 (Start time: 02:44:09 PM, 31 December 2020) Total Marks: 1

The solution set of the constraint: 2x+3y>12, will lie in -------- bisected by the line:2x+3y=12.

Select the correct option

- origin side of half plane
- opposite to the origin side in half plane
- both half planes
- first quadrant only

Click to Save Answer & Move to Next Question

Question # 27 of 30 (Start time: 02:44:52 PM, 31 December 2020) Total Marks: 1

Which of the following method is used to find an optimal solution of a LP by assuming an initial solution, which later on modified through well defined steps?

Select the correct option

- Graphical
- Algebraic(Simplex)

Click to Save Answer & Move to Next Question

Question # 24 of 30 (Start time: 02:41:02 PM, 31 December 2020) Total Marks: 1

If the cost function in terms of economic order quantity is C = Q^2-400Q, then the minimum cost of inventory per unit of time is ---------- units.

Select the correct option

- 800
- 400
- 200
- 398

Click to Save Answer & Move to Next Question

Question # 23 of 30 (Start time: 02:40:11 PM, 31 December 2020)

Total Marks: 1

For a LP problem say;Max:z=x+y,under the constraints x,y>=0, the feasible region would be--------.

Select the correct option

- empty
- all xy-plane
- all the first quadrant
- point(0,0)

Click to Save Answer & Move to Next Question

Question # 21 of 30 (Start time: 02:39:02 PM, 31 December 2020)
Total Marks: 1

In the relation of finding the expected time of an activity, most likely time is weighted more than the other optimistic and pessimistic times and these exist in the ratio of ----------.

Select the correct option

- 6:1
- 2:1
- 4:1
- 3:1

Click to Save Answer & Move to Next Question

Question # 20 of 30 (Start time: 02:37:42 PM, 31 December 2020) Total Marks: 1

If we increase the inventory, then the shortage cost will decrease and the carrying cost will ----------.

Select the correct option

- remain constant
- increase
- also decrease
- uncertain

Click to Save Answer & Move to Next Question

Question # 19 of 30 (Start time: 02:35:53 PM, 31 December 2020)
Total Marks: 1

Under which of the following condition, the Economic Order Quantity (EOQ) of Manufacturing Model with allowed Shortage can be transformed into the same Model without Shortage?

Select the correct option

- When Setup Cost is zero
- When Setup Cost is infinity
- When Shortage Cost is infinity
- When Shortage Cost is zero

Click to Save Answer & Move to Next Question

Question # 16 of 30 (Start time: 02:30:26 PM, 31 December 2020) Total Marks: 1

If someone asks you how much time you require to reach your VU campus from your house, you may say that it may take 1 to 1½ hours. Then you meant which of the following variable?

Select the correct option

- Discrete
- Complex
- Random
- Continuous

Click to Save Answer & Move to Next Question

Question # 17 of 30 (Start time: 02:32:10 PM, 31 December 2020) Total Marks: 1

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '8' and '9' days durations respectively, start earlier simultaneously on 2nd day, then the next activity containing 'n' as head event can't start until the entering activity_____is completed.

Select the correct option

- o (l,n)
- o (m,n)
- o (m,l)
- o (l,m)

Click to Save Answer & Move to Next Question

Question # 18 of 30 (Start time: 02:33:35 PM, 31 December 2020) Total Marks: 1

Which of the following is uncertain in PERT?

Select the correct option

- Activity Cost
- Activities' completion times
- Resources associated with each activity
- Activities' precedence relation

Click to Save Answer & Move to Next Question

Question # 14 of 30 (Start time: 02:27:39 PM, 31 December 2020) Total Marks: 1

While solving a Linear Programming problem, the first step is ----------.

Select the correct option

- to obtain alternate solutions
- to obtain basic feasible solutions
- formulation of the problem
- model building

Click to Save Answer & Move to Next Question

BC180200983: ZULFIQAR ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 07:54 AM, 31 December 2020

Question # 29 of 30 (Start time: 08:28:53 AM, 31 December 2020)

Total Marks: 1

Which of the following event has no immediate successor in any network flow diagram?

Select the correct option

- Head Event
- Critical Event
- Non-Critical Event
- Tail Event

Question # 15 of 30 (Start time: 02:28:58 PM, 31 December 2020)
Total Marks: 1

Which of the following is the meaning of 'Linear'?

Select the correct option

- Relationship between the variables is a straight line
- Relationship between the variables is a curvilinear
- Relationship between the variables is discrete line
- Relationship between the variables is finite or infinite

Click to Save Answer & Move to Next Question

Question # 2 of 30 (Start time: 01:13:23 PM, 31 December 2020) Total Marks: 1

When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to delay the job having ----------.

Select the correct option

- dummy branch
- largest free float
- largest total float
- largest independent float

1:12 PM ... 50.0 K/s 43

Quiz
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MC180403689: SALMAN ALI

Time Left 90 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 1 of 30 (Start time: 01:12:48 PM, 31 December 2020)

Total Marks: 1

While solving a network flow problem by using PERT, the area under the Normal Distribution curve gives ----------.

Select the correct option

- Variance of the project length
- Standard Deviation of project length
- Probability of completion
- Expected time of completion

Click to Save Answer & Move to Next Question

1:14 PM
Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 3 of 30 (Start time: 01:14:45 PM, 31 December 2020)

Total Marks: 1

Non-feasible solution———— associated with a given linear programming problem.

Select the correct option

- satisfy all the constraints
- does not satisfy all the constraints
- satisfy least one of the constraint
- does not satisfy least one of the constraint

Click to Save Answer & Move to Next Question

1:16 PM
Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 4 of 30 (Start time: 01:16:12 PM, 31 December 2020)

Total Marks: 1

Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

Select the correct option

- CPM Times
- PERT Times

Click to Save Answer & Move to Next Question

1:16 PM 16.7 K/s 42

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 5 of 30 (Start time: 01:16:46 PM, 31 December 2020)

Total Marks: 1

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

- Maximization
- Minimization
- Inflection
- Average

Click to Save Answer & Move to Next Question

1:17 PM ••• 16.6 K/s 42

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 6 of 30 (Start time: 01:17:53 PM, 31 December 2020)

Total Marks: 1

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is --------proportional to the quantity ordered.

Select the correct option

- directly
- inversely

Click to Save Answer & Move to Next Question

1:21 PM
Quiz
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MC180403689: SALMAN ALI

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 9 of 30 (Start time: 01:21:46 PM, 31 December 2020)

Total Marks: 1

While solving a LP by Simplex method, which of the following corresponds to corner points of the feasible region in Graphical method?

Select the correct option

- non-feasible solution
- basic feasible solution
- non-basic feasible solution
- non-negative solutions

Click to Save Answer & Move to Next Question

1:20 PM ••• 279 B/s 42

Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 8 of 30 (Start time: 01:20:26 PM, 31 December 2020)

Total Marks: 1

The objective function and ——— together form a Linear Programming problem.

Select the correct option

- a set of constraints
- unrestricted variables
- feasible region
- optimal solution

Click to Save Answer & Move to Next Question

◻ ○ ◁

1:19 PM ••• 14.0 K/s 42

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 7 of 30 (Start time: 01:19:10 PM, 31 December 2020)

Total Marks: 1

If we increase the inventory, then the shortage cost will ------ ------.

Select the correct option

- remain constant
- increase
- decrease
- uncertain

Click to Save / Answer & Move to Next Question

1:24 PM

Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 12 of 30 (Start time: 01:24:14 PM, 31 December 2020)

Total Marks: 1

In the Purchasing Model with Shortages the Cost Function 'C(S,Q)' can also be expressed as function of ----------.

Select the correct option

- (t,D)
- (Q,D)
- (S,t)
- (S,D)

Click to Save / Answer & Move to Next Question

1:23 PM ••• 4.4 K/s 41

Quiz
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MC180403689: SALMAN ALI

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 11 of 30 (Start time: 01:23:35 PM, 31 December 2020)

Total Marks: 1

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '7' and '8' days durations respectively, start earlier simultaneously on 4th day, then the next activity containing 'n' as head event can't start until the entering activity ---------- is completed.

Select the correct option

- (m,n)
- (l,n)
- (m,l)
- (l,m)

Click to Save Answer & Move to Next Question

◻ ○ ◁

1:29 PM ••• 15.6 K/s 40

Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 16 of 30 (Start time: 01:29:21 PM, 31 December 2020)

Total Marks: 1

Which of the following is true about the Manufacturing model without Shortage?

Select the correct option

- Demand rate = Replacement rate
- Demand rate > Replacement rate
- Demand rate < Replacement rate
- None of the above choices

Click to Save Answer & Move to Next Question

1:30 PM
Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 17 of 30 (Start time: 01:30:14 PM, 31 December 2020)

Total Marks: 1

In the context of Wilson's model, at time t = 0, the quantity 'Q' of the product is-------- in the inventory.

Select the correct option

- Minimum
- Average
- Maximum
- Zero

Click to Save / Answer & Move to Next Question

1:30 PM ••• 0 B/s 39

Quiz
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MC180403689: SALMAN ALI

Time Left 82 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 18 of 30 (Start time: 01:30:44 PM, 31 December 2020)

Total Marks: 1

Convex set is the collection of all ————.

Select the correct option

- ○ optimal solutions
- ● feasible solutions
- ○ non-feasible solutions
- ○ negative solutions

Click to Save Answer & Move to Next Question

1:33 PM ••• 12.9 K/s 39

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 23 of 30 (Start time: 01:33:51 PM, 31 December 2020)

Total Marks: 1

Which of the following event has no immediate successor in any network flow diagram?

Select the correct option

- Head Event
- Critical Event
- Non-Critical Event
- Tail Event

Click to Save Answer & Move to Next Question

1:33 PM
15.7 K/s 39
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 22 of 30 (Start time: 01:33:18 PM, 31 December 2020)

Total Marks: 1

For an industrialist, optimization means ----------.

Select the correct option

- maximizing its profit only
- minimizing its cost only
- both maximizing its profit and minimizing its cost
- change the business strategy and looking forward new customers

Click to Save / Answer & Move to Next Question

1:32 PM 423 B/s 39

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 20 of 30 (Start time: 01:32:09 PM, 31 December 2020)

Total Marks: 1

The solution set of the constraint: 2x+3y>12, will lie in ——— bisected by the line:2x+3y=12.

Select the correct option

- origin side of half plane
- opposite to the origin side in half plane
- both half planes
- first quadrant only

Click to Save Answer & Move to Next Question

1:34 PM
144 K/s 39
Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 24 of 30 (Start time: 01:34:20 PM, 31 December 2020)

Total Marks: 1

The Simplex method is a———— for solving Linear Programming problem.

Select the correct option

- graphical procedure
- analytical procedure
- probabilistic procedure
- computational procedure(Algorithm)

Click to Save Answer & Move to Next Question

1:38 PM ••• 102 B/s 38

Quiz
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MC180403689: SALMAN ALI

Time Left 79 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 27 of 30 (Start time: 01:37:49 PM, 31 December 2020)

Total Marks: 1

Which of the type of path will be critical in network flow diagram?

Select the correct option

- ○ Shortest path from starting node to end node
- ○ Any path from starting node to end node
- ○ Longest path from starting node to end node
- ○ Any path from starting node to end node with minimum cost

Click to Save Answer & Move to Next Question

1:37 PM
Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 26 of 30 (Start time: 01:37:06 PM, 31 December 2020)

Total Marks: 1

In the PERT, the expected time is classified as a———— variable.

Select the correct option

- complex valued
- fix valued (constant)
- random
- continuous

Click to Save Answer & Move to Next Question

1:35 PM ••• 14.4 K/s 38

Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 25 of 30 (Start time: 01:35:50 PM, 31 December 2020)

Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the -------- of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

- demand
- profit
- cost
- transportation

Click to Save Answer & Move to Next Question

1:39 PM ••• 199 B/s 37

Quiz
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MC180403689: SALMAN ALI

Time Left 83 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 28 of 30 (Start time: 01:39:15 PM, 31 December 2020)

Total Marks: 1

Under which of the following condition, a mathematical Program would be non-linear?

Select the correct option

- If least Objective function is non-linear
- If both Objective function and constraints are non-linear

Click to Save Answer & Move to Next Question

1:40 PM
234 B/s 37
Quiz
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MC180403689: SALMAN ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 29 of 30 (Start time: 01:40:29 PM, 31 December 2020)

Total Marks: 1

While finding the Critical Path, in the phase of Late Start and Late Finish, we start from ---------- node, the computations are performed from right to left up to the beginning event.

Select the correct option

- first
- arbitrary
- middle
- last

Click to Save Answer & Move to Next Question

1:41 PM ••• 10.7 K/s 37

Quiz
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MC180403689: SALMAN ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:12 PM, 31 December 2020

Question # 30 of 30 (Start time: 01:41:20 PM, 31 December 2020)

Total Marks: 1

By which of the following method, any complex linear programming problem can be handled?

Select the correct option

- Graphical method
- Simplex method
- Dual method
- Degenerate method

Click to Save Answer & Move to Next Question

Question # 3 of 30 (Start time: 01:44:36 PM, 31 December 2020)

Total Marks: 1

By putting the Replacement Rate 'R?Infinity' and Shortage Cost 'C4?Infinity' in the Economic Order Quantity(EOQ) of Manufacturing Model with allowed Shortage, then the result is ----------

Select the correct option

- EOQ of Wilson's Model
- EOQ of Manufacturing Model without Shortage
- EOQ of Purchasing Model with allowed Shortage
- Invariant

Question # 2 of 30 (Start time: 01:43:21 PM, 31 December 2020) Total Marks: 1

In the Dynamic Order Quantity problem if the ratio of Setup and Carrying Costs is '500' and the Demands of 2nd and 3rd months are 50 and 130 respectively then which of the following is true about the 2nd month's requirment?

Select the correct option

- 2nd month demand can be included in 1st month
- 2nd month demand will have to fulfill in 2nd month
- 2nd month demand can be included in 3rd month

Question # 4 of 30 (Start time: 01:45:35 PM, 31 December 2020) Total Marks: 1

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Question # 5 of 30 (Start time: 01:46:56 PM, 31 December 2020)

Total Marks: 1

If the variance of the project length = 25, expected and scheduled times are 40 and 36 respectively, then the Standard Normal Variable of the given project is----------.

Select the correct option

- 0.80
- -0.80
- 0.16
- -0.16

Question # 8 of 30 (Start time: 01:51:51 PM, 31 December 2020)

Total Marks: 1

Excess of inventories are undesirable due to the following reasons;

Select the correct option

- It locks up the capital of the enterprise.
- It includes the expenses of stores, equipment, personnel, insurance etc..
- It may not lead to high volume of output instead it might hamper the production.
- All above reasons are equivalent

MC190401605: SIDRA ATTA

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 7 of 30 (Start time: 01:50:24 PM, 31 December 2020)

Total Marks: 1

If the cost function in terms of economic order quantity is C = Q^2-400Q, then the minimum cost of inventory per unit of time is ------- units.

Select the correct option

- 800
- 400
- 200
- 398

Question # 6 of 30 (Start time: 01:48:38 PM, 31 December 2020)

Total Marks: 1

Which of the following would be the objective of the yield per minute in a chemical process which depends upon the temperature ‘x’ and pressure ‘y’?

Select the correct option

- Maximization
- Minimization
- Inflection
- Average

MC190401605: SIDRA ATTA

Time Left 76 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 9 of 30 (Start time: 01:52:58 PM, 31 December 2020) Total Marks: 1

In a Project Management, if the Critical activities of a network are delayed then --------.

Select the correct option

- ○ project finish time will have to extend
- ○ project cost will increase
- ○ more resources have to employed
- ● All above choices are equivalent

Question # 10 of 30 (Start time: 01:54:31 PM, 31 December 2020)

Total Marks: 1

A Critical Path in a network flow diagram ------.

Select the correct option

- is unique
- may at most two
- depends on number of dummies
- may be multiple

Question # 11 of 30 (Start time: 01:55:39 PM, 31 December 2020)

Total Marks: 1

In the ABC analysis, the items are classified into three categories with respect their----------.

Select the correct option

- cost value
- demand value
- turn over value
- mark up value

MC190401605: SIDRA ATTA

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 13 of 30 (Start time: 01:59:19 PM, 31 December 2020)

Total Marks: 1

Which of the following method is used to find an optimal solution of a LP by assuming an initial solution, which later on modified through well defined steps?

Select the correct option

- Graphical
- Algebraic(Simplex)

Click to Save Answer & Move to Next Question

Question # 12 of 30 (Start time: 01:57:34 PM, 31 December 2020) Total Marks: 1

Which of the following event has no immediate successor in any network flow diagram?

Select the correct option

- Head Event
- Critical Event
- Non-Critical Event
- Tail Event

Question # 14 of 30 (Start time: 02:00:21 PM, 31 December 2020) Total Marks: 1

Which of the following is the is major assumed difference between the models of Purchasing and Manufacting without shortage ?

Select the correct option

- Replacement rate
- Shortage
- Demand
- Item, Setup and Holding Costs

Question # 15 of 30 (Start time: 02:01:10 PM, 31 December 2020)

Total Marks: 1

If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the time between the consecutive orders is -------- months.

Select the correct option

- 02
- 2.50
- 0.50
- 1.50

Question # 16 of 30 (Start time: 02:02:31 PM, 31 December 2020) Total Marks: 1

While solving a LP by Simplex method, which of the following corresponds to corner points of the feasible region in Graphical method?

Select the correct option

- non-feasible solution
- basic feasible solution
- non-basic feasible solution
- non-negative solutions

MC190401605: SIDRA ATTA

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 17 of 30 (Start time: 02:03:36 PM, 31 December 2020)

Total Marks: 1

Which of the following is the meaning of 'Linear'?

Select the correct option

- Relationship between the variables is a straight line
- Relationship between the variables is a curvilinear
- Relationship between the variables is discrete line
- Relationship between the variables is finite or infinite

Question # 18 of 30 (Start time: 02:04:58 PM, 31 December 2020) Total Marks: 1

While solving a Linear Programming problem by graphical method, the non-negativity constraints will restrict the feasible region(solution space) in --------quadrant.

Select the correct option

- 1st
- 2nd
- 3rd
- 4th

Question # 20 of 30 (Start time: 02:07:54 PM, 31 December 2020) Total Marks: 1

Time by which an activity can expand with out affecting subsequent activity is _____ .

Select the correct option

Reload Math Equations

- O F=$J_E - i_E - D$
- O F=$J_E - i_E + D$
- O F=$J_E + i_E - D$
- O F=$-J_E + i_E + D$

Click to Save Answer & Move to Next Question

Question # 19 of 30 (Start time: 02:07:01 PM, 31 December 2020) Total Marks: 1

For any activity (i,j), if , a) Earliest start time of 'i' = Latest finish time of 'i', b) Earliest start time of 'j' = Latest finish time of 'j', c) difference of Earliest start times of events 'i' and "j = difference of Latest finish times of events 'i'and 'j' = time to complete the job, then the activity (i,j) is said to be -------.

Select the correct option

- dummy
- critical
- non-Critical
- non of the above

Question # 22 of 30 (Start time: 02:09:45 PM, 31 December 2020)

Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the ------ of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

- demand
- profit
- cost
- transportation

MC190401605: SIDRA ATTA

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 21 of 30 (Start time: 02:08:42 PM, 31 December 2020)

Total Marks: 1

Constraints: x>2, x=2 and x<2 are equivalent to ----------.

Select the correct option

- |x|=2
- x=2
- |x|= −2
- x= −2

Question # 23 of 30 (Start time: 02:11:18 PM, 31 December 2020) Total Marks: 1

The purpose of maintaining the inventory is to balance the inter-related ----------.

Select the correct option

- times
- costs
- demands
- items

Question # 25 of 30 (Start time: 02:12:46 PM, 31 December 2020) Total Marks: 1

While finding the critical path in a network flow diagram, if an activity (i,j) starts and finish late on 3rd and 9th day respectively, then it will be completed in ---------days.

Select the correct option

- Three days
- Twelve days
- Fifteen days
- Six days

Question # 1 of 30 (Start time: 02:11:47 PM, 31 December 2020) Total Marks: 1

In a development project, if an activity (m,n) of seven days duration, starts late on 3rd day then which of the following will be its latest finish time?

Select the correct option

- 3rd day
- 4nd day
- 10th day
- 21th day

MC190401605: SIDRA ATTA

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 24 of 30 (Start time: 02:11:53 PM, 31 December 2020)

Total Marks: 1

The amount of the inventory will always taken to be ---------

Select the correct option

- Negative
- Positive

Click to Save Answer & Move to Next Question

Question # 2 of 30 (Start time: 02:12:58 PM, 31 December 2020) Total Marks: 1

Which of the following distribution gives the probability of completing the assigned project by considering the expected time of any activity as random variable?

Select the correct option

- Binomial
- Poisson
- Beta
- Normal

Click to Save Answer & Move to Next Question

Question # 26 of 30 (Start time: 02:13:46 PM, 31 December 2020) Total Marks: 1

Which of the following quantity will vary in case of Dynamic Order Quantity Model?

Select the correct option

- Setup Cost
- Carrying Cost
- Item Cost
- Demand

MC190401605: SIDRA ATTA

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 01:43 PM, 31 December 2020

Question # 27 of 30 (Start time: 02:14:51 PM, 31 December 2020)

Total Marks: 1

Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

Select the correct option

- CPM Times
- PERT Times

Click to Save Answer & Move to Next Question

Question # 4 of 30 (Start time: 02:16:17 PM, 31 December 2020) Total Marks: 1

Which of the following is first step to model a linear programming problem?

Select the correct option

- Identifying the objective function
- Identifying the non-negative constraints
- Identifying the unknown decision variables
- Identifying all the restrictions

Click to Save Answer & Move to Next Question

Question # 28 of 30 (Start time: 02:15:45 PM, 31 December 2020) Total Marks: 1

If we increase the inventory, then the shortage cost will ————————.

Select the correct option

- remain constant
- increase
- decrease
- uncertain

Question # 3 of 30 (Start time: 02:14:15 PM, 31 December 2020) Total Marks: 1

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Click to Save Answer & Move to Next Question

Question # 29 of 30 (Start time: 02:47:39 PM, 31 December 2020) Total Marks: 1

In a development project, if an activity (m,n) of seven days duration, starts late on 5rd day then which of the following will be its latest finish time?

Select the correct option

- 5th day
- 7th day
- 12th day
- 5th or 7th day

Click to Save Answer & Move to Next Question

Question # 28 of 30 (Start time: 02:45:41 PM, 31 December 2020)
Total Marks: 1

In which of the following model, the Carrying Cost per unit of time is reduced in the ratio (1-D/R):1 ?

Select the correct option

- Purchasing Model without Shortage
- Manufacturing Model without Shortage
- Purchasing Model with allowed Shortage
- Manufacturing Model with allowed Shortage

Click to Save Answer & Move to Next Question

Question # 30 of 30 (Start time: 02:48:32 PM, 31 December 2020)
Total Marks: 1

While solving a network flow problem by PERT, which of the following type of time will be used to measure the length of Critical Path?

Select the correct option

- Pessimistic
- Expected
- Most Likely
- Optimistic

Click to Save Answer & Move to Next Question

Question # 25 of 30 (Start time: 02:42:30 PM, 31 December 2020)
Total Marks: 1

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is --------- proportional to the quantity ordered.

Select the correct option

- directly
- inversely

Click to Save Answer & Move to Next Question

Question # 26 of 30 (Start time: 02:44:09 PM, 31 December 2020) Total Marks: 1

The solution set of the constraint: 2x+3y>12, will lie in -------- bisected by the line:2x+3y=12.

Select the correct option

- origin side of half plane
- opposite to the origin side in half plane
- both half planes
- first quadrant only

Click to Save Answer & Move to Next Question

Question # 27 of 30 (Start time: 02:44:52 PM, 31 December 2020) Total Marks: 1

Which of the following method is used to find an optimal solution of a LP by assuming an initial solution, which later on modified through well defined steps?

Select the correct option

- Graphical
- Algebraic(Simplex)

Click to Save Answer & Move to Next Question

Question # 24 of 30 (Start time: 02:41:02 PM, 31 December 2020) Total Marks: 1

If the cost function in terms of economic order quantity is C = Q^2-400Q, then the minimum cost of inventory per unit of time is ---------- units.

Select the correct option

- 800
- 400
- 200
- 398

Click to Save Answer & Move to Next Question

Question # 23 of 30 (Start time: 02:40:11 PM, 31 December 2020)

Total Marks: 1

For a LP problem say;Max:z=x+y,under the constraints x,y>=0, the feasible region would be--------.

Select the correct option

- empty
- all xy-plane
- all the first quadrant
- point(0,0)

Click to Save Answer & Move to Next Question

Question # 21 of 30 (Start time: 02:39:02 PM, 31 December 2020)
Total Marks: 1

In the relation of finding the expected time of an activity, most likely time is weighted more than the other optimistic and pessimistic times and these exist in the ratio of ----------.

Select the correct option

- 6:1
- 2:1
- 4:1
- 3:1

Click to Save Answer & Move to Next Question

Question # 20 of 30 (Start time: 02:37:42 PM, 31 December 2020) Total Marks: 1

If we increase the inventory, then the shortage cost will decrease and the carrying cost will ----------.

Select the correct option

- remain constant
- increase
- also decrease
- uncertain

Click to Save Answer & Move to Next Question

Question # 19 of 30 (Start time: 02:35:53 PM, 31 December 2020)
Total Marks: 1

Under which of the following condition, the Economic Order Quantity (EOQ) of Manufacturing Model with allowed Shortage can be transformed into the same Model without Shortage?

Select the correct option

- When Setup Cost is zero
- When Setup Cost is infinity
- When Shortage Cost is infinity
- When Shortage Cost is zero

Click to Save Answer & Move to Next Question

Question # 16 of 30 (Start time: 02:30:26 PM, 31 December 2020) Total Marks: 1

If someone asks you how much time you require to reach your VU campus from your house, you may say that it may take 1 to 1½ hours. Then you meant which of the following variable?

Select the correct option

- Discrete
- Complex
- Random
- Continuous

Click to Save Answer & Move to Next Question

Question # 17 of 30 (Start time: 02:32:10 PM, 31 December 2020) Total Marks: 1

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '8' and '9' days durations respectively, start earlier simultaneously on 2nd day, then the next activity containing 'n' as head event can't start until the entering activity_____is completed.

Select the correct option

- o (l,n)
- o (m,n)
- o (m,l)
- o (l,m)

Click to Save Answer & Move to Next Question

Question # 18 of 30 (Start time: 02:33:35 PM, 31 December 2020) Total Marks: 1

Which of the following is uncertain in PERT?

Select the correct option

- Activity Cost
- Activities' completion times
- Resources associated with each activity
- Activities' precedence relation

Click to Save Answer & Move to Next Question

Question # 14 of 30 (Start time: 02:27:39 PM, 31 December 2020) Total Marks: 1

While solving a Linear Programming problem, the first step is ----------.

Select the correct option

- to obtain alternate solutions
- to obtain basic feasible solutions
- formulation of the problem
- model building

Click to Save Answer & Move to Next Question

BC180200983: ZULFIQAR ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 07:54 AM, 31 December 2020

Question # 29 of 30 (Start time: 08:28:53 AM, 31 December 2020)

Total Marks: 1

Which of the following event has no immediate successor in any network flow diagram?

Select the correct option

- Head Event
- Critical Event
- Non-Critical Event
- Tail Event

Question # 15 of 30 (Start time: 02:28:58 PM, 31 December 2020)
Total Marks: 1

Which of the following is the meaning of 'Linear'?

Select the correct option

- Relationship between the variables is a straight line
- Relationship between the variables is a curvilinear
- Relationship between the variables is discrete line
- Relationship between the variables is finite or infinite

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Total Marks: 1

Question # 28 of 30 (Start time: 10:07:57 PM, 31 December 2020)

While solving a Linear Programming problem by Simplex method, we convert all the constraints'inequalities into_____.

Select the correct option

- Equations with positive RHS
- Equations with negative RHS
- opposite inequalities
- nothing but keep those strictly the same

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29% 10:09 PM

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MC190400926: AMJAD ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 3 of 30 (Start time: 10:09:08 PM, 31 December 2020) Total Marks: 1

In which of the following model, the Carrying Cost per unit of time is reduced in the ratio (1-D/R):1 ?

Select the correct option

- Purchasing Model without Shortage
- Manufacturing Model without Shortage
- Purchasing Model with allowed Shortage
- Manufacturing Model with allowed Shortage

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 09:38 PM, 3

Question # 29 of 30 (Start time: 10:09:10 PM, 31 December 2020)

If an activity has non-zero value of total float such that it can be further delayed to the length of slack without delaying the project, then it is said to be ------

Select the correct option

- dummy
- critical
- non-Critical
- non of the above

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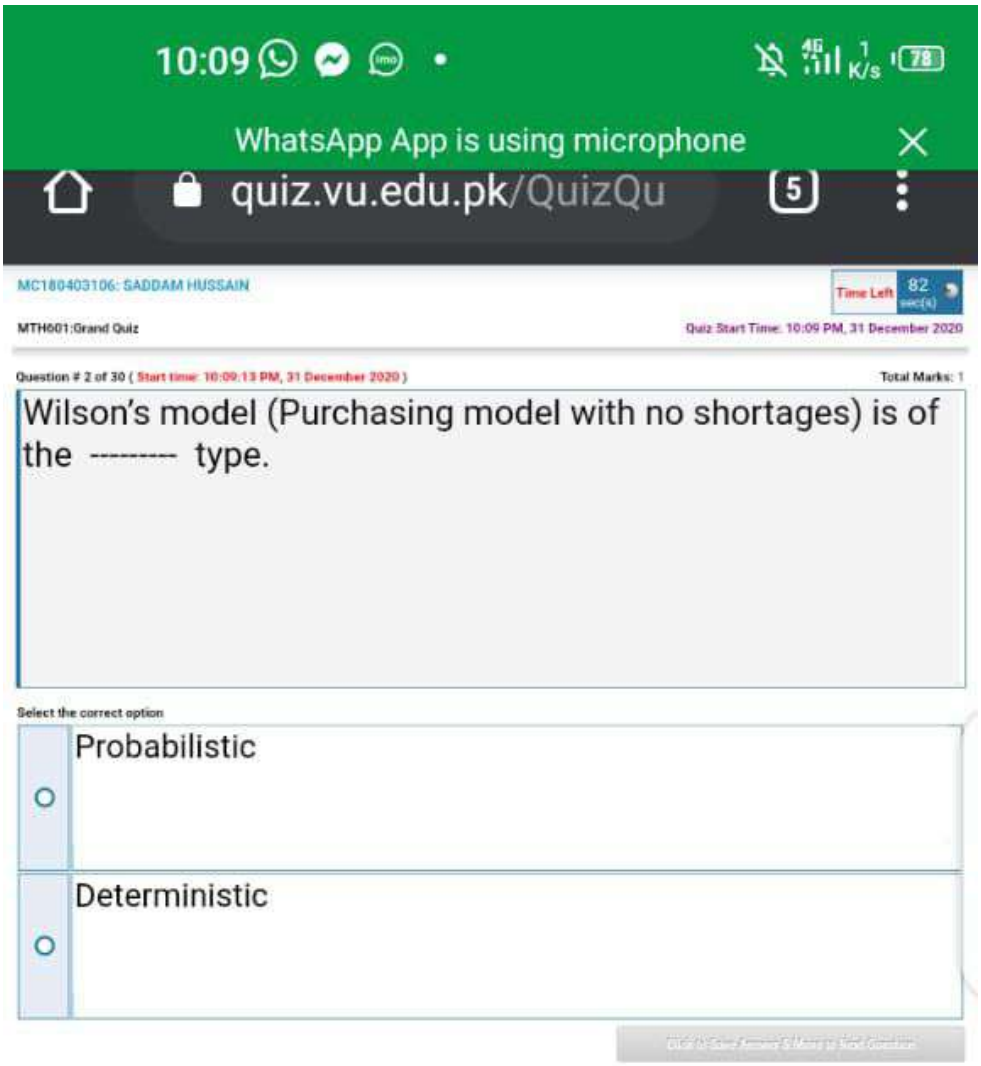
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MC180403106: SADDAM HUSSAIN
Time Left 82
MTH601:Grand Quiz
Quiz Start Time: 10:09 PM, 31 December 2020
Question # 2 of 30 (Start time: 10:09:13 PM, 31 December 2020)
Total Marks: 1
Wilson's model (Purchasing model with no shortages) is of the -------- type.
Select the correct option
Probabilistic
Deterministic

10:10
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MC180403106: SADDAM HUSSAIN

Time Left 85 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 3 of 30 (Start time: 10:09:54 PM, 31 December 2020)

Total Marks: 1

A Critical Path in a network flow diagram --------.

Select the correct option

- is unique
- may at most two
- depends on number of dummies
- may be multiple

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MC180403106: SADDAM HUSSAIN

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 4 of 30 (Start time: 10:10:21 PM, 31 December 2020)

Total Marks: 1

Which of the following distribution gives the probability of completing the assigned project by considering the expected time of any activity as random variable?

Select the correct option

- Binomial
- Poisson
- Beta
- Normal

Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 09:38 PM, 31 December 2020

Question # 30 of 30 (Start time: 10:10:23 PM, 31 December 2020)

Total Marks: 1

If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the time between the consecutive orders is ------ – months

Select the correct option

- 02
- 2.50
- 0.50
- 1.50

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12/31/2020

MC180403106: SADDAM HUSSAIN

Time Left 83 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

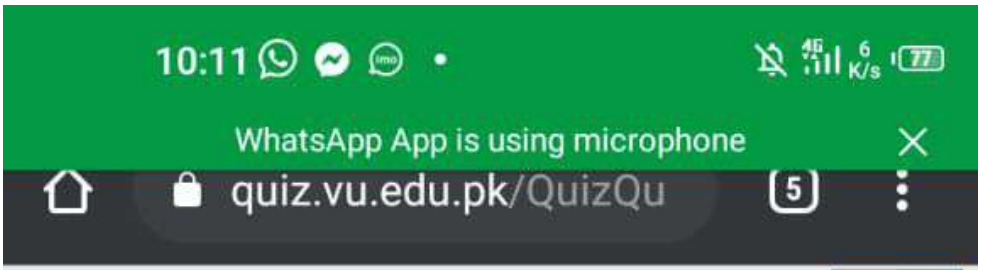
Question # 5 of 30 (Start time: 10:10:50 PM, 31 December 2020)

Total Marks: 1

Linear Programming is the backbone of Operations Research models due to---------.

Select the correct option

- linearity of objective function and constraints
- its fantastic computational efficiency
- non-negativity of decision variables
- unrestricted signs of decision variables



MC180403106: SADDAM HUSSAIN

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MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 6 of 30 (Start time: 10:11:28 PM, 31 December 2020)

Total Marks: 1

In the Dynamic Order Quantity problem, the Demands of all succeeding months can be included in the demand of preceding one provided that ----------.

Select the correct option

- Item and Holding Costs are balanced
- Shortage and Setup Costs are balanced
- Setup and Holding Costs are balanced
- Shortage and Item Costs are balanced

10:12
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MC180403106: SADDAM HUSSAIN

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 7 of 30 (Start time: 10:12:11 PM, 31 December 2020)

Total Marks: 1

For ABC analysis, in the Random Sample of some items, the ranks to the items are assigned with respect to their ----------.

Select the correct option

- lowest annual unit cost
- largest annual unit cost
- lowest annual usage value
- largest annual usage value

26% 10:12 PM

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MC190400926: AMJAD ALI

Time Left 66 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 4 of 30 (Start time: 10:11:41 PM, 31 December 2020)

Total Marks: 1

Except the economic order quantity, the value of Q* will result in --------.

Select the correct option

- lower cost
- average cost
- higher cost
- zero cost

Click to Save Answer & Move to Next Question

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MC180403106: SADDAM HUSSAIN

Time Left 84 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 8 of 30 (Start time: 10:12:55 PM, 31 December 2020)

Total Marks: 1

Which of the following property assures the contribution of all variables in the Objective function and their requirements in constraints?

Select the correct option

- Deterministic
- Additivity
- Divisibility
- Proportionality

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MC180403106: SADDAM HUSSAIN

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MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

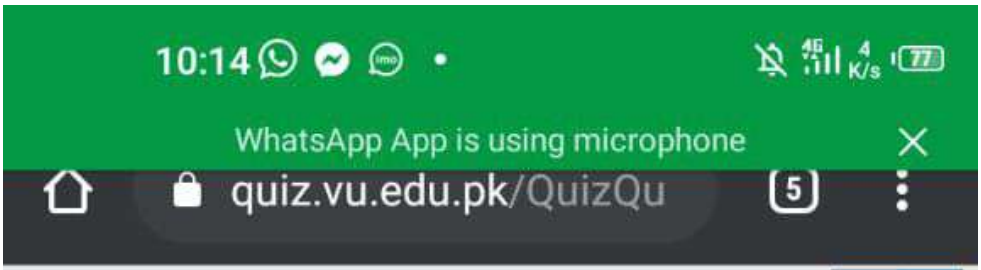
Question # 9 of 30 (Start time: 10:13:39 PM, 31 December 2020)

Total Marks: 1

While identifying the Critical Path of a network flow diagram, the Late Start and Late Finish phase confirms that project start time is_____.

Select the correct option

- 5th day
- zero
- arbitrary
- infinity



MC180403106: SADDAM HUSSAIN

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MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 10 of 30 (Start time: 10:14:07 PM, 31 December 2020)

Total Marks: 1

Which of the following cost of an item varies directly with the size of inventory as well as the time the item held in stock?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

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MC180403106: SADDAM HUSSAIN

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 11 of 30 (Start time: 10:14:32 PM, 31 December 2020)

Total Marks: 1

While identifying the Critical Path of a network flow diagram, the Late Start and Late Finish phase confirms that project start time is ----------.

Select the correct option

- serial
- zero
- infinity
- arbitrary

Save Answer & Move to Next Question

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MC190400926: AMJAD ALI

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MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

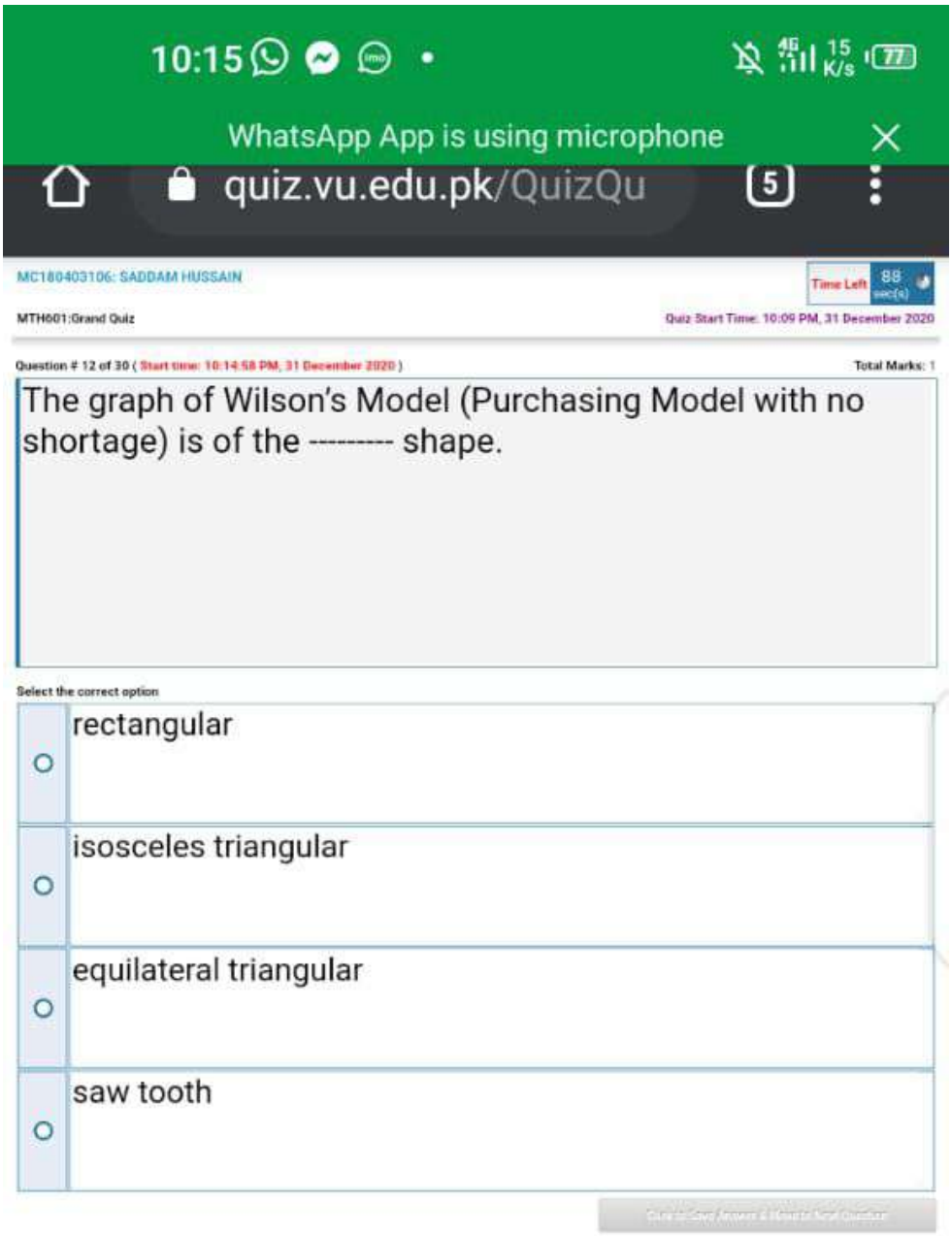
Question # 5 of 30 (Start time: 10:14:11 PM, 31 December 2020) Total Marks: 1

Which of the following category of items in ABC analysis needs special attention by the management?

Select the correct option

- A category
- B category
- C category
- All categories need equal and fair attention

Click to Save Answer & Move to Next Question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MC180403106: SADDAM HUSSAIN
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MTH601:Grand Quiz
Quiz Start Time: 10:09 PM, 31 December 2020
Question # 12 of 30 (Start time: 10:14:58 PM, 31 December 2020)
Total Marks: 1
The graph of Wilson's Model (Purchasing Model with no shortage) is of the --------- shape.
Select the correct option
rectangular
isosceles triangular
equilateral triangular
saw tooth

MC180403106: SADDAM HUSSAIN

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

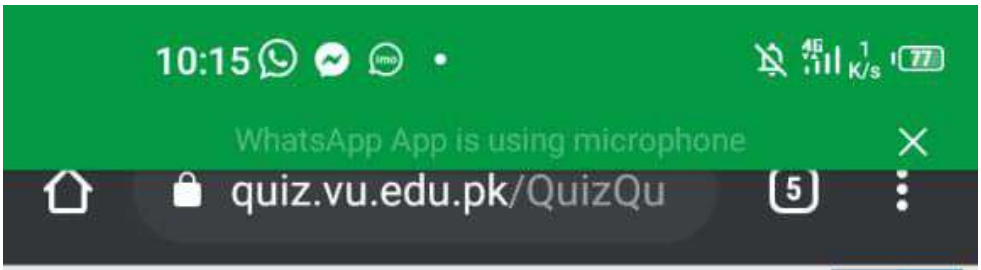
Question # 13 of 30 (Start time: 10:15:16 PM, 31 December 2020)

Total Marks: 1

One of the properties of Linear Programming Model is that -----------.

Select the correct option

- it would not have any non-positive decision variable
- the objective function should be linear
- the relationship between problem variables and constraints must be linear
- All other choices are equally important



MC180403106: SADDAM HUSSAIN

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MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 15 of 30 (Start time: 10:15:36 PM, 31 December 2020)

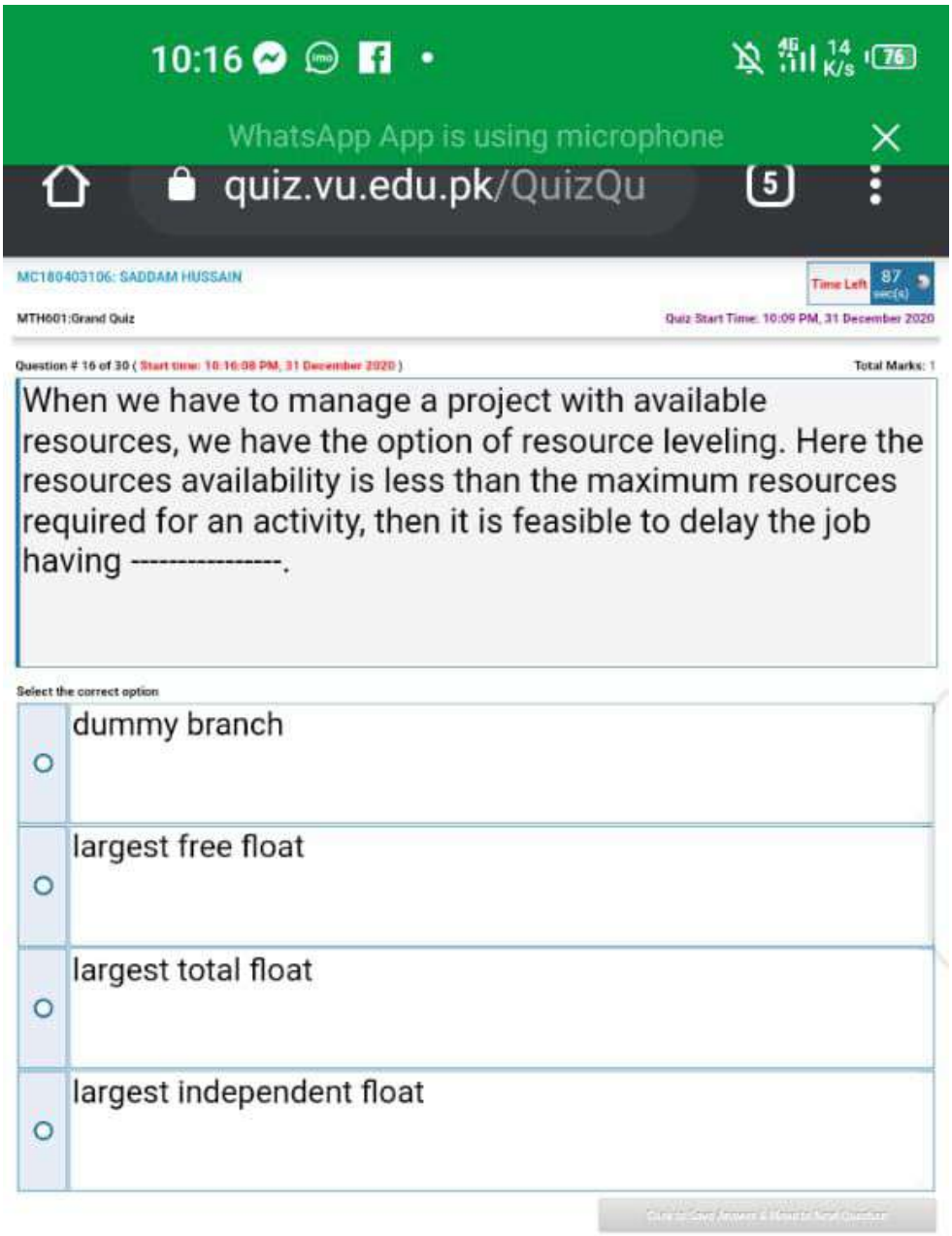
Total Marks: 1

Which of the following would be the objective of the cost per unit of producing certain cameras?

Select the correct option

- (●) Maximization
- ( ) Minimization
- ( ) Inflection
- ( ) Average

Click to Save Answer & Move to Next Question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MC180403106: SADDAM HUSSAIN
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MTH601:Grand Quiz
Quiz Start Time: 10:09 PM, 31 December 2020
Question # 16 of 30 (Start time: 10:16:08 PM, 31 December 2020)
Total Marks: 1
When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to delay the job having ----------.
Select the correct option
dummy branch
largest free float
largest total float
largest independent float

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MC190400926: AMJAD ALI

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MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 6 of 30 (Start time: 10:16:17 PM, 31 December 2020)

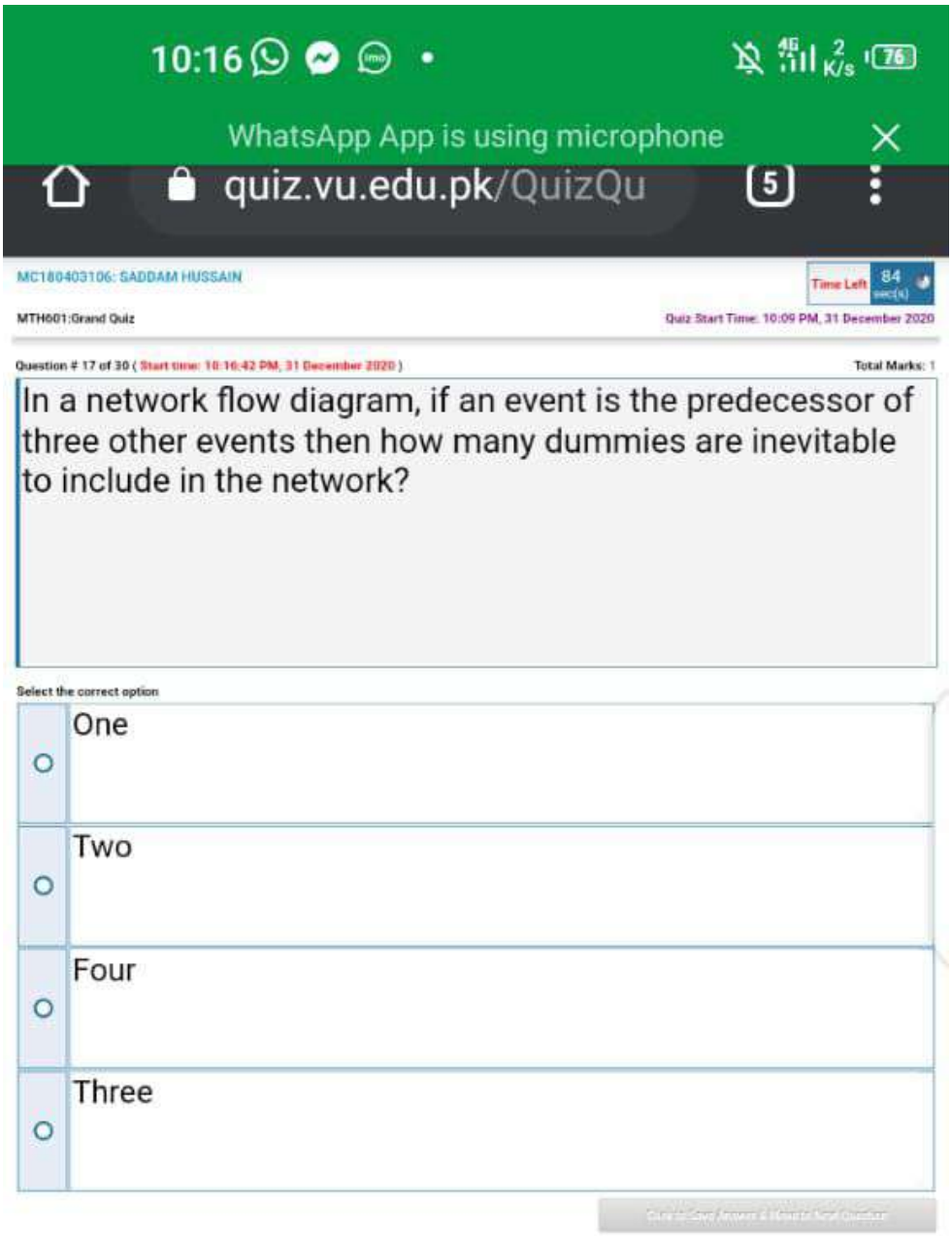
Total Marks: 1

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '8' and '9' days durations respectively, start earlier simultaneously on 2nd day, then the next activity containing 'n' as head event can't start until the entering activity_____is completed.

Select the correct option

- (l,n)
- (m,n)
- (m,l)
- (l,m)

Click to Save Answer & Move to Next Question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MC180403106: SADDAM HUSSAIN
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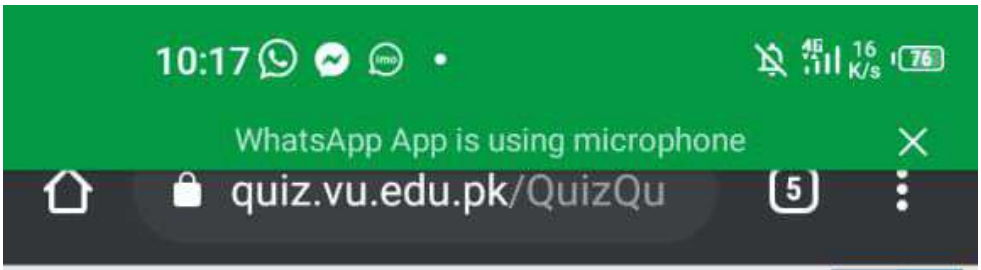
MTH601:Grand Quiz
Quiz Start Time: 10:09 PM, 31 December 2020
Question # 17 of 30 (Start time: 10:16:42 PM, 31 December 2020)
Total Marks: 1
In a network flow diagram, if an event is the predecessor of three other events then how many dummies are inevitable to include in the network?
Select the correct option
One
Two
Four
Three



MC180403106: SADDAM HUSSAIN

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 18 of 30 (Start time: 10:17:14 PM, 31 December 2020)

Total Marks: 1

In PERT, the possible variation in activity times can be measured from ---------- of the corresponding Beta Distribution.

Select the correct option

- Variance
- Mean
- Expected Time
- Standard Deviation

22% 10:17 PM

Quiz
https://quiz.vu.edu.pk

MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 7 of 30 (Start time: 10:17:20 PM, 31 December 2020)

Total Marks: 1

In a network flow diagram, two jobs (i,j) and (i,k) of '9'and '6' days duration leaves the node 'i' then which of the following will be Late start time for 'i', if it is provided that both (i,j) and (i,k) finish late on 12th and 8th day respectively?

Select the correct option

- 6th day
- 2nd day
- 3rd day
- 1st day

Click to Save Answer & Move to Next Question

10:17
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MC180403106: SADDAM HUSSAIN

Time Left 84 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 19 of 30 (Start time: 10:17:40 PM, 31 December 2020)

Total Marks: 1

In a network flow diagram, if two jobs 'a(i,j)' and 'b(i,k)' of '4' and '3' days duration, finish late simultaneously on 7th day, then on which of the following day the event 'i' will finish earlier?

Select the correct option

- 3rd
- 4th
- 1st
- 6th

MC180403106: SADDAM HUSSAIN

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 20 of 30 (Start time: 10:18:21 PM, 31 December 2020)

Total Marks: 1

In a Project Management, if the Critical activities of a network are delayed then ---------.

Select the correct option

- project finish time will have to extend
- project cost will increase
- more resources have to employed
- All above choices are equivalent

MTH601:Grand Quiz

Time Left 71 sec(s)

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 1 of 30 (Start time: 10:18:14 PM, 31 December 2020)

Total Marks: 1

If the annul demand of a product is 10000 items, and its set up and inventory costs are 200 and 100 respectively then its economic order quantity is ---------- provided that shortage is fulfilled INSTANTANEOUSLY.

Select the correct option

- 200 units
- 100 units
- 20000 units
- 40000 units

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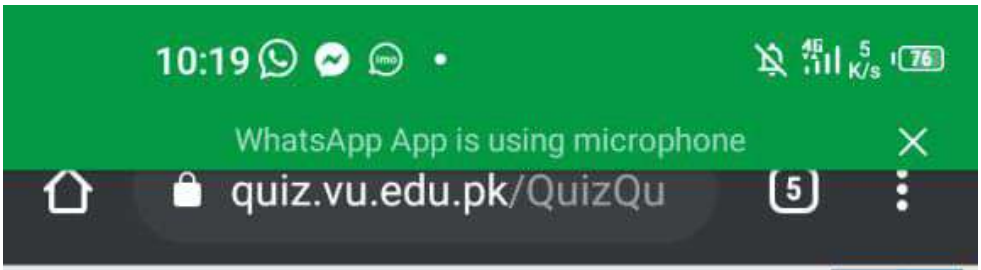


MC180403106: SADDAM HUSSAIN

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

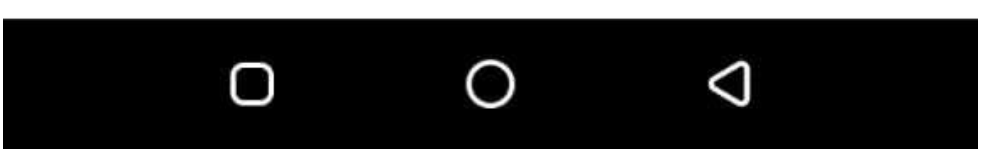
Question # 21 of 30 (Start time: 10:18:59 PM, 31 December 2020)

Total Marks: 1

For an industrialist, optimization means ----------.

Select the correct option

- maximizing its profit only
- minimizing its cost only
- both maximizing its profit and minimizing its cost
- change the business strategy and looking forward new customers



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MC180403106: SADDAM HUSSAIN

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 23 of 30 (Start time: 10:19:36 PM, 31 December 2020)

Total Marks: 1

The purpose of maintaining the inventory is to balance the inter-related ----------.

Select the correct option

- times
- costs
- demands
- items

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 Decembe

Question # 2 of 30 (Start time: 10:19:46 PM, 31 December 2020)

Total Ma

For a development project, if its standard deviation of project length = 8, expected and scheduled duration of the complete project are 50 and 45 days respectively, then which of the following will be the Standard Normal Variable of the project?

Select the correct option

- -1.7677
- -0.625
- 0.625
- 1.7677

Type here to search

10:20 PM
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WhatsApp 10:21 PM
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Question # 8 of 30 (Start time: 10:19:39 PM, 31 December 2020) Total Marks: 1

In PERT, the possible variation in activity times is measured from Standard Deviation which is ----------of the difference between Pessimistic and Optimistic times.

Select the correct option

- one sixth
- one fourth
- one third
- one fifth

Click to Save Answer & Move to Next Question

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MC180403106: SADDAM HUSSAIN

Time Left 44 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 27 of 30 (Start time: 10:20:28 PM, 31 December 2020)

Total Marks: 1

Which of the following Probabilistic time in PERT has the same analogical meaning of Deterministic time (time to complete any activity) in CPM?

Select the correct option

- Expected
- Optimistic
- Pessimistic
- Most Likely

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https://quiz.vu.edu.pk

MC190400926: AMJAD ALI

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 9 of 30 (Start time: 10:21:27 PM, 31 December 2020) Total Marks: 1

While solving a Linear Programming problem by graphical method, the non-negativity constraints will restrict the feasible region(solution space) in ---------- quadrant.

Select the correct option

- 1st
- 2nd
- 3rd
- 4th

Click to Save Answer & Move to Next Question

MTH601 Grand Quiz

Time Left

Quiz Start Time: 10:18 PM, 31 Decem

Question # 3 of 30 (Start time: 10:21:17 PM, 31 December 2020)

Total

In a development project, if an activity (q,r) of five days duration, starts late on 2nd day then which of the following will be its latest finish time?

Select the correct option

- 3rd day
- 4th day
- 7th day
- 10th day

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MC200201228: WAQAR AHMED

Time Left 90 sec(s)

MTH501 Grand Quiz

Quiz Start Time: 10:21 PM, 31 December 2020

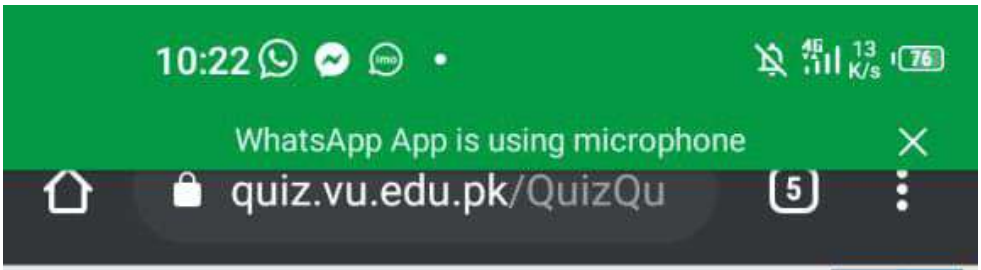
Question # 1 of 30 (Start time: 10:21:25 PM, 31 December 2020)

Total Marks: 1

While finding the Critical Path, in the phase of Early Start and Early Finish computations, we start from -------- event and proceed calculations from left to right up to the final event.

Select the correct option

- arbitrary
- first
- middle
- last



MC180403106: SADDAM HUSSAIN

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

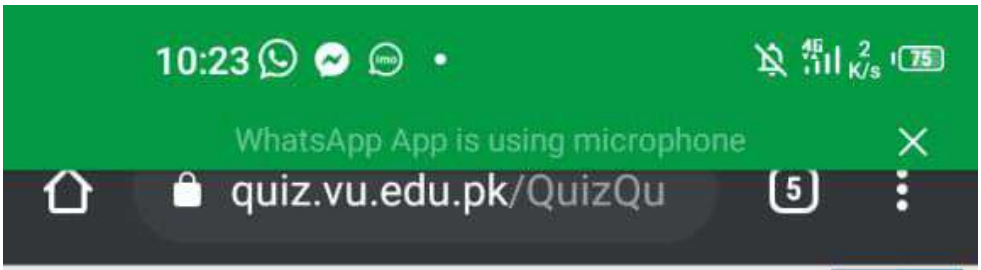
Question # 28 of 30 (Start time: 10:22:21 PM, 31 December 2020)

Total Marks: 1

Non-feasible solution---------- associated with a given linear programming problem.

Select the correct option

- satisfy all the constraints
- does not satisfy all the constraints
- satisfy least one of the constraint
- does not satisfy least one of the constraint



MC180403106: SADDAM HUSSAIN

Time Left 84 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 29 of 30 (Start time: 10:22:57 PM, 31 December 2020)

Total Marks: 1

The word 'Linear' in Linear programming problem describes ---------- between two or more variables.

Select the correct option

- that a Correlation exists
- a recurrence relation
- a direct proportionate relation
- an inverse proportionate relation

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 Decer

Question # 4 of 30 (Start time: 10:22:49 PM, 31 December 2020)

Total

If the cost function in terms of economic order quantity is C = Q^2-400Q, then the minimum cost of inventory per unit of time is -------- units.

Select the correct option

- 800
- 400
- 200
- 398

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MC200201228: WAQAR AHMED Time Left 78 sec(s)

MTH501 Grand Quiz Quiz Start Time: 10:21 PM, 31 December 2020

Question # 2 of 30 (Start time: 10:22:49 PM, 31 December 2020) Total Marks: 1

Solution region of the constraints: '2x+3y>12' and '2x+3y<12' will be the--------.

Select the correct option

- straight line:2x+3y=12
- empty set
- all first quadrant
- point(6,4)

Click to Save Answer & Move to Next Question

MC180403106: SADDAM HUSSAIN

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:09 PM, 31 December 2020

Question # 30 of 30 (Start time: 10:23:43 PM, 31 December 2020)

Total Marks: 1

In which of the following model, the Carrying Cost per unit of time is reduced in the ratio
(1-D/R):1 ?

Select the correct option

- Purchasing Model without Shortage
- Manufacturing Model without Shortage
- Purchasing Model with allowed Shortage
- Manufacturing Model with allowed Shortage

19% 10:24 PM

Quiz
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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 10 of 30 (Start time: 10:23:59 PM, 31 December 2020)

Total Marks: 1

Excess of inventories are undesirable due to the following reasons;

Select the correct option

- It locks up the capital of the enterprise.
- It includes the expenses of stores, equipment, personnel, insurance etc..
- It may not lead to high volume of output instead it might hamper the production.
- All above reasons are equivalent

Click to Save Answer & Move to Next Question

10:24 60

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MC200201228: WAQAR AHMED

Time Left 86 sec(s)

MTH501 Grand Quiz

Quiz Start Time: 10:21 PM, 31 December 2020

Question # 3 of 30 (Start time: 10:24:20 PM, 31 December 2020) Total Marks: 1

If the demand of an item is occurred in the period of stock out and it is satisfied at the beginning of the next cycle then such type of problem will discuss under which of the following?

Select the correct option

- Purchasing Model with shortages
- Purchasing Model with no shortages

Save & Get Answer & View Next Question

MTH601:Grand Quiz

Time Left 54 sec(s)

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 5 of 30 (Start time: 10:24:20 PM, 31 December 2020)

Total Marks: 1

Under which of the following condition the economic order quantity in case of Purchasing Model with Shortage can be transformed into Purchasing Model with no Shortage?

Select the correct option

- When shortage Cost tends to zero
- When Shortag Cost tends to infinity
- When Carrying Cost tends to zero
- When Carrying Cost tends to infinity

Type here to search

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MTH601:Grand Quiz

Time Left 54 sec(s)

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 5 of 30 (Start time: 10:24:20 PM, 31 December 2020)

Total Marks: 1

Under which of the following condition the economic order quantity in case of Purchasing Model with Shortage can be transformed into Purchasing Model with no Shortage?

Select the correct option

- When shortage Cost tends to zero
- When Shortag Cost tends to infinity
- When Carrying Cost tends to zero
- When Carrying Cost tends to infinity

Type here to search

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MTH601 Grand Quiz

Quiz Start Time: 10:18

Question # 6 of 30 (Start time: 10:25:51 PM, 31 December 2020)

Solution region of the constraints: '2x+3y>12' and '2x+3y<12' will be the------.

Select the correct option

- straight line 2x+3y=12
- empty set
- all first quadrant
- point(6,4)

Type here to search

MTH601 Grand Quiz

Quiz Start Time: 10:18

Question # 6 of 30 (Start time: 10:25:51 PM, 31 December 2020)

Solution region of the constraints: '2x+3y>12' and '2x+3y<12' will be the------.

Select the correct option

- straight line 2x+3y=12
- empty set
- all first quadrant
- point(6,4)

Type here to search

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MC200201228: WAQAR AHMED

Time Left 83 sec(s)

MTH501 Grand Quiz

Quiz Start Time: 10:21 PM, 31 December 2020

Question # 4 of 30 (Start time: 10:25:46 PM, 31 December 2020) Total Marks: 1

In a network flow diagram, two jobs (m,n) and (m,o) of '11'and '8' days duration leaves the node 'm' then which of the following will be Late start time for 'm', if it is provided that both (m,n) and (m,o) finish late on 14th and 10th day respectively?

Select the correct option

- 6th day
- 3rd day
- 2nd day
- 1st day

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Quiz
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MC190400926: AMJAD ALI

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 11 of 30 (Start time: 10:26:12 PM, 31 December 2020)

Total Marks: 1

If the economic order quantity and the annual demand of a product are 2500 and 5000 units respectively, then WITHOUT any shortage the number of orders are _____.

Select the correct option

- 02
- 125
- 500
- 30

Click to Save Answer & Move to Next Question

MTH601 Grand Quiz

Quiz Start Time: 10:18

Question # 6 of 30 (Start time: 10:25:51 PM, 31 December 2020)

Solution region of the constraints: '2x+3y>12' and '2x+3y<12' will be the------.

Select the correct option

- straight line 2x+3y=12
- empty set
- all first quadrant
- point(6,4)

Type here to search

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MC200201228: WAQAR AHMED

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MTH501 Grand Quiz

Quiz Start Time: 10:21 PM, 31 December 2020

Question # 5 of 30 (Start time: 10:27:30 PM, 31 December 2020) Total Marks: 1

In a development project, if an activity (m,n) of seven days duration, starts late on 3rd day then which of the following will be its latest finish time?

Select the correct option

- 3rd day
- 4nd day
- 10th day
- 21th day

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Quiz
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MC190400926: AMJAD ALI

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MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 12 of 30 (Start time: 10:27:47 PM, 31 December 2020)

Total Marks: 1

Which of the following cost for an item is independent of the quantity ordered or produced?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Click to Save Answer & Move to Next Question

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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 13 of 30 (Start time: 10:29:03 PM, 31 December 2020) Total Marks: 1

Which of the following would be the objective of the yield per minute in a chemical process which depends upon the temperature ‘x’ and pressure ‘y’?

Select the correct option

- Maximization
- Minimization
- Inflection
- Average

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz

Question # 8 of 30 (Start time: 10:28:54 PM, 31 December 2020)

In the ABC analysis, the items are classified into three categories with respect their----------.

Select the correct option

- cost value
- demand value
- turn over value
- mark up value

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MC190400926: AMJAD ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 14 of 30 (Start time: 10:29:53 PM, 31 December 2020)

Total Marks: 1

One of the properties of Linear Programming Model is that ----------.

Select the correct option

- it would not have any non-positive decision variable
- the objective function should be linear
- the relationship between problem variables and constraints must be linear
- All other choices are equally important

Click to Save Answer & Move to Next Question

MTH601 Grand Quiz

Quiz Start Time: 10:18 PM

Question # 9 of 30 (Start time: 10:30:26 PM, 31 December 2020)

Which of the following would be the objective of the yield per minute in a chemical process which depends upon the temperature 'x' and pressure 'y'?

Select the correct option

- Maximization
- Minimization
- Inflection
- Average

18% 10:32 PM

Quiz
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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 15 of 30 (Start time: 10:31:54 PM, 31 December 2020)

Total Marks: 1

Which of the following will be annual optimum cost of a product if the number of annual orders WITHOUT any shortage is 25 and the total cost of one cycle is 10,000?

Select the correct option

- Rs.400
- Rs. 10025
- Rs.9975
- Rs. 250000

Click to Save Answer & Move to Next Question

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MC200201228: WAQAR AHMED

Time Left 79 sec(s)

MTH501 Grand Quiz

Quiz Start Time: 10:21 PM, 31 December 2020

Question # 6 of 30 (Start time: 10:29:13 PM, 31 December 2020) Total Marks: 1

If weights are 2, 5 and 20 for optimistic, most likely and pessimistic time estimates, respectively, then select the most suitable answer.

Select the correct option

Reload Math Equations

- $t_e = \frac{t_o - 4t_m + t_p}{6}$
- $t_e = 9$
- $S_t = 4$
- $V_t = 9$

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MC200201228: WAQAR AHMED

Time Left 67 sec(s)

MTH501 Grand Quiz

Quiz Start Time: 10:21 PM, 31 December 2020

Question # 7 of 30 (Start time: 10:30:26 PM, 31 December 2020) Total Marks: 1

Which of the type of path will be critical in network flow diagram?

Select the correct option

- (●) Shortest path from starting node to end node
- ( ) Any path from starting node to end node
- ( ) Longest path from starting node to end node
- ( ) Any path from starting node to end node with minimum cost

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz

Question # 10 of 30 (Start time: 10:31:58 PM, 31 December 2020)

A Critical Path in a network flow diagram ------.

Select the correct option

- is unique
- may at most two
- depends on number of dummies
- may be multiple

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31

Question # 11 of 30 (Start time: 10:33:29 PM, 31 December 2020)

While finding the Critical Path, in the phase of Late Start and Late Finish, we start from last node and the computations are performed from ------ up to the beginning event.

Select the correct option

- right to left
- left to right
- top to bottom
- bottom to top

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MC200201228: WAQAR AHMED

Time Left 83 sec(s)

MTH501 Grand Quiz

Quiz Start Time: 10:31 PM, 31 December 2020

Question # 7 of 30 (Start time: 10:32:26 PM, 31 December 2020)

Total Marks: 1

Which of the type of path will be critical in network flow diagram?

Select the correct option

- (●) Shortest path from starting node to end node
- ( ) Any path from starting node to end node
- ( ) Longest path from starting node to end node
- ( ) Any path from starting node to end node with minimum cost

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31

Question # 11 of 30 (Start time: 10:33:29 PM, 31 December 2020)

While finding the Critical Path, in the phase of Late Start and Late Finish, we start from last node and the computations are performed from ------ up to the beginning event.

Select the correct option

- right to left
- left to right
- top to bottom
- bottom to top

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MC190400926: AMJAD ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 16 of 30 (Start time: 10:34:11 PM, 31 December 2020)

Total Marks: 1

While solving a Linear Programming problem, the first step is ---------.

Select the correct option

- to obtain alternate solutions
- to obtain basic feasible solutions
- formulation of the problem
- model building

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start

Question # 12 of 30 (Start time: 10:35:01 PM, 31 December 2020)

For a LP problem say:Max:z=x+y,under the constraints x,y>=0, the feasible region would be-------.

Select the correct option

- empty
- all xy-plane
- all the first quadrant
- point(0,0)

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Quiz
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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 17 of 30 (Start time: 10:35:15 PM, 31 December 2020)

Total Marks: 1

Which is the following reason is essential for an enterprise to have an inventory?

Select the correct option

- It helps in maintaining economy by absorbing some of the fluctuations when the demand of an item fluctuates or is seasonal.
- It helps in minimizing the loss due to deterioration, obsolescence, damage etc.
- Take advantage of price discounts by bulk purchasing.
- All above reasons are equivalent.

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start

Question # 12 of 30 (Start time: 10:35:01 PM, 31 December 2020)

For a LP problem say:Max:z=x+y,under the constraints x,y>=0, the feasible region would be-------.

Select the correct option

- empty
- all xy-plane
- all the first quadrant
- point(0,0)

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MC200201228: WAQAR AHMED

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MTH501 Grand Quiz

Quiz Start Time: 10:21 PM, 31 December 2020

Question # 11 of 30 (Start time: 10:35:13 PM, 31 December 2020)

Total Marks: 1

If a firm produces its own goods instead of purchasing the same from an outside source, then the cost of resetting the equipment for production may referred to ---------

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

MTH601:Grand Quiz

Quiz Start

Question # 12 of 30 (Start time: 10:35:01 PM, 31 December 2020)

For a LP problem say:Max:z=x+y,under the constraints x,y>=0, the feasible region would be-------.

Select the correct option

- empty
- all xy-plane
- all the first quadrant
- point(0,0)

MTH601:Grand Quiz

Quiz Start

Question # 12 of 30 (Start time: 10:35:01 PM, 31 December 2020)

For a LP problem say:Max:z=x+y,under the constraints x,y>=0, the feasible region would be-------.

Select the correct option

- empty
- all xy-plane
- all the first quadrant
- point(0,0)

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Quiz
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MC190400926: AMJAD ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 18 of 30 (Start time: 10:36:55 PM, 31 December 2020)

Total Marks: 1

In the graph of Purchasing Model with shortages, the area above the horizontal axis represents ----------.

Select the correct option

- set up cost of item
- shortage cost of item
- purchase cost of item
- carrying cost of item

Click to Save Answer & Move to Next Question

Grand Quiz

Quiz Start Time: 10:18

Question # 14 of 30 (Start time: 10:38:05 PM, 31 December 2020)

If the point (6,t) lies in the feasible region associated with constraint 2x+3y>=12, then minimum value of 't' would be-------

Select the correct option

- 0
- 1
- 3
- 4

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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 19 of 30 (Start time: 10:38:55 PM, 31 December 2020)

Total Marks: 1

Which of the following method follows iterative procedure to solve a Linear Programming problem?

Select the correct option

- Algebraic (Simplex)
- Graphical

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Question # 15 of 30 (Start time: 10:39:37 PM, 31 December 2020)

Modeling of a Linear Programming problem from a real life world is a/an ----- .

Select the correct option

- art
- science
- engineering
- technology

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MC190400926: AMJAD ALI

Time Left 84 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 20 of 30 (Start time: 10:40:08 PM, 31 December 2020)

Total Marks: 1

For any activity (i,j), if , a) Earliest start time of 'i' = Latest finish time of 'i', b) Earliest start time of 'j' = Latest finish time of 'j', c) difference of Earliest start times of events 'i' and "j = difference of Latest finish times of events 'i'and 'j' = time to complete the job, then the activity (i,j) is said to be -------.

Select the correct option

- dummy
- critical
- non-Critical
- non of the above

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM

Question # 16 of 30 (Start time: 10:40:54 PM, 31 December 2020)

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is --------proportional to the quantity ordered.

Select the correct option

- directly
- inversely

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM

Question # 16 of 30 (Start time: 10:40:54 PM, 31 December 2020)

If the ordered quantity of a commodity is 'q', then the ordering cost per unit is " C2/q " which is --------proportional to the quantity ordered.

Select the correct option

- directly
- inversely

Question # 1 of 30 (Start time: 10:41:15 PM, 31 December 2020) Total Marks: 1

Solution region of the constraints: '2x+3y>12' and '2x+3y<12' will be the---------.

Select the correct option

- straight line:2x+3y=12
- empty set
- all first quadrant
- point(6,4)

Click to Save Answer & Move to Next Question

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Quiz
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MC190400926: AMJAD ALI

Time Left 87 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 21 of 30 (Start time: 10:41:20 PM, 31 December 2020)

Total Marks: 1

In a Linear Programming model, any constraint contains at least ----- variable.

Select the correct option

- three
- two
- one
- arbitrary

Click to Save Answer & Move to Next Question

Question # 2 of 30 (Start time: 10:42:11 PM, 31 December 2020) Total Marks: 1

The parameters like demand and ordering quantity cost are quite -------- in the Probabilistic inventory models.

Select the correct option

- certain
- uncertain

Click to Save Answer & Move to Next Question

18% 10:43 PM

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MC190400926: AMJAD ALI

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 22 of 30 (Start time: 10:42:34 PM, 31 December 2020) Total Marks: 1

If weights are 2, 9 and 20 or optimistic, most likely and pessimistic time estimates, respectively, then select the most suitable answer.

Select the correct option

Reload Math Equations

- $t_e = \frac{t_o + 4t_m + t_p}{6}$
- $t_e = 9$
- $S_t = 4$
- $V_t = 9$

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10

Question # 17 of 30 (Start time: 10:42:25 PM, 31 December 2020)

If the point (t,6) lies in the feasible region associated with constraint:2x+3y<=12, then the maximum value of 't' would be-------.

Select the correct option

- 0
- -1
- -3
- -4

MTH601:Grand Quiz

Quiz Start Time: 10

Question # 17 of 30 (Start time: 10:42:25 PM, 31 December 2020)

If the point (t,6) lies in the feasible region associated with constraint:2x+3y<=12, then the maximum value of 't' would be-------.

Select the correct option

- 0
- -1
- -3
- -4

MTH601:Grand Quiz

Quiz Start Time: 10

Question # 17 of 30 (Start time: 10:42:25 PM, 31 December 2020)

If the point (t,6) lies in the feasible region associated with constraint:2x+3y<=12, then the maximum value of 't' would be-------.

Select the correct option

- 0
- -1
- -3
- -4

18% 10:44 PM

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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 23 of 30 (Start time: 10:43:36 PM, 31 December 2020)

Total Marks: 1

When we have to manage a project with available resources, we have the option of resource leveling. Here the resources availability is less than the maximum resources required for an activity, then it is feasible to divert the resources to ----------.

Select the correct option

- non-critical activities
- dummy activities
- head or tail activities
- critical activities

Click to Save Answer & Move to Next Question

Question # 3 of 30 (Start time: 10:43:40 PM, 31 December 2020) Total Marks: 1

In the PERT, the expected time is classified as a---------- variable.

Select the correct option

- complex valued
- fix valued (constant)
- random
- continuous

Click to Save Answer & Move to Next Question

Question # 4 of 30 (Start time: 10:44:26 PM, 31 December 2020) Total Marks: 1

In the Dynamic Order Quantity problem, the Demands of all succeeding months can be included in the demand of preceding one provided that ----------.

Select the correct option

- Item and Holding Costs are balanced
- Shortage and Setup Costs are balanced
- Setup and Holding Costs are balanced
- Shortage and Item Costs are balanced

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 Decem

Question # 18 of 30 (Start time: 10:43:56 PM, 31 December 2020)

Total

In a network flow diagram, if an event is the predecessor of three other events then how many dummies are inevitable to include in the network?

Select the correct option

- One
- Two
- Four
- Three

Type here to search

10:44 PM
12/31/2020

Quiz List | Quiz

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MC200202162: KHIZRA AFZAL

Time Left 85 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:43 PM, 31 December 2020

Question # 1 of 30 (Start time: 10:43:27 PM, 31 December 2020) **Total Marks:** 1

A linear programming problem can be converted into a Transportation problem by restricting the coefficients 'aij' to have the values----------.

Select the correct option

- Zero only
- One only
- either '0' or '1'
- arbitrarily

Click to Save Answer & Move to Next Question

Type here to search

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31/12/2020

17% 10:45 PM

Quiz
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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 24 of 30 (Start time: 10:44:39 PM, 31 December 2020)

Total Marks: 1

Which of the following property assumed that the coefficients of decision variables in a Linear Programming model are known with certainty?

Select the correct option

- Deterministic
- Additivity
- Proportionality
- Divisibility

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 Decem

Question # 18 of 30 (Start time: 10:43:56 PM, 31 December 2020)

Total

In a network flow diagram, if an event is the predecessor of three other events then how many dummies are inevitable to include in the network?

Select the correct option

- One
- Two
- Four
- Three

Type here to search

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MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 Decem

Question # 18 of 30 (Start time: 10:43:56 PM, 31 December 2020)

Total

In a network flow diagram, if an event is the predecessor of three other events then how many dummies are inevitable to include in the network?

Select the correct option

- One
- Two
- Four
- Three

Type here to search

10:44 PM
12/31/2020

Quiz Start Time: 10:18 P

Question # 19 of 30 (Start time: 10:45:28 PM, 31 December 2020)

In CPM each activity has one deterministic time while in PERT each activity has ----------probabilistic time/times.

Select the correct option

- three
- two
- also one
- no

WhatsApp 10:46 PM
2309 messages from 50 chats

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MC190400926: AMJAD ALI

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 25 of 30 (Start time: 10:45:55 PM, 31 December 2020)

Total Marks: 1

In a network flow diagram, if an event is the predecessor of three other events then how many dummies are inevitable to include in the network?

Select the correct option

- One
- Two
- Four
- Three

Click to Save Answer & Move to Next Question

Quiz Start Time: 10:18 P

Question # 19 of 30 (Start time: 10:45:28 PM, 31 December 2020)

In CPM each activity has one deterministic time while in PERT each activity has ----------probabilistic time/times.

Select the correct option

- three
- two
- also one
- no

Question # 5 of 30 (Start time: 10:46:12 PM, 31 December 2020) Total Marks: 1

Which of the following times are directly related to the activity cost that is by reducing the activity duration, the direct cost of the corresponding activity is increased?

Select the correct option

- CPM Times
- PERT Times

Click to Save Answer & Move to Next Question

Question # 6 of 30 (Start time: 10:46:57 PM, 31 December 2020)
Total Marks: 1

In the graph of the Purchasing Model with shortages, the area under the horizontal axis represents ----------.

Select the correct option

- carrying cost of item
- set up cost of item
- shortage cost of item
- purchase cost of item

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31

Question # 20 of 30 (Start time: 10:46:59 PM, 31 December 2020)

If weights are 1, 5 and 13 or optimistic, most likely and pessimistic time estimates, respectively, then select the most suitable answer.

Select the correct option

Reload Mat

- $t_e = \frac{t_o + 4t_m + t_p}{6}$
- $t_e = 4$
- $S_t = 2$
- $V_t = 5$

Question # 7 of 30 (Start time: 10:48:26 PM, 31 December 2020) Total Marks: 1

While solving a problem by any of Linear Programming method, an objective function is to be ----------.

Select the correct option

- maximized
- minimized
- optimized
- standardized

Click to Save Answer & Move to Next Question

17% 10:49 PM

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MC190400926: AMJAD ALI

Time Left 79 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 26 of 30 (Start time: 10:48:22 PM, 31 December 2020)

Total Marks: 1

The task which is executed by the usage of resources and time is called_____. Select correct option.

Select the correct option

- node
- event
- project
- activity

Click to Save Answer & Move to Next Question

TH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 21 of 30 (Start time: 10:48:30 PM, 31 December 2020)

Total Marks: 1

While estimating the time for a PERT activity, the estimator will give the three times estimates. When these three estimates are plotted on a graph, the probability distribution that we get is closely associated with ----------Distribution curve.

Select the correct option

- Poisson
- Chi
- Binomial
- Beta

10:48 PM

TH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 21 of 30 (Start time: 10:48:30 PM, 31 December 2020)

Total Marks: 1

While estimating the time for a PERT activity, the estimator will give the three times estimates. When these three estimates are plotted on a graph, the probability distribution that we get is closely associated with ----------Distribution curve.

Select the correct option

- Poisson
- Chi
- Binomial
- Beta

10:48 PM

TH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 21 of 30 (Start time: 10:48:30 PM, 31 December 2020)

Total Marks: 1

While estimating the time for a PERT activity, the estimator will give the three times estimates. When these three estimates are plotted on a graph, the probability distribution that we get is closely associated with ----------Distribution curve.

Select the correct option

- Poisson
- Chi
- Binomial
- Beta

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Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 27 of 30 (Start time: 10:49:13 PM, 31 December 2020)

Total Marks: 1

In the ABC analysis, the items are classified into three categories with respect their----------.

Select the correct option

- cost value
- demand value
- turn over value
- mark up value

Click to Save Answer & Move to Next Question

Question # 8 of 30 (Start time: 10:49:38 PM, 31 December 2020) Total Marks: 1

Which of the following relation is correct for the Standard Deviation of an activity times having optimistic, pessimistic and most likely values as 4, 8 and 6 days respectively?

Select the correct option

- 0.666
- 01
- 0.147
- 7.111

Click to Save Answer & Move to Next Question

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MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 28 of 30 (Start time: 10:49:49 PM, 31 December 2020)

Total Marks: 1

The parameters like demand and ordering quantity cost are quite -------- in the Probabilistic inventory models.

Select the correct option

- certain
- uncertain

Click to Save Answer & Move to Next Question

10:49

quiz.vu.edu.pk/QuizQuesti

BC180204824: ZAHERIA HABIB

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:37 PM, 31 December 2020

Question # 13 of 30 (Start time: 10:49:20 PM, 31 December 2020)

Total Marks: 1

The mathematical technique which is used to solve a wide class of problems such as allocating scarce resources among competitive activities is known as --------.

Select the correct option

- Classical model
- Mathematical Model
- Descriptive model
- Linear Programming model

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Question # 22 of 30 (Start time: 10:50:02 PM, 31 December 2020)

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

MTH601:Grand Quiz

Question # 22 of 30 (Start time: 10:50:02 PM, 31 December 2020)

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Question # 9 of 30 (Start time: 10:50:17 PM, 31 December 2020)
Total Marks: 1

Holding Cost is ---------- proportional to the quantity to be kept in stock and the time to which the inventory is maintained.

Select the correct option

- directly
- inversely

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Question # 22 of 30 (Start time: 10:50:02 PM, 31 December 2020)

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Question # 10 of 30 (Start time: 10:50:54 PM, 31 December 2020) Total Marks: 1

Which of the following cost of an item vary directly with its lack of quantity and delaying time ?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

Click to Save Answer & Move to Next Question

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MC190400926: AMJAD ALI

Time Left 88 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:05 PM, 31 December 2020

Question # 29 of 30 (Start time: 10:51:24 PM, 31 December 2020)

Total Marks: 1

Which of the following is the major objective of the ABC analysis of inventory?

Select the correct option

- To fulfill the demand of items
- To replace the required items in proper time
- To control the inventories
- To classify the items in different categories

Click to Save Answer & Move to Next Question

Question # 11 of 30 (Start time: 10:51:44 PM, 31 December 2020) Total Marks: 1

A single cycle consists of two phases in case of "Manufacturing without Shortage ", then which of the following is the first phase?

Select the correct option

- Withdrawl of items from inventory at constant rate in per unit of time
- The inventory is assembled at constant rate per unit of time

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2

Question # 23 of 30 (Start time: 10:51:33 PM, 31 December 2020)

Total Marks

Which of the following would be the objective of the efficiency of a certain air conditioning system depending upon temperature 'x', pressure 'y', moisture 'w' and cross-sectional area of outlet 'u'?

Select the correct option

- Minimization
- Inflection
- Average
- Maximization

Type here to search

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MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2

Question # 23 of 30 (Start time: 10:51:33 PM, 31 December 2020)

Total Marks

Which of the following would be the objective of the efficiency of a certain air conditioning system depending upon temperature 'x', pressure 'y', moisture 'w' and cross-sectional area of outlet 'u'?

Select the correct option

- Minimization
- Inflection
- Average
- Maximization

Type here to search

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12/31/2020

Question # 12 of 30 (Start time: 10:52:24 PM, 31 December 2020) Total Marks: 1

While finding the Critical Path, in the phase of Early Start and Early Finish computations, we start from first event and proceed calculations from left to right up to the ---------- event.

Select the correct option

- starting
- middle
- last
- arbitrary

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2

Question # 23 of 30 (Start time: 10:51:33 PM, 31 December 2020)

Total Marks

Which of the following would be the objective of the efficiency of a certain air conditioning system depending upon temperature 'x', pressure 'y', moisture 'w' and cross-sectional area of outlet 'u'?

Select the correct option

- Minimization
- Inflection
- Average
- Maximization

Type here to search

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Question # 30 of 30 (Start time: 10:52:55 PM, 31 December 2020) Total Marks: 1

Which of the following would be the objective of the efficiency of a certain air conditioning system depending upon temperature ‘x’, pressure ‘y’, moisture ‘w’ and cross-sectional area of outlet ‘u’?

Select the correct option

- Minimization
- Inflection
- Average
- Maximization

Click to Save Answer & Move to Next Question

Quiz Start Time: 10:

Question # 24 of 30 (Start time: 10:53:05 PM, 31 December 2020)

Except the economic order quantity, the value of Q* will result in ------.

Select the correct option

- lower cost
- average cost
- higher cost
- zero cost

Quiz Start Time: 10:

Question # 24 of 30 (Start time: 10:53:05 PM, 31 December 2020)

Except the economic order quantity, the value of Q* will result in ------.

Select the correct option

- lower cost
- average cost
- higher cost
- zero cost

Question # 13 of 30 (Start time: 10:53:43 PM, 31 December 2020) Total Marks: 1

Which of the following correspond to the economic or physical quantities which are under the control of operations researcher while modeling a real life situation into a linear program?

Select the correct option

- Objective function
- Constraints
- Decision variables
- Linear function

Click to Save Answer & Move to Next Question

Quiz Start Time: 10:

Question # 24 of 30 (Start time: 10:53:05 PM, 31 December 2020)

Except the economic order quantity, the value of Q* will result in ------.

Select the correct option

- lower cost
- average cost
- higher cost
- zero cost

Question # 14 of 30 (Start time: 10:54:40 PM, 31 December 2020) Total Marks: 1

In the Graphical method, we determine -------- -------.

Select the correct option

- feasible solution from among all optimal solutions.
- optimal solution from among all feasible solutions.
- basic variables from among all non-basic variables.
- non-basic variables from among all basic variables.

Click to Save Answer & Move to Next Question

Question # 25 of 30 (Start time: 10:54:36 PM, 31 December 2020)

If we increase the inventory, then the shortage cost will decrease and the carrying cost will ----------.

Select the correct option

- remain constant
- increase
- also decrease
- uncertain

Question # 25 of 30 (Start time: 10:54:36 PM, 31 December 2020)

If we increase the inventory, then the shortage cost will decrease and the carrying cost will ----------.

Select the correct option

- remain constant
- increase
- also decrease
- uncertain

Question # 15 of 30 (Start time: 10:55:53 PM, 31 December 2020)
Total Marks: 1

For ABC analysis, in the Random Sample of some items, the ranks to the items are assigned with respect to their ----------.

Select the correct option

- lowest annual unit cost
- largest annual unit cost
- lowest annual usage value
- largest annual usage value

Click to Save Answer & Move to Next Question

MTH601 Grand Quiz

Quiz Start Time: 10:18 PM

Question # 26 of 30 (Start time: 10:56:07 PM, 31 December 2020)

Non-feasible solution---------- associated with a given linear programming problem.

Select the correct option

- satisfy all the constraints
- does not satisfy all the constraints
- satisfy least one of the constraint
- does not satisfy least one of the constraint

Type here to search

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MTH601 Grand Quiz

Quiz Start Time: 10:18 PM

Question # 26 of 30 (Start time: 10:56:07 PM, 31 December 2020)

Non-feasible solution---------- associated with a given linear programming problem.

Select the correct option

- satisfy all the constraints
- does not satisfy all the constraints
- satisfy least one of the constraint
- does not satisfy least one of the constraint

Type here to search

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MTH601 Grand Quiz

Quiz Start Time: 10:18 PM

Question # 26 of 30 (Start time: 10:56:07 PM, 31 December 2020)

Non-feasible solution---------- associated with a given linear programming problem.

Select the correct option

- satisfy all the constraints
- does not satisfy all the constraints
- satisfy least one of the constraint
- does not satisfy least one of the constraint

Type here to search

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Question # 16 of 30 (Start time: 10:57:21 PM, 31 December 2020) Total Marks: 1

In a network flow diagram, if an event is the predecessor of three other events then how many dummies are inevitable to include in the network?

Select the correct option

- One
- Two
- Four
- Three

Click to Save Answer & Move to Next Question

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MC200201045: BILAL DIN

Time Left 75 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 27 of 30 (Start time: 10:57:38 PM, 31 December 2020) Total Marks: 1

While applying Simplex method to a LP of minimization type, we proceed stepwise from one basic feasible solution to another in such a way that the objective function always -------- its value.

Select the correct option

- increases
- decreases
- does not change
- oscillate

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MC200201045: BILAL DIN

Time Left 75 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 27 of 30 (Start time: 10:57:38 PM, 31 December 2020) Total Marks: 1

While applying Simplex method to a LP of minimization type, we proceed stepwise from one basic feasible solution to another in such a way that the objective function always -------- its value.

Select the correct option

- increases
- decreases
- does not change
- oscillate

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MC200201045: BILAL DIN

Time Left 75 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 27 of 30 (Start time: 10:57:38 PM, 31 December 2020) Total Marks: 1

While applying Simplex method to a LP of minimization type, we proceed stepwise from one basic feasible solution to another in such a way that the objective function always -------- its value.

Select the correct option

- increases
- decreases
- does not change
- oscillate

Question # 17 of 30 (Start time: 10:58:40 PM, 31 December 2020) Total Marks: 1

Time by which an activity can expand without affecting the overall duration of the project is _____ .

Select the correct option

Reload Math Equations

- T=$J_L - i_E$+D
- T=$J_L - i_E$−D
- T=$J_L + i_E$+D
- T=$-J_L - i_E$+D

Click to Save Answer & Move to Next Question

10:59

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MC190400160: ZAHID MUKHTAR

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 1 of 30 (Start time: 10:59:14 PM, 31 December 2020)

Total Marks: 1

For ABC analysis, in the Random Sample of some items, the ranks to the items are assigned with respect to their ----------.

Select the correct option

- lowest annual unit cost
- largest annual unit cost
- lowest annual usage value
- largest annual usage value

Question # 28 of 30 (Start time: 10:59:10 PM, 31 December 2020)

The task which is executed by the usage of resources and time is called_____. Select correct option.

Select the correct option

- node
- event
- project
- activity

Type here to search

10:59 PM
12/31/2020

Question # 18 of 30 (Start time: 10:59:21 PM, 31 December 2020) Total Marks: 1

While finding the Critical Path, in the phase of Early Start and Early Finish computations, we start from -------- event and proceed calculations from left to right up to the final event.

Select the correct option

- arbitrary
- first
- middle
- last

Click to Save Answer & Move to Next Question

Question # 28 of 30 (Start time: 10:59:10 PM, 31 December 2020)

The task which is executed by the usage of resources and time is called_____. Select correct option.

Select the correct option

- node
- event
- project
- activity

Type here to search

10:59 PM
12/31/2020

Question # 28 of 30 (Start time: 10:59:10 PM, 31 December 2020)

The task which is executed by the usage of resources and time is called_____. Select correct option.

Select the correct option

- node
- event
- project
- activity

Type here to search

10:59 PM
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Question # 28 of 30 (Start time: 10:59:10 PM, 31 December 2020)

The task which is executed by the usage of resources and time is called_____. Select correct option.

Select the correct option

- node
- event
- project
- activity

Type here to search

10:59 PM
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Question # 19 of 30 (Start time: 11:00:00 PM, 31 December 2020) Total Marks: 1

In the graph of Purchasing Model with shortages, the area above the horizontal axis represents ----------.

Select the correct option

- set up cost of item
- shortage cost of item
- purchase cost of item
- carrying cost of item

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Question # 29 of 30 (Start time: 11:00:41 PM, 31 December 2020)

The solution set of the constraint 2x+3y>12, will lie in ------- bisected by the line 2x+3y=12.

Select the correct option

- origin side of half plane
- opposite to the origin side in half plane
- both half planes
- first quadrant only

Question # 20 of 30 (Start time: 11:00:44 PM, 31 December 2020) Total Marks: 1

In a network flow diagram, if two jobs 'a(l,n)' and 'b(m,n)' of '8' and '9' days durations respectively, start earlier simultaneously on 2nd day, then the next activity containing 'n' as head event can't start until the entering activity_____is completed.

Select the correct option

- (l,n)
- (m,n)
- (m,l)
- (l,m)

Click to Save Answer & Move to Next Question

MTH601:Grand Quiz

Question # 29 of 30 (Start time: 11:00:41 PM, 31 December 2020)

The solution set of the constraint 2x+3y>12, will lie in ------- bisected by the line 2x+3y=12.

Select the correct option

- origin side of half plane
- opposite to the origin side in half plane
- both half planes
- first quadrant only

11:01

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MC190400160: ZAHID MUKHTAR

Time Left 90 sec(s)

MTH001:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 2 of 30 (Start time: 11:01:25 PM, 31 December 2020)

Total Marks: 1

Which of the following cost of an item varies directly with the size of inventory as well as the time the item held in stock?

Select the correct option

- Item cost
- Set-up cost
- Inventory holding cost
- Shortage Cost

MTH601:Grand Quiz

Question # 29 of 30 (Start time: 11:00:41 PM, 31 December 2020)

The solution set of the constraint 2x+3y>12, will lie in ------- bisected by the line 2x+3y=12.

Select the correct option

- origin side of half plane
- opposite to the origin side in half plane
- both half planes
- first quadrant only

Question # 21 of 30 (Start time: 11:01:46 PM, 31 December 2020) Total Marks: 1

In a Linear Programming model, any constraint contains at least ----- variable.

Select the correct option

- three
- two
- one
- arbitrary

Click to Save Answer & Move to Next Question

MC200201045: BILAL DIN

Time Left 69 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 30 of 30 (Start time: 11:02:12 PM, 31 December 2020)

Total Marks: 1

In a network flow diagram, two jobs (i,j) and (i,k) of '9'and '6' days duration leaves the node 'i' then which of the following will be Late start time for 'i', if it is provided that both (i,j) and (i,k) finish late on 12th and 8th day respectively?

Select the correct option

- 6th day
- 2nd day
- 3rd day
- 1st day

Question # 22 of 30 (Start time: 11:02:25 PM, 31 December 2020) Total Marks: 1

In which of the following model, the Carrying Cost per unit of time is reduced in the ratio
(1-D/R):1 ?

Select the correct option

- Purchasing Model without Shortage
- Manufacturing Model without Shortage
- Purchasing Model with allowed Shortage
- Manufacturing Model with allowed Shortage

Click to Save Answer & Move to Next Question

MC200201045: BILAL DIN

Time Left 69 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:18 PM, 31 December 2020

Question # 30 of 30 (Start time: 11:02:12 PM, 31 December 2020)

Total Marks: 1

In a network flow diagram, two jobs (i,j) and (i,k) of '9'and '6' days duration leaves the node 'i' then which of the following will be Late start time for 'i', if it is provided that both (i,j) and (i,k) finish late on 12th and 8th day respectively?

Select the correct option

- 6th day
- 2nd day
- 3rd day
- 1st day

Question # 23 of 30 (Start time: 11:03:15 PM, 31 December 2020) Total Marks: 1

Which of the following property ensures that the decision variables can be divided into any fractional levels so that the rational(fractional) values for the decision variables are permitted?

Select the correct option

- Deterministic
- Additivity
- Proportionality
- Divisibility

Click to Save Answer & Move to Next Question

11:03

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MC190400160: ZAHID MUKHTAR

Time Left 89 sec(s)

MTH001:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 3 of 30 (Start time: 11:03:25 PM, 31 December 2020)

Total Marks: 1

In a development project, if an activity (m,n) of seven days duration, starts late on 5rd day then which of the following will be its latest finish time?

Select the correct option

- 5th day
- 7th day
- 12th day
- 5th or 7th day

Click to Save Answer & Move to Next Question

Question # 24 of 30 (Start time: 11:04:03 PM, 31 December 2020) Total Marks: 1

Which of the following method follows iterative procedure to solve a Linear Programming problem?

Select the correct option

- Algebraic (Simplex)
- Graphical

Click to Save Answer & Move to Next Question

11:04

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MC190400160: ZAHID MUKHTAR

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 4 of 30 (Start time: 11:04:41 PM, 31 December 2020)

Total Marks: 1

For a development project, if its Standard Normal variable = 1.38, expected and the scheduled durations of the project are 100 and 110 days respectively, then the Standard Deviation in the project length is ----------.

Select the correct option

- 7.246
- -7.246
- -52.50
- 52.50

Click to Save Answer & Move to Next Question

Question # 25 of 30 (Start time: 11:05:16 PM, 31 December 2020)
Total Marks: 1

In a network flow diagram, two jobs (i,j) and (i,k) of '9'and '6' days duration leaves the node 'i' then which of the following will be Late start time for 'i', if it is provided that both (i,j) and (i,k) finish late on 12th and 8th day respectively?

Select the correct option

- 6th day
- 2nd day
- 3rd day
- 1st day

Click to Save Answer & Move to Next Question

11:05

17.4 K/s

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3

MC190400160: ZAHID MUKHTAR

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 5 of 30 (Start time: 11:05:58 PM, 31 December 2020)

Total Marks: 1

The parameters like demand and ordering quantity cost are quite --------- in the Probabilistic inventory models.

Select the correct option

- certain
- uncertain

Check Your Answer & Move to Next Question

Question # 26 of 30 (Start time: 11:06:45 PM, 31 December 2020) Total Marks: 1

While finding the critical path in a network flow diagram, if an activity (i,j) starts and finish late on 3rd and 9th day respectively, then it will be completed in ----------days.

Select the correct option

- Three days
- Twelve days
- Fifteen days
- Six days

Click to Save Answer & Move to Next Question

11:07

8.3 K/s

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3

MC190400160: ZAHID MUKHTAR

Time Left 86 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 6 of 30 (Start time: 11:07:26 PM, 31 December 2020)

Total Marks: 1

While solving a Linear programming problem, we find ----------number of basic feasible solution.

Select the correct option

- even
- odd
- infinite
- only finite

Click to Save Answer & Move to Next Question

11:08

17.3 K/s

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3

MC190400160: ZAHID MUKHTAR

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 7 of 30 (Start time: 11:08:13 PM, 31 December 2020)

Total Marks: 1

In the Purchasing Model with Shortages the Cost Function 'C(S,Q)' can also be expressed as function of ----------.

Select the correct option

- (t,D)
- (Q,D)
- (S,t)
- (S,D)

Click to Save Answer & Move to Next Question

Question # 27 of 30 (Start time: 11:08:15 PM, 31 December 2020) Total Marks: 1

In resource leveling when two or more jobs compete for same resources, first try to allocate an activity, which is of --------- duration and next to the activity which having next --------- duration.

Select the correct option

- most likely, pessimistic
- minimum , maximum
- maximum, minimum
- optimistic, most likely

Click to Save Answer & Move to Next Question

11:09

268 B/s

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3

MC190400160: ZAHID MUKHTAR

Time Left 89 sec(s)

MTH601:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 8 of 30 (Start time: 11:09:48 PM, 31 December 2020)

Total Marks: 1

The straight line associated with the constraint:2x+3y <12, will meet y-axis at a distance of-------- units from origin.

Select the correct option

- 2
- 3
- 4
- 6

Click to Save Answer & Move to Next Question

Question # 28 of 30 (Start time: 11:10:04 PM, 31 December 2020)
Total Marks: 1

By which of the following method, any complex linear programming problem can be handled?

Select the correct option

- Graphical method
- Simplex method
- Dual method
- Degenerate method

Click to Save Answer & Move to Next Question

Question # 29 of 30 (Start time: 11:11:29 PM, 31 December 2020) Total Marks: 1

Which of the fiollowing is an intermediate step to model a linear programming problem?

Select the correct option

- Identify the objective function
- Identify the non-negative restrictions and constraints
- Identify the unknown decision variables
- None of these

Click to Save Answer & Move to Next Question

11:11

308 B/s

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3

MC190400160: ZAHID MUKHTAR

Time Left 90 sec(s)

MTH001:Grand Quiz

Quiz Start Time: 10:59 PM, 31 December 2020

Question # 9 of 30 (Start time: 11:11:38 PM, 31 December 2020) Total Marks: 1

Time by which an activity can expand with out affecting subsequent activity is _____ .

Select the correct option

Reload Math Equations

- $F=f_E - i_E - D$
- $F=f_E - i_E+D$
- $F=f_E - i_E - D$
- $F=-f_E + i_E+D$

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Question # 30 of 30 (Start time: 11:12:21 PM, 31 December 2020) Total Marks: 1

Objective of inventory control is to balance between the advantage of having inventories and the --------- of carrying them to arrive at an optimal level of inventories to minimize the total inventory cost.

Select the correct option

- demand
- profit
- cost
- transportation

Click to Save Answer & Move to Next Question